navify®

Roche

navify POC Operations

User Guide
Publication version
Software version

Publication information

Publication version	Software version	Revision date	Change description
1.0	1.0	July 2018	First version

Publication version	Software version	Revision date	Change description
6.0	3.0	March 2024	Data encryption and personal data disclosure security information updated. Disposal and data retention security messages added. The product name updated to **navify**® POC Operations. Table setting options and advanced and quick search filter options added. Export functionality on the device management screen added. New section describing how to synchronize multiple devices created. New section describing how to activate and deactivate multiple material lots at once created. New section describing how to activate and deactivate multiple operators at once created. New section describing how to export operators created. New section describing how to add comments to patient results created. Devices deactivation for license usage for another device added. Function for accepting or rejecting linearity results added. Function for using alphanumeric values to define ranges for tests added.

Publication version	Software version	Revision date	Change description
			Function of only adding operators if they are part of the same site as the assignee added. User information in the import file for RFID tag, network ID and certificate stages updated.

Publication version	Software version	Revision date	Change description
6.1	3.1	December 2024	Deep links added. The filter criteria updated. Viewing QC result details updated with the Levey-Jennings chart added. Accepting or rejecting a QC result updated. Material lot expiry warnings added. Deactivating a lot automatically added. Viewing comments on patient results added. Selecting multiple patient results added. Adding a comment to a patient result updated. Reviewing patient result details updated. Accepting unreviewed patient results from the result list updated. Adding a patient updated. List of reports updated. Retention period for sent and unsent patient results added. Reference range updated.

Publication version	Software version	Revision date	Change description
6.2	3.2	June 2025	Result-reprocessing updated. Importing devices updated. Connecting a **cobas**® **pulse** system to a Wi-Fi network added. Observation time frame updated. In-stage checks updated. Device import added.
6.3	3.3 or higher	October 2025	Connecting a **cobas**® **pulse** system to a Wi-Fi network updated. User Activity Report added. System alarm assignment added. Instrument Alarm Definition updated. Importing devices added. List of reports updated. Observation time frame added.
7.0	3.4 or higher	February 2026	What is new in publication version 7.0 (p. 10)

Revision history

Edition notice

This publication is intended for users of the **navify**® POC Operations application.

Every effort has been made to ensure that all the information contained in this publication is correct at the time of publishing. However, the manufacturer of this product may need to update the publication information as output of product surveillance activities, leading to a new version of this publication.

Where to find information

The ***User Guide*** contains all information about the product, including the following:

- Routine operation
- Safety
- Troubleshooting information
- Software reference
- Configuration information
- Background information

The ***Host Interface Manual*** provides references for implementing connections from external laboratory systems (LIS) to the product.

General attention

To avoid incorrect results, ensure that you are familiar with the instructions and safety information.

- Pay particular attention to all safety notices.
- Always follow the instructions in this publication.
- Do not use the software in a way that is not described in this publication.
- Store all publications in a safe and easily retrievable place.

Software updates are done by Roche Service representatives.

Training

Do not carry out operation tasks or maintenance actions unless you have received training from Roche Diagnostics. Leave tasks that are not described in the user documentation to trained Roche Service representatives.

Images

The screenshots in this publication have been added exclusively for illustration purposes. Configurable and variable data in screenshots, such as tests, results, or path names visible therein must not be used for laboratory purposes.

Warranty

Any customer modification to the system renders the warranty or service agreement null and void.

For conditions of warranty, contact your Roche sales representative or refer to your warranty contract partner.

Copyright

License information

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Open Source and Commercial Software

Portions of the **navify**® POC Operations application might include one or more Open Source or Commercial Software programs. For copyright and other notices and licensing information regarding such Software programs included with **navify**®POC Operations please see the Licenses folder in **navify**® POC Operations installation.

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Feedback

Every effort has been made to ensure that this publication fulfills the intended use. All feedback on any aspect of this publication is welcome and is considered during updates. Contact your Roche representative, should you have any such feedback.

Languages

The application is displayed in the language defined for each user ID in the **navify**® POC Operations application.

Disclaimer

This publication may contain information and references to products that may not be available in your country or sold by Roche Diagnostics. This publication makes no claims regarding the use or performance of those products. Selected third-party products are available for use with the **navify**® POC Operations application.

The complete list of Roche Diagnostics and third-party products available for use with the **navify**® POC Operations application in the U.S. can be found at `http://usdiagnostics.roche.com`.

Other Roche Diagnostics or third-party products listed within the software or manuals which are not listed at `http://usdiagnostics.roche.com` are not available for use with the **navify**® POC Operations application in the U.S.

Cybersecurity

It is important to note that **navify**® POC Operations is not intended to be used for patient diagnosis or treatment decisions. Such decisions should be made on the basis of the results displayed on the POC instrument. Users of the **navify**® POC Operations system should pay particular attention to cybersecurity. In particular, ensure that an attacker is not able to modify results in **navify**® POC Operations nor on the way to any downstream systems that are being used in patient diagnosis or treatment decisions (e.g. a LIS or EHR).

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Preface

Intended use

navify® POC Operations is a data management software system providing governance and connectivity to Point of Care devices centralizing the management of professional Point of Care testing service.

navify® POC Operations supports workflow configuration of connected Point of Care devices, management of device related data, operators, test orders and patient demographic data including results.

These data are automatically received, stored and evaluated based on user-configurable criteria and sent to the customers Health Information Systems.

The professional personnel can review and audit these data and perform other data management functions.

This product is not intended for the diagnosis, screening, monitoring or treatment of patients.

Symbols and abbreviations

Product names

Except where the context clearly indicated otherwise, the following product names and descriptors are used.

Product name	Descriptor
Advanced Operator Management	module
navify® POC Operations	application

Product names

Symbols used in the publication

Symbol	Explanation
	Glossary: Alphabetical list of terms that occur in a document and their definitions.
	Index: Organized list of information in a document.
	Main Table of Content: List of chapters and sections organized in the order in which they appear.

Symbol	Explanation
	Cross-reference to another topic
	Table, used in table titles and cross-references to tables
√xy	Equation, used in cross-references to equations
</>	Code example, used in code titles and cross-references to codes
	Tip, used for extra information on correct use or for useful hints
	Extra information within a task
→	Result of an action within a task
	Frequency of a task
	Duration of a task
	Materials that are required for a task
	User profile
	Prerequisites of a task
	Figure. Used in figure titles and cross-references to figures.

Symbols used in the publication

Symbols used on the product

Symbol	Explanation
GTIN	Global Trade Item Number
SN	Serial number
	Date of manufacture

Symbol	Explanation
	Manufacturer
Rx ONLY	For USA: Caution: Federal law restricts this device to sale by or on the order of a physician.
UDI	Unique device identifier

Symbols used on the product

Abbreviations

The following abbreviations are used.

Abbreviation	Definition
ANSI	American National Standards Institute
API	Application programming interface
LDAP	Lightweight Directory Access Protocol
LIS	Laboratory information system
LMS	Learning management system

Abbreviations

What is new in publication version 7.0

Safety information	Additional information about the ciphers described in the PSP Use of Cryptography Standard was added.
Automated user Import log	Additional information on the Automatic User Import functionality on which fields can be deleted.
About the intended users	The table of the user group was updated.
About other reports	Report field descriptions was added for the control lot statistics.
Retention period for sent and unsent patient results	Additional information on how the retention period is calculated was added.
Comment assignment	The length limitation for the **Marked as erroneous by operator (comment)**alarm was updated.
Use secure SMTP	The default value was corrected.
Connecting a cobas® pulse system to a Wi-Fi network	The procedure for connecting a **cobas ® pulse** system to a Wi-Fi network by scanning a QR code was updated.
Result validation	Additional information was added to ensure clarity in multi-timezone environments.
Advanced operator management	**Advanced operator management** chapter added. The following topics have been updated/ added:

Training settings
To export details from an operator
To export details from multiple operators
To delete trainings
Trainer app

Related concepts

- Safety information for IT managers (p. 15)
- Automated user import (p. 357)
- About the intended users (p. 33)
- About other reports (p. 200)
- Comment Assignment (p. 394)
- Result validation (p. 172)
- Training settings (p. 127)
- Application for trainers (p. 169)

Related tasks

- Connecting a cobas® pulse system to a Wi-Fi network (p. 59)
- To export details of an operator (p. 152)
- To export details from multiple operators (p. 154)
- To delete trainings (p. 168)

Related reference

- Retention period for sent and unsent patient results (p. 261)
- Use Secure SMTP (p. 270)

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Safety

Safety information

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Safety classifications

The safety information applies to the system and is listed in accordance with ANSI Z535.6-2011. The Danger safety classification to indicate a situation that results in injury or death is not applicable to this system and is therefore not listed.

⚠ WARNING

Warning

- Indicates a potentially dangerous situation which, if ignored, may lead to fatal or severe injuries.

⚠ CAUTION

Caution

- Indicates a potentially dangerous situation which, if ignored, may lead to injuries and/or damage to property.

NOTICE

NOTICE

- Indicates a message not related to personal injury.

System safety

Failure to observe the safety information may result in incorrect results, data corruption, and data losses.

⚠ CAUTION

Incorrect or corrupt data due to unauthorized access

Data security is breached if unauthorized users have access to your user name and password.

- Use strong passwords.
- Always enter your password unobserved.
- Do not write down your password.
- Never write down the password in a contact form, in an address book or in a file on the computer.
- Do not disclose your password to anyone. Roche never asks you for your password.
- If you ever disclose your password to anyone, change it immediately after.
- If you think anyone else has access to your account, contact your Roche Service representative immediately.
- Do not share your password.
- Do not share user accounts.

NOTICE

Access to wrong device

Delayed troubleshooting or unnecessary maintenance of a device accessed remotely.

- When you access a device remotely, check that the IP address displayed corresponds to the one configured for the device.

⚠ CAUTION

Unavailability of the system

Unavailability of the system due to network or power failure can lead to delayed or incorrect results, or the loss of data.

- Ensure that the system is always running.
- Ensure that the power supply is never interrupted.
- Regularly check the connectivity to the instruments.
- Regularly check that the server is running.
- It is recommended to use UPS (uninterruptible power supply) software.

⚠ CAUTION

Blocked operator access to POC devices

If the application is compromised by a malicious attacker, operator access to connected POC devices can be blocked. To avoid delayed results due to blocked operator access, do the following:

- Assess the impact of availability of each connected POC device on patient safety. If necessary, implement suitable risk control measures, or manually managed offline devices.

⚠ CAUTION

Loss of unsaved changes due to log off

When logging off from the application's start page, unsaved changes are lost.

- Ensure that all changes are saved before logging off from the application.

Safety information for IT managers

The IT manager of your organization should ensure that the following safety measures are implemented.

Data security

⚠ CAUTION

Monitor the system for suspicious activity and report suspected compromise

If you find any of the typical signs of malicious software or unauthorized access to the system (unexpected warning messages, files, or log entries like multiple failed logon attempts; significantly degraded user interface performance; seemingly random crashes of the system; automated typing of text; and so on), the following recommendations are essential:

- Physically disconnect the system from the network.
- Contact the IT responsible in your organization to report and verify the finding.
- Mistrust results produced while the system has been compromised.
- Contact your Roche Service representative to initiate the system recovery.

⚠ CAUTION

Data loss or unavailability of the system due to malicious software or unauthorized system access

Malicious software or unauthorized system access can result in data loss or system unavailability.

To avoid infection by malicious software or the unauthorized access and misuse of the system, the following recommendations are essential:

- Make sure other computers and services on the network (for example, the LIS, archiving share, backup share, or service) are properly secured and protected against malicious software and unauthorized access.
- Restrict physical access to the system and server and all attached IT infrastructure (computer, cables, network equipment, and so on).
- Store related removable media in a safe place.
- Ensure that all device components maintaining private data are physically secure (i.e. require a physical key to be accessed or removed).
- Check all external storage devices, like CDs, DVDs, or USB mass storage devices with an antivirus software (on another computer) to ensure that they are free of malicious software, before using them on the system.
- Implement security and authentication measures for removable media (i.e. USB flash drives, CD-Rom/DVD, tapes, etc.).

- Use state-of-the-art security mechanisms (e.g. WPA2 EAP) to protect Wi-Fi connections.
- Allow only defined users to read and write encryption keys (IIS App config / Windows Server Hardening).
- Allow only internet access to trusted web sites and web services.
- Enable Data Execution Prevention for all applications (Windows Server).
- Make sure to use secure channels to download software updates of the system.

⚠ CAUTION

Unauthorized access to the system configuration

Unauthorized access to the system configuration (administration interface, configuration store, query packages) can lead to system failure, corrupted data, and unauthorized disclosure of protected health information.

- Make sure that your Wi-Fi connections are properly secured.

If supported, use EAP security for Wi-Fi connections.

⚠ CAUTION

Unauthorized access to confidential data

Unencrypted communication can lead to unauthorized access to confidential data.

- Use encrypted communication channels from the application.
- Contact IT support of your institution when a security warning related to the digital certificate is displayed.

⚠ CAUTION

Sensitive data - data tampering in transmission

Unprotected Wi-Fi connections between instruments and the application can lead to unauthorized data access and tampering.

- Make sure that your Wi-Fi connections are properly secured.
- If supported, use EAP security for Wi-Fi connections.

⚠ CAUTION

Malfunctions and incorrect results due to third-party software

The installation of any third-party software that has not been approved by Roche Diagnostics may lead to malfunctions.

- Do not install any unapproved software.

⚠ CAUTION

Data loss and disaster recovery

Unexpected disasters, such as hard disk failures, system software failures or natural disasters such as fires or floods can occur.

- The Customers IT department must assemble a disaster recovery kit to protect the entire system.
- Ensure that backups are made regularly (ideally ever day) and stored securely.

NOTICE

Data transmission is not secured

Unsecured communication can compromise the integrity and confidentiality of the transmitted data. It is the customer's responsibility to ensure that the network infrastructure provides appropriate protection for the transmitted data.

- Such protection can be achieved by system level tunneling protocols or other means of network encryption.

NOTICE

Data is not encrypted

- Data in the database is not generally encrypted. It is the customer's responsibility to ensure that access to the data storage and backup files is permitted only to authorized entities.
- It is recommended to implement disk encryption to prevent unauthorized access by either using the implemented feature or an alternative encryption solution.

NOTICE

Disposal of media, devices, and assets

Ensure that all media, devices, and assets are disposed of safely according to local privacy and security regulations.

- Dispose of these items through designated collection facilities appointed by government or local authorities.

Laboratory workflows

Failure to observe the following safety information may result in incorrect results, data corruption, and data losses.

CAUTION

System behavior and updates

- The application has been proven to work under the specified operating systems.
- You must perform regular maintenance actions on the server PC, such as hard disk defragmentations, virus checking and checking of system logs in order to prevent the system from becoming unstable or losing performance.
- Emergency software patches can be deployed via remote service if available.
- Software updates must be performed by your Roche Service representative.
- If you wish to have the software updated, contact your Roche Service representative.

Privacy protection

CAUTION

Unauthorized disclosure and distribution of personal data

- Do not enter personal data of patients in free text fields.
- Ensure that user rights to view and export personal data of patients are restricted to authorized users only.
- Ensure that sample IDs, patient IDs and visit IDs do not contain personal data.

CAUTION

Data retention

- Ensure that retention policies are adhered to in accordance to applicable local regulations for data retention.
- Personal data has to be removed in accordance to the data minimization principle.

Operation

Overview of the application

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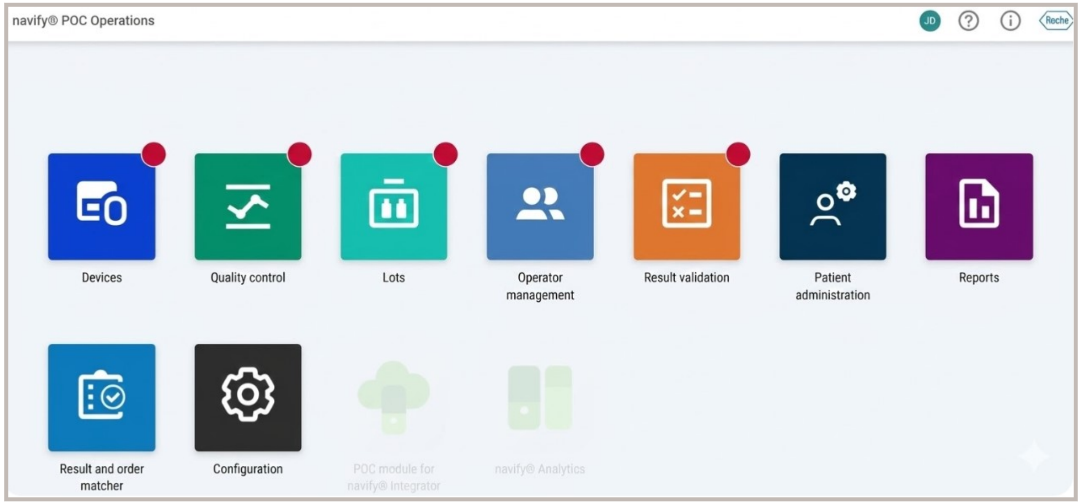
About the application

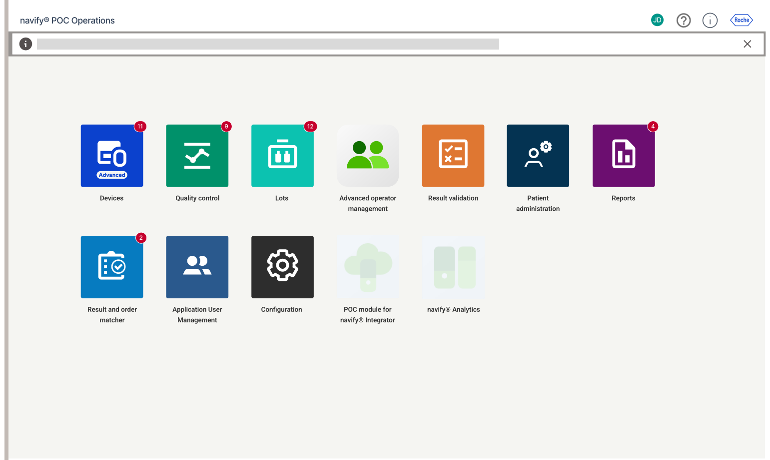
You log on to the application via a web browser or in the **navify**® Portal via single sign-on (SSO).

After logging on, in the top bar, you can see the user profile RD button with the user's initials next to the Roche logo.

From the home screen, you can access one of the following modules:



If you enable the **Advanced operator management** module, a message is displayed, indicating that only the application user profiles are managed using the **Application user management** module. The other existing operator features are now available in the **Advanced operator management** module.



Module	Menus	Main tasks
Devices	Devices	Devices
	Settings	Settings
Quality control	Quality control	
	Proficiency results	
Lots	Lots	
	Settings	
Operator management	Operators	Operators
	Certifications	Certifications
Advanced operator management		
Result validation		Result validation
Patient administration	Patients	Patients
	Visits	Visits
Reports	Signable reports	QC reports
	Other reports	Other reports

Module	Menus	Main tasks
Result and order matcher		About the Result and order matcher
Configuration	System	General settings
	Organization	Organization module components
	Devices	Configuration
	Quality control	QC Test Mapping
POC module for navify ® Integrator		
navify® Analytics		

You must have a user profile with rights to access the application. The default user profiles are the following:

- IT manager
- Nurse educator
- POCT administrator
- POCC
- Respiratory therapist
- Service

- Valid user account with access rights to the application
- Computer or tablet with supported web browser

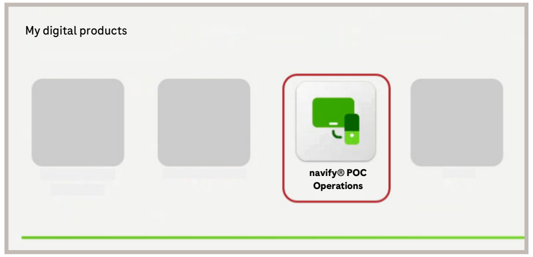
Related tasks

- Logging off from the application
- Logging on to the application via SSO (p. 23)

Related reference

- List of technical requirements (p. 30)

Logging on to the application via SSO



1 Open the **navify**® Portal under `navifyportal.roche.com`

2 Choose the **Login** button to open the logon screen.

3 Enter your credentials and choose the **Sign in** button.

4 Under **My digital products**, choose the **navify**® Portal icon to open the application.

navify® Portal

Logging off from the application

Auto logoff

If you are inactive, you are automatically logged off from the application after a time period that is configured in the general settings (default value = 90 minutes).

If the value is set to 0, the *auto logoff* is disabled.

If you are logged on via the **navify**® Portal and are automatically logged off from the application, you can log on again by choosing the SSO button without entering your credentials. The session in the **navify**® Portal has to be active in order for the SSO to function.

1 In the global information area, choose the RD button ("RD" refers to the user's initials, i.e. the button will be labelled differently for different users according to their initials).

2 Choose the **Log off** button.

3 In the web browser, close the tab.

Related tasks

- About the application (p. 20)

Related reference

- Session Expiration Time (Minutes) (p. 267)

Configuring personal settings

You can choose the language in which you want to use the application, and you can choose the date, time, and number formats.

Non-Western languages

To enable the use of non-Western languages, you must install the font “Arial Unicode MS Regular” on the server hosting the application.

1 In the global information area, choose the RD button.

2 Choose the **Edit settings** button.

3 In the **Personal settings** dialog box, select your preferred settings for user interface language and time, date, and number formats.

4 Choose the **Save** button.

Related tasks

- Changing your password (p. 24)

Changing your password

The new password can only consist of letters (a to z, and A to Z) and numbers (0 to 9). No special characters or country-specific letters are supported.

Password requirements

A valid password must by default contain the following:

- Number of characters: 8
- Number of uppercase characters: 1
- Number of digits: 1

Change the default password at the first logon.

1 In the global information area, choose the RD button.

2 Choose the **Change password** button.

3 Do the following:

- Enter your current password.
- Enter the new password.
- To confirm the new password, enter it again.
- Choose the **Change password** button.

Related tasks

- Configuring personal settings (p. 24)

Searching and filtering

You can search and filter the data in each module of the application.

For the search and filter functions, the following applies:

- A “ . ” period or “ , ” comma character is included in the search as it is.
- To search by date, the following applies:
  - You must enter the full date in the date format configured in the personal settings.
    Example: MM/dd/yyyy.
  - If you enter part of the date including the delimiter, no records are found.
    Example of incorrect date: MM/dd
  - If you enter part of the date without the delimiter, the application performs a numeric search and finds all records fitting the search.
    Example: MM finds all the records that include the numbers “MM” , e.g., in the month or the year, or even in another column, if no option from a drop-down list is defined.

NOTICE

Selecting multiple filter criteria

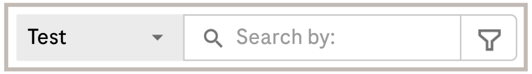
An error may occur if too many filter criteria are selected.

▶ Reduce the number of selected filter criteria.

Configuring personal settings

The numbers shown might use a different symbol for the decimal point than the system is set to use.

▶ To search for records



1 To search for records, do the following:
- If available, choose an option from the drop-down list.
- In the search field, enter the search term.
- Press Enter.

→ The number of results found is displayed in the header.

The search term is displayed under the header.

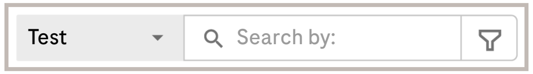
The entered search term is highlighted in blue in the results.

The search term is automatically added to the filter.

2 To clear the search, do the following:
- Choose the ✕ button of the search term.
- Alternatively, choose the **Clear all** button.

→ All records are displayed again.

▶ To filter for records

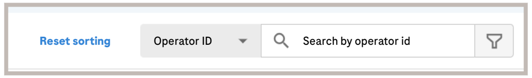


1 To filter for records, do the following:

- Next to the search field, choose the button.
- In the dialog box, enter the filter criteria.
- Choose the **Search** button.

The number of items entered is displayed next to the criteria name in the **Filter**dialog box. When criteria are selected from a drop-down list, the number of selected criteria is displayed.

> → The number of results found is displayed in the header.
>
> The selected filter criteria are displayed under the header.
>
> The entered filter criteria are highlighted in blue in the results.



2 To sort the results using a column name, choose a column.

> Choosing the column name again changes the sorting to reverse alphabetical order. The sorting is indicated by a small arrow next to the column name by which the table is sorted.
>
> The **Reset sorting** button becomes available when the table is sorted by choosing a column name. Reset to the default view by choosing the **Reset sorting** button.

> → The sorting changes from default to alphabetical order.

3 To remove filter criteria, do the following:

- To remove a single filter criterion, choose the ✕ button of the criterion.
- To remove all filter criteria, choose the **Clear all** button.

> → The results are updated automatically.

▶ **To delete a date from a date field**

1 Choose the date field containing the date you want to delete.

→ The date picker is displayed.

2 Choose the date field again.

→ A cursor is displayed in the field.

3 Press the Backspace or Delete key to delete the date.

Advanced and quick search filters

To narrow down the search, advanced and quick search filters can be applied to the devices list or table.

Quick search

Search from a specific device, parameter, or location.

Advanced search

To narrow your search further, use the search filter ( ) which provides additional search criteria including date range, levels, and parameters.

> If more than one option is chosen for one filter criterion, results for all chosen options will be displayed.
>
> If several filter criteria are applied, only results matching all filter criteria will be displayed.

NOTICE

Selecting multiple filter criteria

An error may occur if too many filter criteria are selected.

▶ Reduce the number of selected filter criteria.

Adjusting the table settings

To adjust the table settings, choose the ⚙ icon.

> On the **By site** and **By type** tab, the icon is visible after choosing a site or type respectively.

In the **Table settings** window, you can adjust the following:

- **View settings**
- **Display density**
- **Column settings**

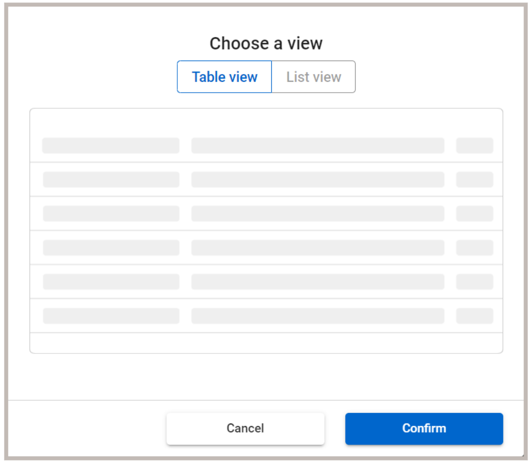
NOTICE

Display inconsistency

The images below may appear differently from those in your application.

▶ To avoid confusion, ensure you are using the same view or display settings as referenced.

View settings

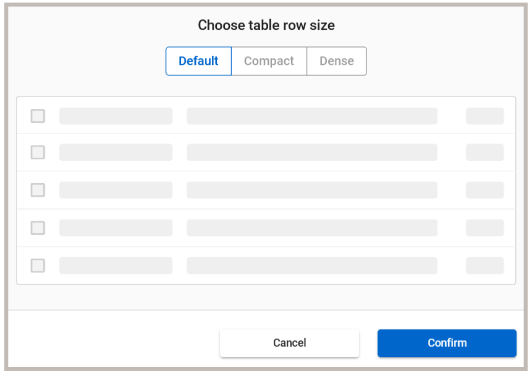


In **View settings**, you can choose one of two displays:

- **Table view**
- **List view**

> **Display density** and **Column settings** can not be adjusted in the **List view**.

Display density



In **Display density** you can choose one of three table row sizes:

- **Default**
- **Compact**
- **Dense**

Column settings

In **Column settings**, you can adjust the displayed columns using the following options:

- Making the table settings screen bigger by choosing the ⛶ icon.
- Reorder the columns by drag and drop using the ⠿ icon.

- Hiding or showing a column by checking or unchecking the box next to the column name.
- Resetting to the default view by choosing the **Reset view** button.

> The applied adjustments will be saved for the user currently logged on and the columns will be displayed accordingly as default at every logon until adjusted again.

Alphabetical sorting

You can sort the items in a table by choosing a column name. Choosing the column name changes the default sorting to alphabetical order. Choosing the column name again changes the sorting to reverse alphabetical order.

The sorting is indicated by a small arrow next to the column name by which the table is sorted.

The **Reset sorting** button becomes available when the table is sorted by choosing a column name. Reset to the default view by choosing the **Reset sorting** button.

About the User Assistance

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About the menu

1 In the global information area, choose the button to display the User Guide.

2 Choose the button to display the About box.

> The **About** box is displayed.
>
> In the **About** box, information like the installed software version, the unique device identifier (UDI), or the serial number of the **navify**®POC Operations installation is displayed.
>
> > The serial number is only displayed in the **About** box, when it is entered in the **General settings** option **Serial number**.

Specifications

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List of technical requirements

The computer or tablet where you access the application should comply with the following requirements.

Tablet resolution

The minimum resolution to display the application properly on the tablet screen is 1024*768 pixels in landscape mode.

Software requirements

The application is compatible with the following web browsers.

Device	Operating system	Web browser
Computer	Microsoft Windows The web application is best viewed when setting the size of text, apps and other items to 100% in your Windows settings.	Microsoft Edge Google Chrome (recommended) Do not access the application with Microsoft Edge or Google Chrome using Compatibility View. The web application is best viewed using the current browser version of Google Chrome and Microsoft Edge. If you are using an out-of-date browser, you may not experience the full effect when viewing the site or get some unhandled errors.
Tablet	iOS and iPadOS	Safari
Tablet	Android	Google Chrome

Software requirements

You must configure the browser to enable the use of cookies.

Connection

For the application to work, all used devices and workstations must be securely connected to the internal network of your institution. If you are using the application remotely from outside the customer network, the device or workstation must be connected to the internal network through VPN.

> Make sure that the used client system is set to the same date and time as the **navify**® POC Operations server to avoid access issues.

Predefined configurations

The application comes with predefined configurations that are intended to facilitate the configuration tasks of Roche Service during the installation and configuration of this software.

⚠ CAUTION

Predefined user configurations for Roche Service

The installation of any third-party software that has not been approved by Roche Diagnostics may lead to malfunctions.

▶ Take the predefined configurations for Roche Service as a reference only. Do not use them as default without further configuration process.

List of supported barcodes

The application allows you to scan device and operator barcodes using the camera of your tablet or a barcode reader connected to your computer.

The application supports the following types of *barcode*:

Barcode type	Example
Aztec code	
Codabar	B2030405060D

Bar-code type	Example
Code 128	0192837460
Code 39	*0121212120*
Code 93	0147246980
GS1 Data-Matrix	
EAN-13	1 234567 890128 12>
GS1	(01)23321450

Barcode type	Example
Interleaved	003432556800
PDF417	
QR code	

Supported barcodes

About the intended users

navify® POC Operations is an application developed for the following user groups:

User group	Tasks
POCC (POC coordinator) (Laboratory Professional, healthcare professional)	▪ Managing devices ▪ Managing operators ▪ Managing materials ▪ Managing QC ▪ Managing patients ▪ Managing patient results ▪ Managing medical orders (primarily for nurse backup, not a primary use function)
Nurse[1] / Nurse educator	▪ Managing operators ▪ Managing medical orders

1 User definitions may vary or be country-specific. In the U.S. market, order matching is performed by a respiratory therapist.

User group	Tasks
IT manager	Managing network and security configurations including the following: ▪ Network and security configurations ▪ WLAN device configuration security type EAP ▪ EAP settings management (EAP settings and EAP settings assignment) ▪ WLAN device configuration security type EAP
Roche Service representative	▪ Installing the application ▪ Customizing the application

Deep links to the navify® Analytics for Point of Care dashboards

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Deep links to the navify® Analytics for Point of Care dashboards

The deep links are designed to display specific dashboard within **navify**® POC Operations, providing insights into targeted data.

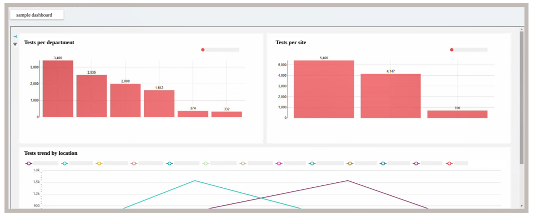
To access the deep links, log on to the application via the **navify**® Portalor via the single sign-on (SSO) in the **navify**® POC Operationsapplication.

For more information, refer to the following procedure:

About the application

Displaying the deep links to navify® Analytics for Point of Care

The following procedure shows an example of how to display the deep links via the **Operator management** screen.

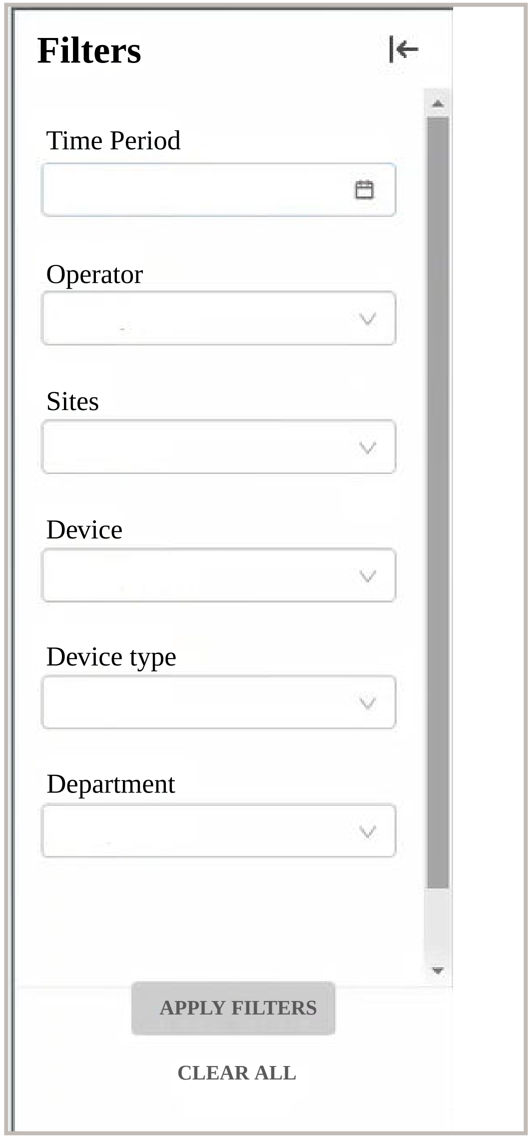


1 Choose **Operator management** > **Operators**.

2 Choose the desired operator to open the **Operator details**.

→ The deep links to **navify**® Analytics for Point of Care are displayed in the **Analytics** dialog box.

3 Choose the desired deep links displayed to view the specific dashboard.



4 Optionally, to filter for operators, do the following:

- Choose the filter icon to open the filter criteria sidebar.
- In the dialog box, review the default filters that are automatically applied and make any necessary adjustments.
- Choose the **Apply filters** button.

→ The dashboards are displayed.

5 Repeat the previous steps for each deep link that you want to display.

Deep links overview

The table below provides an overview of all the deep links available in the different screens of the application.

List of screens that includes deep links.

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Operator managem	**Operator details** page	Filtered by opera		Filtered by opera	Filtered by opera	Filtered by opera	Filtered by opera

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
ent		tor ID		tor ID	tor ID	tor ID	tor ID

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Operators (View by site)	Filtered by site		Filtered by site	Filtered by site	Filtered by site	Filtered by site	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
	ID		ID	ID	ID	ID	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Operators (By loc	Filtered by locat		Filtered by locat	Filtered by locat	Filtered by locat	Filtered by locat	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
ation)	ion ID		ion ID	ion ID	ion ID	ion ID	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Operators (View by name/All operators)	Dashboard withou		Dashboard withou	Dashboard withou			

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
	tfilter		tfilter	tfilter			

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Devices	**Devices** (View by site)	Filtered by site	Filtered by site			Filtered by site	Filtered by site

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
		ID	ID			ID	ID

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Devices (By locat	Filtered by locat	Filtered by locat			Filtered by locat	Filtered by locat	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
ion)	ion ID	ion ID			ion ID	ion ID	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Devices (View by name)	Dashboard withou	Dashboard withou					

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
	tfilter	tfilter					

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Devices (View by type)	Dashboard withou	Dashboard withou					

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
	tfilter	tfilter					

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Devices (By type)	Filtered by type (	Filtered by type (			Filtered by type (	Filtered by type (	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
	device type)	device Type)			device Type)	device Type)	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Device details page	Filtered by devic	Filtered by devic			Filtered by devic	Filtered by devic	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
	e (serial number)	e (serial number)			e (serial number)	e (serial number)	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Signable reports	**QC list** (By families/type)					Filtered by type (	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
						deviceType)	

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
QC result details					Filtered by devic		

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
					e (serial number)		

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
Result validati							Dashboard withou

Landing page		Issues	Device performance	Operator statistics	Operator certification	QC insights	Patient test statistics
on							tfilter

Device management

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Devices

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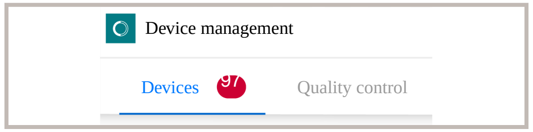
About device status

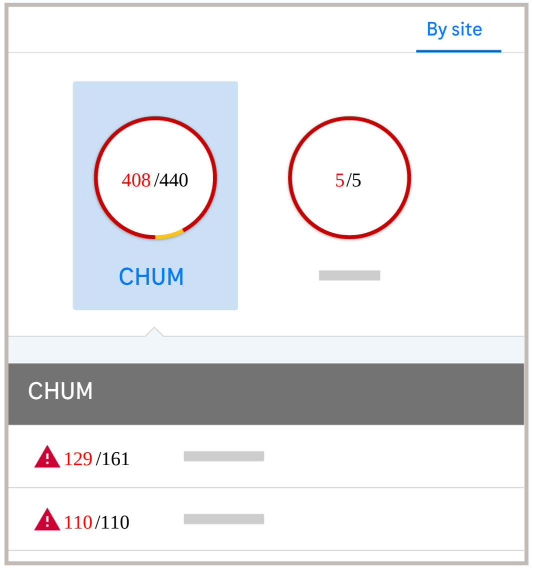
In the application, you can find the following device status information.

Global device status



In the global information area, the number next to the **Devices** tab indicates the total number of devices with errors across all sites.

Device status per site and location

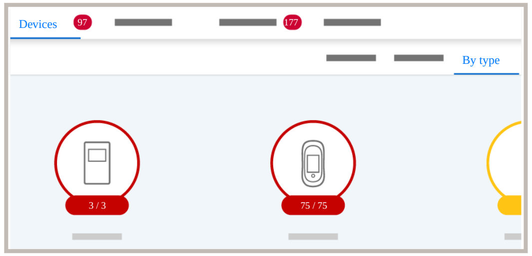


On the **By site** tab, the following device status information is provided:

- For each site, the circular chart visualizes the percentage of devices with errors (red), warnings (yellow), and no alerts (gray).
- The 2 numbers inside the chart reflect the number of devices with errors (red), and the total number of devices in that site (gray).
- The number of devices with errors (red) and the total number of devices are additionally displayed for each location of the site.
- An additional circular chart is displayed for unassigned, active devices. Unassigned devices are counted as errors.

> Variable tab name
>
> ▶ If you have access only to 1 site, the tab is called **By location**.
>
> ▶ If you have access to multiple sites, the tab is called **By site**.

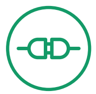
Device status per device type

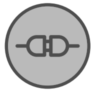


On the **By type** tab, the circular chart visualizes for each device type the percentage of devices with errors (red), warnings (yellow), and no alerts (gray). The 2 numbers below the device icon reflect the number of devices with errors and the total number of devices of that device type.

Connection and QC status

On the device detail panel, the following status icons visualize the connection status and QC status of a device.

Icon	Status	Description
	Connected	The last connection time of the device is within the configured time range (by default, 24 hours).

Icon	Status	Description
	Disconnected	The last connection time of the device is outside the configured time range (by default, 24 hours).
	Unavailable	No information is available about the last connection time.
	Disabled	The setting of the connection status has been disabled.

List of connection status icons

Icon	Status	Description
	OK	The last passed QC run of any level was performed within the configured time frame (by default, 12 hours).
	Error	The last QC run of any level has failed. A new lot has been created in the application for the first time and the **Application wide** settings are set to **Manual material lot distribution**
	Warning	No QC run has been performed within the configured time range (by default, 12 hours).
	Disabled	The setting of the QC status has been disabled.

List of QC status icons

Locked and stopped status

On the device detail panel, the following status icons and status texts visualize if a device is partially locked, locked, or stopped.

Icon	Description
	The device is partially locked.
	The device is locked.
	The device is stopped.

List of status icons and status texts

License status

The application detects unlicensed devices. On the device detail panel, unlicensed devices are marked with an

icon.

You cannot perform actions on unlicensed devices.

Station type devices

The device type "Station" is not a device as such, it is a concept used to link device download locations to organizational locations (as identified by their IP address or host names) e.g. identify devices docked outside their assigned location.

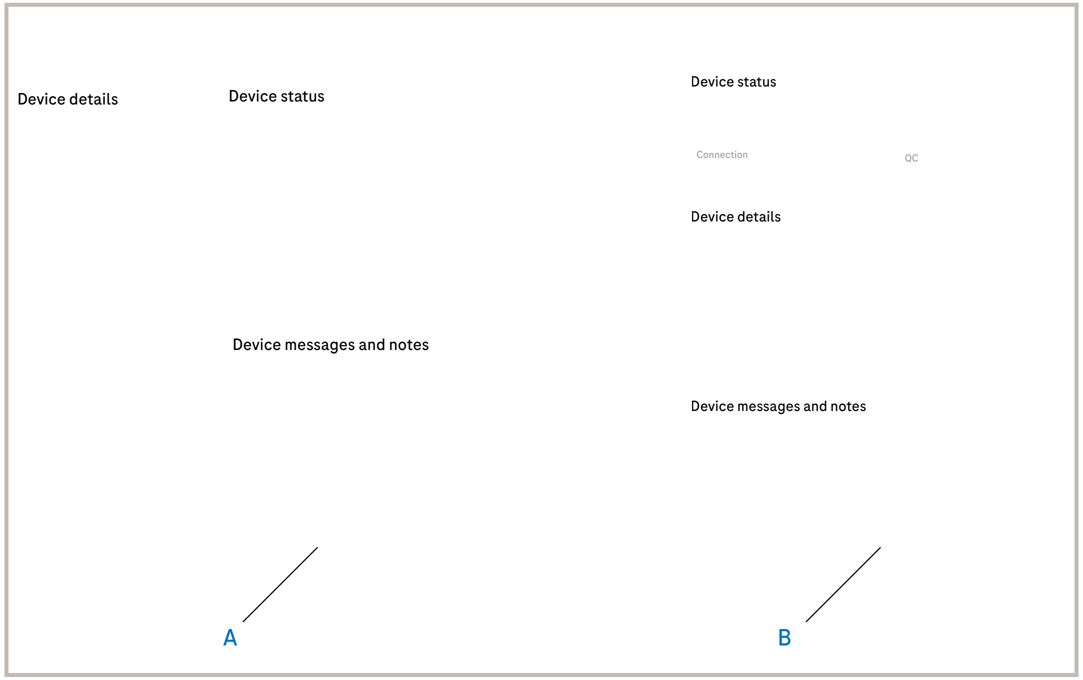
Due to the availability (or non-availability) of network information and instrument usage, the concept of stations can only be used in certain cases. For information on whether your installation can make use of stations, please contact a Roche Service representative.

Searching for a device

You can search for a device by its location, name, or type.

Device information

Device information screen in list view:



Device information screen in table view	Device information screen in list view

The application provides the following information about a device.

Device status:

- Connection status
- QC status

Device details:

- The available details depend on the device type and whether Rili-BÄK QC is used.

Device messages and notes:

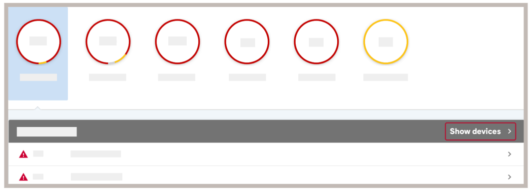
- You can view the device messages and comment the messages by adding a note.

▶ To search for a device by location

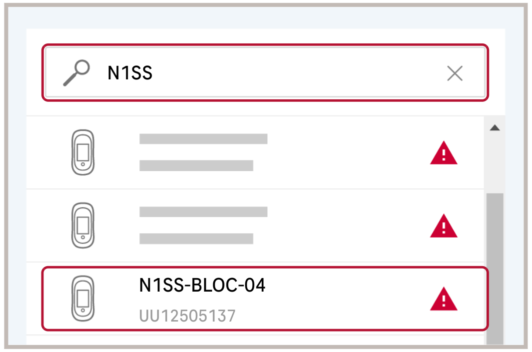
1 Do one of the following:

- If you have access to only 1 site, choose **Devices** > **Devices** > **By location**.
- If you have access to multiple sites, choose **Devices** > **Devices** > **By site**.

2 If you have access to multiple sites, choose a site.



3 Choose a location or choose the **Show devices** button to display all devices of that location.

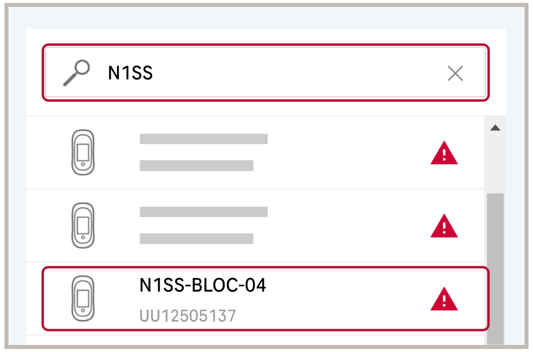


4 Do any of the following:

- Search the device in the list.
- Enter the device name or serial number in the search field.

To display active and inactive devices, Toggle the **Active devices**button to **All devices**. To display only active devices again, choose the toggle button once more.

To search for a device by name



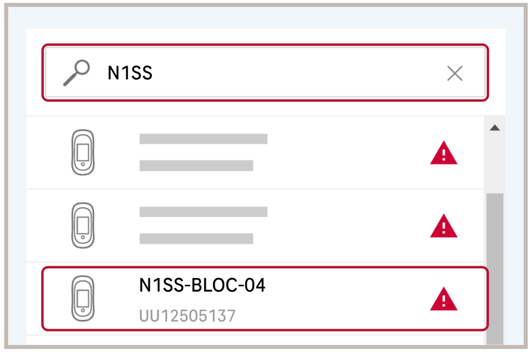
1 Choose **Devices** > **Devices** > **By name**.

2 Do any of the following:

- Search the device in the list.
- Enter the device name or serial number in the search field.

To display active and inactive devices, Toggle the **Active devices**button to **All devices**. To display only active devices again, choose the toggle button once more.

To search for a device by type



1 Choose **Devices** > **Devices** > **By type** and choose a device type.

2 Do any of the following:

- Search the device in the list.
- Enter the device name or serial number in the search field.

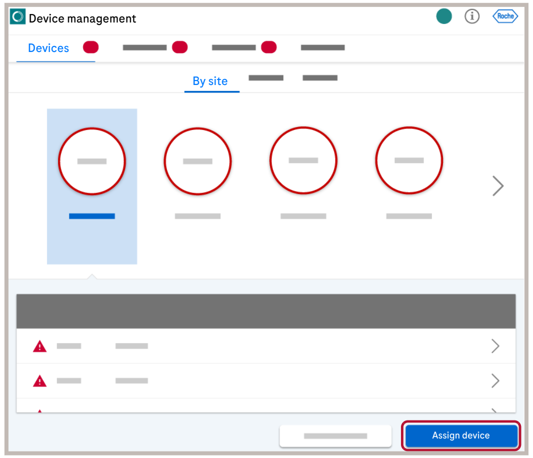
To display active and inactive devices, Toggle the **Active devices**button to **All devices**. To display only active devices again, choose the toggle button once more.

Assigning a device

When a device is docked the first time, it is automatically added to the application but has no location assigned.

In the application, you can view all unassigned devices and assign them to a location or use them to replace another device.

You can assign active and inactive devices to a location. If you assign an inactive device, the device stays inactive.

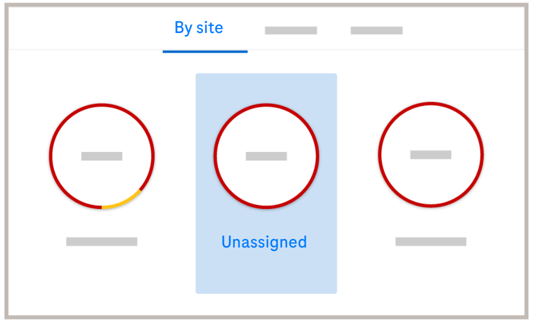


1 Open **Devices** > **Devices**.

2 Choose one of the sorting options: **By site**, **By type** or **By name**.

3 Choose the **Assign device** button.

Assign deviceMore actions



4 Alternatively, choose **Devices** > **Devices**. On the **By site** tab, open the **Unassigned** chart and choose the **Assign device** button.

5 Choose the device you want to assign and choose the **Next** button.

6 Enter or edit the following device details (**Device type**) , **Device name**, **IP address (optional)**and **Serial number**). Choose the **Next** button.

7 Choose a location and choose the **Next** button.

8 Ensure that the pre-selected default configuration is correct, or choose the correct one, and choose the **Finish** button.

9 Enter a reason for the device assignment and choose the **Confirm** button.

Importing devices

You can automatically import devices when connecting the device for the first time.

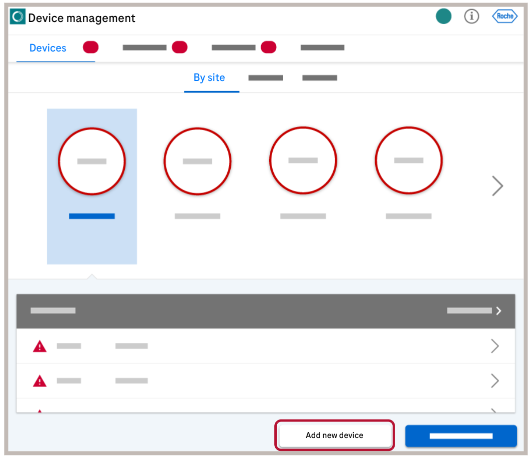
Also refer to the documentation of the individual instrument drivers.

CAUTION

Duplication of serial number

Connecting a new device with a serial number that is already used by an existing device leads to a false association between patient results and devices, which can delay results.

- If the serial number of an existing device needs to be changed, ensure that both the serial number and the device name are updated. Updating only the serial number may prevent correct device recognition and causes a synchronization timeout.
- Check whether the new device has a unique serial number before connecting it.
- Do not connect a new device that does not have a unique serial number.
- On connected devices, do not use a serial number longer than 16 characters. This action creates an unassignable device that hides all of its results from view, without displaying an error message.



1 Choose **Devices** > **Devices**.

2 Choose one of the sorting options: **By site**, **By type** or **By name**.

3 Choose the **Add new device** button.

Add new deviceMore actions

4 Complete the assignment wizard.

Assigning a device

5 Choose the **Confirm** button.

Exporting devices

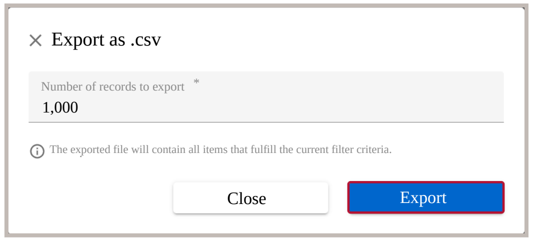
A list of devices can be exported as a CSV file.

In addition to the current filter criteria in the devices list, you can only limit the export by number of records to export. To refine the export, modify the filter criteria.

> Depending on the number of devices, the export may take a long time. Exporting more devices than the default number (1000) may lead to an error.

About the export file

The export file contains the devices that fulfill the current filter criteria in the devices tab.



1 Open **Devices** > **Devices**.

2 Choose one of the sorting options: **By site**, **By type** or **By name**.

3 Choose the **Export devices** button.

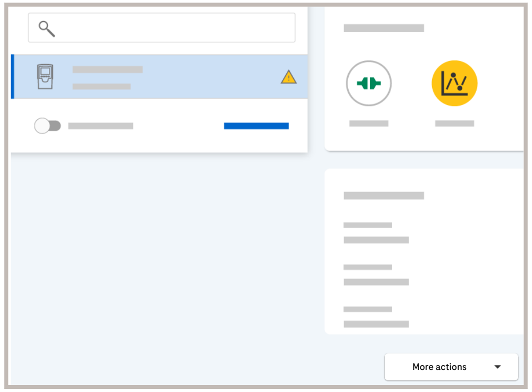
Export devicesMore actions

4 Optionally, in the **Export as .csv** dialog box, enter the number of devices to be exported.

5 In the **Export as .csv** dialog box, choose the **Export** button.

> → A CSV file containing the list of devices is downloaded. On iOS devices, the exported file is attached automatically to an email for sending it.

Synchronizing a device



If you synchronize a device, the application sends the latest information stored in the database about patient, operator, and configuration to the selected device. The kind of information that can be sent depends on the device type.

1 Choose **Devices** > **Devices**.

2 Choose a device, and choose the **Synchronize** button.

Synchronizing muliple devices

Synchronizing multiple devices is only possible in table view.

1 Choose **Devices** > **Devices**.

2 Choose the devices you want to synchronize by choosing the check box next to it.

Inactive devices can be synchronized with inactive devices only.

3 Choose the **Synchronize** button.

> The synchronization request was successful if the message **Multiple devices will be synchronized.** appears in the upper right-hand corner of the screen.

Deactivating a device

You *deactivate* a device in the following situations:

- To prevent the use of a defective or incorrectly configured device.
- To free the license for a new device.

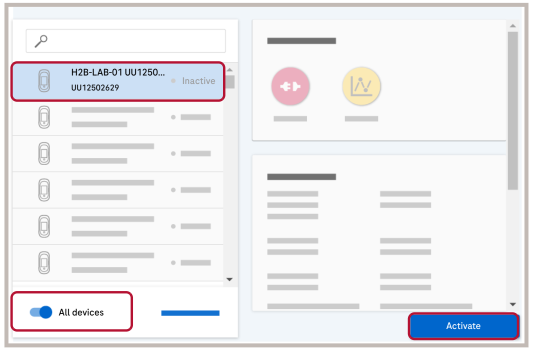
1 Choose **Devices** > **Devices**.

2 Select an active device and choose **Deactivate**.

DeactivateBy siteBy nameBy type

3 Enter a reason for the deactivation and choose the **Confirm** button.

Activating a device



1 Choose **Devices** > **Devices**.

2 Switch on the **Active devices** toggle button, select an inactive device, and choose the **Activate** button.

ActivateBy siteBy nameBy type

3 Enter a reason for the activation and choose the **Confirm** button.

Connecting a **cobas**® pulse system to a Wi-Fi network

You can connect a **cobas**® pulse system to the Wi-Fi network by scanning a QR code containing the network credentials.

- Wi-Fi network configuration information
- DMS information
- **navify**® POC Operations gateway information
- **cobas**® pulse system

1 Choose **Devices** > **Devices** > **By type** > **cobas pulse**.

2 Choose the device you want to configure.

3 Choose **More actions** > **Scan QR codes**.

→ The QR codes with the device's network credentials are displayed.

4 If the QR codes cannot be generated, ensure that the configuration is complete. Otherwise, skip to step 7.

→ A pop-up window appears, allowing you to enter a a static IP address or to select an EAP configuration for the device. An EAP configuration must be created beforehand to be available for selection.

5 If you completed step 5, choose the **Save** button.

The QR codes are displayed.

SSIDs containing UTF-8 characters (e.g., Chinese SSIDs) can only be read by devices with firmware versions 02.02.02 and newer. Make sure to update the device with firmware version 02.02.02 or newer.

To avoid a QC lockout, only use freshly generated QR codes, as any date/time adjustment exceeding 90 minutes will automatically reset the QC schedule of the device.

6 Scan all the displayed QR codes using the **cobas® pulse** system, and move from a QR code to another by choosing the **Next** button.

7 Once all the QR codes are scanned, choose the **Close** button.

Locking a device

You can lock a device remotely from the application. Lock functionality is a device-specific and depends on driver availability. Locking a device may not apply to all devices.

Locking can also be done from the device itself.

Locking a device remotely from the application can take a while. The device is locked only after the next synchronization between the application and the device.

- User rights for locking and locking devices
- Device driver supports device locking and unlocking

1 Choose **Devices** > **Devices**.

2 Choose a device and choose the **More actions** > **Lock device** button.

The device is locked and cannot be used anymore. The device must be unlocked first to be used again. Unlocking can be done from the device itself (if supported by the device type) or from the application.

Unlocking a device

You can unlock a locked device remotely from the application. Lock functionality is a device-specific and depends on driver availability. Unlocking a device may not apply to all devices.

Unlocking can also be done from the device itself.

Unlocking a device remotely from the application can take a while. The device is unlocked only after the next synchronization between the application and the device.

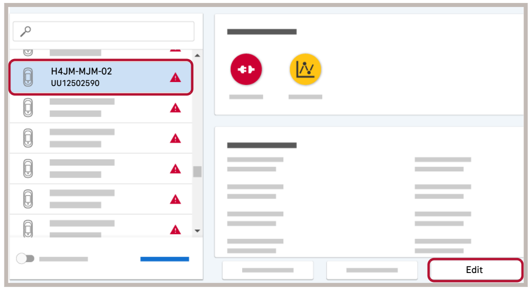
- User rights for locking and locking devices
- Device driver supports device locking and unlocking

1 Choose **Devices** > **Devices**.

2 Choose a locked device and choose the **More actions** > **Unlock device** button.

The device is unlocked and can be used again.

Editing a device



You can edit the following device information:

Device details:

- Name
- Serial number
- *Host name*/ IP address

Location

Configuration

NOTICE

Be careful when changing device configuration

Changing device configuration can result in loss of data and malfunctioning of the application.

▶ For device configuration changes, contact your Roche Service representative.

1 Choose **Devices** > **Devices**.

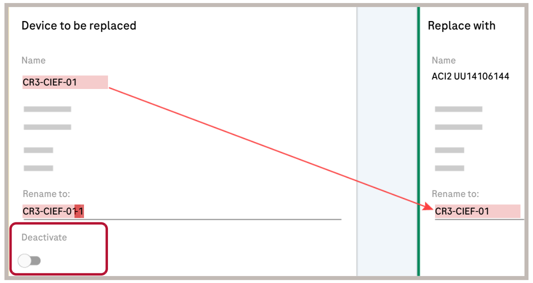
2 Choose a device and choose the **Edit** button.

EditBy siteBy nameBy type

3 Edit the desired information and choose the **Save** button.

4 Enter a reason for the change and choose the **Confirm** button.

Replacing a device



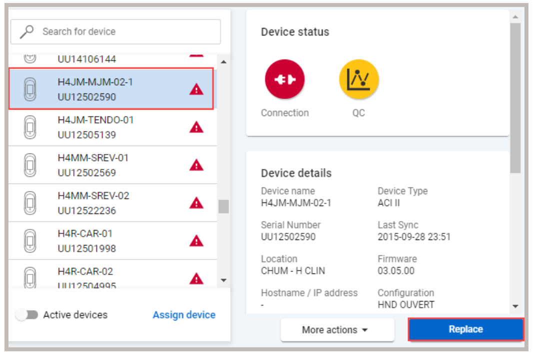
If you replace a device, consider the following:

By default, the new device inherits the name of the device it replaces and the replaced device is renamed by addition of the suffix “-1”.

- You can rename the replaced device.
- You can rename the new device.

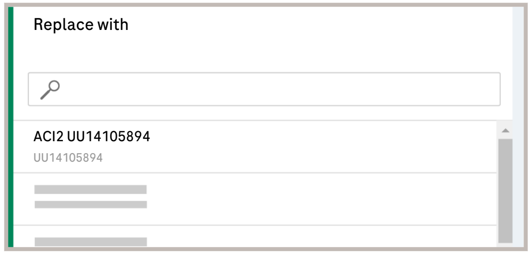
You can deactivate the device being replaced. By default, the replaced device remains active.

Active devices continue to consume licenses. To make a license available to be used by other devices, the device being replaced has to be deactivated.



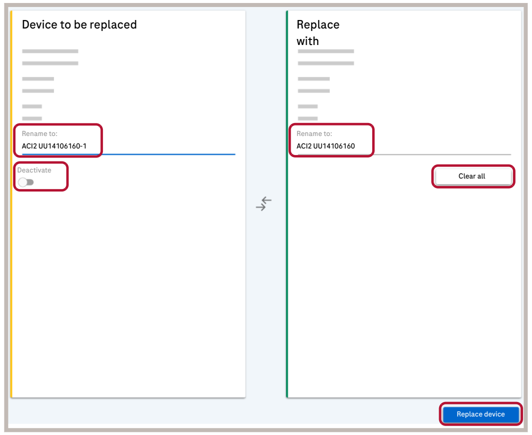
1 Choose **Devices** > **Devices**.

2 Choose the device you want to replace and choose the **Replace** button.



3 In the **Replace with** area, choose the new device.

- If you want to select a different device, choose the **Clear all** button.
- If you know the name or serial number of the device, use the search field to find it.
- When using the application on a mobile platform, you can scan the serial number barcode of the device.



4 Optional:

- Rename both the device you want to replace and the new device. If you do not rename the new device, it inherits the name of the device to be replaced.
- Deactivate the device to be replaced.

5 Choose the **Replace device** button.

6 Enter a reason for the replacement and choose the **Confirm** button.

Viewing device events and adding notes

The following device events are logged in the application.

Device messages: If an event or error occurs, the device sends a predefined message to the application.

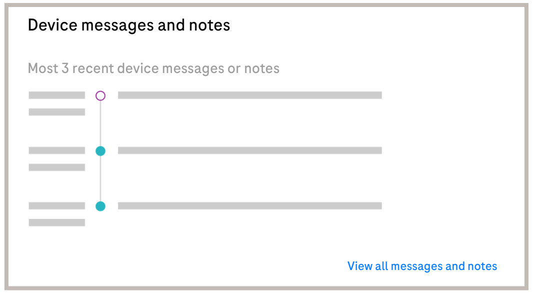
Device message examples:

- Cold boot start occurred
- Instrument firmware error: operator database
- Instrument battery empty, charging required.

Operator notes: To perform an action on a device via the application, you must enter a reason for the intended change. The application logs this entry as an “`operator note`”.

For each device, you can view the logged device messages and operator notes.

You can add notes to device messages, and you can delete logged operator notes. You can also add individual notes.

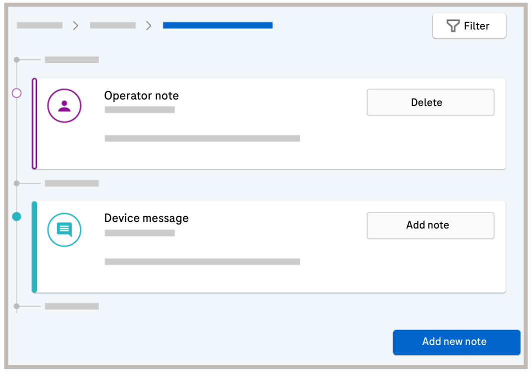


1 Choose **Devices** > **Devices**.

2 Choose a device with *Last message / note*.

Device messages and notes

3 To view all messages and notes, choose the **View all messages and notes** button.



4 Do any of the following:

- To add a note to a device message, choose the **Add note** button, enter the note, and choose the **Confirm** button.
- To add a note, choose the **Add new note** button, enter the note, and choose the **Confirm** button.
- To delete an operator note, choose the **Delete** button, and choose the **Confirm** button.

Blood gas devices

Sorting the blood gas overview

You can configure how the blood gas devices are sorted in the blood gas overview.

By default, the blood gas devices are sorted by system status severity in the following sort order:

- Locked
- Stopped
- With error
- Partially locked
- With warning
- OK (no error or warning)

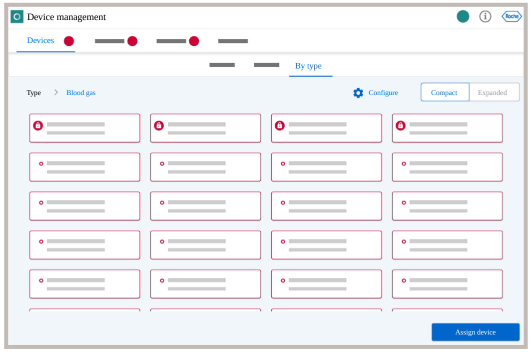
But you can also configure one of the following custom sort orders:

- By device name
- By device type
- By location
- In fixed sort order

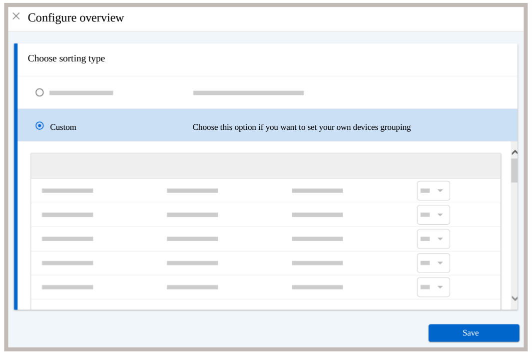
The custom sort order is retained, and applies application-wide and is user-specific.

1 Choose **Devices** > **Devices** > **By type**

2 Choose the **Show blood gas overview** button.



3 Choose the **Configure** button.



4 To configure a custom sort order, choose the **Custom** option.

5 To sort by device name, device type, or location, do the following:

- Choose the corresponding column header.
- To toggle between ascending and descending sort order, choose the column header again.

→ An arrow marks the sort column and the sort order.

6 To sort the blood gas devices in a fixed sort order, assign sort positions from the **Position** drop-down lists.

7 Choose the **Save** button.

→ The **Confirmation** dialog box informs you how the sorted blood gas devices will be displayed.

8 In the **Confirmation** dialog box, choose the **Confirm** button.

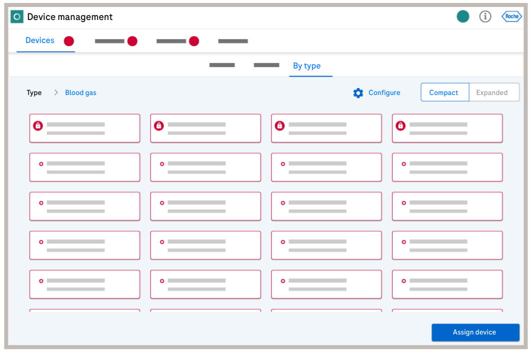
→ The blood gas overview is sorted accordingly.

Viewing additional information on a blood gas device

For certain blood gas devices, the application can display information sent by the device about the overall device status, sensors, parameters, *auto QC*, paper supply, *fluid pack*, and connection status. You can find more detailed information in the corresponding device User Guide.

1 Choose **Devices** > **Devices** > **By type**

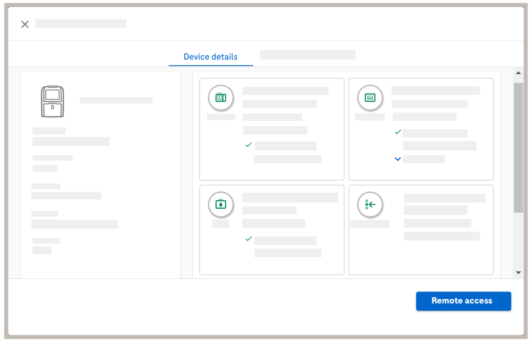
2 Choose the **Show blood gas overview** button.



3 To toggle between the expanded and the compact blood gas overview, choose the **Expanded** or **Compact** buttons.

4 To view more details about a device, choose a device from the blood gas overview.

→ The **Device details** tab is displayed including information about sensors, parameters, auto QC, paper supply, fluid pack, and connection status.



5 To view the **Remote access** button, choose one of the devices.

→ The **Remote access** button will appear in the lower right corner of the new tab.

6 To display the device messages and notes, choose the **Device messages and notes** tab.

Auto QC remainingShow levelscobas b 123

→ The 3 most recent device messages or operator notes are displayed.

7 To view all device messages and notes, choose the **View all messages and notes** button.

→ The **Device messages and notes** screen is displayed.

Laboratory-established deviation

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Laboratory-established deviation

Rili-BÄK Table B 1 specifies an allowed deviation for the tests listed in this table. These tests are referred to as Rili-BÄK tests in the **navify®** POC Operations application. For tests which are not listed in Table B 1, laboratory-established deviations (LED) can be defined. The LED functionality of the **navify®** POC Operations application automatically calculates and proposes LEDs for non-Rili-BÄK tests. The calculation is using the formula defined in the Rili-BÄK regulation. A proposed LED is always associated with a combination of instrument, test, material lot (combination) and level. If no LED is defined, the standard QC range is applied.

Workflow overview

The following paragraphs provide an overview of the steps and scenarios involved in establishing and managing proposed LEDs.

Accept and activate proposed LED

Even if an LED is proposed automatically, this does not mean that the LED is already applied. For a proposed LED to be applied, it needs to be active. A proposed LED only becomes active once it is accepted. The application provides two possibilities to accept proposed LEDs:

- Manual acceptance
- Automatic acceptance

Accepting a proposed LED

Review proposed LED

If a proposed LED has not been accepted automatically, it can be rejected, deleted or manually updated. You may also trigger a recalculation or accept an LED manually.

If a proposed LED has already been accepted automatically, it can still be rejected or manually updated. It is also possible to trigger a recalculation.

For more information on reviewing proposed LEDs, refer to the following procedures:

Viewing LED details
Accepting a proposed LED
Rejecting a proposed LED
Editing an LED manually
Recalculating an LED
Deleting a proposed or rejected LED, or an LED with No proposal possible status

No proposal available

If no LED proposals are displayed, check whether you need to adjust the calculation rules under **Configuration** > **System** > **General Settings**.

LED calculation principles

QC results or calculation rules changed for existing LED proposal

If the QC result set or the calculation rules associated with an existing LED proposal are changed, an automatic recalculation is triggered.

Automatic recalculation upon changes on QC result set or calculation rules
No automatic recalculation for active LEDs

LED handling overview

The following table provides an overview of the interrelationship between the states of an LED and the possible user actions.

Actions	Accepted	Proposal	No proposal possible	Declined	Deleted
Recalculate	✓	✓	✓	✗	✗
Accept	✗	✓	✗	✗	✗

Actions	Accepted	Proposal	No pro-posal possible	Declined	Deleted
Reject	✓	✓	✓	✗	✗

Actions	Accepted	Proposal	No pro-posal possible	Declined	Deleted
Delete	✗	✓	✓	✓	✓

Actions	Accepted	Proposal	No pro-posal possible	Declined	Deleted
Add manually	✗	✗	✓	✗	✗

Actions	Accepted	Proposal	No pro-posal possible	Declined	Deleted
Update manually	✓	✓	✗	✓	✓

LED calculation principles

Prerequisites for automatic calculation

For an automatic LED proposal to be calculated, the following requirements have to be met:

- Material and QC lot need to be defined.
- The set of QC results for the instrument QC test, material lot (combination) and level contains at least 15 calendar days having accepted QC results. Depending on the settings (defined under **Configuration** > **System** > **General Settings**), these 15 days may need to be within the last control cycle or in the current control cycle.

QC results used for calculation

- Only one QC result per calendar day is used.
- The 15 calendar days do not have to be consecutive.

The settings concerning the rules about which results are used for calculation are defined under **Configuration** > **System** > **General Settings**.

- An LED proposal is displayed only after a completed control cycle (after the end of the month), if the selected calculation method is **Most recent completed control cycle results only**.
- An LED proposal is displayed as soon as 15 days within the current control cycle with accepted QC results are available even if the control cycle is not finished, if the selected calculation method is **Current control cycle results/non-control cycle results also**.

For a more detailed description of the calculation rule settings, see the corresponding description under **Configuration** > **System** > **General Settings**:

- QC result set used for automatic LED proposal calculation
- QC result (per calendar day) for LED proposal calculation
- Numeric value of n (for LED proposal calculation)

Automatic recalculation upon changes on QC result set or calculation rules

An already existing LED proposal may be deleted and recalculated automatically. Two scenarios trigger deletion followed by recalculation:

- If the underlying QC result set has changed that already has an LED proposal associated with it (i.e. there is a new completed control cycle or the current control cycle now contains additional QC results).
- If calculation rules are changed which are already applied to LED proposals.

After deletion, a new LED proposal is automatically calculated based on the current settings.

No automatic recalculation for active LEDs

No automatic recalculation upon change of the QC result set or calculation rule is being triggered, if the related LED has already been accepted and is active or is rejected.

LED proposal: largest possible

When a LED proposal is generated, the largest possible LED is calculated. The largest possible LED is the LED which, if used to calculate an effective QC, would lead to the widest QC range not exceeding the actual QC range. In the List area, the **% of QC Range** column is used to indicate how large the proposed LED is in relation to the largest possible LED value.

Calculation of **Max. LED**

Max. LED is the maximal allowed LED in percent for the selected test, derived from the QC range.

The formula used to calculate the **Max. LED**:

ABS(1-(minRange/Target)*100)

The numerical precision of the **Max. LED** (and consequently of the proposed LEDs) is defined as follows:

- Number of digits before the decimal separator of the target value plus the number of decimal digits of the system test.
- Leading zeros before the decimal separator are not counted.

Example: QC range: 98.0-100.0. Target: 99.0. Significant digits:

3Max LED [%]=(1-98/99)*100=(0.010)*100=1.0

Viewing LED details

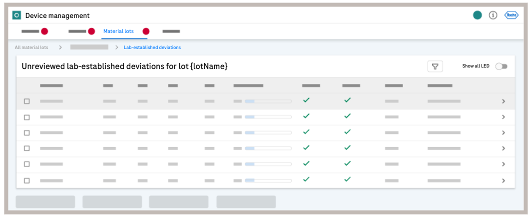
The LED details can be viewed irrespective of an LED status. The details can also be viewed for deleted LED proposals.

1 Choose **Lots**.

2 From the desired QC lot, choose the ⋮ button and choose the **Show LEDs** option.

Show lab-established deviations
If no LEDs exist for a lot, the text **No lab-established deviations available**is displayed without a further link.

→ The **Lab-established deviations** screen is displayed.



3 On the **Lab-established deviations** screen, choose the › button for an LED.

4 Optionally, switch on the **Show all LED** toggle button and/or filter the list ( ).

% of QC Range

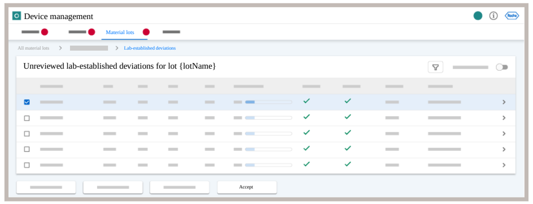
→ The **Lab-established deviations** details are displayed.

5 View the LED details.

Accepting a proposed LED

To become active, proposed LEDs must be accepted. Acceptance can be done automatically for all LEDs or manually for individual LEDs.

▶ To accept a proposed LED manually



1 Choose **Devices** > **Lots**.

2 From the desired QC lot, choose the ⋮ button and choose the **Show LEDs** option.

Show lab-established deviations
If no LEDs exist for a lot, the text **No lab-established deviations available**is displayed without a further link.

→ The **Lab-established deviations** screen is displayed.

3 From the list, select one or more LEDs you want to accept.

4 Alternatively, choose one LED from the list, to display the LED details.

5 Choose the **Accept** button.

▶ To accept proposed LEDs automatically

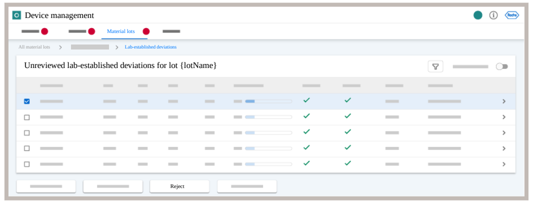
1 Choose **Configuration** > **System** > **General settings**.

2 Choose the **Apply proposed LEDs automatically** setting and choose the **Edit** button.

3 From the **Current Value** drop-down list, choose the **Yes** option.

4 Choose the **Save** button.

Rejecting a proposed LED



1 Choose **Devices** > **Lots**.

2 From the desired QC lot, choose the ⋮ button and choose the **Show LEDs** option.

Show lab-established deviations
If no LEDs exist for a lot, the text **No lab-established deviations available**is displayed without a further link.

→ The **Lab-established deviations** screen is displayed.

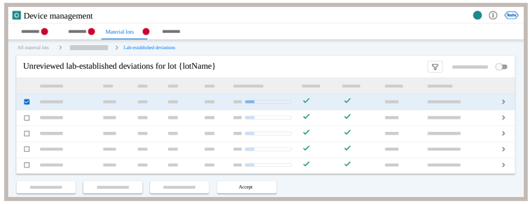
3 From the list, select one or more LEDs you want to reject.

4 Alternatively, choose one LED from the list, to display the LED details.

5 Choose the **Reject** button.

6 On the dialog box, enter a comment and choose the **Reject** button.

Editing an LED manually



1 Choose **Devices** > **Lots**.

2 From the desired QC lot, choose the ⋮ button and choose the **Show LEDs** option.

Show lab-established deviations
If no LEDs exist for a lot, the text **No lab-established deviations available**is displayed without a further link.

→ The **Lab-established deviations** screen is displayed.

3 On the **Lab-established deviations** screen, choose the › button for a LED.

4 Optionally, switch on the **Show all LED** toggle button and/or filter the list ( ).

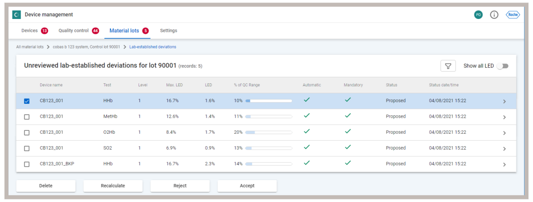
→ The **Lab-established deviations** details are displayed.

5 To edit an LED manually, choose the **Edit LED** option.

6 Enter an LED value and choose the **Confirm** button.

Max. LED

Recalculating an LED



1 Choose **Devices** > **Lots**.

2 From the desired QC lot, choose the ⋮ button and choose the **Show LEDs** option.

Show lab-established deviations
If no LEDs exist for a lot, the text **No lab-established deviations available**is displayed without a further link.

→ The **Lab-established deviations** screen is displayed.

3 From the list, select the LED you want to recalculate.

4 Alternatively, choose one LED from the list, to display the LED details.

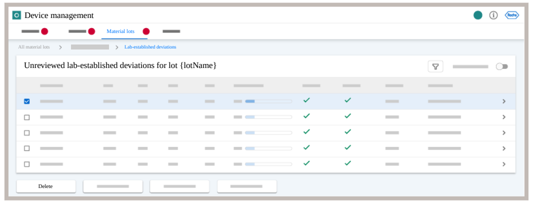
5 Choose the **Recalculate** button.

6 In the **Recalculation criteria** dialog box, enter the following information:

- **Recalculation criteria** - Choose a recalculation criteria from the drop-down list.
- **n (every n-th result number of the day)**: Enter the result number of the day that is used for the LED calculation. This option is only available, when the **n-th result** option is select as recalculation criteria.
- **K (Extension factor)** - Enter the extension factor. By default, the extension factor is set to 3, as defined in the Rili-BÄK regulations. If needed, a different extension factor can be entered for recalculation. The extension factor is used to define an LED that is large enough to provide an effective QC range that is wider than just the target value but does not exceed the actual QC range.

Recalculation criteriaGeneral Settings

Deleting a proposed or rejected LED, or an LED with **No proposal possible** status



1 Choose **Devices** > **Lots**.

2 From the desired QC lot, choose the ⋮ button and choose the **Show LEDs** option.

Show lab-established deviations
If no LEDs exist for a lot, the text **No lab-established deviations available** is displayed without a further link.

→ The **Lab-established deviations** screen is displayed.

3 From the list, select one or more LED's you want to delete.

4 Alternatively, choose one LED from the list, to display the LED details.

5 To delete a proposed or rejected LED, choose the **Delete** button.

6 Choose the **Confirm** button.

→ The entry for a rejected LED is now marked as deleted. User ID, date and time are stored.

The entry for a proposed LED, or LED with **No proposal possible** status, is now deleted permanently.

Settings

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User specific

Configuring the connection interval of a device type

You can configure the connection interval (timeout) of a device type. This configuration is stored in your user profile and is not reset when logging off.

Connection interval

The connection interval refers to the time period within which a device must make at least one successful connection with the application.

If the device fails to connect within this interval, the *connection status* of the device switches to "`error`".

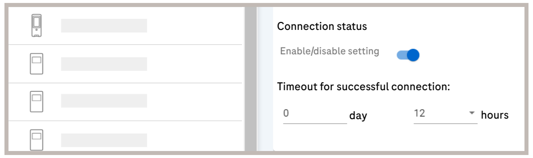
⚠ WARNING

Switching off the connection status setting

When the toggle button for the connection status setting is switched off, the connection status will not be displayed for the devices.

▶ Make sure that the connection status toggle button is switched on.

Connection and QC status



1 Choose **Devices** > **Settings**.

2 Choose a device type.

3 Switch on the **Connection status** > **Enable/ disable setting** toggle button to enable the setting.

4 Enter the time period for the connection timeout.

Configuring the QC timeout of a device type

You can configure the QC timeout of a device type. This configuration is stored in your user profile and is not reset when logging off.

QC timeout

The QC timeout refers to the time period within which a passed QC must be performed on the device.

- If the last passed QC has been performed within the QC timeout period, the QC status is "`OK`" (green).
- If no passed QC has been performed within the QC timeout period, the QC status switches to "`Warning`" (yellow).

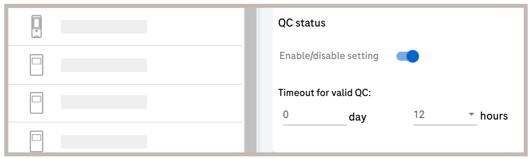
⚠ WARNING

Switching off the QC status setting

When the toggle button for the QC status setting is switched off, the QC status will not be displayed for the devices.

▶ Make sure that the QC status toggle button is switched on.

Connection and QC status



1 Choose **Devices** > **Settings**.

2 Choose a device type.

3 Switch on the **Connection status** > **Enable/ disable setting** toggle button to enable the setting.

4 Enter the time period for the QC timeout.

Quality control

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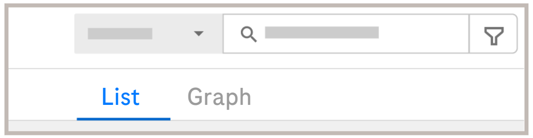
QC view options

The QC results can be displayed by device types. If a device type is selected, the results can be displayed as a list or graph.



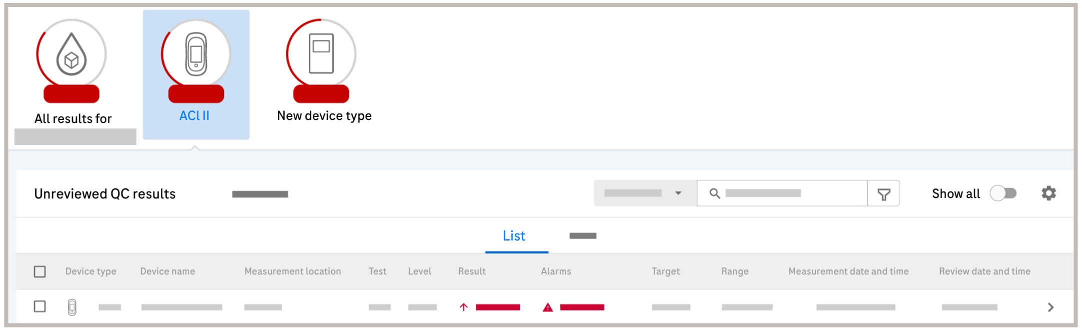
- To switch to the graph view, choose the **Graph** option.
- To return to the list view, choose the **List** option.
- E. g. urinalysis and molecular results are only displayed as a list.

About QC results

The results are ordered by date and time. Only QC results from your sites are displayed.

QC results panel

When you select a test family, the application displays a list of results for it. You can view the results for all devices or for a specific device type.



By default, only unreviewed results are displayed. To see all results, switch on the **Show all** toggle button.

When *all* QC results from a device type are *older than 2 months*, the **QC families** tab does not display the icon of this device type on top of the selected QC family.

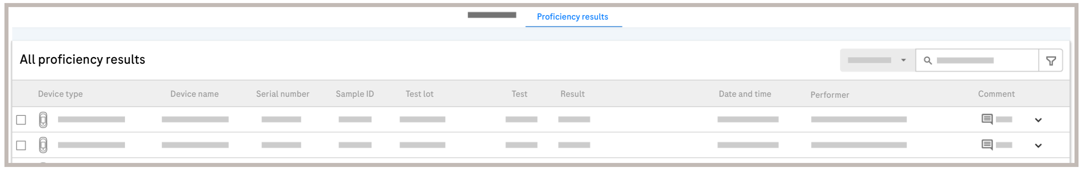
Proficiency results

Proficiency tests refer to tests performed on quality control samples with unknown target values provided by an external agency such as a regulatory authority. The tests are performed without knowing the target values.

The test results are reported to the agency to allow an independent assessment of the laboratory's QC procedures.

The proficiency results are stored in the application. You can view the results, search for individual results by a filter, and add comments to the results.

Viewing proficiency results



Related tasks

- [Searching for QC results (p. 79)]
- [Viewing QC result details (p. 80)]
- [Viewing proficiency results (p. 84)]

Searching for QC results

To narrow the search for QC results, you can apply filters.

By default, QC results from the last 2 months are displayed. To view older QC results, define the date range in the filter.

Manually created QC families are not shown on the **QC families** tab.

Quick search

Search for QC results from a specific device, parameter, or location.

Advanced search

To narrow your search further, use the search filter ( ) which provides additional search criteria including date range, levels, and parameters.

NOTICE

Selecting multiple filter criteria

An error may occur if too many filter criteria are selected.

▶ Reduce the number of selected filter criteria.

Viewing QC result details

▶ To view QC result details from the result list

1 Choose the **Quality control** module and select a test family.

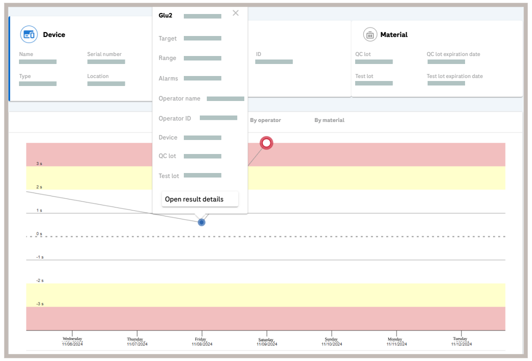
2 Choose a QC result from the result list.

- To search for a QC result from a specific device type, select the respective device icon (for example: ACI II).
- To search for a specific QC result, use the filter.
- To display also reviewed results, choose the **Show all** button.

3 To view the next QC result details, choose the **Next result >** button in the upper right corner.

< Previous result

▶ To view QC result details from the Levey-Jennings chart



1 Choose **Devices** > **Quality control** > **QC families** and choose a test family.

2 Choose a QC result.

3 To view the result details, choose a data point on the Levey-Jennings chart.

→ A pop up is displayed with the result details.

4 To view all the details, choose the **Open result details** button.

Viewing the Levey-Jennings chart

Accepting or rejecting QC results from the result list

Viewing QC data alarms

If the validation of a QC result has failed, you can view details about the violated rules.

1. Choose the **Quality control** module and select a test family.
2. Choose a QC result from the list.
   - To search for a QC result from a specific device type, select the respective device icon (for example: ACI II).
   - To search for a specific QC result, use the filter.
   - To display also reviewed results, choose the **Show all** button.
3. From the **Result details** group box, choose the **Show more** button.

   → The violated rules are displayed.

Viewing the Levey-Jennings chart

Depending on the test family, QC results can be displayed as a *Levey-Jennings chart* or in table format.

- Blood gas results can be displayed in chart or table format.
- Molecular and urinalysis results can only be displayed in table format.
- Results from all other test families can only be displayed in chart format.

Only 2 months of QC results, starting from the most recent result, are displayed in the chart.

Alphanumeric results

Actual *alphanumeric results* cannot be displayed on the chart. Instead, only their date and time are indicated on the chart.

QC level and result status

The followings icons reflect QC level and result status.

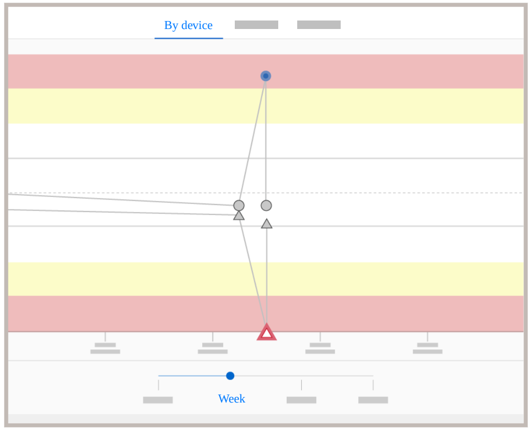
	Lev-el 1	Lev-el 2	Lev-el 3	Lev-el 4	Level 5 and high-er[2]
Current result	○	△	□	◇	☆

2 The level 5 icons are also used to represent QC levels higher than 5.

	Level 1	Level 2	Level 3	Level 4	Level 5 and higher[2]
Selected result					
To review					
Reviewed: rejected					
Reviewed: accepted					
Automatically validated (no issue or warning only)					

Graphic representation of QC level and result status

[2] The level 5 icons are also used to represent QC levels higher than 5.



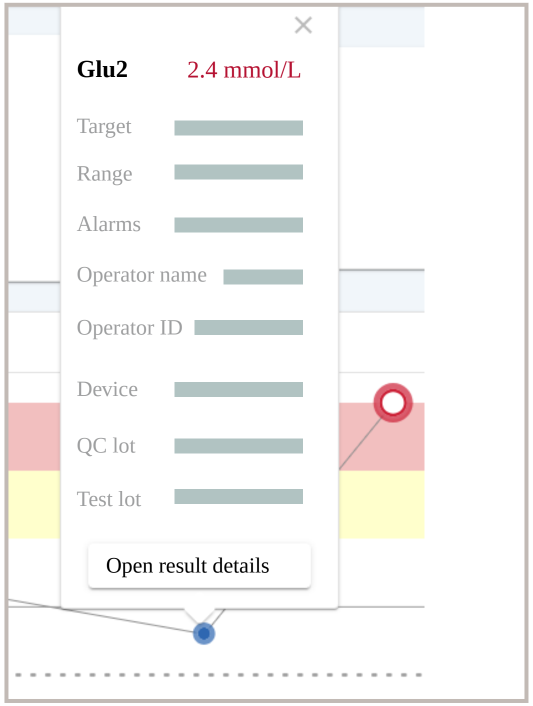
1 Choose the **Quality control** module and select a test family.

2 Choose a QC result from the list.

- To search for a QC result from a specific device type, select the respective device icon (e.g. ACI II).
- To search for a specific QC result, use the filter.
- To display also reviewed results, choose the **Show all** button.

3 On the chart, choose any of the following options to display the results:

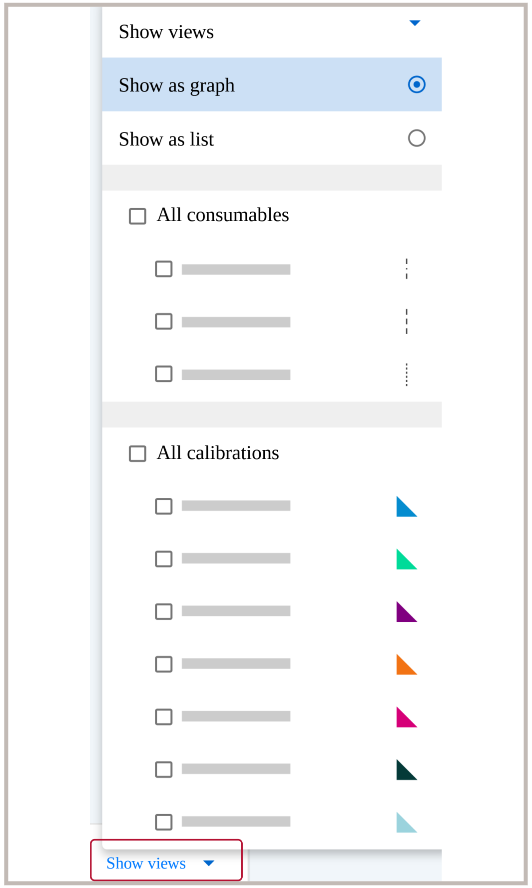
- **By device**: the chart displays all QC results created with the selected device.
- **By operator**: the chart displays all QC results created by the selected operator.
- **By material**: the chart displays all QC results created with the selected material.



4 To view the result details, choose a data point on the chart.

→ A pop up is displayed with the result information.

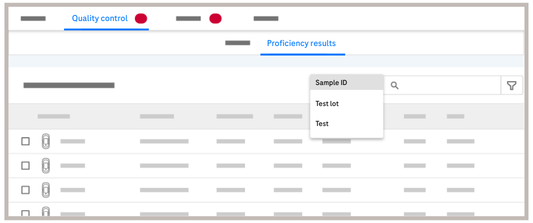
5 To view all the details, choose the **Open result details** button.



6 For certain blood gas results, the **Show views** drop-down list is available providing you the following additional options:

- To display the results in a table, choose the **Show as list** button.
- To display the result in a graph, choose the **Show as graph** option.
- Select the type of consumables and calibrations to be displayed on the graph.

Viewing proficiency results



1 Choose **Quality control** > **Proficiency results**.

2 To search for results based on sample ID, test lot, or test, do the following:

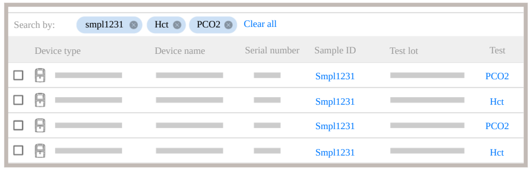
- From the drop-down list, select the desired option.
- Enter the sample ID, test lot, or test name in the search field, and press Enter.

→ The search result is displayed. The entered search term is highlighted in blue.

3 To further narrow your search, choose the button and define the search filter as follows:

- Select the device type.
- Enter the sample ID and press Enter. You can enter multiple IDs.
- Enter the test lot and press Enter. You can enter multiple test lots.
- Define the measurement date range.
- To clear the filter, choose the **Clear all** button.
- Choose the **Search** button.

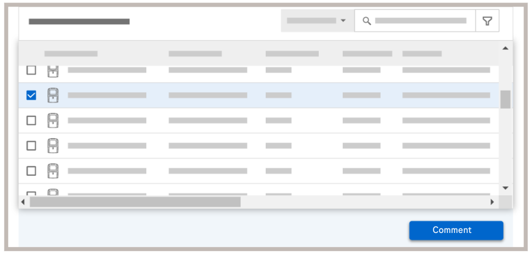
→ The search result is displayed.



4 View the search result.

- The filter criteria are displayed in the header. If you remove a criterion, the search result is automatically updated.
- To clear the filter, choose the **Clear all** button.

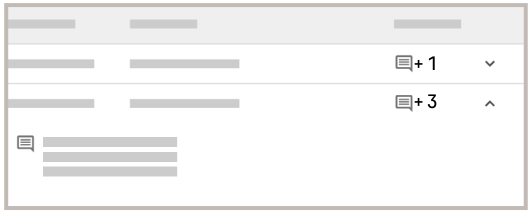
“Smpl1231”



5 To add a comment, select the result, and choose the **Comment** button.

6 On the dialog box, enter the comment and choose the **Add** button.

→ The comment is added.



7 To view alarms or the comments added to a result, choose the button.

Accepting or rejecting QC results from the result list

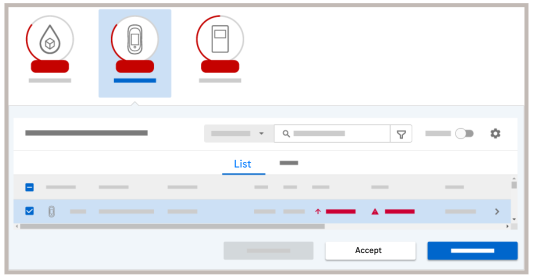
You can manually accept or reject QC results which have not yet been automatically accepted.

If a QC result is accepted, it is taken into account in the validation of the related patient results. Only accepted QC results are included in statistics. Rejected QC results are not included in statistics.

You reject a QC result when the measurement has not been performed correctly. Rejected results are considered to contain no information on device quality.

You can optionally enter a comment at any time, regardless of its status.

▶ To accept a QC result



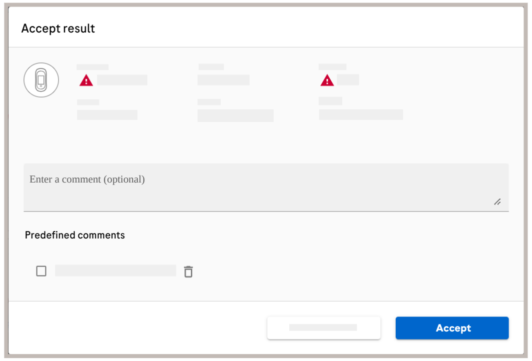
1 Choose the **Quality control** module and select a test family.

- To search a QC result from a specific device type, select the respective device icon (for example: ACI II).
- To search for a specific QC result, use the filter.

2 Select the check box next to the QC result you wish to accept and choose the **Accept** button.

Result details
Already rejected QC results can also be selected, but the action of accepting results only applies to pending and on-hold results.

The **Accept**button is disabled unless there is a pending or on-hold result within the selection.



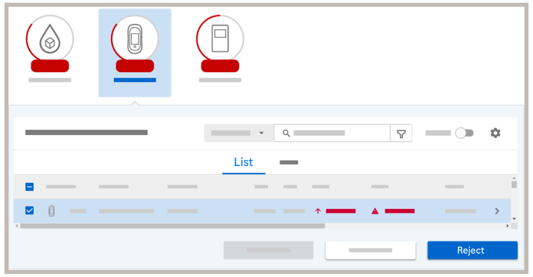
3 In the dialog box, optionally enter a comment or choose one of the **Predefined comments**.

Save as a predefined comment

4 Choose the **Accept** button.

→ Accepted results are automatically sent to the HIS/ LIS.

▶ To reject a QC result



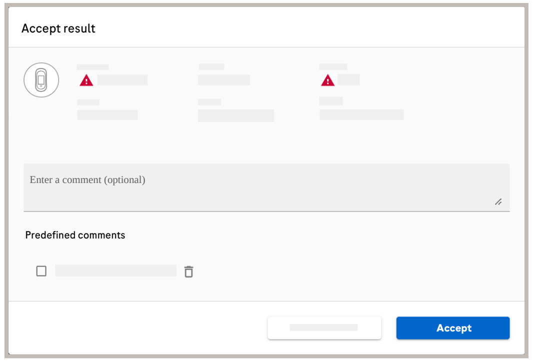
1 Choose the **Quality control** module and select a test family.

- To search a QC result from a specific device type, select the respective device icon (for example: ACI II).
- To search for a specific QC result, use the filter.

2 Select the check box next to the QC result you wish to reject and choose the **Reject** button.

Result details
Already accepted QC results can also be selected, but the action of rejecting results only applies to pending and on-hold results.

The **Reject**button is disabled unless there is a pending or on-hold result within the selection.



3 In the dialog box, optionally enter a comment or choose one of the **Predefined comments**.

Save as a predefined comment

4 Choose the **Reject** button.

→ The QC result is rejected. Rejected QC results are not sent to the HIS/LIS.

Exporting QC results

You can export a list of QC results as CSV file.

In addition to the current filter criteria in the QC results list, you can only limit the export by number of results to be exported. To refine the export, modify the filter criteria.

> Depending on the number of results, the export may take a long time. Exporting more results than the default number (5000) may lead to an error.

About the export file

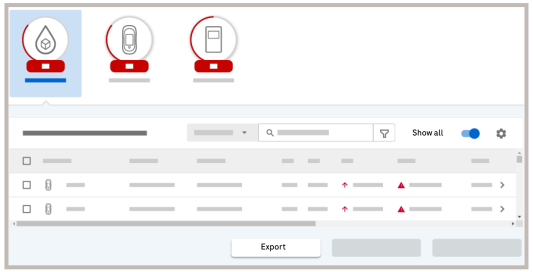
The export file contains the QC results that fulfill the current filter criteria in the QC results list.

The exported file contains the complete QC result information, including details that are not displayed in the result list, e.g., the device serial number.

There are the following differences between the display of information in the QC results list and in the export file:

- QC results that have status “`Unreviewed`” in the QC results list have status “`NotReviewed`” in the export file.
- QC results that have status “`Accepted`” in the QC results list have status “`AutoAccepted`” or “`ManuallyAccepted`” in the export file, depending on who or what accepted the result.
- QC results that have status “`Rejected`” in the QC results list have status “`ManuallyRejected`” in the export file.

- For automatically accepted QC results, “`system`” is listed as the reviewer ID in the export file.
- Depending on your local and the server settings, the date and time formats may differ between the QC results list and the export file.



1 Choose the **Quality control** module and select a test family.

2 Optionally, switch on the **Show all** toggle button and/or filter the result (▽) list as described in

Searching and filtering

.

3 Choose the **Export** button.

4 Optionally, in the **Export as .csv** dialog box, enter the number of QC results to be exported.

Depending on the number of QC results, the export may take a long time. Exporting more than the default number (5000) may lead to an error.

5 In the **Export as .csv** dialog box, choose the **Export** button.

→ A CSV file containing the list of results is downloaded. On iOS devices, the exported file is attached automatically to an email for sending it.

Exporting proficiency results

You can export a list of proficiency results as CSV file.

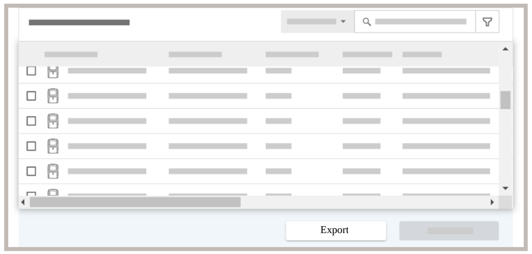
In addition to the current filter criteria in the proficiency results list, you can only limit the export by number of results to be exported. To refine the export, modify the filter criteria.

> Depending on the number of results, the export may take a long time. Exporting more results than the default number (1000) may lead to an error.

About the export file

The export file contains the proficiency results that fulfill the current filter criteria in the proficiency results list.

The exported file contains the complete proficiency result information, including details that are not displayed in the result list, e.g., the device serial number.



1 Choose **Quality control** > **Proficiency results**.

2 Optionally, filter the result list as described in

Searching and filtering

.

3 Choose the **Export** button.

4 Optionally, in the **Export as .csv** dialog box, enter the number of proficiency results to be exported.

Depending on the number of proficiency results, the export may take a long time. Exporting more than the default number (1000) may lead to an error.

5 In the **Export as .csv** dialog box, choose the **Export** button.

→ A CSV file containing the list of results is downloaded. On iOS devices, the exported file is attached automatically to an email for sending it.

Effective QC ranges of Rili-BÄK tests

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About effective QC ranges of Rili-BÄK tests

If you follow Rili-BÄK rules, you can view the effective QC ranges of QC tests for which the allowed deviation is specified in the Rili-BÄK tables and for QC tests for which a lab-established deviation is applied.

Rili-BÄK

Rili-BÄK quality control mode enables institutions to perform quality control procedures accordingly.

About effective QC ranges

In **Quality controlEffective QC ranges**, you can view the effective QC ranges, i.e. the QC validity interval of an instrument test expressed as QC range, irrespective of whether the validity interval is defined by a column 3a deviation, a lab-established deviation (LED), or a QC range.

An effective QC range is displayed for a material lot, level, and test combination, if the following conditions are met:

- The material lot (combination) is active.
- The material lot (combination) and level is used by a device.
- A QC range or numerical target value is defined.
- The system test is active

Effective QC ranges are not used by the application itself; the application evaluates the validity of a QC result always using the defined validity method. Effective QC ranges can be used to align instrument settings with the application's validity interval.

The following information is provided:

Column header	Description
Device type	Type of device used to perform the QC.
Device name	Name of device used to perform the QC.
Test lot	Lot number of the test material used for the QC test.
Material name	Name of test material used for the QC test.
Control lot	Control material lot
Test	(Abbreviated) name of the test with which the alarm is associated.
Level	QC level number
Min	Lower range limit of the effective QC range
Max	Upper range limit of the effective QC range
Unit	Measurement unit
Derived from	The source of the effec-tive QC range (column 3a, actual QC range, LED)

Column header	Description
Valid date and time	Date and time when the effective QC range came into effect.

Calculation of lower and upper range limits

The following example explains how the application calculates the lower and upper limits for the "effective QC range".

The calculation of the range limits depends on the defined precision (number of decimal places) of the test result, and the allowed limits of deviation from the target value (Column 3a limit).

Example calculation

For a given test (i.e. tHb for **cobas b** 123 instrument), the following settings are defined:

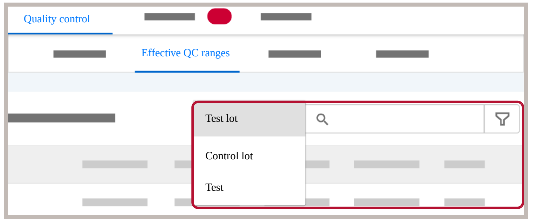
Parameter	Value	Screen
Number of decimal places	1	**Configuration** > **System** > **Test definition**
Column 3a limit	4,0	**Configuration** > **Quality control** > **RiliBÄK test assignment**
Range type	%	

If the instrument sends a target value of 7.2, the application calculates the values for the lower and upper effective QC range limits according to the following algorithm.

With an allowed deviation of 4.0%, **navify**®POC Operations uses 4.04999% (deviation not rounding up to 4.1) and then rounds the upper limit down and the lower limit up as follows:

```
Min.: 7,2 - 7,2*4.04999% = 6,908401
=> 7.0 (round lower limit up) Max.:
7,2 + 7,2*4.04999% = 7.491599 => 7,4
(round upper limit down)
```

Viewing effective QC ranges



1 Choose **Devices** > **Quality control** > **Effective QC ranges**.

2 To search for results based on test lot, control lot, or test, do the following:

- From the drop-down list, select the desired option.
- Enter the test lot, control lot, or test name and press Enter.

→ The search result is displayed. The entered search term is highlighted in blue.

3 To further narrow your search, choose the button and define the search filter as follows:

- Select one or multiple device types from the drop-down list.
- Select one or multiple device names from the drop-down list.
- Enter the test lot and press Enter. You can enter multiple test lots.
- Enter the control lot and press Enter. You can enter multiple control lots.
- Enter the test name and press Enter. You can enter multiple test names.
- Select the QC levels.
- Select from where the range is derived from.
- Select the date and time range for your search.
- To clear the filter, choose the **Clear all** button.
- Choose the **Search** button.

→ The search result is displayed.

4 View the search result.

- The filter criteria are displayed in the header. If you remove a criterion, the search result is automatically updated.
- To clear the filter, choose the **Clear all** button.

Exporting effective QC ranges

You can export a list of effective QC ranges as a CSV file.

In addition to the applied filter criteria in the effective QC ranges list, you can limit the export by number of effective QC ranges to be exported. To refine the export, modify the filter criteria.

> Depending on the number of effective QC ranges, the export may take a long time. Exporting more effective QC ranges than the default number (1000) may lead to an error.

About the export file

The export file contains the effective QC ranges that fulfill the applied filter criteria in the effective QC ranges list.

The exported file contains the complete effective QC ranges information, including details that are not displayed in the result list.

1. Choose **Quality controlEffective QC ranges** and apply filter criteria if needed.
2. Choose the **Export** button.
3. Optionally, in the **Export as .csv** dialog box, enter the number of effective QC ranges to be exported.

   Depending on the number of effective QC ranges, the export may take a long time. Exporting more than the default number (1000) may lead to an error.
4. In the **Export as .csv** dialog box, choose the **Export** button.

> → A CSV file containing the list of results is downloaded. On iOS devices, the exported file is attached automatically to an email for sending it.

RMSE Rili-BÄK alarm monitor

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RMSE Rili-BÄK alarm monitor

The RMSE alarm monitor allows you to view and acknowledge RMSE Rili-BÄK alarms.

RMSE

Root mean square error of measurement.

RMSE alarms and severity levels

The following RMSE alarms exist:

RMSE alarm	Severity	Visualization of severity
QC result on hold	Warning	⚠
Running RMSE outside allowed deviation/ range	Warning	
RMSE outside allowed deviation/range	Error	⚠

RMSE alarm	Severity	Visualization of severity
RMSE repeatedly outside allowed deviation/ range	Severe error	

> **For blood gas devices**
>
> When the predefined value for empty QC results is set to the default “-” or another qualitative value and the QC result is accepted and released, QC results with an empty result value (e.g. “-”) are included in the RMSE warning calculation as if they would have a value of 0.

Searching and viewing RMSE alarms

In the **RMSE alarm monitor** screen, you can search for RMSE alarms by applying a filter.

The following information is provided:

Column header	Description
Device type	Type of device for which the alarm was generated.
Device name	Name of device for which the alarm was generated.
Test lot	The test, QC lot, or combination of test lot and QC lot, of the alarmed time frame.
Test	(Abbreviated) name of the test with which the alarm is associated.
Level	QC level

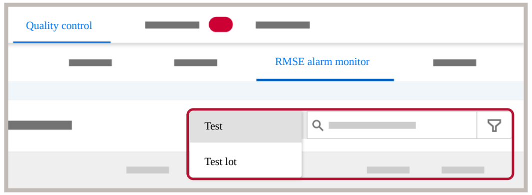
Column header	Description
Alarm	Rili-BÄK RMSE alarm that was generated.
Date and time	The time frame during which the alarm was generated.
Calculated	The calculated RMSE deviation associated with the RMSE alarm. If there is a QC result pending/on hold alarm, the field remains empty.

Column header	Description
Allowed	The allowed RMSE deviation associated with the RMSE alarm. For RMSE alarms triggered for the current control cycle, the value defined under **Configuration** > **System** > **General settings** > **Quality Control (Rili-BÄK)** > **Extension (in percent) of the allowed deviation/range for running RMSE** is applied as a percentage deviation to the allowed RMSE deviation used to evaluate control cycles upon completion. If there is a QC result pending/on hold alarm, the field remains empty.
Method	The method that was used to check the RMSE deviation.

Column header	Description
Acknowledged on	Date and time when the Rili-BÄK alarm was acknowledged.
Acknowledged by	User who acknowledged the Rili-BÄK alarm.
Comment	The calculated RMSE deviation associated with the RMSE alarm. If there is a QC result pending/on hold alarm, the field remains empty.

RMSE alarm monitor column headers

1 Choose **Quality controlRMSE alarm monitor**.



2 To search for results based on test or test lot, do the following:

- From the drop-down list, select the desired option.
- Enter the test name or test lot and press Enter.

→ The search result is displayed. The entered search term is highlighted in blue.

3 To further narrow your search, choose the button and define the search filter as follows:

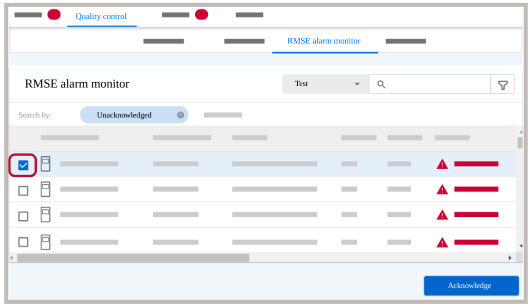
- Select the severity level. You can select multiple levels.
- Select the device. You can select multiple devices.
- Enter the test name and press Enter. You can enter multiple tests.
- Enter the test lot and press Enter. You can enter multiple test lots.
- Select the status: acknowledged and/or Unacknowledged
- Select the date and time range for your search.
- To clear the filter, choose the **Clear all** button.
- Choose the **Search** button.

→ The search result is displayed.

4 View the search result.

- The filter criteria are displayed in the header. If you remove a criterion, the search result is automatically updated.
- If you want to view the comments associated with the RMSE alarm, choose the ⌄ button to expand the row.
- To clear the filter, choose the **Clear all** button.

Acknowledging an RMSE alarm



1 Choose **Quality control** > **RMSE alarm monitor**, and filter for unacknowledged alarms.

2 Select the check box of the desired alarm and choose the **Acknowledge** button.

3 On the dialog box, enter a mandatory comment and choose the **Confirm** button.

Exporting RMSE alarms

You can export a list of RMSE alarms as a CSV file.

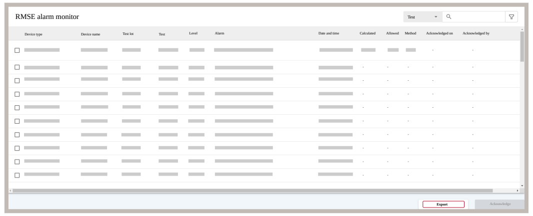
In addition to the applied filter criteria in the RMSE alarm list, you can limit the export by number of RMSE alarms to be exported. To refine the export, modify the filter criteria.

> Depending on the number of RMSE alarms, the export may take a long time. Exporting more RMSE alarms than the default number (1000) may lead to an error.

About the export file

The export file contains the RMSE alarms that fulfill the applied filter criteria in the RMSE alarm list.

The exported file contains the complete RMSE alarm information, including details that are not displayed in the result list, e.g., the alarm severity.



1 Choose **Quality control** > **RMSE alarm monitor** and apply filter criteria if needed.

2 Choose the **Export** button.

3 Optionally, in the **Export as .csv** dialog box, enter the number of RMSE alarms to be exported.

Depending on the number of RMSE alarms, the export may take a long time. Exporting more than the default number (1000) may lead to an error.

4 In the **Export as .csv** dialog box, choose the **Export** button.

> → A CSV file containing the list of results is downloaded. On iOS devices, the exported file is attached automatically to an email for sending it.

Material lots

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Searching for a lot

To find a lot, you can either search by lot number or material name, or use a filter combining multiple search criteria.

1 Choose the **Lots** module.

2 You can search the material lots by lot number or by material name.

Searching and filtering

Show inactive

3 Select the desired lot and perform the desired action (view details, edit, or deactivate).

Clear all

Related tasks

- Viewing lot details (p. 100)

Viewing lot details

The following lot information is provided:

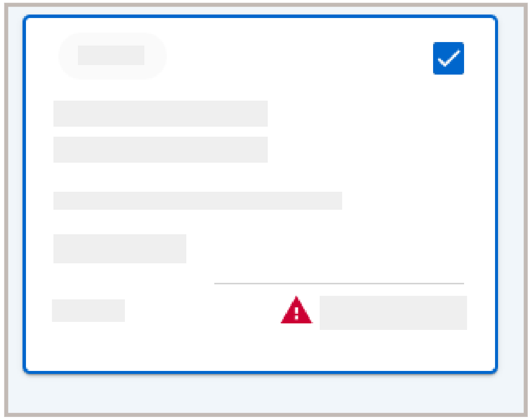
General information:

- Device type
- Lot number
- Material name
- Material long name
- Material type
- Lot provider
- Activation and expiry date
- Assigned locations
- Lot version (for device types supporting lot versions only)

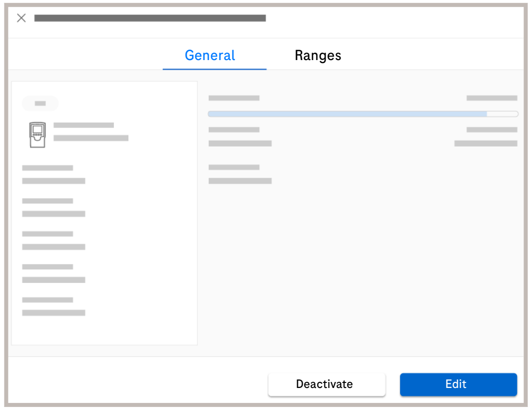
QC and linearity ranges/target values and allowed deviations

1 Choose the **Lots** module.

- To display also inactive lots, choose the **Show inactive** button.
- To view the material lots in list view instead of card view, choose the ▤ button.



2 Choose the check box of the desired lot and choose the **Show details** option at the bottom of the screen.



3 Switch between the **General** and **Ranges** tabs, to view the desired information.

Reviewing linearity results

Use this feature to review linearity runs by means of linearity charts, including target values and allowed deviations. For the identification of linearity runs, the application groups all linearity results of the same device done on the same day using the same lots into a chart.

The screen provides a list view to show all linearity charts that can be generated. It allows you to review all linearity test results in a systematic approach. The review is documented by system, and individual comments can be added to each of the linearity charts.

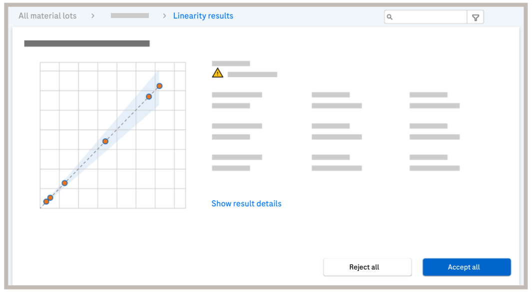
> The review of linearity results is only available after the day you performed the results.

1 Choose the **Lots** module.

- To display also inactive lots, choose the **Show inactive** button.
- To view the material lots in list view instead of card view, choose the ▤ button.

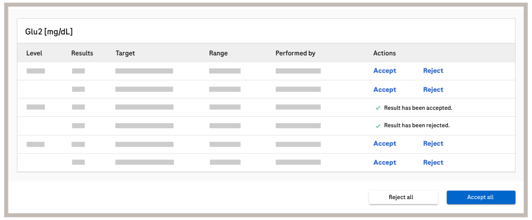
2 Filter the material lots by type by choosing the ⛉ button and choosing **Linearity lot** under **Material type**.

3 Choose the check box of the desired lot and choose the **Show details** button, or choose directly the **Show linearity results** button shown at the bottom of the screen.



4 Review the linearity results of individual devices performed on a specific day and do any of the following:

- To filter the results, choose the ⛉ button. Enter the filter criteria and choose the **Search** button.
- To accept the linearity results, choose the **Accept all** button, optionally enter a comment, and choose the **Confirm** button.
- To reject the linearity results, choose the **Reject all** button, optionally enter a comment, and choose the **Confirm** button.
- To view the result details, choose the **Show result details** button.



5 To accept or reject individual linearity results one at a time, choose the **Accept** or **Reject** button, respectively. Optionally, enter a comment and then choose the **Confirm** button.

> → The system will display in the edited row the message **Result has been accepted.** or **Result has been rejected.**accordingly.

6 The result details are displayed.

Creating a lot

You can create a new lot and assign ranges.

Depending on the instrument's measurement technology, QC ranges/QC target values are maintained with either test lots, QC lots or a combination thereof.

For device types supporting lot versions, absolute glucose deviations are maintained in **Devices** > **Settings** > **Application wide**.

Configuring absolute glucose deviations per lot version for device types supporting lot versions

For some materials the possibility to create lots in the application has been disabled to prevent lot information from being incomplete; e.g. **Accu-Chek® Inform II** glucose test strip lots can only be added on the instruments using the code key reader to ensure that the code key information is part of the lot information stored in the application.

CAUTION

Incorrect material lot configuration

Incorrect configuration of material lots can lead to false alerts, incorrect reference ranges, and delayed patient results.

- Ensure that manually entered lot number and expiry date are correct.
- Ensure that ranges are correctly defined.
- Ensure that the lot is assigned to the correct locations.

For specific device types (depending on the used instrument connectivity platform) it is possible to define default ranges. These default ranges are applied if the device does not send actual QC ranges and when processing the first result of a new material lot (combination).

These default range lot numbers need to start with "`DEF-RANGES-`"followed by the material ID.

For example, the lot number of a default range lot for material ID "`RAD90 OB-Q`"would be "`DEF-RANGES-RAD90 OB-Q`".

These default ranges are only applied for matching tests and result units.

- Device types have been defined in **Configuration** > **Devices**.
- Tests have been defined in **Configuration** > **Test definition**.
- Materials have been defined in **Configuration** > **Material definition**.

1 Choose **Devices** > **Lots** and choose the **Add new** button.

2 Select a device type and choose the **Next** button.

3 Enter and choose the details and choose the **Next** button.

4 Optionally, If you want to *activate* the lot and send it to the assigned devices immediately after its creation, switch on the **Lot in use** toggle button.

Lot in use

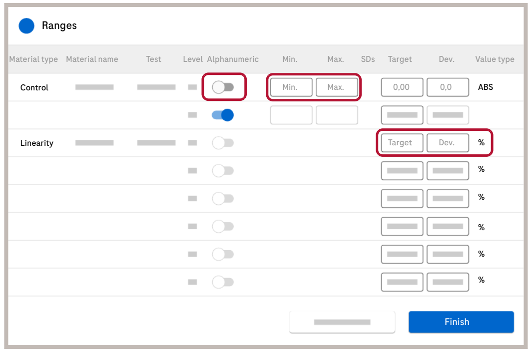
5 Assign the desired locations and choose the **Save** button.

6 On the dialog box, choose the **Confirm** button.

Defining ranges for material lot tests

For some Materials when creating a lot material test, the **Ranges** section is available. In the **Ranges** section you can define the ranges for tests with numeric results and tests with qualitative results.

▶ To define ranges for a lot with numeric results



1 Follow steps 1 to 5 of the
Creating a lot
procedure.

2 Make sure that the **Alphanumeric** toggle button is inactivated (gray).

3 For control material, enter the minimum and maximum values.

> → Target and deviation are automatically calculated. In Rili-BÄK mode, you can only enter the target value for listed tests.

4 For linearity material, either define the target and deviation or enter the minimum and maximum values.

> → Target and deviation are automatically calculated if the minimum and maximum values are entered.
>
> Minimum and maximum values are automatically calculated if the target and deviation are entered.

5 Choose the **Finish** button.

▶ **To define ranges for a lot with qualitative results**

1 Follow steps 1 to 5 of the Creating a lot procedure.

2 Make sure that the **Alphanumeric** toggle button is activated (blue).

3 Enter the target value.

> If multiple target values are entered, they have to be separated by the special character **;** .

4 Choose the **Finish** button.

Editing a lot

You can edit the following information of an existing lot:

- Activation date
- Expiry date
- Assigned locations
- QC ranges/target values

CAUTION

Incorrect material lot configuration

Incorrect configuration of material lots can lead to false alerts, incorrect reference ranges, and delayed patient results.

▶ Ensure that manually entered lot number and expiry date are correct.

▶ Ensure that ranges are correctly defined.

▶ Ensure that the lot is assigned to the correct locations.

1 Choose **Devices** > **Lots**.

- To view the material lots in a table view instead of a card view, choose the button.

2 Choose the check box of the desired lot and choose the **Edit** button at the bottom of the page.

3 Make the desired changes, and choose the **Save** button.

4 On the dialog box, choose the **Confirm** button.

Deactivating a lot

You can manually deactivate an active lot.

▶ To deactivate a lot manually

1 Choose the **Lots** module.

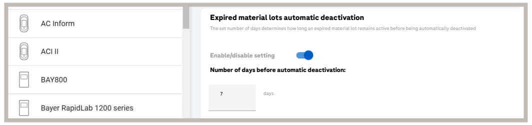
2 Choose the check box of the desired lot and choose the **Deactivate** button at the bottom of the page.

3 In the dialog box, choose the **Confirm** button.

▶ To deactivate a lot automatically

1 Choose **Lots** > **Settings**.

2 Choose a device type.



3 In the **Expired material lots automatic deactivation** dialog box, switch on the toggle button to enable the setting.

4 To deactivate a material lot after the specified period, enter a number of days between 0 and 365. The default value is set to 7 days.

Activating an inactive lot

You can activate an inactive lot.

1 Choose **Lots**.
   - To view the material lots in a table view instead of a card view, choose the ▤ button.

2 To display inactive lots, choose the **Show inactive** toggle button.

3 Choose the check box of the desired inactive lot and choose the **Activate** button at the bottom of the page.

4 On the dialog box, choose the **Confirm** button.

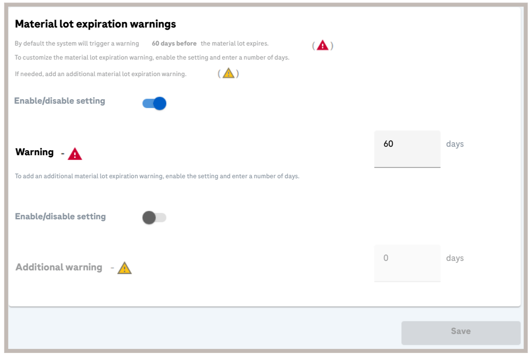
Material lot expiry warnings

You can configure the **Material lot expiration warnings** of a device type.

The **Material lot expiration warnings** refer to the time period before the material lot expires.

- The **red warning icon** ⚠: By default, the system notifies you 60 days before the expiration date.
- The **yellow warning icon** ⚠: This is an additional warning to get an earlier reminder.

These warnings help you to track how many days remain before a material lot expires.



1 Choose **Lots** > **Settings**.

2 Choose a device type.

3 In the **Material lot expiration warnings** dialog box, switch on the toggle button to enable the red warning setting.

> The system notify you with a red warning ⚠ by default 60 days before a material lot reaches its expiration date.

4 To adjust the notification period for the material lot expiration date, enter a number of days between 0 and 365.The default value is set to 60 days.

5 Switch on the toggle button to enable the yellow warning setting.

6 To adjust the notification period, enter a number of days between 0 and 365.

7 Choose the **Save** button.

Configuring lot management

You can configure for a device type that added material lots are activated automatically.

By default, you must activate material lots manually.

Enabling automatic activation affects only material lots that are added afterwards. You must still activate material lots that already exist.

POCC user profile or Service user profile

1 Choose **Lots** > **Settings**.
2 Choose a device type and choose the **Edit** button.
3 Do one of the following:
   - To configure automatic lot distribution, choose the **Automatic material lot distribution** option.
   - To configure manual lot distribution, choose the **Manual material lot distribution** option.
4 Choose the **Save** button.

Configuring absolute glucose deviations per lot version for device types supporting lot versions

For device types supporting lot versions, you can configure the allowed deviation for glucose QC (i.e. how much a QC result is allowed to deviate from the QC target value) per level and QC material version. These allowed deviations are absolute values i.e. defined in the result unit of the glucose test. This setting does not apply to Rili-BÄK QC.

1. Choose **Lots** > **Settings**.
2. Choose a device type and choose the **Edit** button.
3. In the **Absolute glucose deviation** group box, enter the deviation for each level and QC lot version.
4. Choose the **Save** button.

Operator management

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Operators

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Advanced and quick search filters

To narrow down the search, advanced and quick search filters can be applied to the operators list or table.

Quick search

Search from a specific device, parameter, or location.

Advanced search

To narrow your search further, use the search filter ( ) which provides additional search criteria including date range, levels, and parameters.

> If more than one option is chosen for one filter criterion, results for all chosen options will be displayed.
>
> If several filter criteria are applied, only results matching all filter criteria will be displayed.

NOTICE

Selecting multiple filter criteria

An error may occur if too many filter criteria are selected.

- Reduce the number of selected filter criteria.

Searching for operators and viewing operator details

You can search for operators registered in the application and view their details.

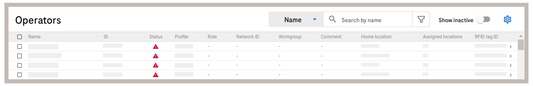
Operators are grouped in tabs according to their certification status.

To search for operators, you can enter an operator name, ID, or RFID tag ID in the search field or browse through the list. You can also search for operators by their certification status (all, upcoming, or expired) on the corresponding tab.

> Also when the certification status is valid, a tab of the operator's certification may display unfulfilled certification criteria because the counter is set back to 0 immediately when the criteria are fulfilled and not when the certification status changes to upcoming again.

▶ To search for an operator by name, ID or RFID tag ID

1 Choose **Operator management** > **Operators** > **By name**.



2 Search for the operator in the list, or enter his/her name, ID (or a part of it), or RFID tag ID (in full) in the search field.

- By default, only active operators are displayed. To display inactive operators, choose the **Show inactive** button.
- To display only operators with upcoming certification status, open the list view and choose the **Upcoming** tab.
- To display only operators with expired certification status, open the list view and choose the **Expired** tab.

3 In the **Certifications** area, choose a device type to view the status of the operator's certifications.

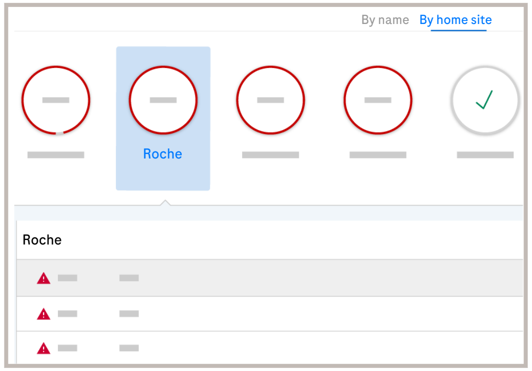
- Certification criteria are displayed as tabs.
- If an operator has no active certification assigned, the **Certifications** area is not displayed.

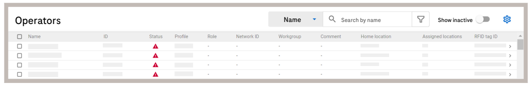
4 In the **Operator details** area, you can view the available operator information such as the assigned locations.

▶ To search for an operator by location

1 Choose **Operator management** > **Operators** > **By home site**.

2 Choose a site and a location.





3 Search for the operator in the list, or enter his/her name or ID (or a part of it) in the search field.

- By default, only active operators are displayed. To display inactive operators, choose the **Show inactive** button.
- To display only operators with upcoming certification status, open the list view and choose the **Upcoming** tab.
- To display only operators with expired certification status, open the list view and choose the **Expired** tab.

4 In the **Certifications** area, choose a device type to view the status of the operator's certifications.

- Certification criteria are displayed as tabs.
- If an operator has no active certification assigned, the **Certifications** area is not displayed.

5 In the **Operator details** area, view the available operator information such as the assigned locations.

Adding an operator

When adding an operator, a wizard guides you through the process.

User ID requirements

The user ID can contain uppercase or lowercase letters (A to Z), numbers (0 to 9), and the additional characters +,/,_,-,$,\,@.

NOTICE

The user ID is case sensitive

Users with the ID "A123" and "a123" are considered as two different users by the application.

▶ It is strongly recommended to use consistent casing (i.e. only uppercase or only lowercase letters) for all user IDs on the application and on all associated devices.

▶ Check the user IDs after import and change the casing if required.

Different device types may have specific limitations or workflow disadvantages regarding the use of uppercase or lowercase letters in user IDs.

Password requirements

By default, a valid password must contain the following:

- Number of characters: 8
- Number of uppercase characters: 1
- Number of digits or symbols: 1

Only locations from the same site as the user performing the action are displayed and can be assigned to the new operator.

1 Open **Operator management** > **Operators** and choose the **Add operator** button.

On the By home site tab, Add operator In list view, the **Add operator** button is located under **More actions**.

2 Follow the wizard and enter or assign the following information:

- **Operator details**: ID, first name, last name, and password are mandatory.
- **Profile**: you can only assign 1 profile per operator.
- **Locations**: you can assign multiple locations.
- **Certifications**: select a certification and do the following:- Define the stage: certified or pending, interim if defined for this specific certification.- Define the certification start date.

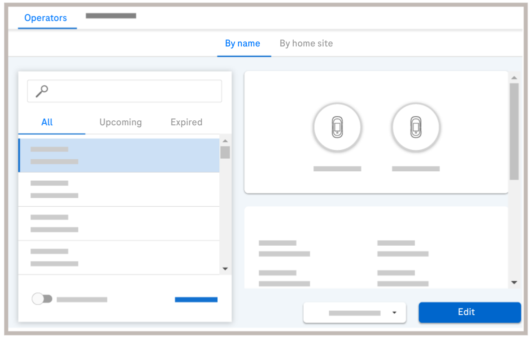
3 Choose the **Finish** button.

4 On the dialog box, choose the **Confirm** button.

Editing an operator

You can edit the operator details, the assigned profile and locations, and you can remove or add certifications.

Only locations from the same site as the user performing the action are displayed and can be assigned to the operator being edited.



1 Choose **Operator management** > **Operators**.

2 Select an operator and choose the **Edit** button.

3 Follow the wizard and edit the desired information.

- Edit the operator details.
- Change the profile.
- Change the assigned location.
- Remove a certification by choosing the **Remove** button twice, or add a certification by choosing the **Add new certification** button.

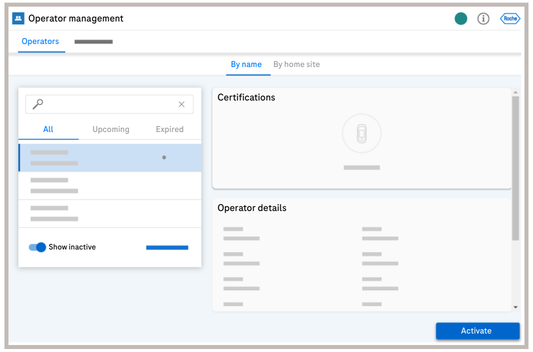
4 If you add a certification, select type, stage, and start date of the certification, and choose the **Add** button.

5 Choose the **Save** button.

6 On the dialog box, choose the **Confirm** button.

Activating and deactivating an operator

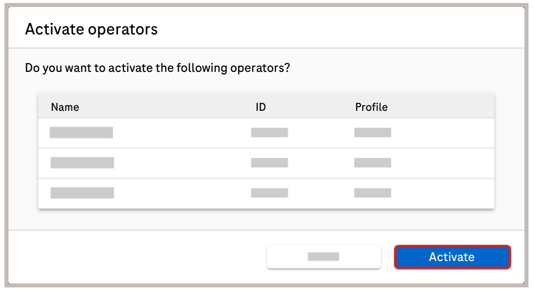
▶ To activate an operator



1 Choose **Operator management** > **Operators**.

2 To display inactive operators, choose the **Show inactive** toggle button.

3 Choose an inactive operator.

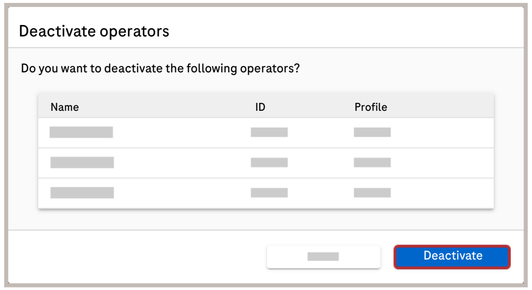


4 Choose the **Activate** button.

> If multiple operators were chosen, a new window listing the chosen operators is displayed.

5 Choose the **Activate** to confirm.

▶ To deactivate an operator



1 Choose **Operator management** > **Operators**.

2 Select an active operator and choose the **Deactivate** button.

In table view, it is possible to choose activate and deactivate multiple operators.

3 Choose the **Deactivate** button.

> If multiple operators were chosen, a new window listing the chosen operators is displayed.

4 Choose the **Deactivate** button.

Resetting the password of an operator

1 Choose **Operator management** > **Operators**.

2 Select an active operator and choose the **Reset password** button.

3 Choose the **Confirm** button.

→ The password is reset to the default password.

Exporting operators

A list of operators can be exported as a CSV file.

In addition to the current filter criteria in the operators list, you can also limit the export to a defined number of records. To refine the export, modify the filter criteria.

Depending on the number of operators, the export may take a long time. Exporting more operators than the default number (1000) may lead to an error.

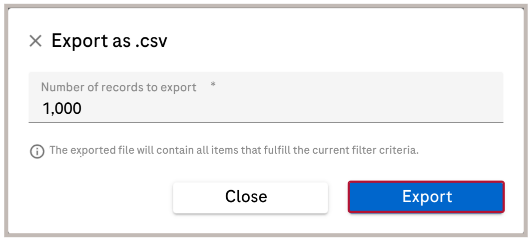
About the export file

The export file contains the operators that match the current filter criteria in the operators list.

1 Choose **Operator management** > **Operators** and choose the **Export operators** button.

By home sitetab,**Add operator**
In list view,the **Add operator**button is located under **More actions**.

2 Optionally, in the **Export as .csv** dialog box, enter the number of operators to be exported.



3 Choose the **Export** button.

→ A CSV file containing the list of operators is downloaded. On iOS devices, the exported file is attached automatically to an email.

Certifications

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Finding and exporting operators needing a certification update

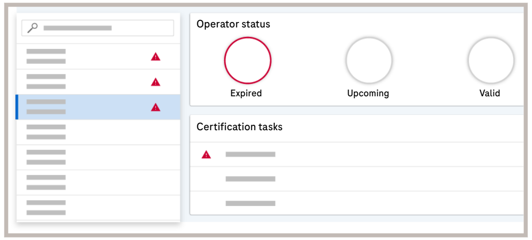
You can export a list of operators with expired or upcoming certification status to an HTML or XLS file, and then send it via email.

When you use the application on an iOS (Apple) tablet, by default the email app adds automatically the application's link and the type of file to the email body. This is not harmful, it is normal behavior.

Before sending the email, delete the link manually.

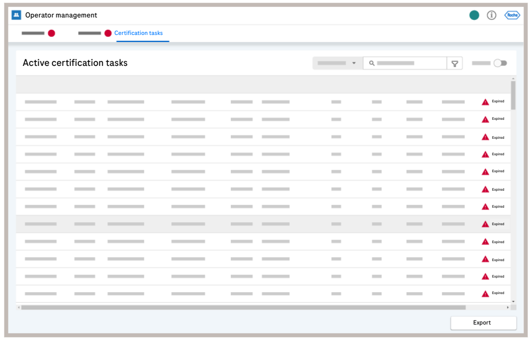
For more information, contact your Roche representative.

Operator status



The operator status of a certification can have one of the following 3 severity levels:

- ⚠ *Expired*: certification tasks under this certification have expired.
- ⚠ *Upcoming:* certifications tasks under this certification expire soon.
  Valid: All certifications tasks under this certification are valid.



1 Choose **Operator management** > **Certifications**.

2 Choose a certification with a ⚠ or ⚠ icon.

3 From the **Certification tasks** dialog box, filter by a criterion for which operators have a status of expired (⚠) or upcoming (⚠).

→ A list with all operators (name and ID) who require training for the selected criterion is displayed.

You can then export the complete list of results based on the applied filter.

4 Choose the **Export** button.

On iOS devices, the exported file is attached automatically to an email for sending it.

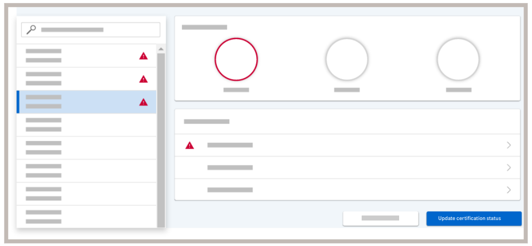
A CSV file containing the list of operators with valid, expired, or upcoming certification status for the selected criterion is downloaded. Operators with valid certification status are only listed when the certification is in pending stage.

Updating the certification status of operators

With the appropriate observer rights, you can update the certification status of operators if the status is not auto-calculated.

After operators have completed a training permitting them to operate a POC device, you can update their certification status in the application.

Observer rights for the operators and criteria for which you want to update the certification status.



1 Choose **Operator management** > **Certifications**.

2 Choose a certification and choose the **Update certification status** button.

Certifications are sorted as follows: 1) status, 2) device type, 3) certification name.

3 Select the criteria you want to update and choose the **Next** button.

A list of operators who can be updated is displayed.

4 From the list of operators, select all the operators you want to certify.

- To search for operators, you can enter an operator name or ID in the search field or browse through the list.

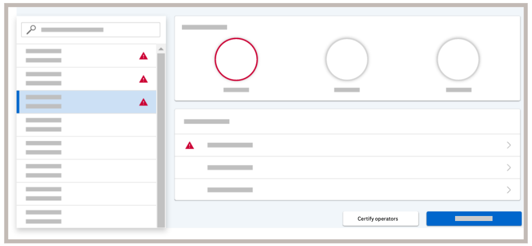
5 After selecting all operators, choose the **Finish** button.

6 On the dialog box, check your selection and choose the **Confirm** button.

Certifying operators

You can certify operators for a POC device.

> Observer rights for the operators you want to certify



1 Choose **Operator management** > **Certifications**.

2 Choose a certification and choose the **Certify operators** button.

3 Select certification start date and stage, and choose the **Next** button.

> → A list of operators who can be certified is displayed.

4 From the list of operators, select all the operators you want to certify.

5 After selecting all operators, choose the **Finish** button.

6 On the dialog box, check your selection and choose the **Confirm** button.

Importing exam results form a third-party LMS

The application can import results exported from a *learning management system* (LMS). This allows you to have exam results of multiple users updated simultaneously.

> To save storage space, delete error files after successful import of the exam results.

NOTICE

Failed LMS exam result imports due to expiring certifications

Expiring certifications and not updated exam tasks in the application can lead to failed imports of LMS exam results.

- In the **LMS import errors** menu, regularly check that the exam name matches the definition in the application.
- Check that the user’s first and last name match the names in the application.
- Check that the exam date and time is in the OTF of the user's certification.
- Check that the exam result values in the Lms.Config file match the values provided by the third-party LMS.

Exam result input file

The following rules apply to the exam results input file:

- The exam results input file must be saved as plain text file (e.g. TXT or CSV file).
- The content of the input file must be UTF-8 encoded.
- All string columns in the input file are case-insensitive.
- The Windows CR/LF line breaking character (i.e., the [Return] key) must be used as row separator.
- The input file must at least contain columns with the following user data (one row per user):

First name column	First name of user as defined in the application.
Last name column	Last name of user as defined in the application.
Course name column	The name of the training course, as listed in the Course Name field of the certification in the application The exam must be a certification task for the corresponding user for successful exam results import.

Exam date column	Date and time when the exam was taken. The exam date must be in the observation timeframe of the corre-sponding cer-tification for successful ex-am results im-port.
Exam re-sult col-umn	Result of the exam

- The formatting of the user data in the input file depends on an LMS configuration file set up by Roche Service.
- The certification task status and certification status are updated with the date and time of LMS file import (not the date and time of the eLearning exam outcome in the LMS file).

- Make sure to use the right formatting in the input file. Ensure with the third-party LMS administrator that data is exported from the LMS with the right formats, especially regarding the exam date.
- Contact Roche Service, if the LMS configuration file needs to be changed.
- Do not use double-quotes to encapsulate spaces or escape the column delimiter. The application does not recognize quoted/escaped field values in the input file.

The following figure shows an example of an input file using the default formatting:

FirstName	LastName	CourseName	ExamDate	ExamResult
BOBBIE	HETRICK	Training 1	2019-05-19 09:52:00	PASS

NOTICE

Incorrect data

Mistakes in the LMS input file can result in erroneous or failed exam results import, leading to incorrect data in the application.

- Make sure that the input file contains the correct user data in the right formatting.
- Check correct import of exam results.

1 Place the exam results input file into the following folder on the application server:
   - *<drive:>\cobasIT1000\Services\Lms\Import*(<drive> = drive on which the application is installed).
   - After successful import, the certification tasks are updated.

2 Check the status of the certifications for the operators in **Operator management** > **Operators**.

Certification tasks

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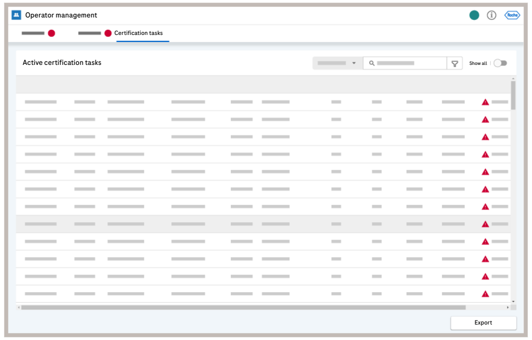
Viewing and exporting the status of certification tasks

Use this component to view the status of the certification tasks for operators who need to maintain their certifications. You can export the list of certification tasks that is displayed on the screen into a CSV file.

About the export file

The export file contains the records that fulfill the current filter criteria in the records list. The exported file contains the complete record information, including details that are not displayed in the list,



1 Choose **Operator management** > **Certification tasks**.

2 Optionally, switch on the **Show all** toggle button and/or filter the result list as described in
Searching and filtering

3 Choose the **Export** button.

4 Optionally, in the **Export as .csv** dialog box, enter the number of records to be exported.

Depending on the number of records, the export may take an extended period of time. Errors may occur if you export more records than the default number (1000).

5 In the **Export as .csv** dialog box, choose the **Export** button.

→ A CSV file containing the list of records is downloaded.

Resetting the exam attempts counter

When failing the predefined number of exam attempts, the counter can be reset.

1 Choose **Operator management** > **Certification tasks**.

2 Choose the **Reset count** button next to the number of test attempts.

→ The attempts counter is set to 0.

Advanced operator management

About the Advanced operator management module

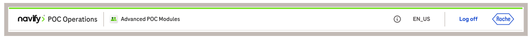
The Advanced operator management is a module of **navify** ®POC Operations for managing part of the professional Point of Care (POC) testing service. It ensures governance and monitoring of operators performing Point of Care testing, as well as managing their adherence to training and regular competency assessment.

Advanced operator management can be used on desktop computers, tablets, and mobile devices.

The main features are the following:

- Automation of adding, updating and deactivating operators through active directory integration or HR system integration, including automation rules managing those operators.
- Comprehensive certification management.
- Digitalisation of competenecy checks and reporting.

Global information area structure



In the global information area, you can access the About box, application settings, system language configuration, and can log off from the system.

About the Settings menu

To access the About box

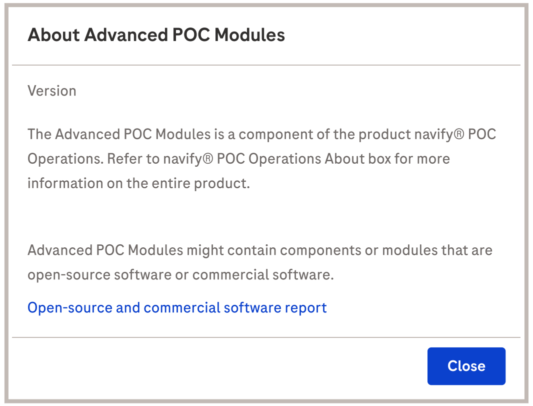
The About box shows software version of the Advanced operator management, and short description of the product. Additional information about the product is available in the **navify**®POC Operations About box.

1 From the global information area, choose the ⓘ button.

About Advanced POC modulesOpen-source and commercial software report

→ The **About Advanced POC modules** dialog box is displayed.

2 To exit the About box, choose the **Close** button.



To access the Settings menu

1 From the global information area, choose the ⚙ button.

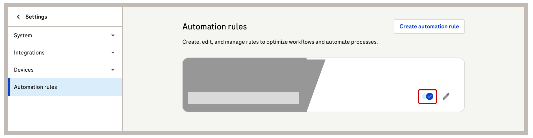
→ The **Settings** menu is displayed.

2 From the **Settings** menu, choose other available settings options.

Automation rules settings

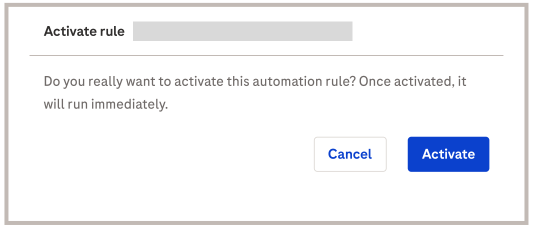
Automation rules allow users to create, edit, and manage rules to optimize workflows and automate processes. These settings display a list of all existing automation rules. Each rule entry includes the rule name, a toggle button to activate or deactivate it, and an edit icon.

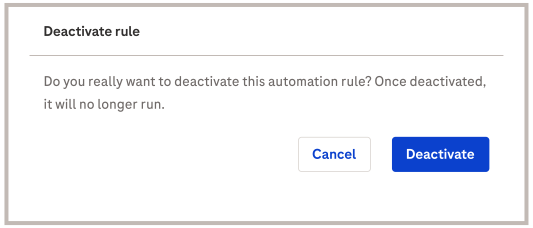
▶ To activate or deactivate an existing rule



1 Choose the toggle button associated with the rule that you want to activate or deactivate.

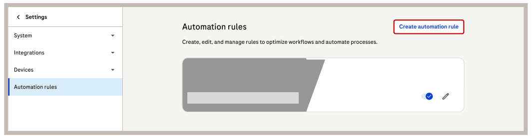
→ Depending on your choice, the **Activate rule** or the **Deactivate rule** screen opens.

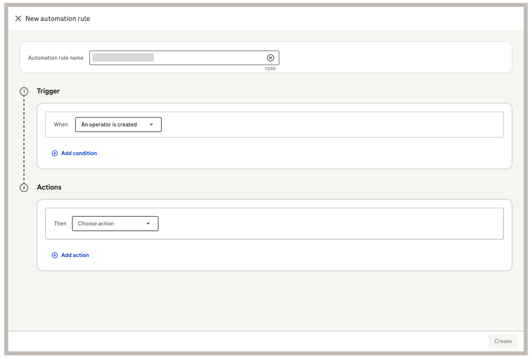




2 Choose the **Activate** or **Deactivate** button, depending on your choice.

▶ To create new automation rules





1 Choose the **Create automation rule** button.

→ The **New automation rule** screen opens.

2 Fill out the **Automation rule name**.

ⓘ All fields are mandatory.

3 The **Trigger** section is automatically preselected and cannot be edited. If necessary, you can add a condition by choosing the **Add condition** button.

ⓘ
- ▶ It is possible only to add conditions that are fully validated before executing the selected actions. To create rules for contradictory conditions, create a separate rule for each one.
- ▶ It is possible to delete all conditions as they are optional. It is also possible to create multiple conditions for the same field.

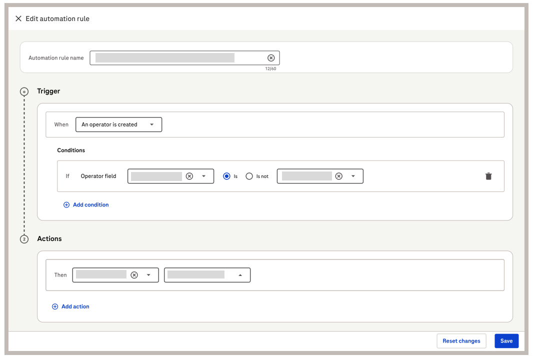
4 From the **Actions** section, choose the action. If necessary, you can add a new action by choosing the **Add action** button.

5 To save the new automation rule, choose the **Create** button.

▶ To edit an automation rule

1 Choose the ✎ button next to the rule that you want to edit.

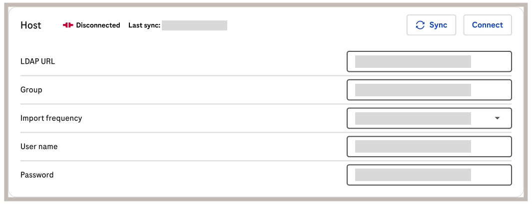
→ The **Edit automation rule** screen opens.



2 Edit all required fields and then choose the **Save** button. To cancel the changes, close the screen, or choose the **Reset changes** button. The **Reset changes** button brings back the last saved configuration.

Active Directory settings

Active Directory settings allow you to configure the synchronization between the middleware and the Active Directory service.



Active Directory settings are used to make sure that user information from Active Directory is correctly imported and mapped to the internal system fields. This process is crucial for authentication, user management, and access control.

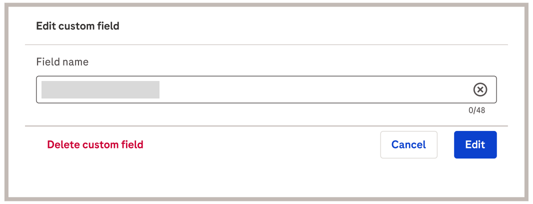
Active Directory stores user information and system information. These settings enable administrators to do the following:

- Connect and authenticate to an Active Directory server via LDAP.
- Set the frequency of data imports from Active Directory.
- Map the mandatory fields of the system, such as badge ID, first names, last names, and password, to their corresponding Active Directory attributes.
- Add additional mappings for other data points in Active Directory.
- Manage and synchronize data from Active Directory to the system.

Custom fields settings

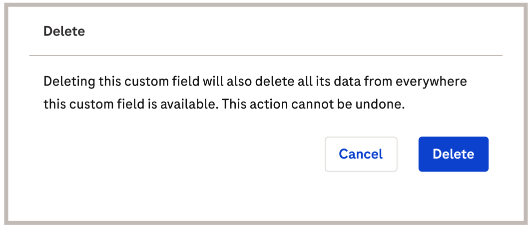
The **Custom fields** settings allow you to create, edit, and manage fields that are displayed in the Advanced operator management module..

Custom fields settings allow you to create additional fields available as optional for each user. You can edit fields, assign them through **All custom fields ({{total}})** and use them as conditions triggers for **Automation rules**



In the **Edit custom field** settings, you can do the following:

- To add a custom field, choose the **Create custom field** button. Enter the custom field name and choose the **Create** button to save the changes.
- To change the name of the existing custom field, choose the field that you want to change. Enter the new name in the **Field name** and choose the **Edit** button to save the changes.
- To delete a field, select the field you want to delete and choose the **Delete custom field** button. The confirmation window opens allowing you to cancel the procedure or confirming it by choosing the **Delete** button.



Training settings

The training settings allow you to manage the training delivery and tracking.

You can enable training locations when setting up a new training using the **Training locations** toggle button.

For the trainings available in the in the application for trainers, you can enable the **In-app training proof of completion** setting. Once enabled, the following options are available for signature collection as proof of completion:

- **Trainer signature**
- **Operator signature**

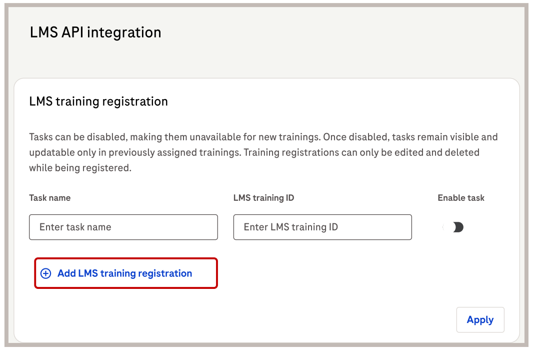
LMS API integration settings

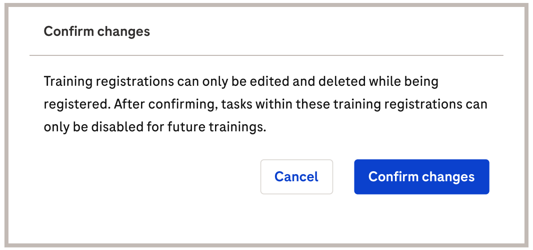
LMS API integration settings is a part of the Training Management settings that manages LMS training registrations and data mapping.

LMS training registration

LMS training registration allows you to manage tasks related to LMS training registration. Tasks can be disabled, making them unavailable for new training sessions. Disabled tasks remain visible and can still be updated for previously assigned trainings. You can edit and delete training registration only while being registered.

▶ To add new LMS training registrations





1 Choose the **Add LMS training registration** button.

2 Fill out the required fields.

3 To save the changes, choose the **Apply** button.

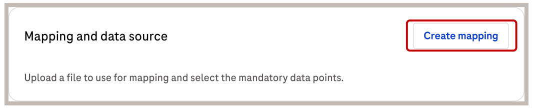
→ The **Confirm changes** dialog box with additional information opens.

4 Choose the **Confirm changes** button.

Mapping and data source

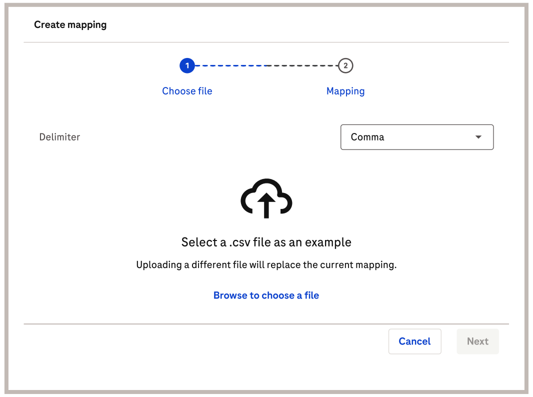
Mapping and data source enables you to manage data mapping. You can upload a file for mapping and choose the required data points.

▶ To map new data source



1 Choose the **Create mapping** button.

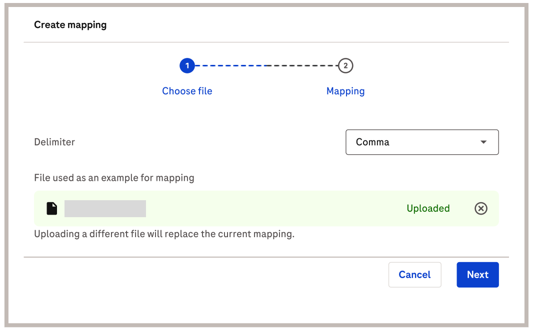
→ The **Create mapping** dialog box opens.



2 From the drop-down list, choose the **Delimiter** parameter.

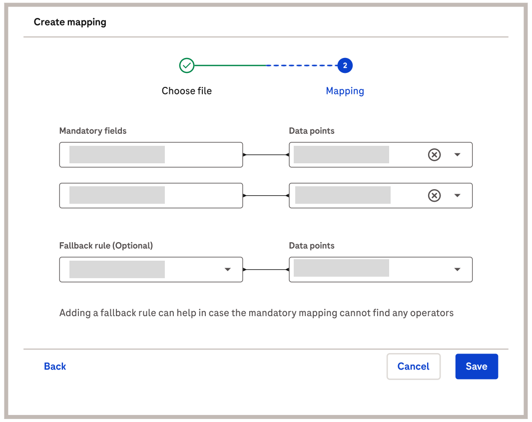
3 To upload a .csv file, choose the **Browse to choose a file** button.

4 Choose the file from your database.



5 After the file is uploaded, choose the **Next** button.

ⓘ The green ribbon indicates that the file is successfully uploaded.



6 Fill the **Mandatory fields** and **Data points**.

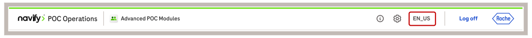
7 To save the changes, choose the **Save** button.

Cookie preferences

You can manage your consent preferences or access to Cookie Notice by choosing the **Cookie preferences** button in the **Settings** screen.

To change the system language

The user interface is displayed in the language defined for each user ID in Advanced POC modules.



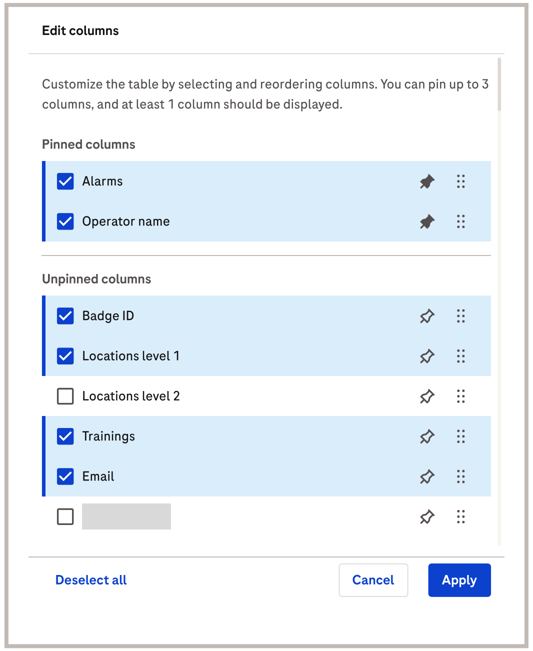
1 In the global information area, choose the user interface language abbreviation.

→ The drop-down list opens with the available language list.

2 Choose your preferred language for the user interface.

→ The user interface language is changed immediately.

Edit columns



To customize the columns, choose the **Edit columns** button.

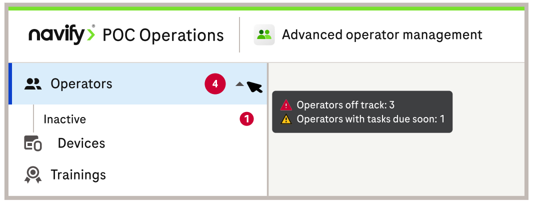
- To hide or show the column, choose or clear the check box next to the column name.
- To reorder columns, choose and hold the icon and drag and drop the item.
- You can pin up to 3 columns, and at least 1 column should be displayed. Once 3 columns are pinned, the rest of the pins become disabled. When unpinning a column, the item will go to the first position of the **Unpinned columns** list.
- To clear all or select all columns, choose the **Select all** button or the **Clear all** button.
- To save the changes, choose the **Apply** button.

At least one column should be visible. If there are not selected columns, the **Apply** button is disabled.

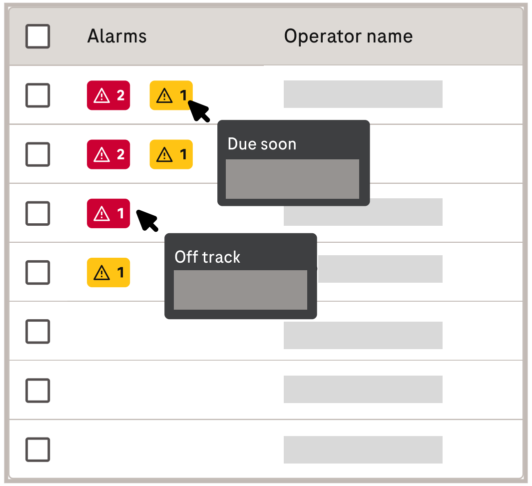
The applied changes are saved for the logged on user as user preferences. Columns are displayed accordingly as default at every logon until edited again.

Operators module

Operators are grouped in tabs according to their status (active/inactive). For easier identification and modification, inactive operators are separated in a different tab. Both operators tabs have the same options and settings. Active operators can also be grouped by applying filters on the list and saving that filtered view for further use.



The **Operators** tab can display notification badges that show the total number of operators with alarms. Alarms indicate **Operators off track** with red badges and **Operators with tasks due soon** with yellow badge. **Operators off track** have a higher priority than **Operators with tasks due soon**. If at least 1 operator has an alarm that indicates **Operators off track**, the notification badge is red. Otherwise, the notification badge is yellow. When hovering over the notification badge, the number of operators associated with each alarm is displayed.



If operators are associated with alarms, their names appear first on the list. In the **Alarms** column, **Operators off track** and **Operators with tasks due soon** are displayed with separate icons. The number within the icon indicates the number of trainings associated with the alarm. If only one type of alarm is present, only 1 icon with the count is displayed. When hovering over the alarm icon, the alarm displays the name of the training that is **Due soon** or operator that is **Off track**.

Filtering options

Filtering options

The Operators table provides the overview of all active and inactive operators. You can filter the displayed list to find specific operators based on various criteria.

The filter bar is located above the table. It contains multiple filter fields that allow the user to narrow down the list of operators. The filter options are:

- **Alarm type**
- **Name / badge ID**
- **Locations level 1**
- **Locations level 2**
- **Trainings**
- custom fields filter

To filter operators by alarm type

You can search for operators registered in the application and view their details by using filtering options.

If there are operators with active alarms, their name will appear automatically first on the list. The operators are sorted first by the type of alarms (first off track, then due soon) and then alphabetically. If 2 operators have the same number of alarms, they will be organized alphabetically.

Off track alarms are represented with the red icon and due soon alarms with the yellow icon. Errors have a higher priority than warnings. The number in the icon represents the number of trainings and/or operators with active alarms. When hovering over the alarm icon, you can see the name of the training with errors or warnings.

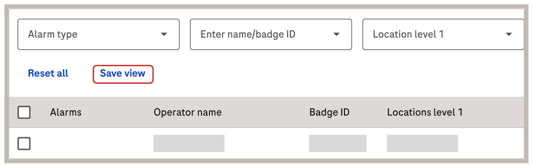
1 Choose the **Alarm type** filter.

2 From the drop-down list, choose the alarm type by choosing the check box next to the alarm type name.

3 Choose the **Apply** button.

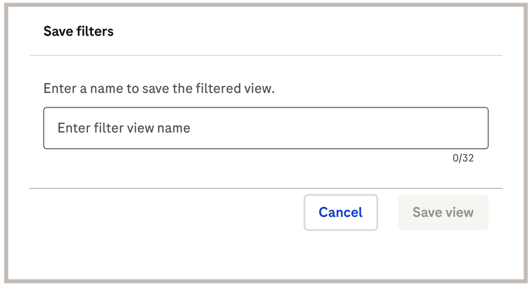
▶ To save the view after filtering

After applying filters, you can save the filtered view. You can access and manage the saved filtered view on the side menu.

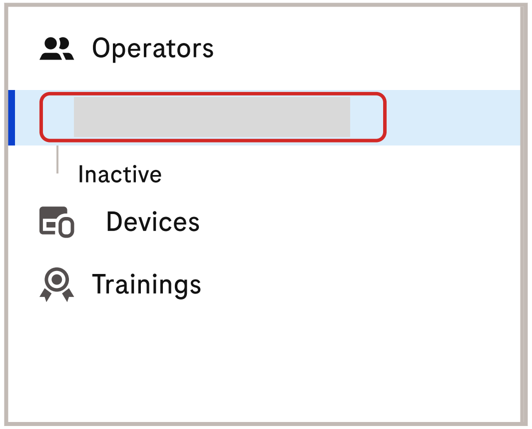


1 After applying filters, choose the **Save view** button.

→ The **Save view** screen is displayed.



2 Enter a name for the filter view.

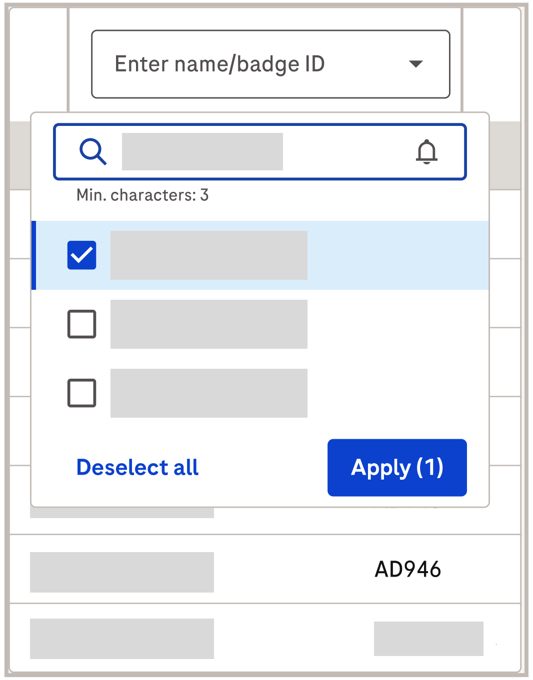


3 Choose the **Save view** button.

→ The filter view is saved on the side menu.

To filter operators by name or ID

You can search for an operator registered in the application and view their details. To search for operators, you can **Enter operator name or badge ID** in the search field of the **Name / badge ID** filter.



1 Choose the **Name / badge ID** filter.

2 Enter the operator name or the badge ID.

3 Choose the operators from the list by choosing the check box next to the operator name.

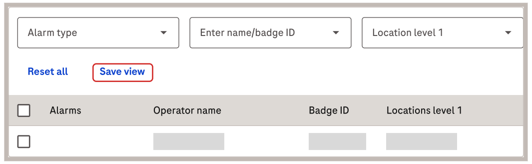
4 Choose the **Apply** button.

The number in brackets on the Apply button indicates how many operators are selected.

→ The list is updated displaying only the filtered operators.

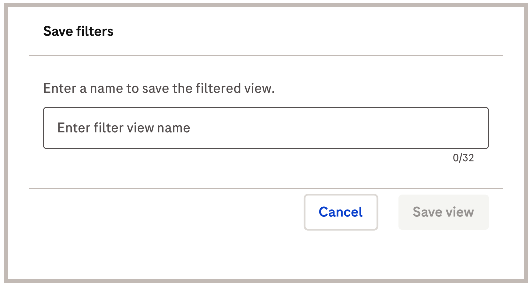
To save the view after filtering

After applying filters, you can save the filtered view. You can access and manage the saved filtered view on the side menu.

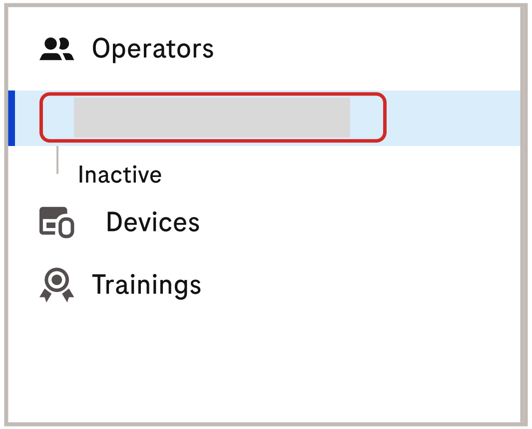


1 After applying filters, choose the **Save view** button.

→ The **Save view** screen is displayed.



2 Enter a name for the filter view.

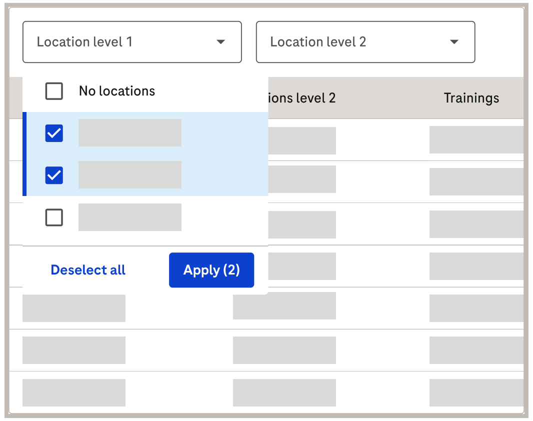


3 Choose the **Save view** button.

→ The filter view is saved on the side menu.

To filter operators by location

You can filter operators by location using one or both location filters: **Location level 1** or **Location level 2**. You can choose between **No locations**, one specific location or multiple locations from the lists.



1 Choose the **Location level 1** filter or the **Location level 2** filter.

2 From the drop-down list, choose the location by choosing the check box next to the location name.

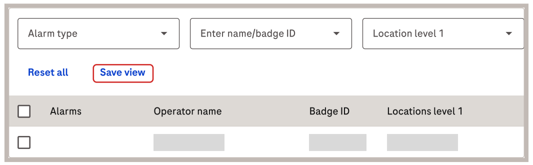
> If there are 10 or more locations on the drop-down list, the **Search** bar is visible. Then, you can search for the location by typing the location name.

3 Choose the **Apply** button.

> → The list is updated showing only filtered results.

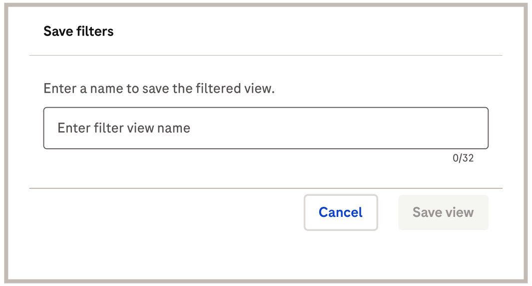
▶ To save the view after filtering

After applying filters, you can save the filtered view. You can access and manage the saved filtered view on the side menu.

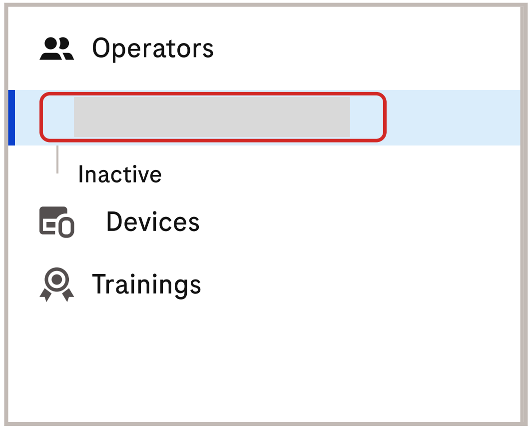


1 After applying filters, choose the **Save view** button.

> → The **Save view** screen is displayed.



2 Enter a name for the filter view.

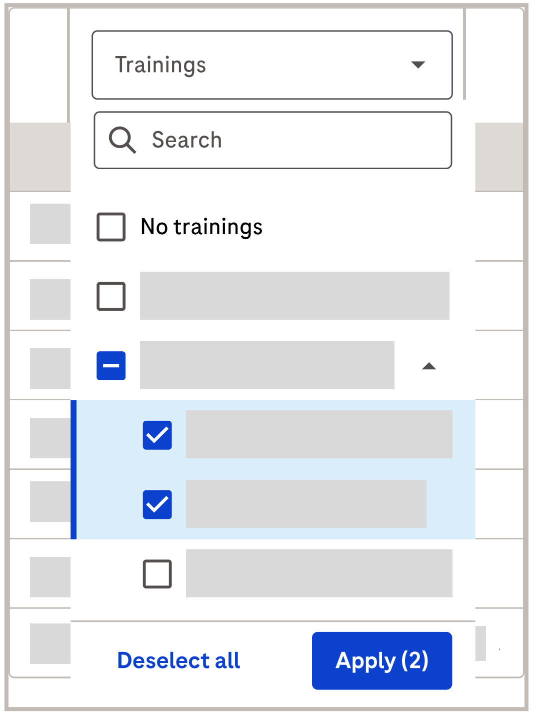


3 Choose the **Save view** button.

> → The filter view is saved on the side menu.

To filter operators by assigned trainings

You can filter operators by the trainings they are assigned to using the **Trainings** filter. The trainings usually consist of different training stages. You can view the training stage by choosing the ▾ next to the training name on the drop-down list.



1 Choose the **Trainings** filter.

2 From the drop-down list, choose the training or just the training stage by choosing the check box next to the training or training stage name.

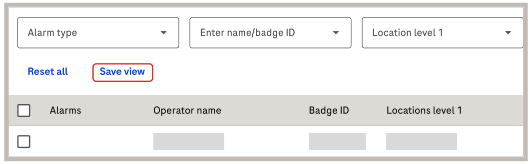
> If there are 10 or more trainings on the drop-down list, the **Search** bar is visible. Then, you can search for the training by typing the training name.

3 Choose the **Apply** button.

> → The list is updated showing only filtered results.

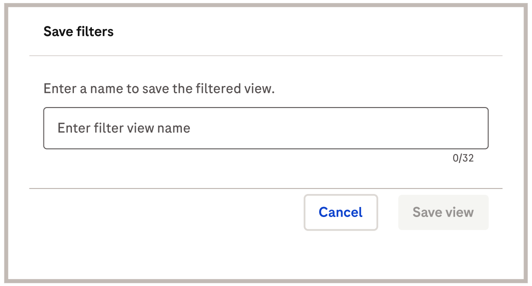
▶ To save the view after filtering

After applying filters, you can save the filtered view. You can access and manage the saved filtered view on the side menu.

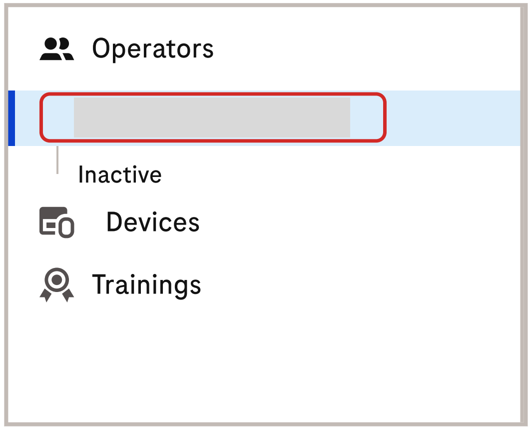


1 After applying filters, choose the **Save view** button.

> → The **Save view** screen is displayed.



2 Enter a name for the filter view.



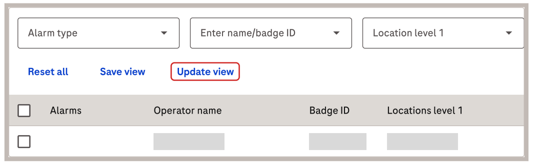
3 Choose the **Save view** button.

> → The filter view is saved on the side menu.

Managing the saved views

After applying filters, you can save the filtered view. You can access and manage the saved filtered view on the side menu.

▶ To update saved filter view



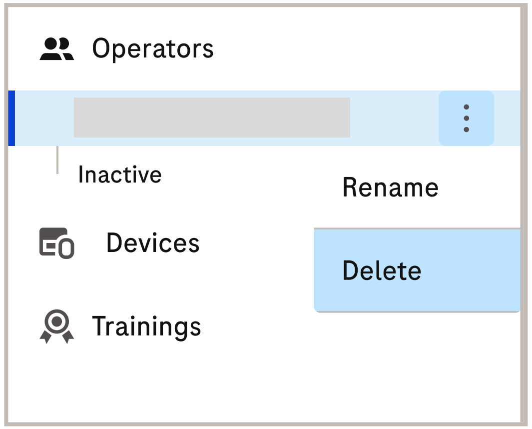
1 From the side menu, choose the saved view that you want to update.

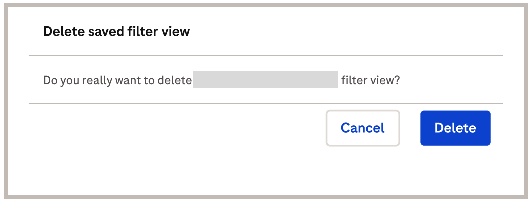
2 Choose and apply new filters that you want on this saved view.

3 Choose the **Update view** button.

> ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

▶ Delete saved filter view





1 On the side menu, hover over the filtered view that you want to delete.

2 Choose the ⋮ button.

> → The drop-down list opens.

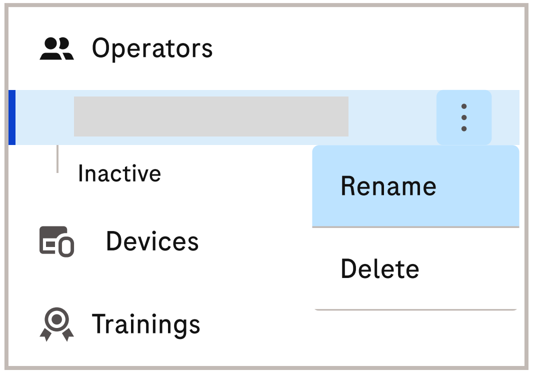
3 From the drop-down list, choose the **Delete** button.

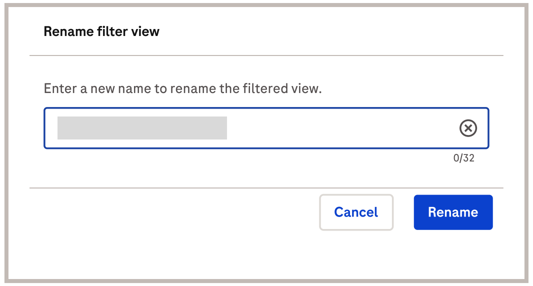
> → The **Delete saved filter view** dialog box opens.

4 Choose the **Delete** button.

> ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

Rename saved filter view





1 On the side menu, hover over the filtered view that you want to rename.

2 Choose the ⋮ button.

> → The drop-down list opens.

3 From the drop-down list, choose the **Rename** button.

> → The **Rename filter view** dialog box opens.

4 Enter a new filter view name.

5 Choose the **Rename** button.

> ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

Viewing operator details

Viewing operator details page

Viewing operator details on the sidebar

You can view details of the operators registered in the application.

To view the operator details, choose the specific operator in the table. The sidebar opens with the available operator details.

The categories displayed in the sidebar are:

- **Status**
- **Email address**
- **Badge ID**

> Other categories are optional and can be added or removed in **Custom fields** settings. When you hover over any of these categories, you can copy its content if needed.

The sidebar displays **Ongoing trainings** and **Locations** assigned to the operator.

> The number in brackets indicates the number of trainings or locations that are assigned to the operator.

The training card displays the name of the training, the training stage to which the operator is assigned, information on the time that operators have to complete the training stage, open tasks if operators have any, and alarms associated with the operators.

The date indicates the due date until the training must be completed or since when the training is overdue. When hovering over the ⓘ icon, the number of remaining days is displayed. A negative number indicates by how many days the training is overdue.

If the training has no due date, the time to complete of the training is displayed as **No due date**. If the training starts in the future, a badge is displayed with the starting date.

When the name of the training stage is too long, hovering over the name displays the complete name.

The training cards are sorted by severity. The cards with higher severity are on the top of the list:

1. Operators off track with trainings due soon
2. Operators off track
3. Trainings due soon
4. Trainings with pending task
5. Trainings without pending tasks
6. Training starting in the future.

If there are open training tasks that need to be performed by the operator, you can change the status of the training task from **Pending** to **Done** after the task is completed.

Below the **Ongoing trainings** list, the sidebar displays **Locations** that are assigned to the operator.

From the sidebar, on the top right, you can access to the following information

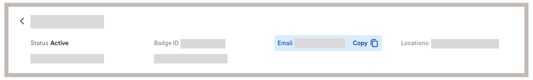
- **Details page** by choosing the ↗ icon
- **More actions** by choosing the ⋮ button.

Related concepts

- Custom fields settings (p. 126)

To view the operator details page

1 Hover over the operator name.



2 Choose the ⤴ button.

→ Details page information is displayed.

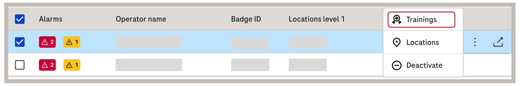
ⓘ You can copy the information for each field, including custom fields, by hovering over the information that you want to copy and choosing the **Copy** button.

Managing trainings

To assign a training to an operator

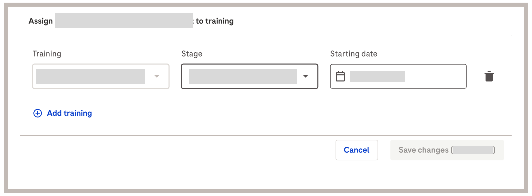
1 To view and manage trainings assigned to the operator, do one of the following:

- Select the operator by choosing the check box at the left side of the operator name. Then choose the **Trainings** button at the right bottom of the page.



- Hover over the operator name on the right side of the related row and choose the ⋮ button. Then choose the **Trainings** button.

→ The **Assign to training** dialog box opens.



2 From the **Assign to training** dialog box, choose the **Add training** button.

3 From the **Training** drop-down list, choose the training program.

4 From the **Stage** drop-down list, choose the training stage.

5 From the date picker, choose the starting date of the training.

6 To assign additional trainings to the operator, choose the **Add training** button.

7 In case of adding additional trainings to the operator, repeat steps 3 to 6.

8 Choose the **Save changes** button.

> The number in brackets indicates the number of new trainings assigned to the operator.

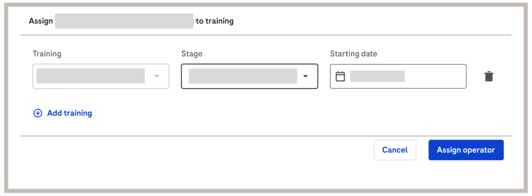
> The green snackbar in the upper right corner confirms the successful completion of the action.

To assign a training to multiple operators

1 Select the operators to which you want to assign a training by choosing the check box at the left side of each operator name.

2 Hover over the **Trainings** button at the right bottom of the page and choose the **Assign** button.

> → The **Assign to training** dialog box opens.



3 From the **Training** drop-down list, choose the training program.

4 From the **Stage** drop-down list, choose the training stage.

5 From the date picker, choose the starting date of the training.

6 To assign additional training programs to the selected operators, choose the **Add training** button.

7 In case of adding additional trainings to the selected operators, repeat steps 3 to 6.

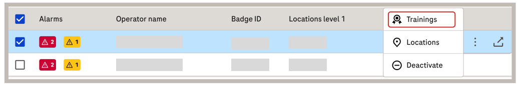
8 To save the changes, choose the **Assign operator** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

To remove a training from an operator

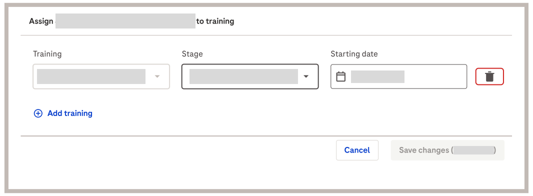
1 To view and manage trainings assigned to the operator, do one of the following:

- Select the operator by choosing the check box at the left side of the operator name. Then choose the **Trainings** button at the right bottom of the page.



- Hover over the operator name on the right side of the related row and choose the ⋮ button. Then choose the **Trainings** button.

→ The **Assign to training** dialog box opens.

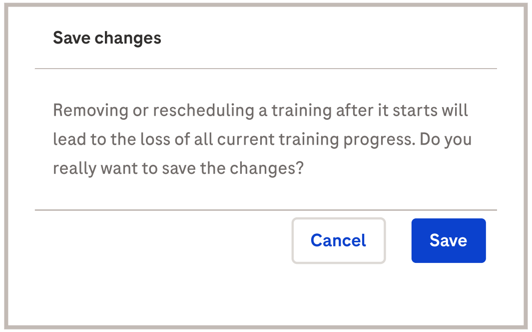


2 Choose the 🗑 button next to the training that you want to remove from the operator.

3 Choose the **Save changes** button.

ⓘ The number in brackets indicates the number of trainings that will be removed from the operator.

→ The **Save changes** dialog box opens.



4 From the **Save changes** dialog box, choose the **Save** button to save the changes.

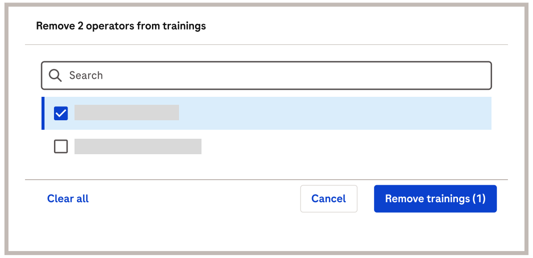
ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

To remove a training from multiple operators

1 Select the operators from which you want to remove a training by choosing the check box at the left side of each operator name.

2 Hover over the **Trainings** button at the right bottom of the page and choose the **Remove** button.

> The **Remove from trainings** dialog box opens.

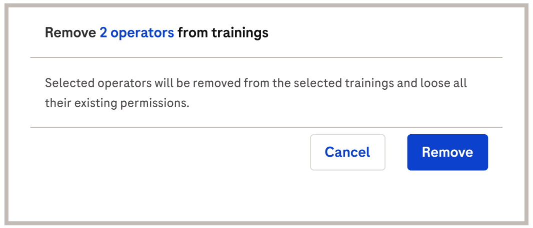


3 Select the training that you want to remove by choosing a check box next to the training name. You can also search for the training by typing the training name in the Search box.

4 Choose the **Remove trainings** button.

> The number in brackets indicates the number of trainings that will be removed from the operators.

> The **Remove from trainings** dialog box opens.



5 Choose the **Remove** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

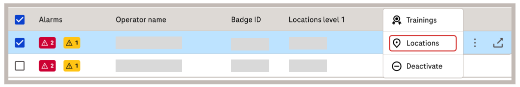
Managing locations

To assign a location to an operator

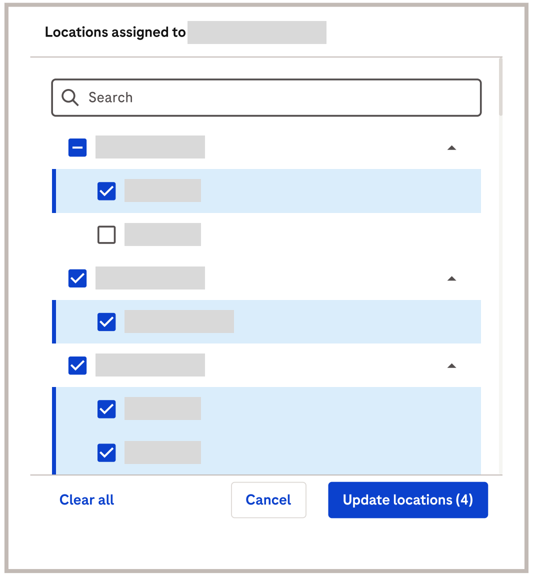
1 To view and manage locations assigned to the operator, do one of the following:

- Select the operator by choosing the check box at the left side of the operator name. Then choose the **Locations** button at the right bottom of the page.

- Hover over the operator name and on the right side of the related row and choose the ⋮ button. Then choose the **Locations** button.



> The **Locations assigned to** dialog box opens.



2 From the **Locations assigned to** dialog box, choose the locations that you want to assign to an operator by choosing a check box at the left side of the location name.

> In the **Locations assigned to** dialog box, the **Locations level 1** list is displayed. Choose the ▾ button on the right to display the **Locations level 2** list.
>
> - If you choose a location from the **Locations level 1** list, all locations on the **Locations level 2** list are automatically selected.
> - To choose a single location, choose it from the **Locations level 2** list. The corresponding **Locations level 1** will be automatically assigned.

3 Choose the **Update locations**.

> The number in brackets indicates the number of new locations that are assigned to the operator.

> The green snackbar in the upper right corner confirms the successful completion of the action.

To assign a location to multiple operators

1 Select the operators to which you want to assign a location by choosing the check box at the left side of each operator name.

2 Hover over the **Locations** button at the right bottom of the page and choose the **Assign** button.

> → The **Assign locations to** dialog box opens.

3 From the **Assign locations to** dialog box, choose the locations that you want to assign to operators by choosing a check box at the left side of the location name.

> In the **Assign locations to** dialog box, the **Locations level 1** list is displayed. Choose the ▾ on the right to display the **Locations level 2** list.
>
> - If you choose a location from the **Locations level 1** list, all locations on the **Locations level 2** list are automatically selected.
> - To choose a single location, choose it from the **Locations level 2** list. The corresponding **Locations level 1** will be automatically assigned.

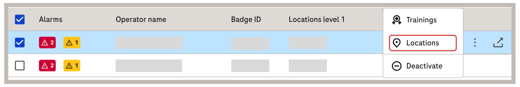
4 Choose the **Assign locations** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

To remove a location from an operator

1 To view and manage locations assigned to the operator, do one of the following:

- Select the operator by choosing the check box at the left side of the operator name. Then choose the **Locations** button at the right bottom of the page.



- Hover over the operator name and on the right side of the related row and choose the **More actions** button. Then choose the **Locations** button.

> → The **Locations assigned to** dialog box opens.

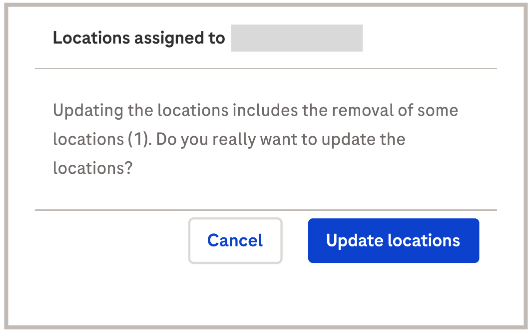
2 From the **Locations assigned to** dialog box, choose the locations that you want to remove from an operator by clearing the check box at the left side of the location name.

> In the **Locations assigned to** dialog box, the **Locations level 1** list is displayed. Choose the ▾ button on the right to display the **Locations level 2** list.
>
> - If you remove the location from the **Locations level 1** list, all locations on the **Locations level 2** list are automatically removed.
> - To remove a single location, remove it from the **Locations level 2** list. The corresponding **Locations level 1** will be automatically removed.

3 Choose the **Update locations** button.

> The number in brackets indicates the number of trainings that will be removed from the operator.

> → The **Locations assigned to** dialog box opens.



4 From the **Locations assigned to** dialog box, choose the **Update locations** button to save the changes.

> The green snackbar in the upper right corner confirms the successful completion of the action.

To remove a location from multiple operators

1 Select the operators from which you want to remove a location by choosing the check box at the left side of each operator name.

2 Hover over the **Locations** button at the right bottom of the page and choose the **Remove** button.

> → The **Remove operators <tooltip>({{number}})</tooltip> from locations** dialog box opens.

3 From the **Remove operators <tooltip>({{number}})</tooltip> from locations** dialog box, choose the locations that you want to remove from operators by choosing the check box at the left side of the location name.

> ⓘ In the **Remove operators <tooltip>({{number}})</tooltip> from locations** dialog box, the **Locations level 1** list is displayed. Choose the ▾ on the right to display the **Locations level 2** list.
>
> - If you remove the location from the **Locations level 1** list, all locations on the **Locations level 2** list are automatically removed.
> - To remove a single location, remove it from the **Locations level 2** list.

4 Choose the **Remove locations ({{number}})** button.

> ⓘ The number in brackets indicates the number of locations that will be removed from the operators.

> → The confirmation dialog box opens.

5 Choose the **Remove** button.

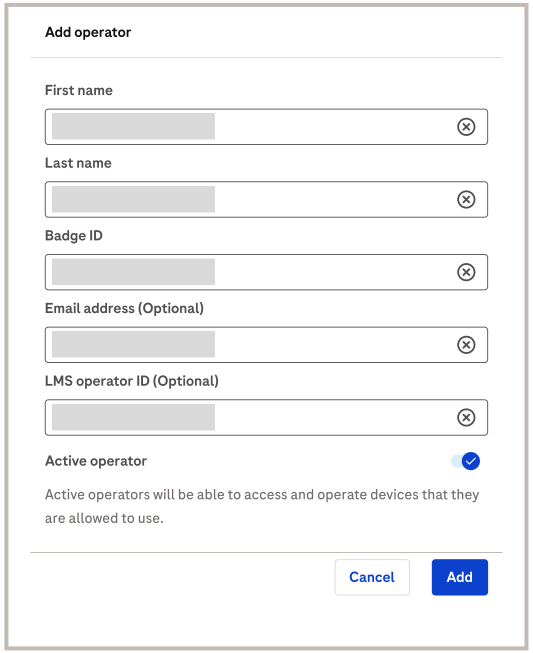
> ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

Managing operators

To add an operator

1 In the **Operators** module, choose the **Add operator** button.

→ The **Add operator** dialog box opens.



2 Follow the wizard and fill out the mandatory fields: **First name**, **Last name** and **Badge ID**.

> All other fields are optional and are available only if added in **Custom fields** settings.

Custom fields settings (p. 126)

3 To activate or deactivate the operator, switch on or off the toggle button at the end of the list.

4 To save the changes, choose the **Add** button

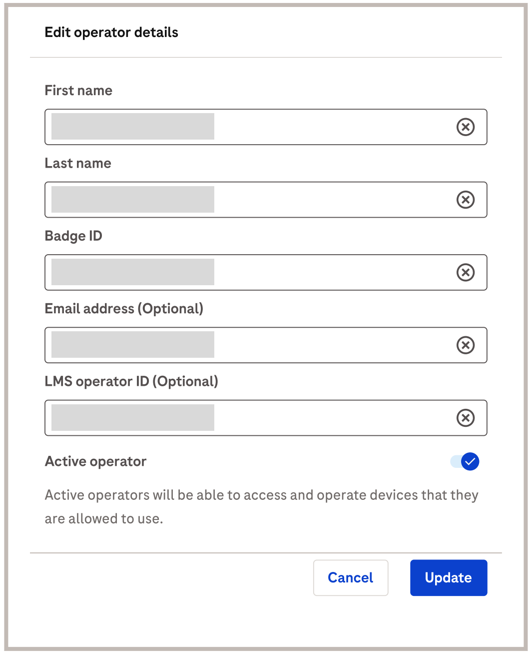
> The green snackbar in the upper right corner confirms the successful completion of the action.

To edit operator details

1 To edit operator details, do one of the following:

- Select the operator whose details you want to edit by choosing the check box at the left side of the operator name. Then choose the **Edit details** button at the right bottom of the page.
- Hover over the operator name on the right side of the related row and choose the ✎ button.

→ The **Edit operator details** dialog box opens.



2 Follow the wizard and edit the desired information.

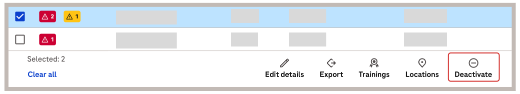
> If the operator was added by Active Directory, some fields are filled out automatically. These fields are disabled and cannot be edited.

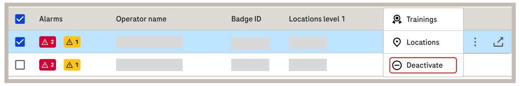
3 To activate or deactivate the operator, switch on or off the toggle button at the end of the list.

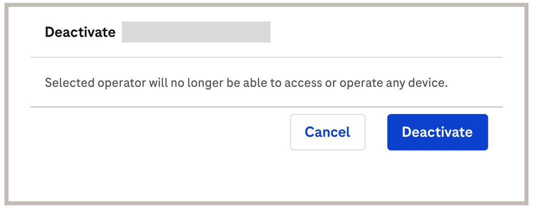
4 Choose the **Update** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

To deactivate an operator







1 To deactivate an operator, do one of the following:

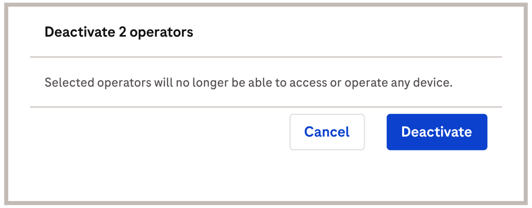
- Select the operator by choosing the check box at the left side of the operator name. Then choose the **Deactivate** button at the right bottom of the page.
- Hover over the operator name and on the right side of the related row and choose the ⋮ button. Then choose the **Deactivate** button.

> The **Deactivate** dialog box opens.

2 Choose the **Deactivate** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

To deactivate multiple operators



1 Select the operators that you want to deactivate by choosing the check box at the left side of each operator name.

2 Choose the **Deactivate** button at the right bottom of the page.

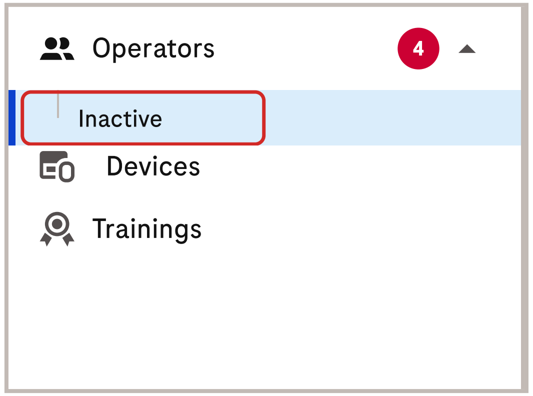
> The **Deactivate operators** dialog box opens.

3 Choose the **Deactivate** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

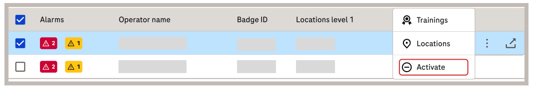
To activate an operator

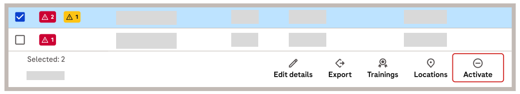
The inactive operators are grouped in the separate tab on the **Operators** side menu.



1 From the side menu, choose the **Inactive** operators.

→ The **Operators** module screen displays the table with all inactive operators.

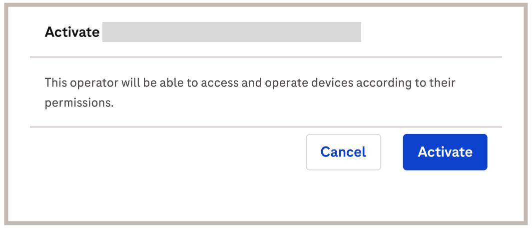




2 To activate an operator, do one of the following:

- Select the operator by choosing the check box at the left side of the operator name. Then choose the **Activate** button at the right bottom of the page.
- Hover over the operator name and on the right side of the related row and choose the **More actions** ⋮ button. Then choose the **Activate** button.

→ The **Activate** dialog box opens.

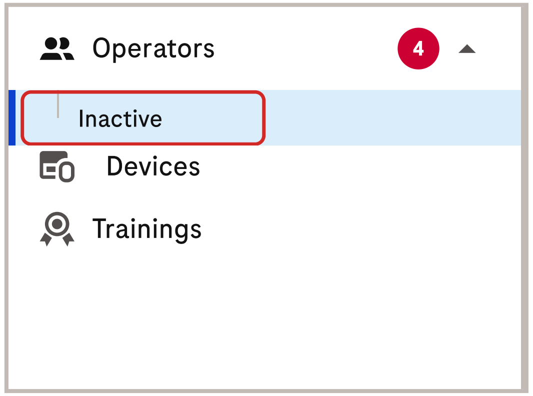


3 Choose the **Activate** button.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

To activate multiple operators

The inactive operators are grouped in the separate tab on the **Operators** side menu.



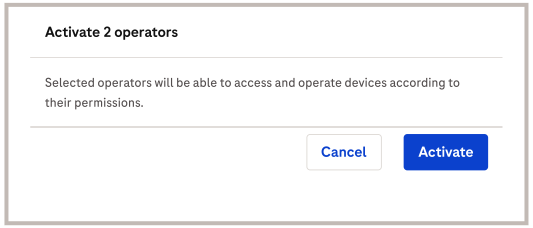
1 Choose the **Inactive** operators from the side menu.

> → The **Operators** module screen displays the table with all inactive operators.

2 Select the operators by choosing the check box at the left side of each operator name.

3 Then choose the **Activate** button at the right bottom of the page.

> → The **Activate operators** dialog box opens.



4 Choose the **Activate** button.

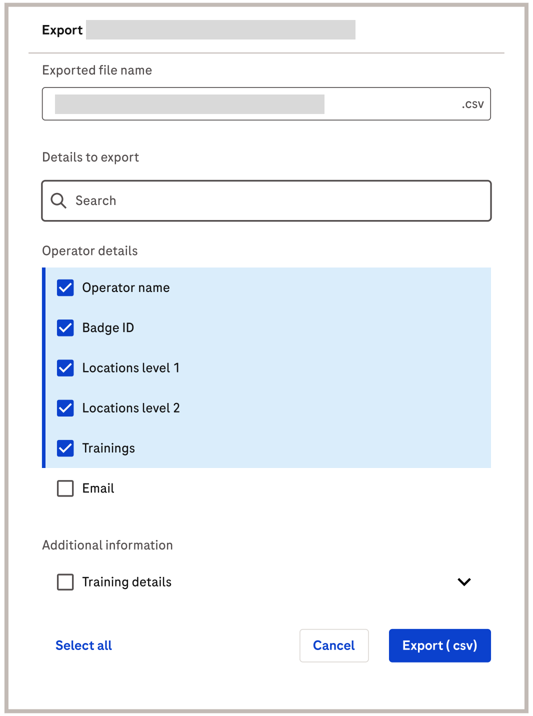
> ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

To export details of an operator

1 To export details of an operator, do one of the following:

- Select the operator by choosing the check box at the left side of the operator name. Then choose the **Export** button at the right bottom of the page.
- At the upper right corner of the page, from the **Export** drop-down list, choose the file you want to export.

> → The **Export** dialog box opens.



2 If you choose the **Current status (CSV)** file, in the **Export** dialog box, enter the **Exported file name**

> The system automatically names the export file using the number of selected operators and the export time stamp. You can change the name of the file.

- Choose the **Operator details** that you want to export by choosing the check box next to the operator details name. You can also search for operator details by entering the operators details name in the **Details to export** search box.
- You can choose the **Additional information** like the **Training details** by choosing the ⌄ on the right.

> → The drop-down list opens with the training details list.

- Choose the training details that you want to export by selecting the check box next to the training details name.
- Choose the **Export (CSV)** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

> → The .csv file is downloaded locally to the device that runs the Advanced POC modules.

3 If you choose the **Training history report (PDF)**, enter the relevant details.

- If relevant, change the value in the **Exported file name** field.

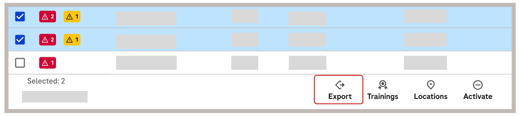
> The system automatically names the export file using the number of selected operators and the export time stamp.

- From the **Period** drop-down list, choose the period of time for the export.
- If applicable, enter a header name in the **Custom header name (Optional)** field.
- Choose the **Operator details** that you want to export by choosing the check box next to the operator details name. To expand the list, choose the ⌄ icon.
- Choose at least one of the following options to include in the report: **Latest training status** and/or **Training details**
- Choose the **Export (PDF)** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

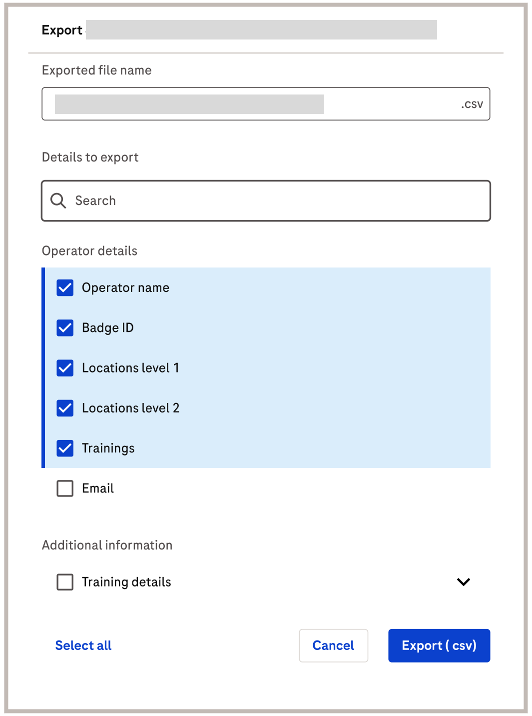
> → The .pdf file is downloaded locally to the device that runs the Advanced POC modules.

To export details from multiple operators



1 To export details from multiple operators, do one of the following:

- Select operators by choosing the check box at the left side of the operator name. Then choose the **Export** button at the right bottom of the page.
- At the upper right corner of the page, from the **Export** drop-down list, choose the file you want to export.



2 If you choose the **Current status (CSV)** file, in the **Export** dialog box, enter the **Exported file name**.

> The system automatically names the export file using the number of selected operators and the export time stamp. You can change the name of the file.

- Choose the **Operator details** that you want to export by choosing the check box next to the operator details name. You can also search for operator details by entering the operators details name in the **Details to export** search box.
- You can choose the **Additional information** like the **Training details** by choosing the ⌄ on the right.

> → The drop-down list opens with the training details list.

- Choose the training details that you want to export by selecting the check box next to the training details name.
- Choose the **Export (CSV)** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

> → The .csv file is downloaded locally to the device that runs the Advanced POC modules.

3 If you choose the **Training history report (PDF)**, enter the relevant details.

- If relevant, change the value in the **Exported file name** field.

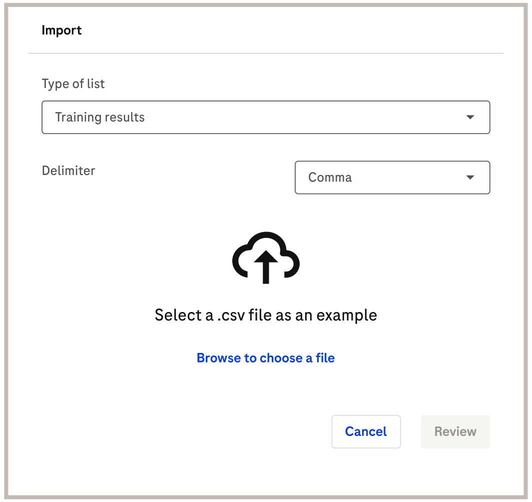
> The system automatically names the export file using the number of selected operators and the export time stamp.

- From the **Period** drop-down list, choose the relevant value.
- If applicable, enter a header name in the **Custom header name (Optional)** field.
- Choose the **Operator details** that you want to export by choosing the check box next to the operator details name. To expand the list, choose the ⌄ icon.
- Choose at least one of the following options to include in the report: **Latest training status** and/or **Training details**.
- Choose the **Export (PDF)** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

> → The .pdf file is downloaded locally to the device that runs the Advanced POC modules.

To manually import training results



1 Choose the **More actions** button at the upper right corner of the page.

2 Choose the **Import** button.

> → The **Import** dialog box opens.

3 On the **Import** dialog box, choose the **Type of list** that you want to import.

4 To determine how the data is presented in .csv file, choose the **Delimiter** from the drop-down list.

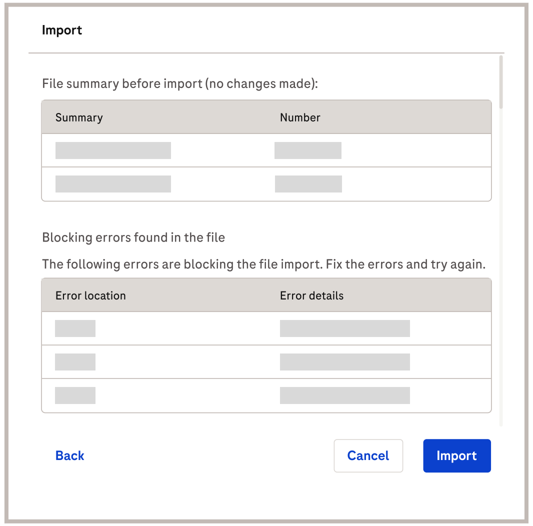
5 To upload the file, choose the **Browse to choose a file** button.

6 From the device local storage, choose the .csv file that you want to import.

> → The **Import** dialog box opens showing the upload status.

7 After the .csv file is uploaded, choose the **Review** button.

> → The import file summary dialog box opens.



8 Review the summary and choose the **Import** button.

> The green snackbar in the upper right corner confirms the successful completion of the action.

To manually synchronize the operators list from cloud to POC module for navify Integrator

1 Choose the ⋮ button at the upper right corner of the page.

2 Choose the **Sync now** button.

3 Wait until the synchronization is completed.

> The green snackbar in the upper right corner confirms the successful completion of the action.

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Trainings screen overview

The **Trainings** screen displays training programs and progress of the operators assigned to that training programs. The user can manage existing programs or create new ones.

The **Trainings** screen provides an overview of all training programs, showing their current status and progress. From the **Trainings** screen, you can manage all existing training programs or create new trainings.

The table displays the following details:

- **Alarms**: This column shows errors and warnings related to the training program. Errors are displayed with a red icon and indicate the number of **Operators off track**. Warnings are displayed with a yellow icon and indicate the number of **Operators with task due soon**. To display the number of operators for each alarm, hover over the icons.
- **Trainings / Training stage**: This column lists the names of the training programs and their respective stages.
- **Time to complete**: This column indicates the time allocated to an operator to complete a training stage.
- **Operators**: Several columns provide operator-related data, including the number of assigned operators and those with pending tasks, no pending tasks, or tasks that have not started.

Viewing training details

The **Trainings** module lists all training programs and their stages, allowing you to manage assigned operators and view stage details.

Each training consists of stages. When you choose the **Trainings** module, the system, by default, expands the table and displays all stages for the respective trainings. You can collapse the table by choosing the ▲ button on the left side of each trainings row.

To view the details for a training stage, choose the corresponding entry in the table.

> Training stages are presented in tabs. Use tabs to switch between training stages.

Training stage assignment view

A Training stages tabs

B Number of operators assigned to the training stage

C Stage duration in days

The training stage assignment view allows you to manage operators assigned to each training stage. This screen displays the following information:

- Training stages tabs displaying the table with assigned operators.

- **Operators assigned to this stage** with the number of operators indicated in brackets.
- **Stage duration** indicating the length of the stage.

You can perform the following actions on the **Trainings** screen:

- **Assign operators**
- **Edit columns**
- Manage **Due date**
- View operator details and manage operators assigned to the training stage.

Edit columns in the Trainings module

You can manage the displayed columns in the table in the Training module.

You can edit table columns only after choosing a trainings from the list. The training stage assignment view displays the table of operators assigned to the corresponding training stage.

This table has the same columns as the table in the Operators module and is the result of the **Custom fields** settings. In the Trainings module, you can manage which columns to display or hide. The column order reflects the one from the Operators module and cannot be edited.

To adjust the table columns, choose the **Edit columns** button.

- To hide or show the column, choose or clear the check box next to the column name.
- To deselect all or select all columns, choose the **Select all** button or the **Clear all** button.
- To save the changes, choose the **Apply** button.

> At least one column should be visible. If there are no selected columns, the **Apply** button is disabled.

The applied changes are saved for the logged on user as user preferences. Columns are displayed accordingly as default at every logon until edited again.

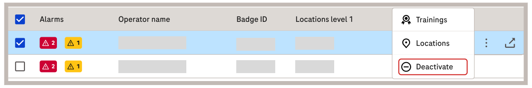
Related concepts

- Custom fields settings (p. 126)

Managing operators in the Trainings module

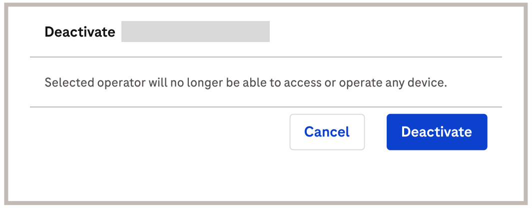
To deactivate an operator in the Training module

You can deactivate an operator by choosing the operator from the training stage assignment table.



1 To deactivate an operator, hover over the operator name on the right side of the related row and choose the ⋮ button. Then choose the **Deactivate** button.

→ The **Deactivate** dialog box opens.

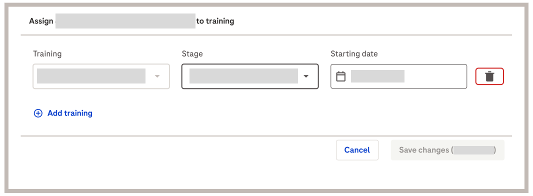


2 Choose the **Deactivate** button.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

To remove a training from an operator in the Training module

You can remove the training from an operator by choosing the operator from the training stage assignment table.



1 To remove a training from the operator, hover over the operator name on the right side of the related row and choose the ⋮ icon. Then choose the **Trainings** button.

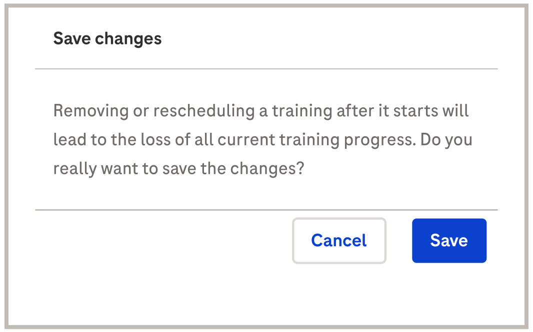
→ The **Assign to training** dialog box opens.

2 Choose the 🗑 button next to the training that you want to remove from the operator.

3 Choose the **Save changes** button.

ⓘ The number in brackets indicates the number of trainings that will be removed from the operator.

→ The **Save changes** dialog box opens.

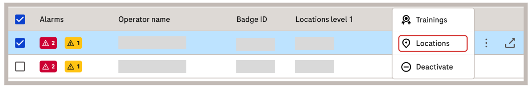


4 From the **Save changes** dialog box, choose the **Save** button to save the changes.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

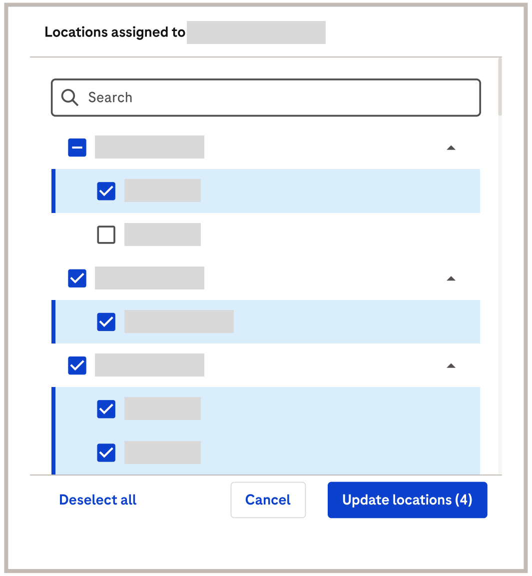
To assign a location to an operator in the Training module

You can assign locations to an operator by choosing the operator from the training stage assignment table.



1 To assign location to the operator, hover over the operator name on the right side of the related row and choose the ⋮ button. Then choose the **Locations** button.

→ The **Locations assigned to** dialog box opens.



2 From the **Locations assigned to** dialog box, choose the locations that you want to assign to an operator by choosing a check box at the left side of the location name.

ⓘ In the **Locations assigned to** dialog box, the **Locations level 1** list is displayed. Choose the ▾ button on the right to display the **Locations level 2** list.

- If you choose a location from the **Locations level 1** list, all locations on the **Locations level 2** list are automatically selected.
- To choose a single location, choose it from the **Locations level 2** list. The corresponding **Locations level 1** will be automatically assigned.

3 Choose the **Update locations**.

> The number in brackets indicates the number of new locations that are assigned to the operator.

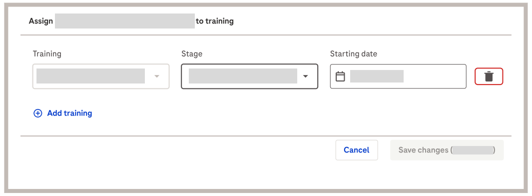
> The green snackbar in the upper right corner confirms the successful completion of the action.

To remove a training from an operator in the Training module

You can remove the training from an operator by choosing the operator from the training stage assignment table.

1 To remove a training from the operator, hover over the operator name on the right side of the related row and choose the ⋮ icon. Then choose the **Trainings** button.

> → The **Assign to training** dialog box opens.

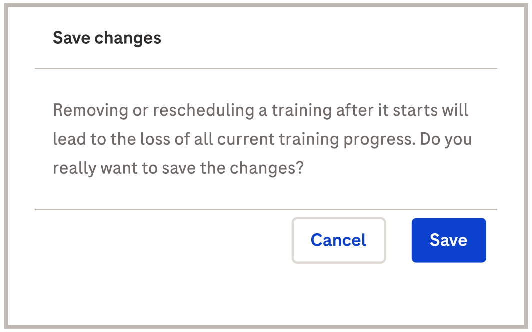


2 Choose the 🗑 button next to the training that you want to remove from the operator.

3 Choose the **Save changes** button.

> The number in brackets indicates the number of trainings that will be removed from the operator.

> → The **Save changes** dialog box opens.

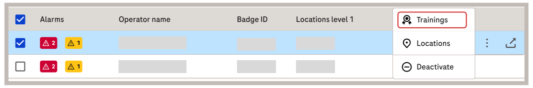


4 From the **Save changes** dialog box, choose the **Save** button to save the changes.

> The green snackbar in the upper right corner confirms the successful completion of the action.

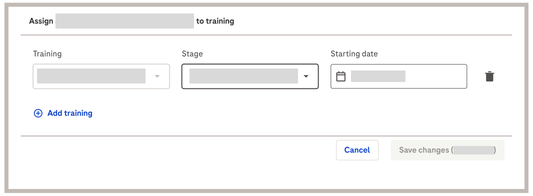
To assign a training to an operator in the Trainings module

You can assign a training to an operator by choosing the operator from the training stage assignment table.



1 To assign a training to the operator, hover over the operator name on the right side of the related row and choose the ⋮ button. Then choose the **Trainings** button.

> → The **Assign to training** dialog box opens.



2 From the **Assign to training** dialog box choose the **Add training** button.

3 From the **Training** drop-down list, choose the training program.

4 From the **Stage** drop-down list, choose the training stage.

5 From the date picker, choose the starting date of the training.

6 To assign additional trainings to the operator, choose the **Add training** button.

7 To add additional trainings to the operator, repeat steps 3 to 6.

8 Choose the **Save changes** button.

> ⓘ The number in brackets indicates the number of new trainings assigned to the operator.

> ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

Viewing operator details in the Trainings module

Trainings stage assignment view displays the table of operators assigned to the corresponding training stage. You can view the details of the operators in the table.

You can view details of the operators listed in the training stage assignment table. Operator details can be displayed on the sidebar or on the operator details page.

To view the operator details on the sidebar, choose the specific operator in the table. The sidebar opens with the available operator details.

To view the operator details page, hover over the operator name and choose the ↗ button. The details page information is displayed.

To learn more about each operator details view, refer to:

Related concepts

- Viewing operator details on the sidebar (p. 140)

Related tasks

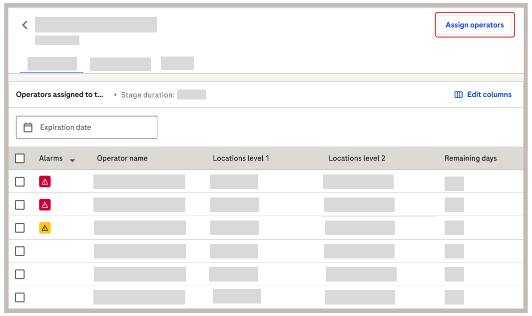
- To view the operator details page (p. 140)

To assign operators to the training

You can a assign single operator or multiple operators to each training stage.

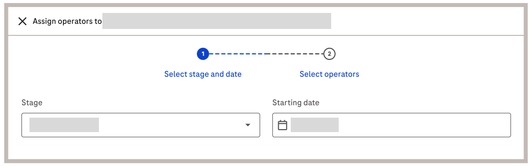
1 In the **Trainings** module, choose the training that you want to assign operators to.

2 Choose the **Assign operators** button.



→ The **Assign operators to** dialog box opens.

3 Choose the **Stage** from the drop-down list.

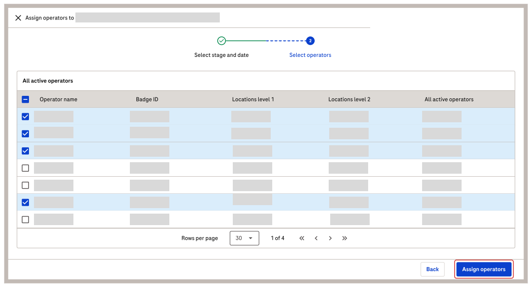


4 Then choose the **Starting date** from the date picker.

5 Choose the **Next** button.

→ The new **Assign operators to** dialog box opens with the list of all active operators.

6 From the list, choose the operators you wan to assign to the training stage.



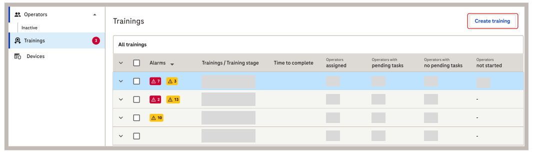
7 Choose the **Assign operators** button.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

Managing trainings in the Trainings module

Creating a training from an existing training

Modify the existing training in the Trainings module by entering the required details. You can modify all the existing fields or add the new ones before creating the final training program.



1 From the **Trainings** module view, choose the **Create training** button.

→ The **Create training** dialog box opens.

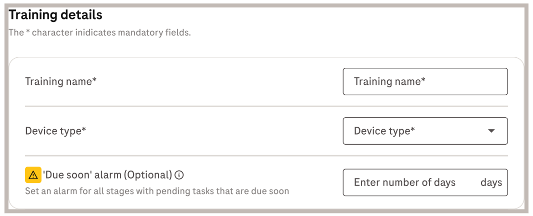
2 From the **Create training** dialog box, choose the **Start from an existing training** option.

→ The existing training programs are available on the **Create training** view.

3 From the **Create training** view, choose the training program that you want to modify.

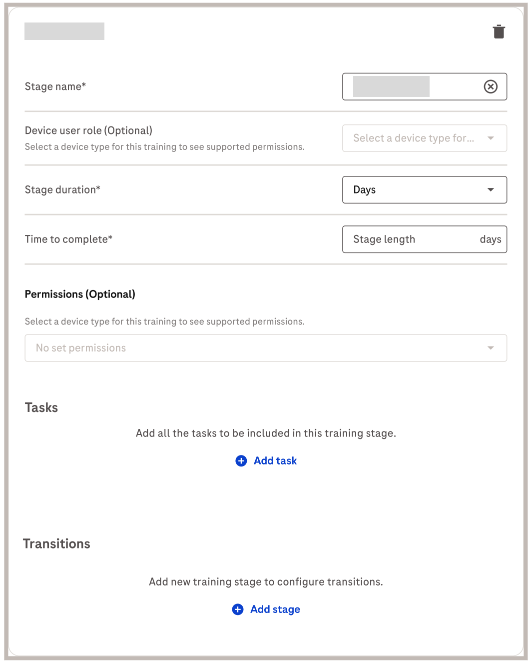
4 Choose the **Next** button.

→ The **Start from an existing training** dialog box opens.



5 In the **Training details** section, modify the required information.

ⓘ Fields marked with * character are mandatory.



6 In the **Stage** section, modify the required information.

ⓘ Fields marked with * character are mandatory.

7 Modify the **Tasks** if necessary.

Tasks **Add task**

8 Modify the **Transitions** if necessary.

9 To modify all other existing training stages, repeat the steps 6–8.

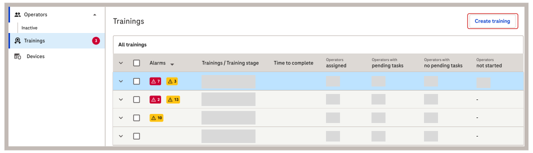
Add stageAdd stageStart from an existing training

10 To save the changes, choose the **Create** button.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

Creating a training from scratch

Create a new training from scratch in the Trainings module by entering the required details. You can configure the training by defining its stages and adding tasks if needed before creating the final training program.

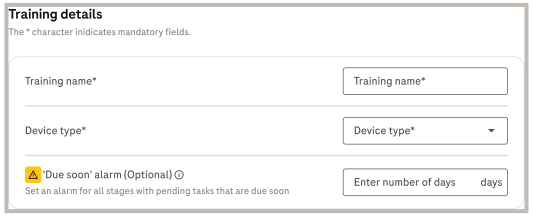


1 From the **Trainings** module view, choose the **Create training** button.

→ The **Create training** dialog box opens.

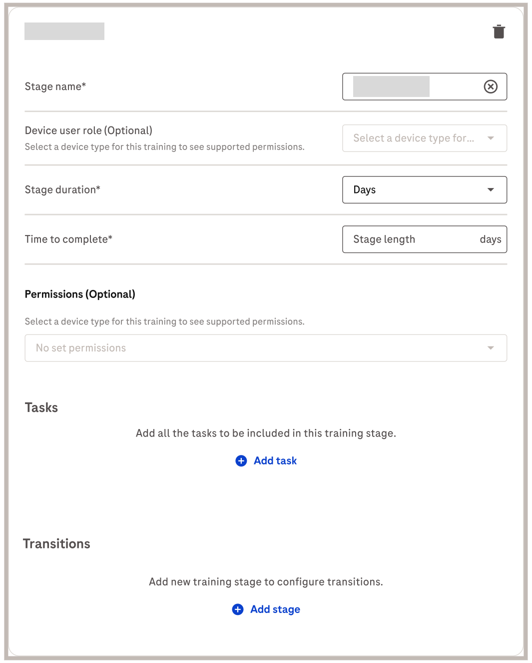
2 From the **Create training** dialog box, choose the **Start from scratch** option. Then choose the **Next** button.

→ The **Start from scratch** dialog box opens.



3 In the **Training details** section, enter the required information.

ⓘ Fields marked with the * character are mandatory.

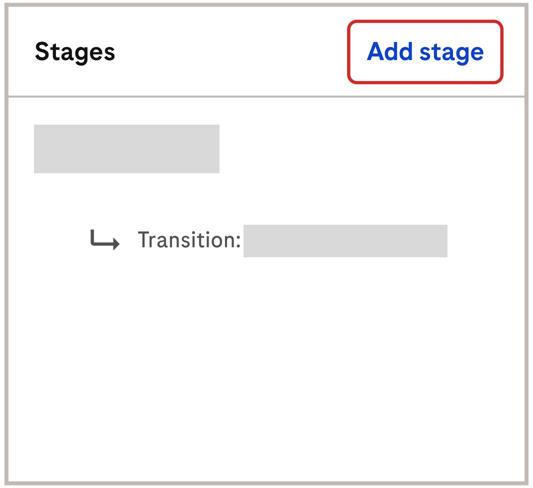


4 In the **Stage** section, enter the required information.

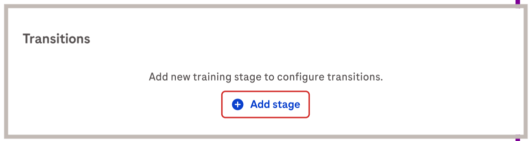
ⓘ Fields marked with * character are mandatory.

- If the training has tasks, add them in the **Tasks** section by choosing the **Add task** button. Then enter the required information.

ⓘ
- **If the training stage has tasks:** Adding tasks automatically creates the necessary transition rules. The operator must complete all assigned tasks to successfully proceed to the next training stage.
- **If the training has no tasks:** The operator moves to the next training stage once the current training stage ends. In this case, you must specify which training stage the operator proceeds to.



5 To add the next training stage, choose the **Add stage** button on the side menu, or the **Add stage** button in the **Transitions** section.



→ The new **Stage** section opens.

6 In the **Transitions** section, from the drop-down lists, choose one of the following:

- If the training has no **Tasks**, from the **Transitions** section, choose the next training stage for the operator. The operator will proceed to this training stage when the previous stage ends.

- If the training has **Tasks**, choose all **Transitions** rules for the operator to proceed to the next training stage.

7 In the **Stage** section, enter the required information.

ⓘ Fields marked with * character are mandatory.

8 If the training has no more stages, choose the **Create** button.

ⓘ To add additional training stages, repeat steps 5–7.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

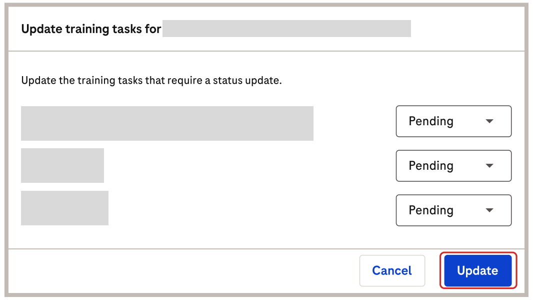
To update training tasks for multiple operators in the Training module

You can update training tasks for multiple operators by choosing operators from the training stage assignment table.

1 Select the operators for which you want to update training tasks by choosing the check box at the left side of each operator name.

2 Choose the **Update tasks** button at the right bottom of the page.

→ The **Update training tasks for** dialog box opens.



3 From the drop-down list, next to the training name, choose the desired option.

4 To save the changes, choose the **Update** button.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

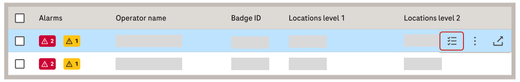
To update training tasks for an operator in the Training module

You can update training tasks for an operator by choosing the operator from the training stage assignment table.

1 To update training tasks for an operator, do one of the following:

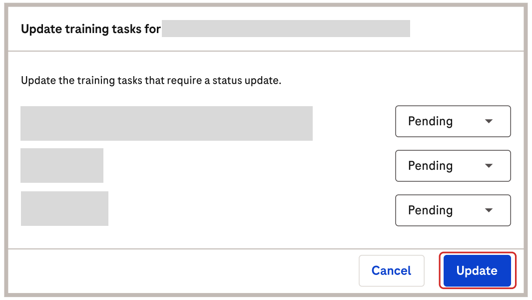
- 

  Select the operator by choosing the check box at the left side of the operator name. Then choose the **Update tasks** button at the right bottom of the page.

- 

  Hover over the operator name on the right side of the related row and choose the ☰ button.

→ The **Update training tasks for** dialog box opens.



2 From the drop-down list, next to the training name, choose the desired option.

3 To save the changes, choose the **Update** button.

ⓘ The green snackbar in the upper right corner confirms the successful completion of the action.

To delete trainings

You can delete single or multiple trainings without assigned operators.

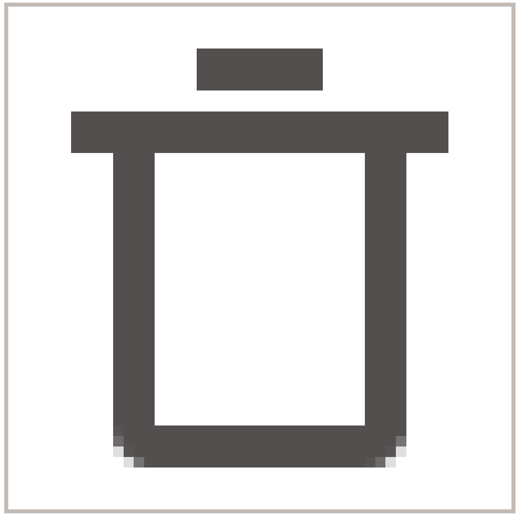
User with admin rights

▶ To delete trainings

1 Access the **Trainings** module view.

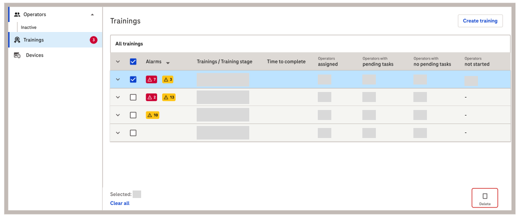
2 To delete a single training, in the displayed list, do one of the following:

- Hover over the relevant training. At the end of the row, choose the displayed



icon.

- Select the relevant training, then choose the **Delete** button.



3 To delete multiple trainings, select the relevant ones. Then choose the **Delete** button.

4 When the confirmation dialog box is displayed, choose the **Delete** button.

→ A green snackbar in the upper right corner confirms the successful deletion of the selected trainings.

Application for trainers

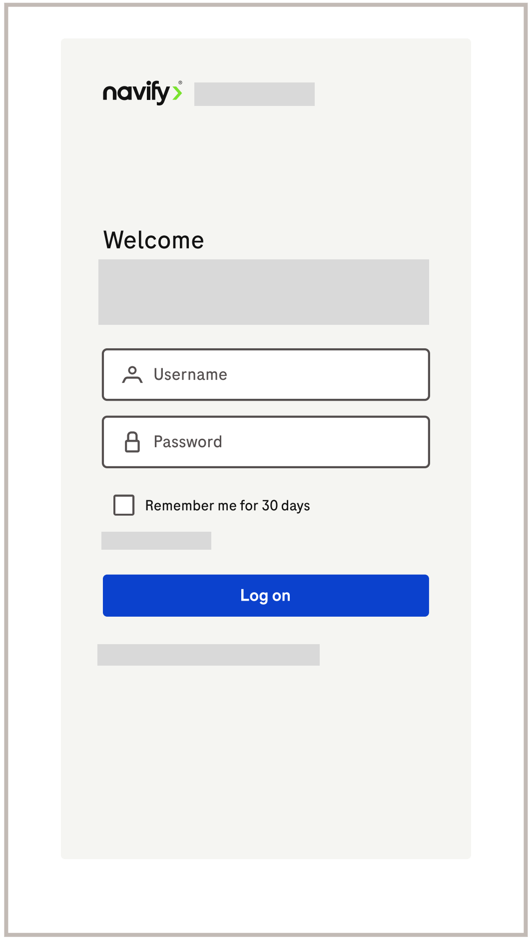
The application for trainers is a web application solution running on portable devices. It provides access to training program details and operators search, ensuring both trainers and operators acknowledge training attendance and task completion.

To access the application for trainers

1 Open the Trainer app using the link provided during the deployment of the **Advanced operator management** module.

2 In the **Username** field, enter your email address.

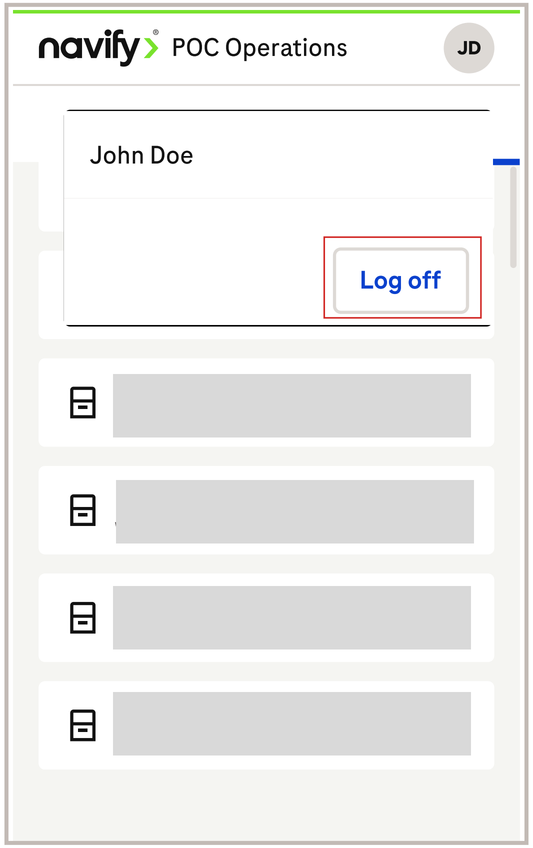
3 In the **Password** field, enter your password.



4 Choose the **Log on** button.

> To save your logon details for the next 30 days, select the **Remember me for 30 days** checkbox.

To log off the application for trainers

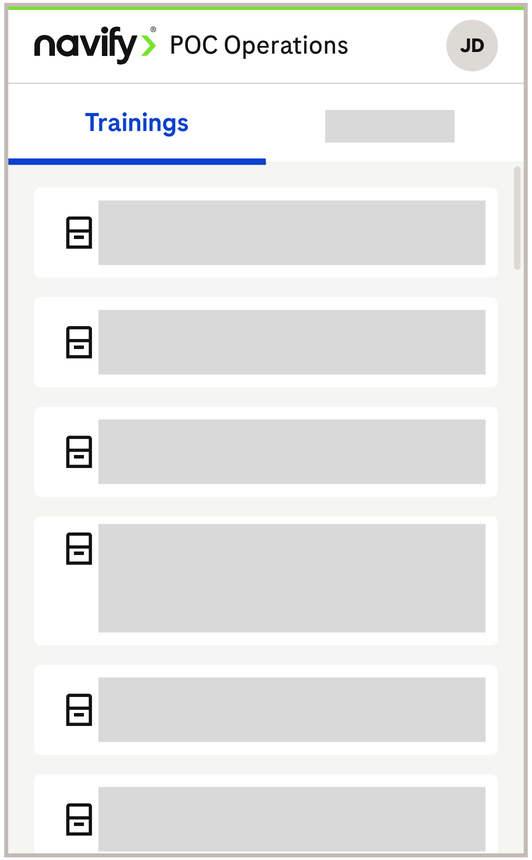


1 In the global information area, choose the icon containing your initials.

2 Choose the **Log off** button.

→ You are redirected to the logon screen.

Checking the trainings in the application for trainers



1 Choose the **Trainings** tab.

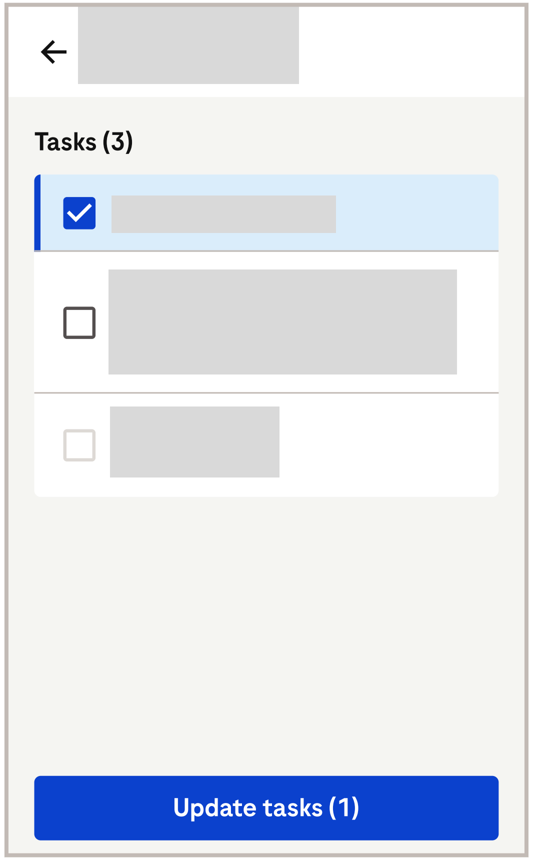
2 From the displayed list, choose the relevant card.

→ You can see the detailed view of that selected training, including all the stages in the training program.

3 Choose the relevant training stage.

→ You can see a list of the associated tasks.

4 Select the check boxes next to the relevant tasks.



5. Choose the **Update tasks** button.
6. Choose the **Next** button.
7. Choose the **Next** button and hand the device to the operator who has completed the training.
8. On the training confirmation screen, the oprator must confirm that the training has been completed using the **Confirm** button.
9. Choose the **Next** button.
10. On the next training confirmation screen, the trainer must confirm that the training has been completed using the **Confirm** button.

Result validation

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Result validation

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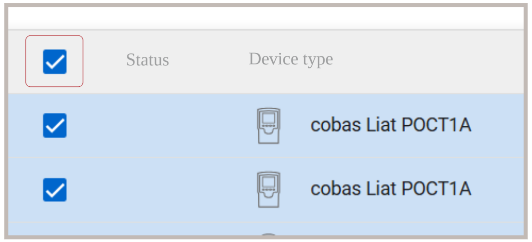
Result validation

The displayed alphanumeric values may not match your configured decimal separator

Upon result reception, the application server will store instrument local time as server local time. If instrument and client are located in a different time zone than the application server, the Date & Time field of the Result Query screen shows the time of a measurement in client time instead of server time. However, the instrument time is stored in the database and sent to the HIS/LIS.

Selecting multiple patient results



To select multiple patient results, choose the check box at the top left of the table.

Only 50 patient results can be selected at a time.

Scroll down and repeat the action as needed to select additional patient results.

Advanced and quick search filters

To narrow down the search, advanced and quick search filters can be applied to the operators list or table.

Quick search

Search from a specific device, parameter, or location.

Advanced search

To narrow your search further, use the search filter ( ) which provides additional search criteria including date range, levels, and parameters.

> If more than one option is chosen for one filter criterion, results for all chosen options will be displayed.
>
> If several filter criteria are applied, only results matching all filter criteria will be displayed.

NOTICE

Selecting multiple filter criteria

An error may occur if too many filter criteria are selected.

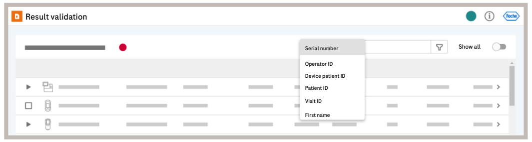
- If an error occurs, reduce the number of selected filter criteria.

Searching for patient results

To find patient results, you can search by device, patient ID, patient ID, visit ID, patient name, or test lot, or by a filter combining multiple search criteria.

Generic patient IDs are not displayed and are substituted by a hyphen.

▶ To search for patient results



1 Choose the **Result validation** module.

2 To search for patient results based on serial number, operator ID, device patient ID. patient ID, visit ID, first name, middle name, last name, second last name, result text starts with, alarms, test lot, test material, or host status, choose the option from the drop-down list.

> → The search option is displayed in the search by field.

3 In the search by field, enter the search term (e.g., the patient ID or a part of it), and press Enter to start the search.

Show all

> → All results associated with the entered search term are displayed in the result list. The entered search term is highlighted in blue.

▶ To search for patient results by using a filter

1 Choose the **Result validation** module.

2 Choose the [filter icon] button.

3 Select any of the filter criteria that is displayed.

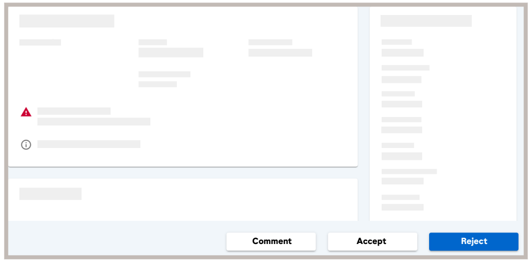
4 Choose the **Search** button to start the search.

5 View the search result.

- The selected filter criteria are displayed in the header. If you remove one, the search result is updated automatically.
- To clear the filter, choose the **Clear all** button.

If more than one filter criterion has been selected, result records which meet all criteria are displayed non-hierarchically.

Adding a comment to a patient result



1 Under **Result validation** > **Unreviewed patient results**, choose the specific patient result to leave a comment on.

2 To display the **Comment** column of the table, drag the scroll bar to the right.

→ The **Result details** screen is displayed.

3 On the **Result details** screen, choose the **Comment** button.

4 Enter a comment or choose from the **Predefined comments**, regardless of the status of the patient result, and choose the **Add** button.

Save as a predefined comment

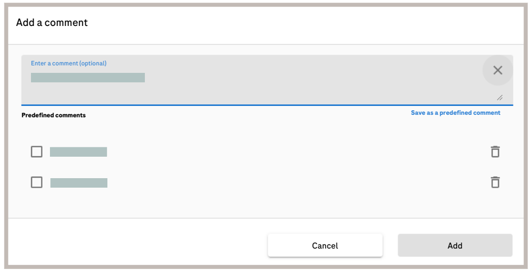
Only some instruments support predefined comments. To clarify whether your instrument supports predefined comments, refer to the respective device driver user manual or ask your Roche Service representative.

→ The message **Comment has been added to the patient sample.** appears in the upper right corner. The comment is visible on the **Result details** screen.

5 To accept or reject the result, proceed with the following:

- To accept the result, choose the **Accept** button.
- To reject the result, choose the **Reject** button.

→ The message **Patient sample has been successfully accepted.**or the message **Patient sample has been successfully rejected.** appears in the upper right corner.



6 To save a comment as a predefined one, enter your comment, and choose the **Save as a predefined comment** button.

→ The comment is added to the list of predefined comments.

Viewing comments on patient results

1 Under **Result validation** > **Unreviewed patient results**, choose the specific patient result to view the comments.

2 To display the **Comments** column of the table, move the scroll bar.

Comments

→ The number of comments for the selected patient result is visible on the **Result details** screen.

3 Choose the **Comments** button to view the details of the comments.

Result details

Reviewing patient result details

1 Choose the **Result validation** module.

2 From the result list, choose a result.

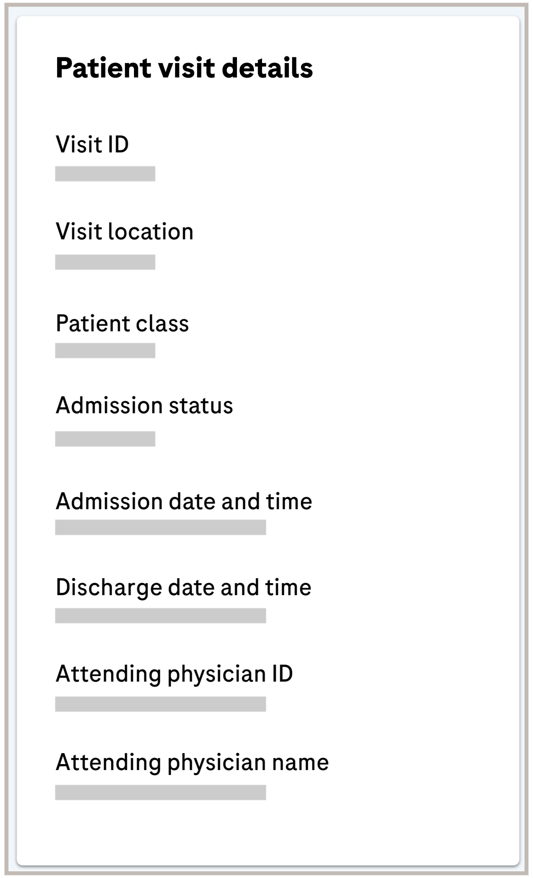
→ The result details are displayed.

3 In the **Result details** group box, you can view the available information about the test result, including the measured value and associated *data alarms*.

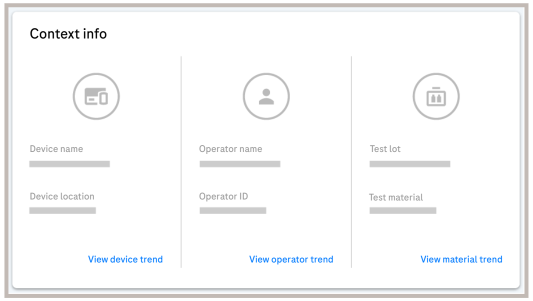
4 In the **Patient details** group box, you can view the patient demographics of the related patient.

5 To change the assigned patient, choose the **Change patient** button. If the intended patient record does not exist yet, choose the **Add patient** button. For more information, refer to the
Adding a patient
procedure.

→ A pop up window appears to enter patient and visit information.



6 In the **Patient visit details** group box, you can view the visit demographics.



7 In the **Context info** group box, you can view additional information about the device, operator, and test lot.

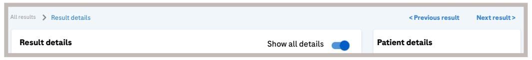
- To view a trend analysis related to the device, operator, or test lot, choose the respective button.

8 You can view the trend analysis for devices issues, operator issues, and material issues by choosing the **View device trend** button.

- When updating the database, only the last 6 months of issue trend data are migrated.
- View trend by week: depending on the browser, the displayed start day (Monday or Sunday) of each week is based on the following settings: - Safari, Chrome and Microsoft Edge: *Browser language setting*

9 To review the result details, you can accept or reject the result.

- To accept the result, optionally enter a comment, and choose the **Accept** button.
- To reject the result, optionally enter a comment, and choose the **Reject** button.



10 To navigate the patient result details, choose the **Next result >** and **< Previous result** buttons in the upper right corner.

Related tasks

- Changing the patient associated with a result (p. 178)
- Accepting unreviewed patient results from the result list (p. 179)
- Rejecting unreviewed patient results from the result list (p. 179)

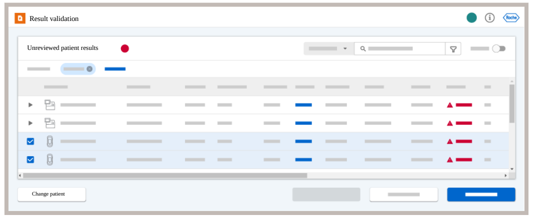
Changing the patient associated with a result

You can change a patient who is incorrectly associated with a test result until the result has been accepted or rejected.

You can change the patient for one or several results directly from the result list. Alternatively, you can change the patient for one result from the result details.

You can add a new patient directly from the change patient workflow. If the patient does not exist, create the patient and then assign the result to the newly created patient.

▶ To change the patient associated with results from the result list



1 Choose the **Result validation** module.

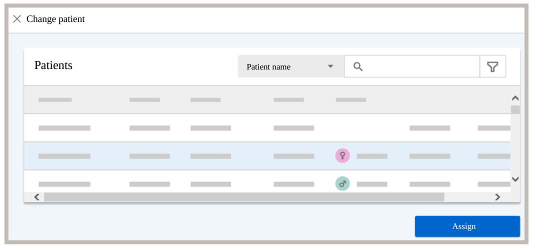
2 Optionally, expand the rows for the device types.

3 Select the check boxes of the unreviewed results for which you want to change the patient.

> The **Change patient** button becomes available.

4 Choose the **Change patient** button.

Change patient



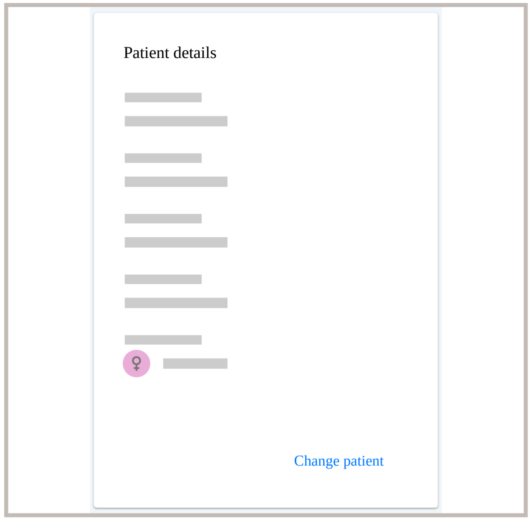
5 From the patient list, choose the correct patient visit, and choose the **Assign** button.

6 On the dialog box, ensure that the patient assignment is correct and choose the **Confirm** button.

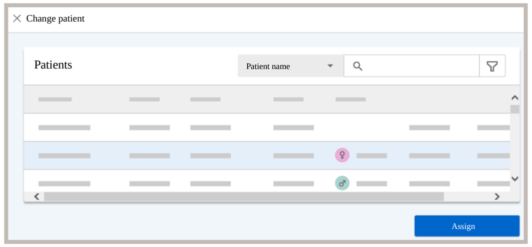
▶ To change the patient associated with a result from the result details

1 Choose the **Result validation** module.

2 Choose the unreviewed result for which you want to change the patient.



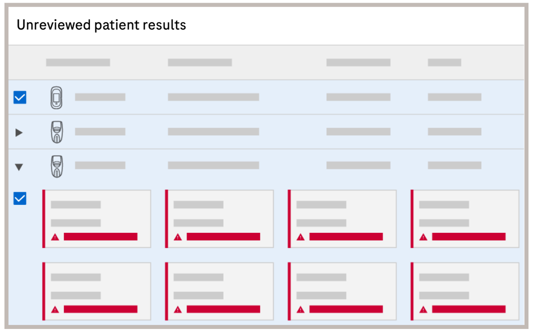
3 From the **Patient details** group box, choose the **Change patient** button



4 From the patient list, choose the correct patient visit, and choose the **Assign** button.

5 On the dialog box, ensure that the patient assignment is correct and choose the **Confirm** button.

Accepting unreviewed patient results from the result list



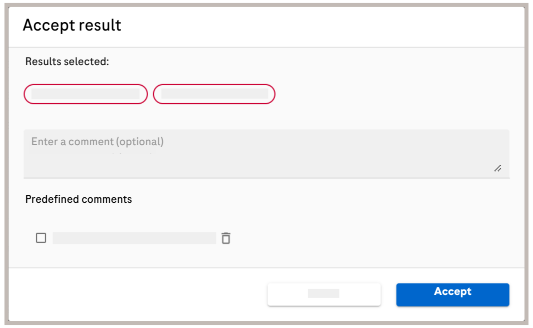
1 Choose the **Result validation** module.

2 To choose the results, do the following:

- To choose results with 1 parameter, choose the check box.
- To choose results with multiple parameters, choose the arrow to expand the row and then select the check box.
- To find a specific result quicker, use the search or filter feature.

3 Choose the **Accept** button.

Result details



4 In the dialog box, optionally enter a comment or choose one of the **Predefined comments**.

Save as a predefined comment

5 Choose the **Accept** button.

Rejecting unreviewed patient results from the result list

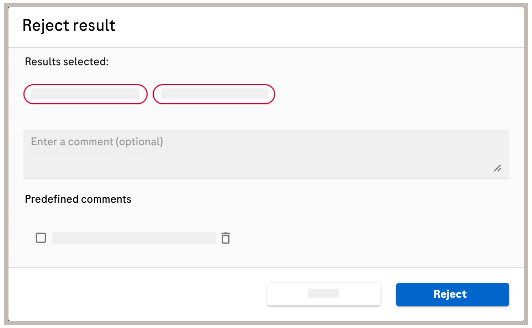
1 Choose the **Result validation** module.

2 To choose the results, do the following:

- To choose results with 1 parameter, choose the check box.
- To choose results with multiple parameters, choose the arrow to expand the row and then select the check box.
- To find a specific result quicker, use the search or filter feature.

3 Choose the **Reject** button.

Result details



4 In the dialog box, enter an optional comment or choose one of the **Predefined comments**.

Save as a predefined comment

5 Choose the **Reject** button.

Resending accepted results to the host

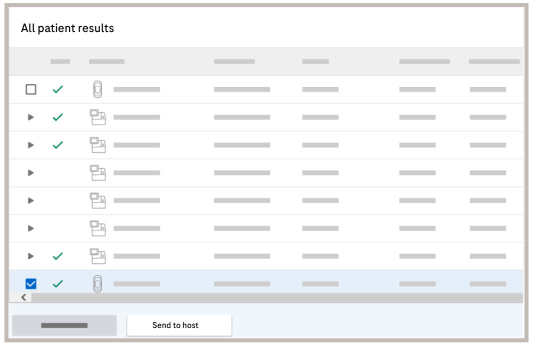
You can manually resend accepted results to the host.

This allows you to resend accepted results, e.g., if automatic sending failed.

You can resend several results from the result list. Alternatively, you can resend one result from the result details.

You cannot resend unreviewed or rejected results to the host.

▶ To resend accepted results to the host from the result list



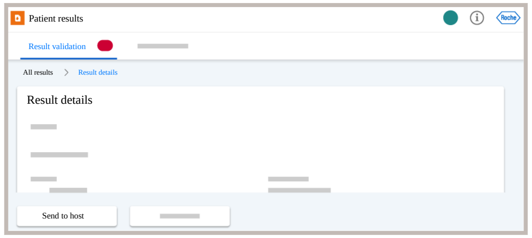
1 Choose the **Result validation** module.
2 Choose the **Show all** toggle button.
3 Optionally, expand the rows for the device types.
4 Select the check boxes of the accepted results that you want to resend to the host.
5 Choose the **Send to host** button.

Send to host

6 In the dialog box, choose the **Send to host** button.

→ In the result list, the **Host date and time** and **Host status** column is updated.

▶ To resend an accepted result to the host from the result details



1 Choose the **Result validation** module.
2 Choose the **Show all** toggle button.
3 From the result list, choose the accepted result that you want to resend to the host.

→ The result details are displayed.

4 On the result details, choose the **Send to host** button.

Send to host

5 In the dialog box, choose the **Send to host** button.

→ In the result list, the **Host date and time** and **Host status** column is updated.

Reviewing the host status

After sending the results to the host, you can check the status of the results in the **Result validation** screen using the **Host status** column.

If the host finds an issue in your results, this is reflected in the **Host status** column. You must then check and perform the corrective actions displayed in the **Host information** column and resend the results to the host.

The host system may find 2 types of issues in your results:

- The patient or visit ID is incorrect.
- The demographic information (i.e name or date of birth) is incorrect

1 Choose the **Result validation** module.

2 Choose the **Show all** toggle button.

→ The **Host status** and **Host information** column are displayed. To find the **Host status** and **Host information** columns, drag the scroll bar to the right.

3 If the host identified an incorrect patient ID or visit ID,

- Undo validation for the result. See Undoing validation for accepted results
- Modify the result according to the **Host information**.
- Accept the result again. See Accepting unreviewed patient results from the result list

→ The result is automatically sent to the host again.

4 If the host identified an incorrect demographic,

- Modify the result according to the **Host information**.
- Send the result to the host again. See Resending accepted results to the host

Undoing validation for accepted results

If configured, you can undo the validation for accepted results again.

You can undo the validation for one or several results from the result list. Alternatively, you can undo the validation of one result from the result details.

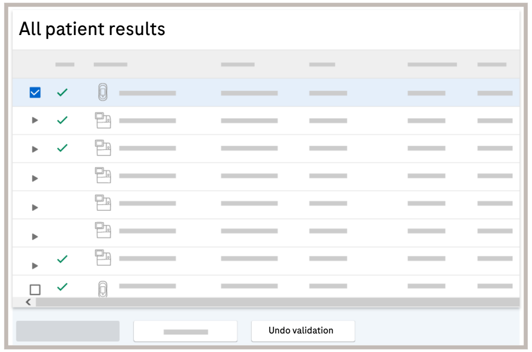
- Undo the validation result enabled in **Configuration** > **General settings** > **Results & Patients** > **Enable undo validation for samples**
- Result was measured in the time period configured for undoing validation in **Configuration** > **General settings** > **Results & Patients** > **Timeframe to undo validation for samples (Hours)**
- Optionally, generation of a new sample ID enabled in **Configuration** > **General settings** > **Results & Patients** > **Generate new sample ID for samples where sample validation was undone**

▶ To undo validation for results from the result list

1 Choose the **Result validation** module.

2 Switch on the **Show all** toggle button.

3 Optionally, expand the rows for the device types.



4 Select the check boxes of the accepted results for which you want to undo validation.

→ The **Undo validation** button becomes available.

5 Choose the **Undo validation** button.

Undo validation

6 Optionally, in the dialog box, enter a comment.

7 In the dialog box, choose the **Undo validation** button.

→ The result is unreviewed again.

In the result list, the **Status** column is updated.

If configured, a new sample ID is generated for the result and displayed in the **Sample ID** column.

▶ To undo validation for a result from the result details

1 Choose the **Result validation** module.

2 Switch on the **Show all** toggle button.

3 From the result list, choose the accepted result for which you want to undo validation.

→ The result details are displayed.

4 On the result details, choose the **Undo validation** button.

Undo validation

5 Optionally, in the dialog box, enter a comment.

6 In the dialog box, choose the **Undo validation** button.

→ The result is unreviewed again.

In the result list, the **Status** column is updated.

If configured, a new sample ID is generated for the result and displayed in the **Sample ID** column.

Exporting results

When exporting a large amount of results, the export may not be immediate. Avoid exporting more than the default amount (1000) of results as this may lead to an error.

About the export file

The export file contains the results that fulfill the current filter criteria in the results list.

The exported file contains the complete result information, including details that are not displayed in the result list, e.g., the device serial number.

There are the following differences between the display of information in the results list and in the export file:

- Results that have status “`Unreviewed`” in the resultslist have status “`Pending`” in the export file.
- Depending on configuration, the date and time formats may differ between the results list and the export file. If you need to change the configuration, contact your Roche Service representative.

1 Choose the **Result validation** module.

2 Optionally, switch on the **Show all** toggle button and/or filter the result ( ) list as described in

Searching and filtering

3 Choose the **Export** button.

4 Optionally, in the **Export as .csv** dialog box, enter the number of results to be exported.

Depending on the number of results, the export may take an extended period of time. Errors may occur if you export more results than the default number (1000).

5 In the **Export as .csv** dialog box, choose the **Export** button.

→ A CSV file containing the list of results is downloaded. On iOS devices, the exported file is attached automatically to an email for sending it.

Patient administration

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Patients

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Patients

Searching for patients and viewing patient details

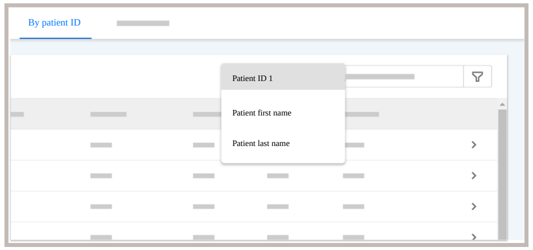
You can search for patients registered in the application and view their details on the **By patient ID** tab.

To find individual patients, search by patient name or patient ID or use the advanced search options by combining multiple search criteria.

You can also search for patients by their visits on the corresponding **By visit ID** tab.

> By default, only basic demographics fields are available for search criteria, filter options and detail views. You can assign additional fields in the configuration module: **Configuration** > **System** > **Demographic assignment**

▶ To search for patients by patient name or patient ID



1 Choose **Patient administration** > **By patient ID**.

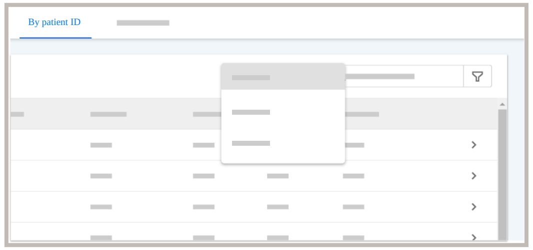
2 Enter the search criteria:

- To search by patient name, choose the **Patient first name** or **Patient last name** option, enter the patient name, and press Enter.
- To search by patient ID, choose the **Patient ID 1** option, enter the patient ID, and press Enter.

3 View the search result.

- The entered search term is highlighted in blue.
- The entered search term is displayed in the search result header.
- To clear the filter, choose the **Clear all** button.

▶ To search for patients by using advanced filter options



1 Choose **Patient administration** > **By patient ID**.

2 Choose the button.

3 Select any of the following filter criteria:

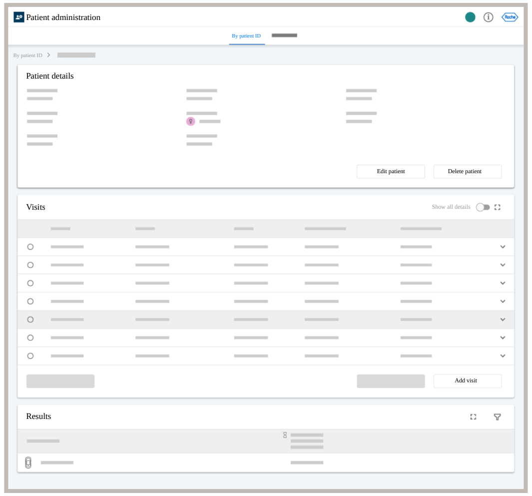
- **Patient ID 1** - Enter a patient ID and press Enter. You can enter multiple patient IDs.
- **Patient first name** or **Patient last name**, or a combination - Enter a patient name and press Enter. You can enter multiple names.
- **Date of birth from** and **Date of birth to** - Use the two options to select a date of birth range. Alternatively, choose the same date for both options if you know the exact date of birth.
- **Gender** - Select the gender from the drop-down list.

Clear all

4 View the search result.

- The selected filter criteria are displayed in the search result header. If you remove one, the search result is updated automatically.
- To clear the filter, choose the **Clear all** button.

▶ To view patient details



1 Choose **Patient administration** > **By patient ID**, and search for patients.

2 Choose the › button for the patient and view the patient and visit details.

3 In the **Patient details** group box, the patient demographics are displayed.

- To edit the patient, choose the **Edit patient** button.
- To delete the patient, choose the **Delete patient** button (requires additional rights).

4 In the **Visits** group box, all patient visits are listed.

- To expand the visit list to full screen mode, choose the icon. To return to normal screen mode, choose the icon.
- To expand visit details, choose the ⌄ button for the visit. Alternatively, enable the **Show all details** option.
- To edit a visit, select the visit and choose the **Edit visit** button.
- To delete a visit, select the visit and choose the **Delete visit** button.
- To add a visit, choose the **Add visit** button.

5 In the **Results** group box, you can find all test results for the selected patient.

- To expand the result list to full screen mode, choose the icon. To return to normal screen mode, choose the icon.
- To filter for specific tests or for a certain measurement date and time, choose the button. Select the desired tests, define the measurement date and time, and choose the **Search** button.
- To view result details, choose a value in the results list.

Clear all

Adding a patient

You can also add a patient and visit details from the **Result validation** screen. In the visit details, you can also add new attending physicians. When adding a patient, a wizard guides you through the process.

Reviewing patient result details

Patient ID and visit ID requirements

The patient ID and visit ID can contain uppercase or lowercase letters (A to Z), numbers (0 to 9), and the additional characters +,/,_,-,$,\,@.

> The user ID is case sensitive. Users with the ID "A123" and "a123" are considered as two different users by the application. Therefore, it is strongly recommended to use consistent casing (i.e. only uppercase or only lowercase letters) for all user IDs on the application and on all associated devices.

Patient first name and last name requirements

The maximum length of the **Patient first name** and **Patient last name** is 30 characters.

1 Choose **Patient administration** > **By patient ID** and choose the **Add patient** button.

Patient administration > **By visit IDAdd patient**

2 Follow the wizard and enter or assign the following information:

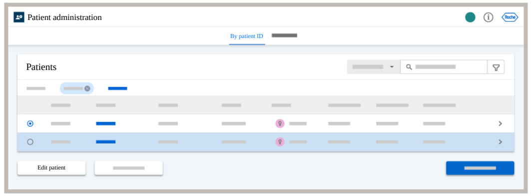
- **Patient information**: **Patient ID 1**, **Patient first name**, **Patient last name**, and **Date of Birth** are mandatory.
- **Visit information**: **Visit ID**, **HIS Location**, **Admission Date/Time**, and **Admission Status** are mandatory.
- **Attending physician**: You can optionally add an existing attending physician to the visit from the drop-down list. If the attending physician is not listed, choose **Add attending physician** and enter the **Physician ID** and **Physician name** in the dialog box.If you choose **Add attending physician** again, you can edit the previous entry. You cannot add another attending physician.

3 Choose the **Finish** button.

4 Choose the **Confirm** button.

Editing a patient

You can edit the patient details.



1 Choose **Patient administration** > **By patient ID**, and search for patients.

2 Select a patient and choose the **Edit patient** button.

3 Follow the wizard and edit the patient details.

4 Choose the **Save** button.

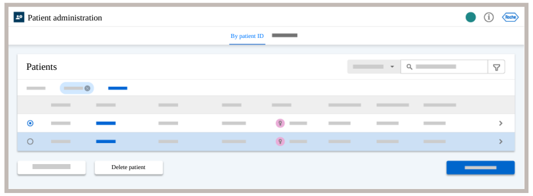
5 Choose the **Confirm** button.

Deleting a patient

You must have additional rights to delete patients.

When you delete a patient, all related data is also deleted:

- Patient visits
- Test results
- Comments and demographics



1 Choose **Patient administration** > **By patient ID**, and search for patients.

2 Select a patient and choose the **Delete patient** button.

3 Choose the **Confirm** button.

Visits

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Visits

Searching for visits and viewing visit details

You can search for patient by their visits in the application and view their details on the **By visit ID** tab.

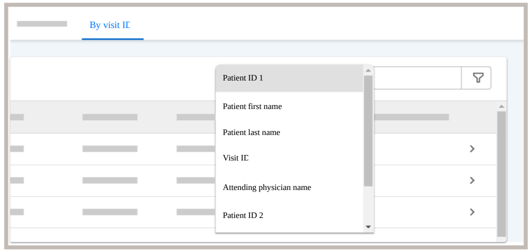
To find individual visits, search by visit ID or use the advanced search options by combining multiple search criteria.

You can also search for patients on the corresponding **By patient ID** tab.

By default, only basic demographics fields are available for search criteria, filter options and detail views. You can assign additional fields in the configuration module: **Configuration** > **System** > **Demographic assignment**

▶ To search for visits by visit ID, patient name, or physician name

1 Choose **Patient administration** > **By visit ID**.



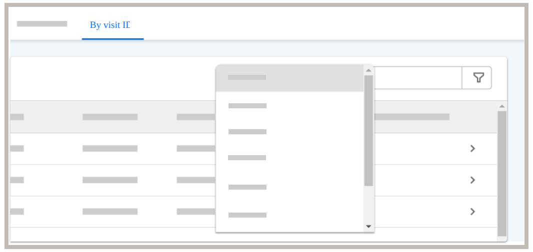
2 Choose your search option:

- To search by visit, choose the **Visit ID** option, enter the visit ID, and press the Enter key.
- To search by physician name, choose the **Attending physician name** option, enter the physician name, and press the Enter key.
- To search by patient name, choose the **Patient first name** or **Patient last name** option, enter the patient name, and press the Enter key.
- To search by patient ID, choose the **Patient ID 1** option, enter the patient ID, and press the Enter key.

3 View the search result.

- The entered search term is highlighted in blue.
- The entered search term is displayed in the search result header.
- To clear the filter, choose the **Clear all** button.

▶ To search for visits by using advanced filter options



1 Choose **Patient administration** > **By visit ID**.

2 Choose the button.

3 Select any of the following filter criteria:

- **Patient ID 1** - Enter a patient ID and press Enter. You can enter multiple patient IDs.
- **Patient first name** or **Patient last name**, or a combination - Enter a patient name and press Enter. You can enter multiple names.
- **Date of birth from** and **Patient last name** - Use the two options to select a date of birth range. Alternatively, choose the same date for both options if you know the exact date of birth.
- **Gender** - Select the gender from the drop-down list.
- **Visit ID** - Enter a visit ID and press Enter. You can enter multiple visit IDs.
- **Location** - Select the location from the drop-down list.
- **HIS location** - Select the HIS location from the drop-down list.
- **Admission date and time** - Use the two options to select a from / to range. Alternatively, choose the same date for both options if you know the exact admission date.
- **Discharge date and time** - Use the two options to select a from / to range. Alternatively, choose the same date for both options if you know the exact discharge date.
- **Attending physician ID** - Select the physician ID from the drop-down list.
- **Attending physician name** - Enter a physician name and press Enter. You can enter multiple names.
- **Admission status** - Select the admission status from the drop-down list.

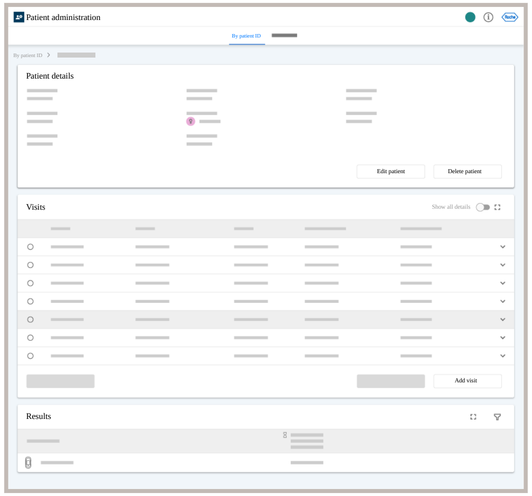
- **Patient class** - Select the patient class from the drop-down list.

Clear all

4 View the search result.

- The selected filter criteria are displayed in the search result header. If you remove one, the search result is updated automatically.
- To clear the filter, choose the **Clear all** button.

▶ To view visit details



1 Choose **Patient administration** > **By visit ID**, and search for visits.

2 Choose the > button for the patient visit to view the patient visit details.

3 In the **Patient details** group box, the patient demographics is displayed.

- To edit the patient, choose the **Edit patient** button.
- To delete the patient, choose the **Delete patient** button.

4 In the **Visits** group box, the selected visit is displayed.

- To expand the visit o full screen mode, choose the ⛶ icon. To return to normal screen mode, choose the ⛶ icon.
- To expand visit details, choose the ⌄ button for the visit. Alternatively, enable the **Show all details** option.
- To edit a visit, select the visit, and choose the **Edit visit** button.
- To delete a visit, select the visit, and choose the **Delete visit** button.
- To add a visit, choose the **Add visit** button.

5 In the **Result** group box, you can find all test results for the selected patient.

- To expand the result list to full screen mode, choose the icon. To return to normal screen mode, choose the icon.
- To filter for specific tests or for a certain measurement date and time, choose the button. Select the desired tests, define the measurement date and time, and choose the **Search** button.
- To view result details, choose a value in the results list.

Clear all

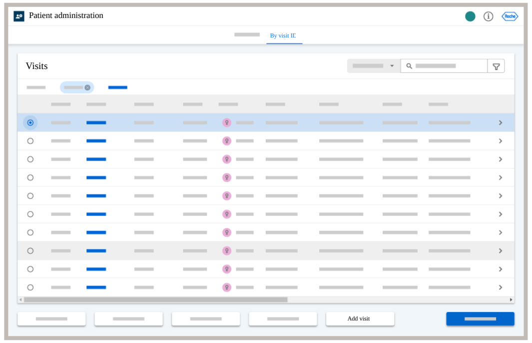
Adding a visit

You can add visits to a patient. You can also create and add new attending physicians.

Patient ID and visit ID requirements

The patient ID and visit ID can contain uppercase or lowercase letters (A to Z), numbers (0 to 9), and the additional characters +,/,_,-,$,\,@.

The user ID is case sensitive. Users with the ID "A123" and "a123" are considered as two different users by the application. Therefore, it is strongly recommended to use consistent casing (i.e. only uppercase or only lowercase letters) for all user IDs on the application and on all associated devices.



1 Choose **Patient administration** > **By visit ID** and search for visits.

2 Select a patient and choose the **Add visit** button.

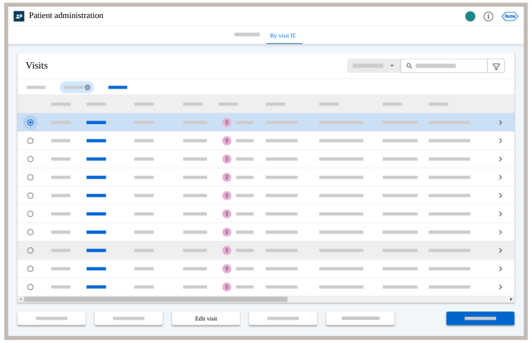
3 In the **Visit information** wizard, enter or assign the following information:

- **Visit ID**, **HIS locations**, **Admission date and time**, and **Admission status** are mandatory.
- **Attending physician**: You can optionally add an existing attending physician to the visit from the drop-down list. If the attending physician is not listed, choose **Add attending physician** and enter the **Physician ID** and **Physician name** in the dialog box.If you choose **Add attending physician** again, you can edit the previous entry. You cannot add another attending physician.

4 Choose the **Finish** button.

5 Choose the **Confirm** button.

Editing a visit



1 Choose **Patient administration** > **By visit ID** and search for visits.

2 Select a patient and choose the **Edit visit** button.

3 In the **Visit information** wizard, edit the information:

- **Visit ID**, **HIS location**, **Admission date and time**, and **Admission status** are mandatory.
- **Attending physician**: You can optionally add an existing attending physician to the visit from the drop-down list. If the attending physician is not listed, choose **Add attending physician** and enter the **Physician ID** and **Physician name** in the dialog box.If you choose **Add attending physician** again, you can edit the previous entry. You cannot add another attending physician.

4 Choose the **Save** button.

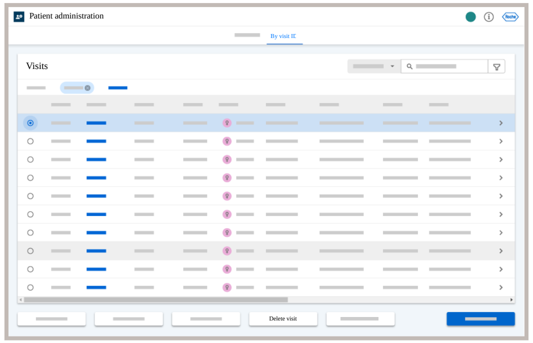
5 Choose the **Confirm** button.

Deleting a visit

When you delete a visit, all visit related data is also deleted:

- Test results
- Comments and demographics

If the last visit of a patient is deleted, the application deletes the patient as well.



1 Choose **Patient administration** > **By visit ID** and search for visits.

2 Select a visit and choose the **Delete visit** button.

3 Choose the **Confirm** button.

Reports

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QC reports

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Creating a QC report

Creating a QC report

To review and *archive* QC results and statistics of your devices, you can create reports for specific devices, QC levels, parameters, materials, and status.

Date, time, and number format

Depending on the browser used, the QC report inherits the date, time, and number format from the following settings:

- Chrome (tablet and desktop): browser language and region settings
- Safari: browser language and region settings
- Microsoft Edge: browser language and region settings

This date format is used for the report creation date displayed in the application and may therefore differ from the format you have selected in the **Personal settings** menu.

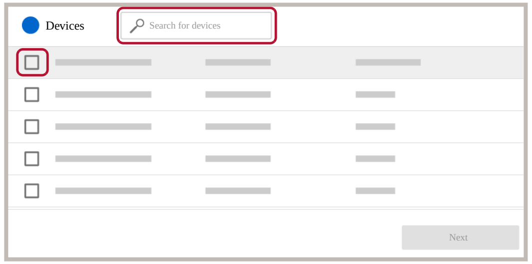
Supported languages

Resources for QC reports can be translated into any language and installed via language package.

If you have selected an untranslated QC report language in the **Personal settings** menu, the QC report is displayed in English.

Configuring personal settings

Creating a QC report by device



1 Choose **Reports** > **QC reports** > **Creation**.

2 From the **QC report by device** window, choose the **Create** button.

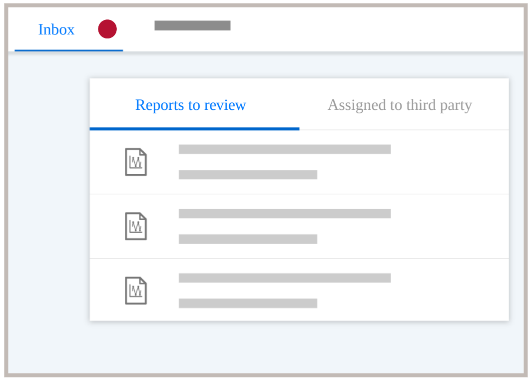
3 Select the devices for which you want to create a report, and choose the **Next** button.

- To find a specific device, enter the device name, serial number, or model name in the search field.
- To include all devices in the report, select the check box next to the **Device name** column header.

4 Select the first and last date to be included in your report, and choose the **Next** button.

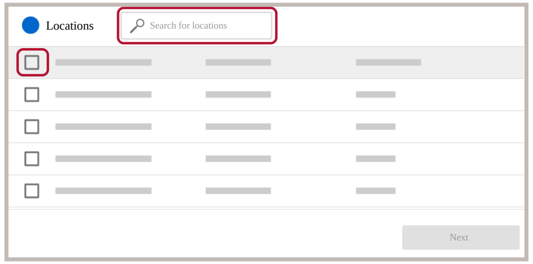
5 Select additional criteria you want to include in your filter, and choose the **Finish** button.

Depending on the amount of data you have selected for your report, the creation of the report may take several minutes.



6 You can find the created report in the **Inbox** tab.

Creating a QC report by location



1 Choose **Reports** > **QC reports** > **Creation**.

2 From the **QC report by location** menu, choose the **Create** button.

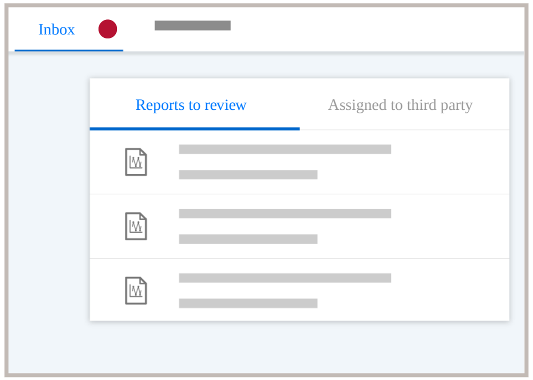
3 Select the locations for which you want to create a report, and choose the **Next** button.

- To find a specific location, enter its name in the search field.
- To include all locations in the report, select the check box next to the **Location** column header.

4 Select the first and last date to be included in your report, and choose the **Next** button.

5 Select additional criteria you want to include in your filter, and choose the **Finish** button.

Depending on the amount of data you have selected for your report, the creation of the report may take several minutes.



6 You can find the created report in the **Inbox** tab.

Reviewing and signing a report

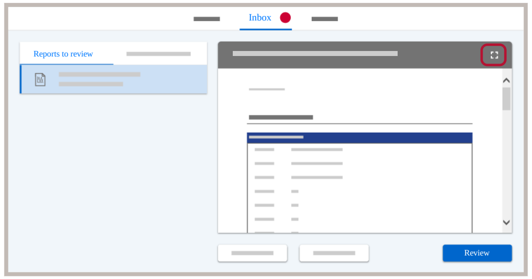
NOTICE

Electronic records and electronic signature

To be able to use the report review and signature process as equivalent substitute for paper records and traditional handwritten signatures:

▶ Follow your local regulations regarding electronic records and electronic signature.

> Reports are created in the language configured in the personal settings of your user.
>
> Numbers and dates are displayed in the format of your web browser.



1 Choose **Reports** > **QC reports** > **Inbox**.

2 On the **Reports to review** tab, choose a report and choose the **Review** button.

3 On the **Review report** dialog box, do the following:

- Optional: Enter an comment.
- Sign with your mouse, finger, or a touch screen stylus. To redo the signature, choose the **Clear** button.
- To add additional reviewers, choose them from the **Reviewer** drop-down list.
- Choose the **Confirm** button.

→ The report is moved to your **Assigned to third party** tab. For the additional reviewers. the report is available in their **Reports to review** tab. Once all reviewers have signed, you can find it in **Reports** > **Archive**.

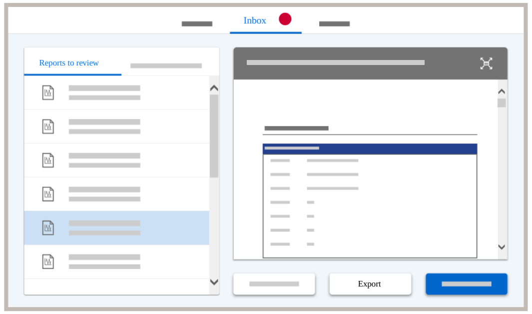
Exporting a report

If you want to use a report outside the application, you can export it as PDF or an Excel file.

When you use the application on an iOS (Apple) tablet, by default the email app adds automatically the application's link and the type of file to the email body. This is not harmful, it is normal behavior.

Before sending the email, delete the link manually.

For more information, contact your Roche representative.



1 Do one of the following:

- If you search for reports with pending signatures, choose **Reports** > **QC reports** > **Inbox**.
- If you search for singed reports, choose **Reports** > **QC reports** > **Archive**.

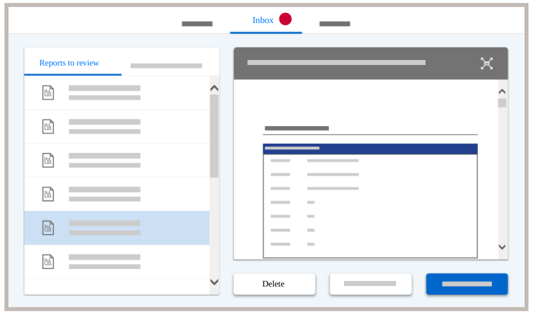
2 Choose the report you want to export and choose the **Export** button.

3 Select the export format (PDF or Excel) and choose the **Confirm** button.

→ The report is downloaded in the selected format to your computer or tablet.

Deleting a report

If you have rights to create reports, you can also delete them, even if reviewers have already signed it.



1 Do one of the following:

- If you search for reports with pending signatures, choose **Reports** > **QC reports** > **Inbox**.
- If you search for singed reports, choose **Reports** > **QC reports** > **Archive**.

2 Choose the report you want to delete and choose the **Delete** button.

3 In the **Confirmation** dialog box, choose the **Delete** button.

→ The report is deleted.

User Activity Report

The User Activity Report allows you to track key actions for one or multiple users.

The report includes:

Login information: A daily count of all user logins, capturing both local and in the **navify**®Portal via single sign-on (SSO).

Result validation: The total number of patient and QC results that each user has manually accepted or rejected.

Tracking the user activity report

You can track the user activity report in **Reports** > **Signable reports** > **User activity report**.

The table includes the following information:

Date: Time of the logon (day/month/year).

User ID: User name and user ID.

Number of logons: Number of logons per day.

Manually accepted QC results: Number of QC results manually accepted.

Manually rejected QC results: Number of QC results manually rejected.

Manually accepted patient results: Number of patient results manually accepted.

Manually rejected patient results: Number of patient results manually rejected.

Other reports

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Other reports

The application provides you a selection of predefined reports you can create.

About other reports

Under **Reports** > **Other reports**, you can select a report from a list of predefined reports.

- You cannot edit a predefined report.
- You cannot create new reports.
- Only records are included in the reports for which you have the required user rights.
- Only reports for installed modules are available.

The available reports serve various purposes, such as paper-based documentation, review of data, or sharing of data with people that have no access to the application.

The report feature allows you to extract information from the application in a standardized report for documentation or information exchange purposes.

The reports provide individual query criteria to customize the information included in the report. When creating a report, you can also define the document setting such as format and password protection.

Report and user interface logic

The “Location” used with result-related reports refers to the location to which the instrument was assigned when the results were sent. The currently assigned location of the instrument may not be the same.

List of reports

Report categories

The reports are grouped by the following categories:

- **All Results** : reports about all types of results
- **Control**: reports about QC related information
- **Corrective**: reports about results for which corrective actions were required
- **Instruments**: reports related to the instruments, instrument data alarms, and instrument maintenance actions
- **Patient**: reports about patient-related information
- **System**: reports about system-related information
- **User**: logs with user-based information

Report types

Under each report category, you can choose from one or multiple report types:

Category	Type	Description
All results	All results (by date)	This report provides a list of all results (Test, QC, Linearity) run over the defined time period, including instrument maintenance and instrument exceptions. This report contains a full log of all results received from an instrument in combination with instruments and user comments related to these results. This report lists all results that match the filter criteria ordered by date.
	All results (by location and instrument)	This type contains the same information as the "`by date`" type result report above, but the results are grouped by location and instrument, and not chronologically by date. If a PDF is slow to generate or fails to generate, choose a specific device in the filter criteria. This is required as the system applies the device filter before the date filter, which successfully reduces the volume of data for processing.
Control	Lot listing (By materials)	This report provides an overview on the materials maintained in the application. If you do not apply any filter criteria, all 3 material types (Test, QC, Linearity) are included in the report grouped into different sections. The total number of lots identified is displayed in the report but only ones having an associated range are displayed in the report list.

Category	Type	Description
	Lot statistics (by instrument type)	This report provides a statistical analysis of the performance of the test lots based on QC runs with the test lots. The report includes rejected as well as pending results. Statistics for QC results are only included in the report if QC lot and test lot remained unchanged during the reported calendar month.
	QC summary (by location and instrument)	This report summarizes the QC results based on a statistical evaluation sorted by location, instrument type, and individual instrument. It allows you to compare QC results between instruments.
	QC summary (by instrument type and location)	This report summarizes QC results based on a statistical evaluation sorted by instrument type and location. It allows you to compare QC results between locations.

Category	Type	Description
	Linearity	This report tests linearity results against linearity target values defined for the material lot. The linearity report displays only results with review status “accepted”. If 2 linearity results have been performed at the same time (i.e., same time stamp down to the minute) and have the same result for the same level, the linearity report displays 1 of the results only. This does not affect the calculated mean or the displayed graph.
	Monthly QC summary (by location)	This report provides a QC summary by “`sending location`” and date (month) for each instrument type, and a comparable statistical analysis for the entire site. The report consists of 2 parts. The first part contains a summary for the entire site and for each month across all locations. The second part contains a summary for each month grouped by the sending location. The report does not include rows with null values in the summary (no results for given instrument type and lot combination).
Control (Rili-BÄK)	LED status history report	
	Rili-BÄK monthly report	

Cate-gory	Type	Description
Correc-tive	Corrective action log	This report only includes results with corrective actions.
Instru-ments	Instrument events log	This report provides a log of the instrument data events that were raised within a certain time period.
Patient	Patient history	This report contains test results for the selected patients. Analytes are listed together with testing dates, their corresponding results, and the visit information. If you apply the filter criteria “`Result alarms`”, “`Alarm group`” and “`Validation status`”, you can query patient results with abnormal values or exceptions. You can print reports of results that have not been validated or have been rejected.
System	Sample count (by instrument type and location)	This report provides the amount of tests performed on individual instruments, sorted by location and instrument type.

Cate-gory	Type	Description
	Test/sample count (by location and instrument type	This report provides statistics on the numbers of performed tests and samples for specific time period and location. You can filter for a single location, all locations, or no location. The location is unknown in the following situations: ▪ *For QC results:* during an update, the sending location could not be identified because there were no patient results for that period. ▪ *For QC and patients results:* the measurement was performed when the instrument was not yet assigned to a location.
	POCT monthly feedback	This report provides information about the site for which the user is currently logged on. Depending on the applied filter criteria, the report lists the results for each sending location, instrument, test, and calendar month.
	Sample count (by panel)	This report provides a summary of all samples (patient, QC,linearity) measured over the defined time period.

Cate-gory	Type	Description
	Test count (by location and Instrument type)	The report provides the number of tests performed per location and per instrument type for a defined time period. You can filter for a single location, all locations, or no location. The location is unknown in the following situations: ▪ *For QC results:* during an update, the sending location could not be identified because there were no patient results for that period. ▪ *For QC and patients results:* the measurement was performed when the instrument was not yet assigned to a location.
User	User certification history	This report documents the history of users and their certifications, requested tasks, and their completion status. Data that has been created before the migration to version 2.1.0 has the completion status Unknown.
	User statistics	This report analyzes QC and patient results per user sorted by sites and instrument types.

Report categories and types

List of filter criteria

Each report type includes unique and common filter criteria. The available filter criteria are explained below:

Filter criterion	Description
Alarm group	Groups of alarms triggered by the result.
Assigned location	Name of location that reported the test results.
Certification name	The type of certification to be used. Certification types are defined and assigned to users by the administrator.
Date range from/to	Results to be included in the report that were created within a specific time period. The start and end date are included.
Include empty results	You can include empty patient result records in the report. By default, empty result records are not included in the report.

Filter criteri-on	Description
Instru-ment name	Name of instrument on which the test was per-formed.
Instru-ment type	The type of instrument on which the test was per-formed.
Last name	Patient's last name (family name)
Linearity lot	Lot number of the linearity material with which the line-arity test have been per-formed.
Linearity material	Long name of the linearity material with which the line-arity tests were performed. For example: **Accu-Chek**® Inform Glucose Linearity Kit
Location	Name of location which re-ported the test results.
Location ID	ID of location which report-ed the test results

Filter criteri-on	Description
Lot	You can filter for lot informa-tion by the lot number of the material for which ranges are maintained. Depending on the material definition, this can be a test or QC ma-terial.
Material activa-tion date from/to	Filter for statistics on materi-al lots that have been acti-vated within a specific time period. The start and end date are included.
Material deacti-vation date from/to	Filter for statistics on materi-al lots that have been deac-tivated within a specific time period. The start and end date are included.
Material ID	ID of test material used to perform the test. For exam-ple: Glucose test strip
Material status	Status (Active, Inactive or Expired) of the test material for which the statistics are calculated.

Filter criteri-on	Description
Material type	Type of material for which the lot information is listed.
Month from/to	Start and end of reporting period
Number of pa-tients	The number of patients for which patient specific statis-tics are reported. You can select from 0 to 5 patients. After selecting the number of patients, you can enter their IDs.
Number of users	The number of generic users for which user specific sta-tistics are reported. You can select from 0 to 5 generic users. After selecting the number of generic users, you can enter their IDs.
Patient ID	Patient identification num-ber (Patient ID 1)

Filter criteri-on	Description
Patient results QC re-sults Linearity results	Choose one (or more) result types to be included in the report.
QC lot	Lot number of the QC or lin-earity material used to per-form the QC test. Does not apply to patient tests.
QC ma-terial	Long name of the QC mate-rial to be included in the re-port or graph. For example: **Accu-Chek® Inform** Glu-cose Control Solution
QC sta-tus	The instrument test QC sta-tus resulting from the QC or linearity result. Does not ap-ply to patient tests.
Report name	The name of the report chosen by the user. If you do not enter a report name, the default name (POCT Monthly Feedback) is used.

Filter criterion	Description
Result alarms	Severity level of alarms triggered for the results
Reviewed by	ID of the user who last changed the validation status of the result.
Sending status	Sending condition of result such as “`Pending`” or “`Sent (unidirectional)`”
Serial number	Serial number of the instrument on which the test was performed.
Site	Name of site which reported the test results.
Test	The abbreviated name of the test.
Test lot	Lot number of the test material used to perform the test such as test strip lot for **Accu-Chek® Inform** instrument family results.

Filter criterion	Description
Test material	Test material (short name) used to perform the test. For example: GLU-T for **Accu-Chek® Inform** Glucose test strips.
Timeframe from/to	Start and end date of the certifications to be included in the report.
User ID	In “`User Certification History`” report: Unique identification number assigned to the selected user. Depending on the number of active users registered in the application, filtering for user ID may take a long time. In other reports: ID of the user who performed the test.
User name	First and last name of the user for whom the statistics is performed.

Filter criterion	Description
User's Home Location	Home location of the user
Validation status	Current validation status of the results: Accepted, On Hold, Pending, Rejected, System Lock* * Automatic validation has created an alarm (applicable to patient results only).
Visit ID	You can filter results for a specific visit ID.
Workgroup	Name of the workgroup

Creating a report

User rights for records to be included in the report

1 Choose **Reports** > **Other reports**.
   - To view also inactive reports, choose the **Show inactive** button.

2 Choose a report from the list and choose the **Create** button.

3 Select the **Criteria** options and choose the **Next** button.

4 Configure the **Document Settings** options:

- **Document format**: Select the document format from the drop-down list. You can create the report in PDF, Word, or Excel format. The following options are only accessible when the report is created in PDF format. All of them are optional.
- **Watermark Text**: Enter a watermark text. This text is displayed as a watermark across all PDF pages.
- **Watermark Font Size**: Enter the font size of the watermark text.
- **Enable Side-by-Side View**: Select the check box to display the left and right report pages side-by-side with continues scrolling enabled.
- **Permissions Password**: Enter a permission password. This protects the PDF report from being edited.
- **Reader Password**: Enter a reader password. This password must be entered to open the PDF report.
- **Enable 128 Bit Encryption**: Select the check box to encrypt the PDF with a 128 bit encryption.
- **Allow Printing**: Select the check box to allow the PDF report to be printed using full quality. If this check box and the **Allow Degraded Printout** check box are unchecked, the document is not allowed to be printed.
- **Allow Content Copying**: Select the check box to allow to copy the content of the PDF report.
- **Allow Assembly**: Select the check box to allow document assembly i.e. the pages of the PDF report can be inserted into another PDF document and pages from other PDF documents can be inserted into this PDF report.
- **Allow Degraded Printout**: Select the check box to allow the document to be printed in degraded mode (i.e. reduced quality). If this check box and the **Allow Printing** check box are unchecked, the document is not allowed to be printed.
- **Allow Content Editing**: Select the check box to allow to edit the content of the PDF report.
- **Embed Fonts**: Select the check box to embed the fonts in the PDF report. Do not change this option.

5 Choose the **Create** button.

Activating or deactivating a report

▶ To activate a report

1 Choose **Reports** > **Other reports** and choose the **Show inactive** button.

2 Choose the report you want to activate and choose the **Activate** button.

▶ To deactivate a report

1 Choose **Reports** > **Other reports**.

2 Choose the report you want to deactivate and choose the **Deactivate** button.

Result and order matcher

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Overview

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About the Result and order matcher

The *Result and order matcher* is designed for respiratory therapists and other caregivers performing blood gas analysis to manage test results for which the application could not find a matching order.

It provides the following functionalities:

Matching results to orders For each unmatched result, a list of matchable orders is provided. You can select an order and match it to the result. The matched result is then sent to the LIS as long as it is validated and accepted in the application.

Sending results without matching order You can send a result to the LIS without a matching order (unsolicited sending). As soon as the result is validated and accepted in the application, it is automatically sent to the LIS.

Discarding results You can discard results. Discarded results are removed from the **Result and order matcher** module and not sent to the LIS. Discarded results are still retained in the application.

Limited access to results The user can view the following results in the **Result and order matcher** module:

- Results from devices assigned to a location (at the time of result reception) that is also assigned to the user.
- Results from devices without an assigned location at the time of result reception.

Ensure that the user has the correct locations assigned in the application.

> If a user has no locations assigned, all results are displayed without any restrictions regarding assigned device locations.

CAUTION

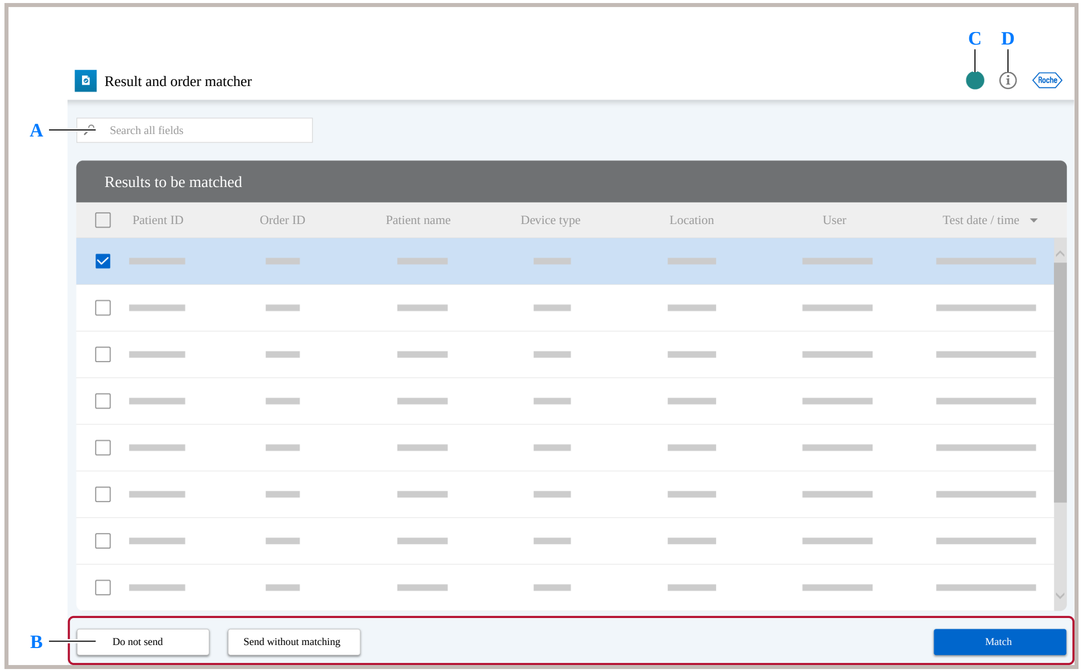
Incorrect patient assignment

The patient visit of a sample may be changed in the **Result and order matcher** module before the sample is sent to the LIS. The visit may be changed to a different visit of the same patient, or a visit of a different patient. Any action performed on that sample in the **Result and order matcher** module before the visit change is undone.

The sample may reappear in the **Result and order matcher** module to be matched or sent as unsolicited again, which can delay the sending of results to the LIS.

▶ Ensure to check the **Result and order matcher** module regularly.

Overview of the main page



A Free-text search field

B Action buttons

C Info button - access to User Assistance and software information

D User button - opens dialog box to log off, change password, and access the **General settings** screen.

User actions

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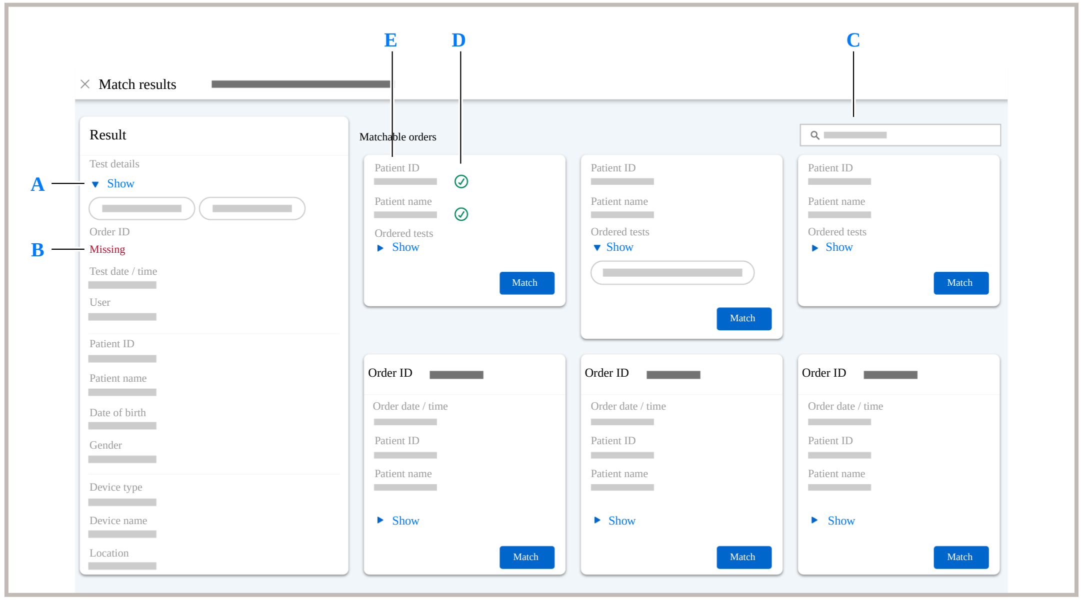
Matching results to orders

Matchable orders are displayed in the **Match results** menu.

Orders from the last 7 days are displayed according to the following ranking:

1. Orders with order ID matching the order ID of the result.
2. Orders with patient ID identical to the primary patient ID of the result. If multiple orders with matching patient ID exist, they are sorted chronologically by order date/time.
3. Remaining orders are listed chronologically by order date/time.



A Drop-down list displays all test results for this sample

B Unavailable information

C Free-text search field to find specific orders

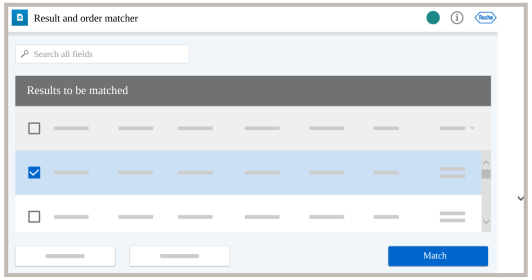
D The green check-mark indicates that this item of the order is equal to the corresponding item of the result.

E If the patient ID of the result is identical to the generic patient ID, "`Unknown`"is displayed in red color.

The generic patient ID is configured in the application.

System > **General settings** > **Results & Patients**

1 Choose the **Result and order matcher** module.



2 Select a result, and choose the **Match** button.

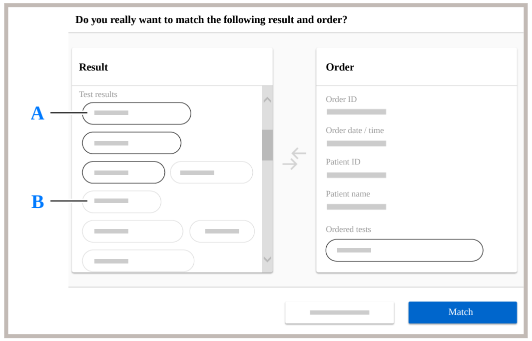
3 CAUTION! If you match a result to a wrong order, patient results sent to the LIS/host are associated with the wrong patient. This mismatch leads to incorrect medical records and billing.

→ Ensure that the order matching is correct.

4 Select an order, and choose the **Match** button.

5 Choose the **Match** button.

(**Results & Patients** > **Result validation**)



A Performed tests ordered by the matched order are displayed in black color

B Performed tests not ordered by the matched order are displayed in gray color

Matching scenarios

If you match a result to an order when the patient of the result differs from the patient of the order, the system acts as follows:

Matching scenario	Action
▪ Patient ID 1 and visit ID of the order exist in the application. ▪ The visit with the visit ID of the order is assigned to the patient with patient ID 1 of the order.	▪ The sample is reassigned to the visit with the visit ID of the order.
▪ Patient ID 1 and visit ID of the order exist in the application. ▪ The visit with the visit ID of the order is assigned to a different patient than the patient of the order.	▪ Error message is displayed - matching is not possible.
▪ Patient ID 1 and visit ID of the order do not exist in the application.	▪ A patient with the patient ID 1 of the order is created. ▪ A visit with the visit ID of the order is created and assigned to the newly created patient ID 1 of the order. The visit is not admitted. ▪ The HIS location of the order is assigned to the newly created visit. If the order contains no HIS location, no HIS location is assigned to the visit. ▪ The sample is reassigned to the visit with the visit ID of the order.

Matching scenario	Action
▪ Patient ID 1 of the order does not exist in the application. ▪ The visit ID of the order exists in the application. ▪ The visit ID of the order belongs to a different patient than the patient of the order.	▪ Error message is displayed - matching is not possible.

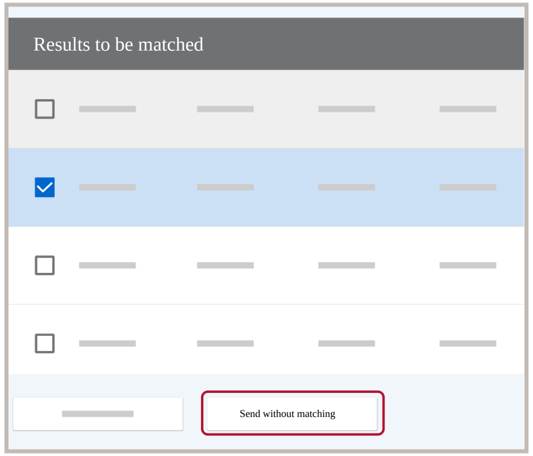
Matching scenario	Action
▪ The patient ID of the order exists in the application. ▪ The visit ID of the order does not exist in the application.	▪ A visit with the visit ID of the order is created and assigned to the patient ID 1 of the order. The visit is not admitted. ▪ The HIS location of the order is assigned to the newly created visit. If the order contains no HIS location, no HIS location is assigned to the visit. ▪ The sample is reassigned to the visit with the visit ID of the order.

Matching scenario	Action
▪ The patient ID 1 of the order exists in the application. ▪ The order does not contain a visit ID.	▪ An auto-generated visit is created which is not admitted. ▪ No HIS location is assigned to the auto-generated visit. ▪ The sample is reassigned to the newly created visit.
▪ The patient ID 1 of the order does not exist in the application. ▪ The order does not contain a visit ID.	▪ A patient with the patient ID 1 of the order is created. ▪ An auto-generated visit is created which is not admitted. ▪ No HIS location is assigned to the auto-generated visit. ▪ The sample is reassigned to the newly created visit.

Matching scenarios

Sending results without matching order

You can send a result to the LIS/host without a matching order (unsolicited sending).



1 Choose the **Result and order matcher** module.

2 CAUTION! If an order is received for a result after sending it to the LIS/host, the order can no longer be matched to the result.

→ Ensure that no order is expected when sending an unsolicited result.

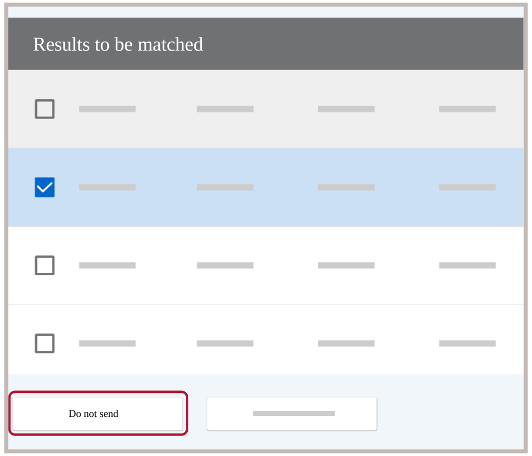
3 Select one or more results, and choose the **Send {results} as unmatched** button.

4 Choose the **Confirm** button.

(**Results & Patients** > **Result validation**)

Discarding results

You can discard results from the **Result and order matcher** module, and not send them to the LIS/host.



1 Choose the **Result and order matcher** module.

2 CAUTION! If an order is received for a result after discarding it, the order can no longer be matched to the result. To be sent to the LIS/host, the respective sample must be repeated.

→ Ensure that no order is expected when discarding a result.

3 Select one or more results, and choose the **Do not send** button.

4 Choose the **Confirm** button.

Result and order matcher

→ If the result was successfully discarded, the following feedback message is displayed: `The results will not be sent.`

Configuration

System

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General settings

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General settings

This component enables you to configure general settings which will influence the behavior of other modules and components within the application.

- System configuration options cannot be added, deactivated or deleted.
- The values defined in this screen affect the behavior of the entire application, therefore careful setting of the parameters is essential.

From this screen, you can modify general setup values to adapt them to the desired Point of Care workflow. They are grouped according to the processes they influence.

NOTICE

Default setting instead of empty value

If an empty value is specified for this general setting (if accepted), **navify®** POC Operations application takes the default value instead.

Exceptions:The general setting **Default password for password reset** takes the empty value.

▶ The general settings **Predefined value for empty QC results** and **Predefined value for empty patient results** do not accept an empty value.

Choose the **Default value** button to reset the parameter to its default value.

Enable Advanced operator management

This parameter enables the **Advanced operator management** module.

Default value No

Possible values Yes

No

Advanced operator management URL

This parameter is used to enter the URL for accessing the **Advanced operator management** module.

Default value <empty>

Possible values free text

> You can only save the parameter if the text has a valid format for a URL.

POC module for navify Integrator

POC module for navify® Integrator URL

This parameter is used to enter the URL to access the POC module for **navify**® Integrator software.

You must refresh the application screen after entering or changing the URL.

Default value <empty>

Possible values free text

> You can only save the parameter if the text has a valid format for a URL.

HCA

Patient data feed (ADT) active

This parameter activates the ADT data processing, which enables the reception of patient information from the hosts and send it to the instruments. When this parameter is set to **Yes**, ADT and order messages received from one or more hosts are processed and the **navify**® POC Operations's patient database is updated accordingly.

To be able to receive order messages from the host, the "`Patient data feed (ADT) active`" setting must be enabled.

Default value **Yes**

Possible values

- **Yes**
- **No**

Delete visit records without result(s) upon discharge

This parameter checks whether discharged patient records shall be deleted.

If the setting is switched to **Yes**, discharged visit records without results will be deleted.

If the setting is switched to **No**, discharged visit records will not be deleted.

> If the discharged visit records without results is the only visit for that patient, the patient record will also be deleted.

Default value **No**

Possible values

- **Yes**
- **No**

Update of non-existing patient creates record

If the host sends an event to update a patient record that did not exist or has been discharged with no test results, the **navify**® POC Operations creates a new patient record with the received information.

If a patient record having test results had been discharged (and hence the discharged patient record still exists), the **navify**® POC Operations still applies demographic field updates to the patient record. If this setting is set to **No**, such an update will not set a discharged patient record to status **Admitted**.

Default value **Yes**

Possible values

- **Yes**
- **No**

NOTICE

Data overload

If this parameter is set to **Yes**, the number of entries in the application database and in the instrument lists can become dangerously large.

- ▶ To avoid the application database and instrument lists from becoming excessively large, ensure this parameter is set to **No** unless absolutely required.

Delay to retry the resend to the Host (Minutes)

This parameter defines the maximum wait time for the **navify**® POC Operations to receive a confirmation from the host, after test results were sent from the application to the host. After this time, if no *acknowledgment* is received, the application sends the test results to the host again.

This parameter is only applicable if the host is configured to send an acknowledge message.

Default value **90** minutes

Send alarms as comments

This parameter allows to enable/disable the possibility to send result alarms to the connected host in form of result comments.
If enabled, the application sends instrument and system alarms that had been activated to this purpose as result comments to the host interface. If disabled, the application does not send any system alarms or instrument alarms.

Default value **Yes**

Possible values

- **Yes**
- **No**

There are two levels for enabling/disabling the sending of these comments:

- Application wide (under **System** > **General settings** > **Send alarms as comments**)
- Instrument specific (under **System alarm assignment** > **Operator comment** and under **Instrument alarm assignment**, check box **Send as comment**)

Instruments

Always send Operator List to instruments

This parameter is used to define whether the operator list is always to be sent to the instruments.

> Also refer to the documentation of the individual instrument drivers as sending an operator list when not selected in the instrument configuration may alter the expected instrument behavior.

When this parameter is enabled, the list of operators is sent to the instruments regardless of the instrument configuration. When this parameter is disabled, the list of operator IDs is only sent to the instruments if their individual configuration requires an operator list.

Default value No

Possible values

- Yes
- No

Always send Patient List to instruments

This parameter is used to define whether the patient list is always to be sent to the instruments.

> Also refer to the documentation of the individual instrument drivers as sending a patient list when not selected in the instrument configuration may alter the expected instrument behavior.

When this parameter is enabled, the list of patients is sent to the instruments regardless of the instrument configuration. When this parameter is disabled, the list of patients is only sent to the instruments if their individual configuration requires a patient list.

Default value No

Possible values

- Yes
- No

Patient Upload: Force daily synchronisation

If this parameter is set to **Yes**, patient lists are synchronized every day.

Default value **Yes**

Possible values

- **Yes**
- **No**

Operator Upload: Always force full upload

If this parameter is set to **Yes**, then whenever a user record is updated, the entire list is uploaded rather than just the changed records.

Default value **No**

Possible values

- **Yes**
- **No**

NOTICE

Data management

Setting this parameter to “`Yes`” may extend the communication time required for the instrument to complete a synchronization cycle. If the instrument communication is interrupted, the instrument can have incomplete lists.

▶ Ensure that the parameter is set to **Yes**

Patient Upload: Always force full upload

If this parameter is set to **Yes**, then whenever a patient record is updated, the entire list is uploaded rather than just the changed records.

Default value No

Possible values
- Yes
- No

Number of characters to read from Hemochron patient ID.

This number represents the patient ID length. This determines how the application extracts the patient ID (right to left) from the number that **ITC Hemochron Jr. Signature** instruments send. This setting applies to **ITC Hemochron Jr. Signature** instruments only, not to other **ITC Hemochron** instruments.

Explanation **ITC Hemochron Jr. Signature** instruments add leading zeros to the Patient ID to fill in the maximum length the instrument supports. The parameter set in this field is intended to cut-off these extra leading zeros.

Patient ID received from the Hemochron instrument	000000034533
Number of ending characters to get from Hemochron's patient identifier	5
Resulting Patient ID stored in the **cobas**® IT 1000 application	34533

Example

⚠ CAUTION

Use this parameter with care

- If your institution supports patient IDs of different lengths, this setting can lead to confusion. e.g.: if there are two patient numbers 01234 and 1234, then setting this length to 5 will result in the second patient ID stored as 01234 instead of 1234, while setting it to 4 will result in the first patient number stored as 1234 instead of 01234.
- The maximum number of characters currently supported by the instrument is 12.
- Setting this parameter to zero prevents the application from storing the Patient ID.

Default value 30 characters. This represents the maximum number of characters supported by the application.

Number of characters to read from Hemochron operator ID

This number represents the user ID length. This determines how the application extracts the user ID (right to left) from the number that **ITC Hemochron Jr. Signature** instruments send. This setting applies to **ITC Hemochron Jr. Signature** instruments only, not to other **ITC Hemochron** instruments.

Explanation **ITC Hemochron Jr. Signature** instruments add leading zeros to the user ID to fill in the maximum length the instrument supports. The parameter set in this field is intended to cut-off these extra leading zeros.

User ID received from the Hemochron instrument	000000034533
Number of ending characters to get from Hemochron's user identifier	5
Resulting user ID stored in the **cobas**® IT 1000 application	34533

Example

CAUTION

Use this parameter with care!

- If your institution supports user IDs of different lengths, this setting can lead to confusion. e.g.: if there are two user IDs 01234 and 1234, then setting this length to 5 will result in the second user ID stored as 01234 instead of 1234, while setting it to 4 will result in the first user ID stored as 1234 instead of 01234.
- The maximum number of characters currently supported by the instrument is 12.
- Setting this parameter to zero prevents the application from storing the user ID.

Default value 12 characters.

The application supports user IDs with up to 20 characters.

navify® Analytics

navify Analytics URL

This parameter is used to enter the URL for accessing the **navify**® Analytics for Point of Care application. After an URL is entered, the icon for the **navify**® Analytics for Point of Care application is enabled on the Home screen.

You must refresh the Home screen after entering or changing the URL.

About the application

Example https://analytics-eu.viewics.com/viewics

Default value <empty>

Possible values free text

You can only save the parameter if the entered URL has a valid format (https://analytics-[eu].viewics.com/[viewics], where the parts in brackets can change).

Organization

Enable authentication via LDAP-based directory service

About LDAP

This setting defines whether user authentication via an LDAP based directory service is active or not for the application.

If the general setting is not active, then the individual user setting is ignored and the user authentication is done against the user password stored in the application.

If the general setting is active, then the individual user setting defines whether a user shall be authenticated against LDAP or the user password stored in the application.

> To enable this functionality the following general settings in **System** > **General settings** must also be configured:
>
> - **LDAP server is a Windows Active Directory server**
> - **LDAP domain name**
> - **LDAP server name or IP address and port**
> - **LDAP search user: Password**
> - **LDAP search user: User name**
> - **LDAP Base DN to use for searches**
> - **LDAP unique search attribute**

Default value **No**

Possible values

- **Yes**
- **No**

Certification Expiration Warning Time (Days)

This parameter sets how many days before certification expiration the system sends a warning email.

The application setting applies to expiring certifications.

The application setting does not apply to auto-recertifying certifications or to non-expiring certifications.

Default value **90**

Possible values “0” to “99” (integer)

Default password for password reset

This parameter defines the default password of the instrument user when the POCC resets the user's password for the instrument.

The parameter only has an effect when the application setting **Reset password to...** is set to **Default Password**.

Default value <empty>

Possible values free text

> The password must comply with the settings that define the minimum length and the type of characters to be used as passwords.

Reset password to...

This parameter defines the options for resetting a user's password for an instrument.

Default value **Empty Password**

Possible values

- **Empty Password**
- **Default Password**

LDAP server is a Windows Active Directory server

This parameter is used to define whether the connected LDAP server is a “Windows Active Directory” or not.

Default value Yes

Possible values

- Yes
- No

LDAP search user: Password

This parameter defines the password of the user that will be used to search the LDAP tree.

Default value <empty>

Possible values free text

LDAP search user: User name

This parameter defines the user name that will be used to search the LDAP tree.

Default value <empty>

Possible values free text

LDAP Base DN to use for searches

This parameter is used to define the LDAP base DN for searches.

Example DC=emea, DC=example, DC=com

Default value <empty>

Possible values free text

LDAP domain name

This parameter is used to define the LDAP domain name.

Default value <empty>

Possible values free text

LDAP server name or IP address and port

This parameter is used to define the server name (or the IP address) and the port of the LDAP server.

Example dc01.emea.example.com:389

Format DNS name:port or IP:port

Default value <empty>

Possible values free text

> To secure the connection to the LDAP server, use port 636 (LDAPS). Port 389 is not secured (LDAP).

LDAP user filter

This parameter supports user groups in LDAP. It is used to define custom LDAP filters to e.g. specify that only members of a specific LDAP group are allowed to access the application. The following filter variables can be used (case sensitive): {UniqueSearchAttribute}, {UserName}, {BaseDN}

Example (&(objectClass=user)(memberof=CN=Group01,OU=OU_Level3,OU=OU_Level2,OU=OU_Level1,DC=cit1k,DC=ch)({UniqueSearchAttribute}={UserName}))

Default value {UniqueSearchAttribute}={UserName}

Possible values free text

LDAP unique search attribute

This parameter is used to define the LDAP unique search attribute depending on the used LDAP server.

The application verifies the inserted user name on logon against the value in the LDAP database for the LDAP unique search attribute.

> The “unique search attribute” specifies a unique identifier of each user record, which therefore makes it appropriate for lookups. As default value for this general setting “sAMAccountName” is used. Unique Search Attribute will be filled with the user name of the LDAP user which is to be queried.

Default value sAMAccountName

Possible values free text

Printer for auto-print of notification letters

This parameter defines the name of the printer used for automatic printing of notification letters.

Default value <empty>

Possible values free text

Recently expired certification timeframe (Days)

This parameter defines how many days after the expiration of a certification a warning (reminder) email is sent.

Default value 60

If you set the value to 0, the status of users whose certifications have expired changes to “`Long term expired`” instead of “`Expired`”.

> The default value of this parameter is changed. When upgrading **cobas**® IT 1000 application version 2.04.01 or lower, the application shows the following behavior:
>
> - An active default value will be changed to the new default value of 60 with the update.
> - An active, differently set value will be kept with the update.

Possible values “`0`” to “`999`” (integer)

Observation timeframe - medium warning timeframe (%)

This parameter defines the “medium warning timeframe” reminder as a percentage of the “observation timeframe”.

The entered value must be greater than or equal the value for the application setting

Observation timeframe - narrow warning timeframe (%) .

Default value 35

Possible values “0” to “99” (integer)

Observation timeframe - narrow warning timeframe (%)

This parameter defines the “narrow warning timeframe” reminder timeframe as a percentage of the “observation timeframe”.

The entered value must be smaller than or equal than the value for the application setting

Observation timeframe - medium warning timeframe (%) .

Default value 15

Possible values “0” to “99” (integer)

Password change period (Days)

This parameter sets how often users must change their password.

Default value 90 days

Password - minimum number of capital letters

This parameter sets the minimum number of uppercase letters that user passwords must contain.

Default value 1 character

Possible values “0” to “99” (integer)

Password - minimum number of digits or symbols

This parameter sets the minimum number of digits or symbols (not letters) that user passwords must contain.

Default value 1 character

Possible values "0" to "99" (integer)

Password - minimum length

This parameter sets the minimum total number of characters that user passwords must have.

Default value 8 characters

Possible values "0" to "99" (integer)

Auto user import: default profile

This parameter sets the user profile for the automated user import.

Automated user import

Default value **Operator**

Possible values Select the user profile from the drop-down list.

Auto user import: default workgroup

This parameter sets the workgroup for the automated user import.

Automated user import

Default value <empty>

Possible values Select the workgroup from the drop-down list.

Auto user import: default certification 1 - certificate type

This parameter sets the certificate type for certification 1 for the automated user import.

Automated user import

Default value <empty>

Possible values Select the certificate type from the drop-down list.

Auto user import: default certification 1 - stage

This parameter sets the certification stage for certification 1 for the automated user import.

Automated user import

Default value <empty>

Possible values Select the certification stage from the drop-down list. Ensure to select Interim only if the respective certificate type supports the use of the Interim stage.

Auto user import: default certification 1 - certified by

This parameter sets the certifier for certification 1 for the automated user import.

Automated user import

Default value <empty>

Possible values free text

Auto user import: default certification 2 - certificate type

This parameter sets the certificate type for certification 2 for the automated user import.

Automated user import

Default value <empty>

Possible values Select the certificate type from the drop-down list.

Auto user import: default certification 2 - stage

This parameter sets the certification stage for certification 2 for the automated user import.

Automated user import

Default value <empty>

Possible values Select the certification stage from the drop-down list. Ensure to select Interim only if the respective certificate type supports the use of the Interim stage.

Auto user import: default certification 2 - certified by

This parameter sets the certifier for certification 2 for the automated user import.

Automated user import

Default value <empty>

Possible values free text

Auto user import: default certification 3 - certificate type

This parameter sets the certificate type for certification 3 for the automated user import.

Automated user import

Default value <empty>

Possible values Select the certificate type from the drop-down list.

Auto user import: default certification 3 - stage

This parameter sets the certification stage for certification 3 for the automated user import.

Automated user import

Default value <empty>

Possible values Select the certification stage from the drop-down list. Ensure to select Interim only if the respective certificate type supports the use of the Interim stage.

Auto user import: default certification 3 - certified by

This parameter sets the certifier for certification 3 for the automated user import.

Automated user import

Default value <empty>

Possible values free text

Auto user import: user identifying value

This parameter sets the value that identifies the user for the automated user import.

The application can identify users in the import file either by the **User ID** or the **Network ID**.

If the **Network ID** is used as identifier, the import file must contain an existing Network ID. Otherwise, the user import fails, and an error is logged in the **Automated user import log** menu.

Automated user import
Automated user import log

Default value **User ID**

Possible values

- **Network ID**
- **User ID**

Auto User Import: Run import in delta or full mode?

This parameter sets the import type for the automated user import.

You can choose between 2 different import modes: *delta* and *full*. The 2 modes differ in how the application updates users who are present in the application but are not included in the import file.

- Delta mode (default): users who are not included in the import file, but are present (and active) in the application are not updated during an import, and remain active.
- Full mode: users who are not included in the import file, but are present (and active) in the application are deactivated during an import.

Automated user import

Default value **Delta**

Possible values

- **Delta**
- **Full**

Auto user import: Exclude users

This parameter defines the users that are excluded from an automatic update ("safelist") during automated user import.

During an automated user import, excluded users are not updated by the information provided in the import file.

Automated user import

Default value <empty>

Possible values free text

Certifications (pending stage): Only send user to the device when no "off instrument" tasks are open

This parameter is used to define whether the user is sent to the device as part of the pending phase tasks.

Default value No

Possible values

- Yes
- No

Weekday for sending of supervisor notifications

This parameter is used to define the day of the week when email notifications shall be sent to the supervisor.

Default value Sunday

Possible values

- Sunday
- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday

Base day for certification date calculation

This parameter is used to define how certification dates from user data import files are calculated.

Default value **Received date**

Possible values

- **Received date**. The system takes the date as it is defined in the import file.
- **First day of the week**. The system replaces the date in the import file with the first day of that week.
- **First day of the month**. The system replaces the date in the import file with the first day of that month.

First day of the week

When the value in **Base day for certification date calculation** is set to **First day of the week**, this setting defines whether the first day of the week is Sunday or Monday.

Default value **Monday**

Possible values

- **Sunday**
- **Monday**

Quality control (Rili-BÄK)

About Quality control (Rili-BÄK)

These settings are only present in installations running in Rili-BÄK mode.

Automatic QC validation for expected values

This parameter defines whether the system should perform QC validation automatically. A QC result is accepted automatically if no alarm associated with this QC result has a severity of **Error**.

If QC management is done on a connected information system, such as an LIS, set this parameter to **Yes**.

If you want to manually validate each QC result, set this parameter to **No**.

Default value **Yes**

Possible values

- **Yes**
- **No**

Apply proposed LED automatically

This parameter is used to define whether or not LEDs proposal which can be accepted are automatically accepted and applied. Acceptable proposals have a numeric proposal which leads to an effective QC range wider than the target value and not exceeding the defined QC range.

If **Yes** is set, proposed LEDs are automatically accepted and applied.

Default value **No**

Possible values

- **Yes**
- **No**

Extension (in percent) of the allowed deviation/range for running RMSE

This setting enables you to extend or reduce the allowed deviation/range that the running RMSE is evaluated against. If it is set, for example, to **-10**, an RMSE warning is raised if the running RMSE for a test is higher than 90% of the allowed deviation (in column 3), i.e. entering -10 in effect gives an advance warning on a potential RMSE violation.

Default value **0** (The default value assumes the defined deviation/range)

Possible values Numeric value to indicate the adjustment in percent to the allowed deviation/range.

Minimum number of QCs required for calculation of running RMSE

RMSE stands for root mean square error of measurement. The mean square is calculated per control cycle. This setting enables you to enter the minimum number of QC tests that are required before a running RMSE is calculated. Only if the number of accepted QC results for a specific instrument, lot (combination) and level not yet included in a previous control cycle exceeds this value, the running RMSE will be calculated.

> RMSE calculation is based on accepted QC results only. Unreviewed QC results are not included in the calculation.
>
> ▶ RMSE is only calculated if enough accepted QC results are available.

Default value 7

Possible values Numeric value to indicate the number of QC results.

Time-triggered routine period check on QC for non-POCT instruments

This setting activates a time-triggered check on whether a second QC test was done 16 hours or less after the first QC of this routine period.

Default value Undefined

Possible values

- Yes
- No
- Undefined

QC result set used for automatic LED proposal calculation

Defines which QC results are used in automatic calculation of LED proposals. If **Most recent completed control cycle results only** is set, an LED proposal is calculated if the latest completed control cycle contains at least 15 days with accepted QC results.

If **Current control cycle results/non-control cycle results also** is set:

- The LED is calculated after having 15 days with accepted QC results, even if the current control cycle is not yet complete.
- If there is a completed control cycle, and the current cycle has less than 15 days with accepted QC results, then the QC results from the most recent completed control cycle are used for LED proposal calculation.
- If for the instrument no RMSE calculation is done (and hence control cycles do not apply), QC results not associated with a control cycle are used. All accepted QC results are used for LED proposal calculation.

Default value **Most recent completed control cycle results only**

Possible values

- **Most recent completed control cycle results only**
- **Current control cycle results/non-control cycle results also**

QC result (per calendar day) for LED proposal calculation

To calculate an LED proposal, one QC result per calendar day (not necessarily consecutive) is used. This parameter is used to define which of the QC results per day is to be used for the LED proposal calculation.

Default value **1st Result**

Possible values

- **1st Result**
- **n-th Result**
- **Last Result**
- **Random Result**

Numeric value of n (for LED proposal calculation)

This parameter is used to define the numeric value of **N** used in the setting **QC result (per calendar day) for LED proposal calculation**. The value must be an integer greater than 0.

Default value **1**

Possible values Any integer > 0

Results & Patients

Predefined value for empty QC results

A character the system uses to indicate that an instrument has sent an empty result for a QC test.

Default value **-** (A dash, or minus sign)

> Empty result values of instruments connected by means of an ICA driver will not be reported in the application. For information on whether a specific instrument type is connected via an ICA driver, contact your Roche Service representative.

> For RMSE calculation: When the predefined value for empty QC results is set to the default "-" or another qualitative value and the QC result is accepted and released, QC results with an empty result value (e.g. "-") are included in the RMSE calculation as if they would have a value of 0.

Predefined value for empty patient results

A character the system uses to indicate that an instrument has sent an empty result for a patient test.

Default value **-** (A dash, or minus sign).

> Empty result values of instruments connected by means of an ICA driver will not be reported in the application. For information on whether a specific instrument type is connected via an ICA driver, contact your Roche Service representative.

Discharge all visits belonging to idle patient record.

This parameter is used to define whether visits of an idle patient (no results done for this patient) will be discharged after a preset time as set by the parameter **Number of days after which a patient record is considered idle**.

When set to **Yes**, visits will be automatically discharged if the idle patient timeframe is reached. This timeframe is defined by the **Number of days after which a patient record is considered idle**.

When set to **No**, visits will not be automatically discharged when idle for a longer period.

> If the visit has results associated with it, the timeframe is calculated from the last received result.
>
> If the visit has no results associated with it, the timeframe is calculated from the admission timestamp.

Default value **No**

Possible value

- **No**
- **Yes**

Enable undo validation for samples

This parameter is used to enable the undoing validation for accepted results in the **Result validation** screen during a preset time as set by the parameter **Timeframe to undo validation for samples (Hours)**.

Default value No

Possible values

- No
- Yes

Generic patient

This parameter is used to set the ID for one or more generic patients.

CAUTION

Generic Patient

- Any ADT message (admit, merge, discharge, pending admit, pending discharge) involving a generic patient will be reported as an error in the application.
- A merge message having the same source and target will be ignored by the application.
- If your current patient ID scheme allows for patient ID 999999, contact Roche Service for assistance with setting a default, generic ID that will not overlap with patient IDs.

Format comma separated list of IDs

Default value 999999

Possible values free text

Number of days after which a patient record is considered idle

This parameter sets after how many days a patient record that is not receiving any test results is considered idle. Visits of an idle patient record will be automatically set to discharged if the value of the setting **Discharge all visits belonging to idle patient record.** is **Yes**.

Default value **30**

Possible values "1" to "9999" (integer)

Instrument Patient ID

The **Instrument Patient ID** is the patient ID sent by the instrument. You can configure which of the patient identifiers stored in the application (Patient ID 1, 2, 3, or Visits ID) is used as Instrument Patient ID.

Default value **Patient ID 1**

Possible values

- **Patient ID 1**
- **Patient ID 2**
- **Patient ID 3**
- **Visit ID**

> This setting is used for instruments connected via an ICA driver.
>
> This setting is used for instruments not connected via an ICA driver, when the
>
> **Instrument Patient ID**
>
> is not set specifically within the instrument's configuration.
>
> For information on whether a specific instrument type is connected via an ICA driver, contact your Roche Service representative.

Patient Class Group I

This parameter is used to set which of the patient classes shall be grouped together.

This group has as default value the patient class “outpatient”.

Default value O

Possible values List of letters, separated by commas

Patient Class Group I - Timeframe (Days)

This parameter is used to define the maximum number of days the visits of the visits class(es) (grouped in **Patient Class Group I**) remain admitted in the application before being automatically discharged by the “Patient Cleaner” background process.

Note: Under the following condition, visits can be discharged already before the maximum number of days has been reached, if:

- The **Discharge all visits belonging to idle patient record.** setting is enabled.
- The value defined in the **Number of days after which a patient record is considered idle** setting is smaller than maximum number of days defined in the **Patient Class Group I** setting.

Default value 2

Possible values “`1`” to “`999`” (integer)

Patient Class Group II

This parameter is used to set which of the patient classes shall be grouped together.

This group has as default value the patient class “inpatient”.

Default value I

Possible values List of letters, separated by commas

Patient Class Group II - Timeframe (Days)

This parameter is used to define the maximum number of days the visits of the visits class(es) (grouped in **Patient Class Group II**) remain admitted in the application before being automatically discharged by the “Patient Cleaner” background process.

Note: Under the following condition, visits can be discharged already before the maximum number of days has been reached, if:

- The **Discharge all visits belonging to idle patient record.** setting is enabled.
- The value defined in the **Number of days after which a patient record is considered idle** setting is smaller than maximum number of days defined in the **Patient Class Group II** setting.

Default value 365

Possible values “1” to “999” (integer)

Patient Class Group III

This parameter is used to set which of the patient classes shall be grouped together.

This group has as default value the patient classes “obstetrics”, “emergency”, “preadmit” and “recurring patient”.

Default value B,E,P,R

Possible values List of letters, separated by commas

Patient Class Group III - Timeframe (Days)

This parameter is used to define the maximum number of days the visits of the visits class(es) (grouped in **Patient Class Group III**) remain admitted in the application before being automatically discharged by the “Patient Cleaner” background process.

Note: Under the following condition, visits can be discharged already before the maximum number of days has been reached, if:

- The **Discharge all visits belonging to idle patient record.** setting is enabled.
- The value defined in the **Number of days after which a patient record is considered idle** setting is smaller than maximum number of days defined in the **Patient Class Group III** setting.

Default value 365

Possible values “`1`” to “`999`” (integer)

Detect repeat results

This parameter allows to set if repeated results shall be automatically detected or not.

Default value **No**

Possible values
- **Yes**
- **No**

Generate new sample ID for samples where sample validation was undone

This parameter is used to define if a result where validation was undone gets a new sample ID if the value of the setting **Enable undo validation for samples** is **Yes**.

Default value **No**

Possible values
- **Yes**
- **No**

Timeframe to undo validation for samples (Hours)

This parameter sets for how many hours accepted results can have their validation undone if the value of the setting **Enable undo validation for samples** is **Yes**.

The time is counted since the result was sent.

Default value 24 hours

Possible values 0-999 hours

Automatic Validation with Tolerance

This parameter allows automatic validation of test results, if they fall within ranges for automatic acceptance. These ranges for automatic acceptance are defined as **Auto accept low** and **Auto accept high** values in the **Configuration** > **System** > **Test definition** > **Reference range** screen.

Default value **Yes**

Possible values

- **Yes**
- **No**

Retention period for sent and unsent patient results

This parameter is used to define the number of days patient results are retained in the system before being automatically deleted or archived.

> The retention period is calculated based on the **Reception date and time** when the software receives the result from the device and not on the **Measurement date and time** when the test is performed.

The number of days for retaining both sent and unsent patient results can be adjusted to meet specific needs.

- **Sent patient results:** The minimum and default retention period is 91 days.
- **Unsent patient results:** The minimum and default retention period is 365 days.

These settings allow flexibility to make sure that the information about unsent (e.g., pending or rejected) results is kept..

Result re-processing

Automatically accept matching results

This parameter is used to automatically assign previously received test results to the correct patient visit, even if the results were received before the corresponding patient or visit information was available.

This setting applies to results received with a Patient ID2, Patient ID3, or Visit ID that were not yet available in the application at the time the results were received. It works in conjunction with the following parameters:

- Matching Time Frame (minutes)
- Postfix for auto generated Patient ID1
- Prefix for auto generated Patient ID1

If the patient or visit information becomes available within the configured Matching Time Frame (minutes), the result will be linked to the corresponding patient or visit. In this case, related patient identification result alarms will have reduced severity, and the application will set the result status to 'Accepted', provided no other result alarms with 'Error' severity are present.

If no matching patient or visit information is received within the configured Matching Time Frame (minutes), the application generates a new patient record using an auto-generated Patient ID1, which may include a defined prefix and/or postfix.

Default value No

Possible value

- No
- Yes

Matching time frame (minutes)

This parameter is used in conjunction with Automatically accept matching results.

Automatically accept matching results

Default value 2,880 minutes

Possible value "0" to "0.999" minutes

Postfix for auto generated Patient ID1

This parameter is used in conjunction with Automatically accept matching results.

Automatically accept matching results

Default value <empty>

Possible value free text

System

Enable automatic patient/visit discharging

This parameter defines whether the admission status of a patient will be automatically changed from admitted to discharged.

If set to **Yes**, visits may be automatically discharged according to **Patient Class Group I - Timeframe (Days)**, **Patient Class Group II - Timeframe (Days)** or **Patient Class Group III - Timeframe (Days)** as well as **Discharge all visits belonging to idle patient record.** together with **Number of days after which a patient record is considered idle**.

If set to **No**, visits will not be automatically discharged.

> You can define 3 different periods for different groups of patient classes. If the period for a visit belonging to such a patient class is reached (based on visit admission timestamp), then this visit will be discharged.

Default value **No**

Possible values

- **No**
- **Yes**

Export log data before purging

This setting defines whether log data shall be exported before it is automatically purged. If enabled, only log data that could be successfully exported from the database will be purged. The automatic purging (and if enabled export) of log data is done automatically (per default) every 24 hours, its configuration also includes the following 3 settings.

> Exported audit log information can be re-imported if needed. For a re-import of exported log information please contact Roche Service.

Default value **Yes**

Possible values

- **Yes**
- **No**

Path for log data export

The path for log data export must be a valid folder where the data to be purged is exported to. This setting is effective only if **Export log data before purging** is set to **Yes**.

> It is strongly recommended to export log data to a hard disk other than those used for database or journal data storage. A mechanism should be put in place to ensure that exported log data is transferred to a permanent medium in regular intervals to prevent the respective hard disk from running out of space.

Default value `\cobasIT1000\Services\Audit\`

Possible values `C:\logbck`

Number of days after which to purge audit trail data

This is the number of days during which audit log data are kept in the database. Audit log data older than the number of days set will be automatically purged from the database.

> The automatic purging of log data can be deactivated entirely by Roche Service. When log data is not automatically purged, an increased growth in database size and decreased log specific performance is expected.

Default value **61**

Possible values One to three digits ("`61`" to "`999`")

Supported character set for instrument communication

This parameter defines what type of character set is used for communicating patient and user demographic data to instruments during a query or when uploading lists. During the communication, the system replaces any characters that are outside the range defined by this parameter.

This parameter does not affect the following features:

- ID fields
- Result upload
- Configuration values

Default value **8-Bit ASCII, replace other**

Possible values

1. **No limitations**. All characters are kept unchanged.
2. **7-Bit ASCII, replace other**. Characters outside the 7-Bit ASCII range are replaced.
3. **8-Bit ASCII, replace other**. Characters outside the 8-Bit ASCII range are replaced.
4. **None, send substitute only**. All communicated characters are replaced by a specified character. This character is determined by service settings.

Enable automatic printing

This parameter defines if automatic printing is enabled.

Default value **No**

Possible values

- **Yes**
- **No**

Enable sending of e-mails

This parameter defines if sending of emails is enabled for the application.

Default value **No**

Possible values

- **Yes**
- **No**

Serial number

This parameter defines the serial number of the **navify®** POC Operations installation. If the serial number is entered, it is displayed in the **About** box.

> The serial number can only be entered by Roche Service.

Default value <empty>

Possible values free text

Session Expiration Time (Minutes)

This parameter sets the period of time the operator can be inactive before the session expires and logon is required again.

If the parameter is set to 0, the auto logoff is disabled.

Default value 90 minutes

Supported character set for host communication

This parameter defines what type of character set is used for host communication. The system replaces any characters during the communication that are outside the range defined by this parameter.

Default value No limitations

Possible values

- **No limitations**. All characters are kept unchanged.
- **7-Bit ASCII, replace other**. Characters outside the 7-Bit ASCII range are replaced.
- **8-Bit ASCII, replace other**. Characters outside the 8-Bit ASCII range are replaced.

ID field character limitation

This parameter allows to restrict ID fields when unicode is active.

Default value No limitation for ID fields

Possible values

- No limitation for ID fields
- 7-Bit ASCII
- 8-Bit ASCII
- Only numeric entry values are valid

Number of days after which to purge device events

This is the number of days for which other log data (by default: device messages) are kept in the database. Other log data older than the number of days set will be automatically purged from the database.

Default value 61

Possible values One to three digit numeric (“0” to “999”)

SMTP password

This parameter defines the password of the user which has to be used for sending emails. This parameter is needed when the SMTP server requires an authentication of a user in order to allow the sending of emails.

Always define an SMTP password.

Default value <empty>

Possible values free text

SMTP username

This parameter defines the user name which has to be used for sending emails.

This parameter is needed when the SMTP server requires an authentication of a user in order to allow the sending of emails.

Always define an SMTP user name.

Default value <empty>

Possible values free text

SMTP server name/IP address and port

This parameter defines the name or the IP address and the port of the SMTP server which has to be used for sending emails.

It is possible to enter a DNS name and the used port or to enter an IP address and the used port.

Format DNS name:port or IP address:port

Default value <empty>

Possible values free text

Use Secure SMTP

This parameter defines whether communication with the SMTP server shall be secured (using SSL/TLS) or not. The application supports implicit and explicit TLS.

Default value No

Possible values

- Yes
- No

Sender address for notification e-mails

This parameter defines the email address of the sender of notification emails.

Format username@domain.TopLevelDomain

Example navify.poc.operations@myhospital.com

Default value <empty>

Possible values free text

Delay of user e-mail notifications on certificate type update (Minutes)

This setting allows to delay the creation of emails for user notifications for the defined number of minutes.

Default value **15**

Possible values “0” to “99” (integer)

The delay does not apply to warnings for a POCC or notifications for a supervisor.

Warnings

Maximum Size of Database (Mb)

This parameter sets the maximum size that the database is expected to have. When free space is expected to be not enough to continue working, a warning email can be sent.

Default value 3600 MB

Warning Size of Database (%)

This parameter sets the percentage of the maximum expected database size that displays a warning to the user when reached. By default, the system can automatically send a warning email when the free space is less than 20%.

Default value 80%

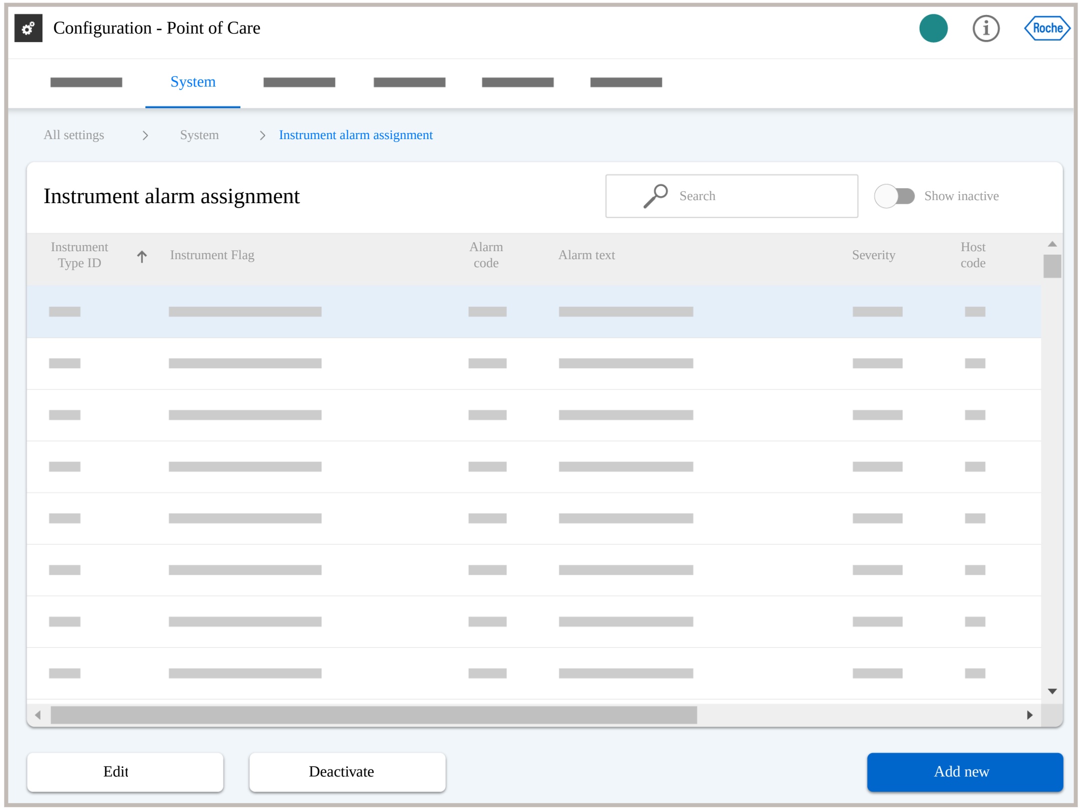
Instrument Alarm Assignment

This screen is used to map the system-predefined alarms with each instrument type and assign them not only an alarm severity, but also a result type (patient or QC).

You must first review the alarms in **Instrument alarm definition**, and then map them in **Instrument alarm assignment**. Alarms mapped as warnings or information flag results, but do not block them.

> Any instrument or system alarm with alarm severity of error will cause a result to be held for validation.

Use this component to correlate instrument alarms with the alarm codes and descriptions displayed in the validation screen, and, if applicable, their corresponding codes in the host transmission.



Tasks

- Creating a new assignment to establish a correlation between system alarms and the respective instrument alarms.
- Querying for the details of existing assignments.
- For each alarm mapping: Enabling/disabling the sending of an alarm as a result comment
- Activating an existing assignment so that it is in use when the application is running.
- Deactivating an assignment so that it is not taken into account when the application is in use.

System Alarm Assignment

This screen is used to map the system-predefined alarms for a result type, **Device type**, and optionally for a specific **Site**. Based on this mapping, the system automatically assigns the corresponding alarm severity.

Alarms are not generated if a mapping has been deactivated.

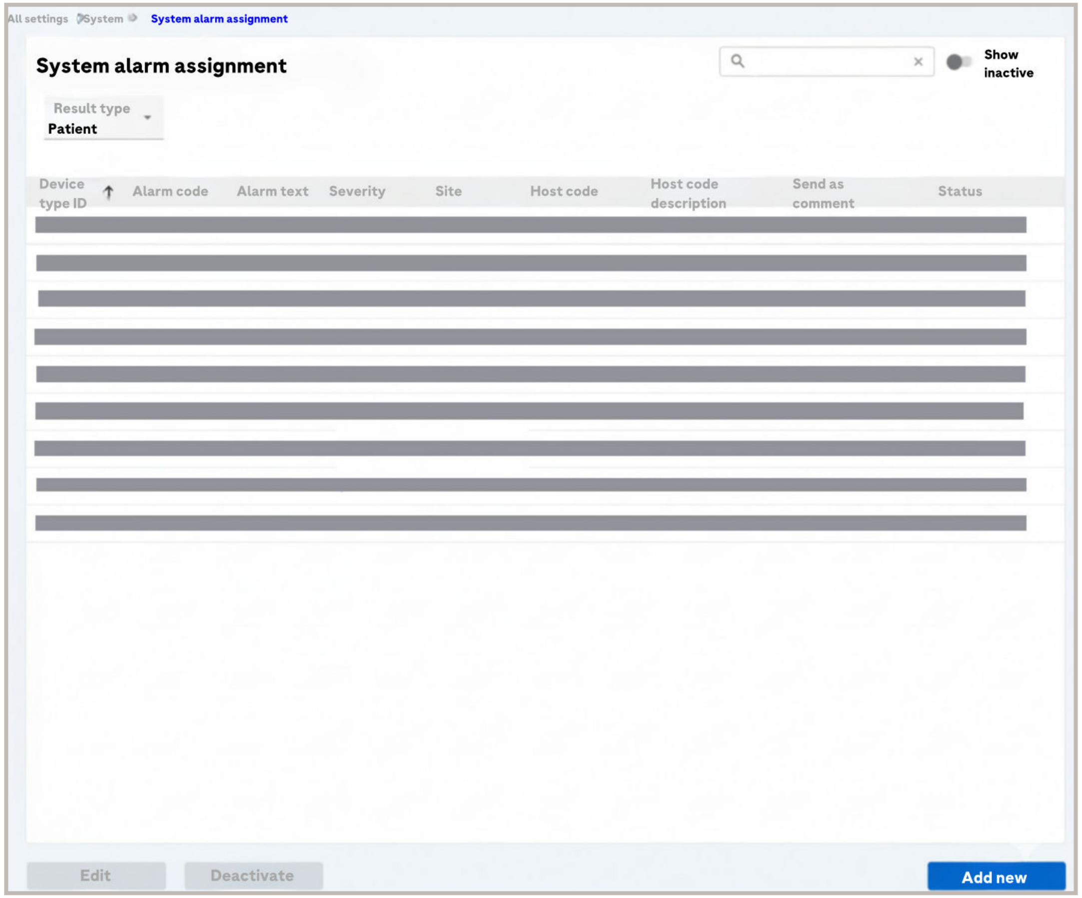
Alarms mapped as warnings or information highlight the results but do not prevent them from being automatically accepted.

> Any instrument or system alarm with alarm severity of error will cause a result to be held for validation.

Use this component to correlate system alarms with the alarm codes and descriptions displayed in the validation screen, and, if applicable, their corresponding codes in the host transmission.

> Alarm types that are not mapped at all for a device type (i.e. do not have an inactive mapping either) can be triggered by the application with their default severity.

Site-specific alarms can be updated with a non-existing site-specific alarm mapping. The application displays an error if you attempst to save a duplicate assignment.



Tasks

- Creating a new assignment to establish a correlation between system alarms, the respective **Device type**, and the specific **Site**.
- Querying for the details of existing assignments.
- For each alarm mapping: Enabling/disabling the sending of an alarm as a result comment
- Activating an existing assignment so that it is in use when the application is running.

- Deactivating an assignment so that it is not taken into account when the application is in use.

Adding a new site specific system alarm

To add a new site specific system alarm

1. Choose **Configuration** > **System** > **Instrument alarm assignment**

   → The window is displayed.

2. Choose **Patient**from the **Result type**drop-down list.

   Patient

3. Choose the **Add new** button to display the **System alarm assignment**window.
4. Choose the **Device type**, **Alarm code**, and **Severity** from the respective drop-down list.
5. In the **Site** drop-down list, choose the site for the new system alarm. The sites are listed by **Site ID**and **Site name**.

   Site

6. Optionally, choose a **Host code** and check the **Send as comment**box.
7. Choose the **Save** button to create the new system alarm assignment.

Instrument Alarm Definition

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Instrument Alarm Definition

There are two categories of *alarms* in the application:

- Instrument alarm Instrument alarms are triggered by an instrument flag (e.g. “result below instrument measurement range”, “Out of instrument linearity limits”).
- System alarm System alarms are triggered by the application (e.g. “Result received from inactive instrument”)

Via the **Editable** column, you can differentiate two types of alarms:

- Alarms that have the check box checked (editable): Only the **Alarm text** can be changed. **Note:** Do not edit alarm texts. The new text is only stored in the database. In the application, the original text is displayed after choosing the **Filter** button, or starting a new session.
- Alarms that have the check box cleared (not editable): Only the **Default severity** can be changed.To change the default severity, double-click the alarm in the **System** > **Instrument alarm definition** window.

Predefined alarms

There are predefined alarms available for use, but you can also create new ones.

Alarm types

Alarm types are used to standardize individual flags sent by different instrument types within the application.

Mapping flags to alarms

You perform the mapping of the defined alarms with the instrument alarms in **Configuration** > **System** > **Instrument alarm assignment** > **Add new**.

Default alarms are not editable. Only those alarms defined by a user can be edited.

Both default and site-specific alarms can be deactivated.

Tasks

- Creating new alarms to be matched with flags sent by the instrument.
- Querying for user-defined alarms for displaying their details.
- Activating an alarm to make it available for other screens.
- Deactivating an alarm to make it unavailable for other screens.
- Modifying user-defined alarm types that are already stored in the database.

List of predefined alarms used as system alarms

List of predefined alarms used as system alarms

Following is a description of all predefined alarms with details of what causes them.

Instrument alarms

Alarm code	Alarm text	Triggering event	Comment	Result type
IAR	**Alarm flag received from instrument**	The related test result received from the instrument was flagged by the instrument.		Patient, QC
IAU	**Unknown alarm flag received from instrument**	The related test result received from the instrument was flagged by the instrument using a flag not set up in the application.	This specific alarm is triggered, when an instrument sends an alarm flag with the result not mapped in **System** > **Instrument Alarm Assignment**. This alarm will be shown as: IAU_< Alarm code sent by the Instrument>, e.g. IAU_2 (2 is the alarm flag sent by the instrument).	Patient, QC
IB	**Result received from locked instrument**	When a result is received from an instrument set to status **Locked**.		Patient, QC

Alarm code	Alarm text	Triggering event	Comment	Result type
IMI	**Test related instrument maintenance due (information)**	A maintenance is due.	This specific alarm is triggered, when a patient result is received and a maintenance with type **Information** is due.	Patient
IMTW	**Test related instrument maintenance due (warning)**	A maintenance is due.	This specific alarm is triggered, when a patient result is received and a maintenance with type **Warning** is due.	Patient
IMT	**Test related instrument maintenance due (error)**	A maintenance is due	This specific alarm is triggered, when a patient result is received and a maintenance with type **Error** is due.	Patient
IMW	**QC related instrument maintenance due (warning)**	A maintenance is due.	This specific alarm is triggered, when a QC result is received and a maintenance with type **Warning** is due.	QC
IMQ	**QC related instrument maintenance due (error)**	A maintenance set up is due.	This specific alarm is triggered, when a QC result is received and a maintenance with type **Error** is due.	QC
INA	**Result received from inactive instrument**	When a result is received from an instrument deactivated in **Devices** > **Devices**.		Patient, QC
IRH	**Result above instrument measurement range**	The result is above the reportable range of the instrument.	For configuration settings, contact the Roche Service.	Patient
IRL	**Result below instrument measurement range**	The result is below the reportable range of the instrument.	For configuration settings, contact the Roche Service	Patient
IUE	**Instrument not docked at home location**	The result was downloaded from an IP address (station) not assigned to the location the instrument is assigned to.	Requires the use of a docking station. Applies only to **Accu-Chek® Inform** instruments.	Patient, QC

Alarm code	Alarm text	Triggering event	Comment	Result type
IUU	**Result unit not defined for this test**	The unit(s) set up for this test in **System** > **Test definition** do not match the unit sent by the instrument.		Patient, QC

Instrument Alarms

QC data alarms (system alarm)

Alarm code	Alarm text	Triggering event	Comment	Result type
MQC	**QC material LOT not valid**	▪ This control material lot did not exist in the application before this result was received ▪ It has not been set to be **In use** yet ▪ A new QC range/target value entry had to be added to be able to process the result. This case occurs frequently for device types where QC ranges/target values are maintained in combination of QC lot and test lot (e.g. the **CoaguChek®** Pro II) when **In use** test and QC lots are used in a previ-		QC

Alarm code	Alarm text	Triggering event	Comment	Result type
		ously unused combination		
MRE	**QC lock (Multirule violation)**	A multirule with severity error (red icon) has been violated.	This alarm will be shown as MRE_< violated multirule code>, e.g. MRE_1x2s	QC (non-Rili-BÄK)
MRW	**QC warning (Multirule violation)**	A multirule with severity warning (yellow icon) has been violated.	This alarm will be shown as MRW_< violated multirule code>, e.g. MRW_1x2s	QC (non-Rili-BÄK)
MTE	**Test material LOT not valid**	This test material lot does not exist in the application, or has expired, is deactivated or has not been reviewed by the user.		Patient, QC
QCW	**Result received during QC warning**	The patient result was received while the related test on this instrument was in QC warning status (multirule violation).		Patient (non-Rili-BÄK)

Alarm code	Alarm text	Triggering event	Comment	Result type
QCE	**Result received during QC lock**	The patient result was received while the related test on this instrument was in QC lock QC (non-Rili-BÄK) - multirule violation.		Patient (non-Rili-BÄK)
QCNR	**No range defined (QC result)**	No ranges are defined for this material in **Devices** > **Lots**.		QC
QCOR	**Value out of range (QC result)**	When a numeric QC result is out of range and when an alphanumeric QC result does not match the target value defined in **Devices** > **Lots**.		Patient (non-Rili-BÄK)
QCQN	**Qualitative result, but defined range is numeric**	The test result is alphanumeric, but the applicable ranges defined for this test in **Devices** > **Lots** are all numeric.		QC

Alarm code	Alarm text	Triggering event	Comment	Result type
QLR	**Result received after completion of control cycle**	The alarm is triggered when a QC result is received after completion of a control cycle.	This QC result is not part of the evaluation of the control cycle. Also, the result is not included in the Rili-BÄK report.	QC (Rili-BÄK mode only)
RBM	**No QC done within required time**	The QC requirements (on number and levels of QC) have not been met within the specified time frame.	Rili-BÄK mode only (deprecated, retained for backwards compatibility)	Patient (Rili-BÄK mode only)
RBS	**No QC schedule defined**	There is no QC schedule defined for this configuration or instrument.	Rili-BÄK mode only (deprecated, retained for backwards compatibility)	Patient (Rili-BÄK mode only)

Alarm code	Alarm text	Triggering event	Comment	Result type
RBE	**QC lock (Rili-BÄK)**	The alarm is triggered when the Rili-BÄK ranges are violated.	Rili-BÄK mode only. In the QC result screen the RBE ID is displayed extended with an ID for the reason: QUAL: Qualitative QC result does not match target value. COL3A: Numeric QC result is outside the 'column 3a' allowed deviation. LED: Numeric QC result is outside the Lab Established Deviation. RNG: Numeric QC result is outside the QC range.	Patient, QC (Rili-BÄK mode only)
RBN1	**No QC done since routine started.**	The alarm is triggered when the Rili-BÄK timing rules are violated by patient results for which no QC has been done since routine started.	Non-POCT instruments only	Patient (Rili-BÄK mode only)

Alarm code	Alarm text	Triggering event	Comment	Result type
RBN2	**No second QC done since routine started.**	The alarm is triggered when the Rili-BÄK timing rules are violated by patient results for which no second QC has been done since routine started.	Non-POCT instruments only	Patient (Rili-BÄK mode only)
RBNL	**No second QC level done since routine started.**	The alarm is triggered when the Rili-BÄK timing rules are violated by patient results for which no second QC level has been done since routine started.	Non-POCT instruments only	Patient (Rili-BÄK mode only)
RBP1	**No QC done in defined interval**	The alarm is triggered when the Rili-BÄK timing rules are violated by patient results for which no QC has been done in the defined interval.	POCT instruments only	Patient (Rili-BÄK mode only)

Alarm code	Alarm text	Triggering event	Comment	Result type
RBP2	**No QC done in defined pre-interval**	The alarm is triggered when the Rili-BÄK timing rules are violated by patient results for which no QC has been done in the defined pre-interval.	POCT instruments only	Patient (Rili-BÄK mode only)
RBPL	**No second QC level done**	The alarm is triggered when the Rili-BÄK timing rules are violated by patient results for which no second QC level has been done in the defined pre-interval.	POCT instruments only	Patient (Rili-BÄK mode only)

Alarm code	Alarm text	Triggering event	Comment	Result type
RBY1	**No second QC done in the last 16 hours.**	The alarm is triggered if over the course of a completed routine period no second QC has been done within 16 hours after the first QC and no active RBN2 alarm mapping exists for patient results of the respective instrument type. **Note:** To prevent unfounded triggering of the alarm, you must allow for the reception and evaluation of the QC results in the application. Therefore, configure the start of the daily routine at least 5 minutes before the time the QC tests are done on the instruments.	Non-POCT instruments only.	N/A (Rili-BÄK mode only)
RBY2	**No second QC level done in the last 16 hours.**	The alarm is triggered if over the course of a completed routine period no second QC using a different QC level has been done within 16 hours after the first QC and no active RBNL alarm mapping exists for patient results of the respective instrument type.	Non-POCT instruments only.	N/A (Rili-BÄK mode only)
RBY3	**No QC done in previous routine period**	The alarm is triggered if over the course of a completed routine period no QC has been done and no active RBN1 alarm mapping exists for patient results of the respective instrument type.	Non-POCT instruments only.	N/A (Rili-BÄK mode only)

QC alarms

Operator data alarms (system alarms)

Alarm code	Alarm text	Triggering event	Comment	Result type
OC	**Operator comment**	The operator performing the measurement has entered/selected a comment during the measurement.		Patient, QC
OCE	**Marked as erroneous by operator (comment)**	The operator performing the measurement has selected an “error comment”.	An “error comment” is a comment defined in **System** > **Comment definition** and has **Mark as erroneous** set in **Devices** > **Comment assignment**.	Patient, QC
OII	**Operator ID contains invalid character**	The operator ID entered during measurement contains an invalid character.		Patient, QC
OM	**Operator ID missing**	No operator ID has been entered on the instrument when the measurement was performed		Patient, QC
ONC	**Operator not certified for this instrument type**	The operator having performed the measurement has no valid certification for this instrument type.		Patient, QC
OOU	**Operator testing outside assigned location(s)**	The operator is not assigned to the location the instrument is assigned to.		Patient, QC
OP	**Operator profile does not support task**	This operator profile has no privileges for task 1 (send results from the instrument).	Retained for backwards compatibility	Patient
OU	**Unknown operator**	The operator ID entered during measurement is not known to the application.		Patient, QC

Operator alarms

Patient data alarms (system alarms)

Alarm code	Alarm text	Triggering event	Comment	Result type
PA	**Unknown Patient**	The patient ID entered during measurement is not known to the application.		Patient
PD	**Patient discharged**	The patient is already discharged.		Patient
PII	**Patient ID contains invalid character**	The patient ID entered during measurement contains an invalid character.		Patient
PN	**Patient not admitted**	The patient is not admitted, but not discharged either, i.e. has an undefined admission status; this is typical for auto-generated visits.		Patient
PU	**Patient not located in instrument location**	The patient is either not assigned to a HIS location or the patient HIS location is not assigned to the location to which the instrument is assigned.		Patient
TG	**Patient ID missing**	No patient ID nor a generic patient ID has been entered during measurement.		Patient

Patient alarms

Validation data alarms (system alarms)

Alarm code	Alarm text	Triggering event	Comment	Result type
TANR	**Qualitative result with numeric range - no auto-validation possible**	The test result is alphanumeric, but the applicable ranges defined for this test in **System** > **Test definition** are all numeric.		Patient
TD	**Delta check violation**	The difference with the last result of this test for this patient was too large.	The check is always done against the oldest result in the specified time frame.	Patient
TNR	**No reference range defined**	No ranges are defined for this test in **System** > **Test definition** > **Reference range**.	Ranges can be set up in **System** > **Test definition** > **Reference range**. They are based upon demographic criteria and their use is therefore only recommended when the application is receiving patient information from the hospital information system.	Patient
TH	**Result above upper reference range**	The result was above the normal range for this test.	Ranges can be set up in **System** > **Test definition** > **Reference range**. They are based upon demographic criteria and their use is therefore only recommended when the application is receiving patient information from the hospital information system.	Patient
TL	**Result below lower reference range**	The result was below the normal range for this test.	Ranges can be set up in **System** > **Test definition** > **Reference range**. They are based upon demographic criteria and their use is therefore only recommended when the application is receiving patient information from the hospital information system.	Patient

Alarm code	Alarm text	Triggering event	Comment	Result type
TPH	**Result above upper critical range**	The result was above the *critical range* for this test.	Ranges can be set up in **System** > **Test definition** > **Reference range**. They are based upon demographic criteria and their use is therefore only recommended when the application is receiving patient information from the hospital information system.	Patient
TPL	**Result below lower critical range**	The result was below the *critical range* for this test.	Ranges can be set up in **System** > **Test definition** > **Reference range**. They are based upon demographic criteria and their use is therefore only recommended when the application is receiving patient information from the hospital information system.	Patient

Alarm code	Alarm text	Triggering event	Comment	Result type
TTH	**Result above upper tolerance range**	The result was above the tolerance range for this test.	Tolerance ranges allow the automatic validation of a specific result range independent of whether the result is marked as out of normal/critical range.	Patient
TTL	**Result below lower tolerance range**	The result was below the tolerance range for this test.	Tolerance ranges allow the automatic validation of a specific result range independent of whether the result is marked as out of normal/critical range.	Patient
TRH	**Result above upper rerun range**	The result was above the rerun range for this test.	Retained for backwards compatibility.	Patient
TRL	**Result below lower rerun range**	The result was below the rerun range for this test.	Retained for backwards compatibility.	Patient

Alarm code	Alarm text	Triggering event	Comment	Result type
TRR	Repeat result detected	The result is a repeated result.	The general setting **Detect repeat results** must be enabled to activate the repeated results detection process. If enabled, for each instrument type configuration it is possible to define in the field **Repeat time interval** the amount of minutes when, if a repeated result is sent, it will be marked as repeated (flagged with system alarm TRR).	Patient

Validation alarms

Other data alarms (system alarms)

Other data alarms (system alarms)

Alarm code	Alarm text	Triggering event	Comment	Result type
MR	Manual Results	The result has been entered manually.	Recommended to be used with connectable instruments. Manual result entry is only possibl	QC

Alarm code	Alarm text	Triggering event	Comment	Result type
			e for QC test results.	
TM	**Result modified by user**	The result has been modified by a user of the application.		Patient

Alarm code	Alarm text	Triggering event	Comment	Result type
ERV	**Empty Result Value**	The related test result has no value.		Patient
ERDT	**Empty result date received from instrument**	The related test result has no date.	Retained for backwards compatibility.	Patient

Test Definition

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Test Definition

Use this component to either add a new *test definition* or to edit parameters such as the **Full test name** or the measurement unit that is used with a test. This is necessary in order to align the settings with the ones used in your organization.

Manually adding new test definitions is only required for offline devices. All connectable device types come with a predefined set of test definitions.

Prerequisites Tubes must have been previously defined.

Related screens The test definitions must be defined before you can use them in the following screens:

- **Configuration** > **Devices** > **Configuration**
- **Configuration** > **System** > **Material definition**
- **Devices** > **Lots**. When changing the units of measurement for a system test, the ranges defined for material lots must be changed accordingly

CAUTION

Test Configuration

- Ensure that test properties are configured correctly, as a wrong configuration may have a serious impact on results interpretation.

CAUTION

Manually entered Test IDs

- Ensure that manually entered Test IDs and Test Abbreviations do not conflict with those that will be included in future software or driver releases.
- **Test ID**: Use only Test IDs in the range of 1000–1999 (should this range be inadequate, contact Roche Service).
- **Test abbreviation**: End each test abbreviation with an underscore '_' character, e.g.'Glu_'

Changing a test abbreviation is not recommended. In case a test abbreviation needs to be modified, contact your Roche Service representative for assistance. Other components of the application may also need to be modified accordingly.

Do not change the test abbreviations "`Glu`" and "`Glu2`" for the **Accu-Chek® Inform** instrument type family.

Field definitions

Test ID	Identification code corresponding to one specific test (number between 0 and 262104)

Version date	Date of the version when the test was created or last modified. Version date must be the current date or later.
Test abbre-viation	Abbreviated name of the test (Max. 12 alphanumeric char-acters).
Full test name	Full name of the test. Test names must be unique (max. 30 alphanumeric charac-ters).
Tube name	Name of the tube used to carry out this specific test.
Tube volume	Sample type volume required to carry out the test (max. 4 numeric digits, reserved for future functionality).

Pri-mary unit	Units in which the results of this test are expressed: %, °C, µmol/L, inch, mL, etc. **Note:** When the primary unit for a result from an **Accu-Chek® Inform** II is mg/dL, the secondary unit defaults to mmol/L. If the instrument measure-ment unit is mmol/L and both the primary as well as the secondary unit configured in **System** > **Test definition** are mmol/L, QC results are flagged with **IUU**. The **IUU** alarm does not appear when the secondary unit is set to mg/dL. **Note:** For certain device types, units should only be changed by your Roche Service representative. Re-view the respective device driver user manual and if necessary contact your Ro-che Service representative

Number of decimals	The number of decimal places to be used in the rounding of test results supplied by the instruments. **Note:** When operating with Rili-BÄK, result ranges, target values and actual results are calculated and reported to the precision defined by this setting. **Note:** When calculating the target value for a QC lot, the application uses 1 decimal digit more than defined for the test.

Secondary unit	Secondary units in which the results are expressed. When the primary unit for a Glu2 test result from an Accu-Chek® Inform II is mg/dl, the secondary unit defaults to mmol/L. **Note:** If the primary unit is “`s`”, the secondary unit must be set to “`sec`” to allow correct processing for Rili-BÄK QC results. **Note:** For certain device types, units should only be changed by your Roche Service representative. Review the respective device driver user manual and if necessary contact your Roche Service representative

Con-version factor	*Configuring the **Conversion factor***:The conversion factor determines how results are converted between the primary and secondary units: *Primary to Secondary conversion:* If a result is received in the primary unit, the application multiplies it by the conversion factor to calculate the result in the secondary unit. Example: If the factor is 10, the primary value is multiplied by 10 to get the secondary value. *Secondary to Primary conversion:*If a result is received in the secondary unit, the application divides it by the conversion factor to calculate the primary unit value. Example: If the factor is 10, the secondary value is divided by 10 (or multiplied by 1/10) to get the primary value. Factor used to determine a secondary unit result from a primary unit result. (Max. 6 characters total – 5 numeric characters +1 decimal separator.) Currently applicable only for **Accu-Chek® Inform** instruments. For other instrument types no unit conversion is done for other than Rili-BÄK QC results. **Note:**if the primary and secondary units are only used for different expressions of the same unit (e.g., sec and s), the conversion factor must be set to 1 to allow correct processing for Rili-BÄK QC results.
Formu-la	Formula applied to calculate a result from the result of other tests, reserved for future functionality.
Max. lineari-ty	Maximum linearity value for the obtained results. The system accepts lower values without alarms (Max. 10 numeric characters.), reserved for future functionality.
Min. lin-earity	Minimum linearity value for the obtained results. The system accepts higher values without alarms (Max. 10 numeric characters.), reserved for future functionality

Delay	(in days) Time lapse before the results are available (Max. 3 numeric characters.), reserved for future functionality.
Catalogue ID	External test identification, reserved for future functionality.
Analytical method	Measuring method used to carry out the test. If defined, each default system test is updated with the information on the analytical method. This field is used in Rili-BÄK QC mode only.

Tasks

- Defining a new tube to specify the tube used for analysis.
- Creating a new test.
- Querying for test information to see the details assigned to a specific test.
- Activating a test so that it is available in other screens supporting test information.
- Deactivating a test so that it is not available in other screens supporting test information.
- Modifying an existing test to change its features, such as the measurement unit.
- Access the **Reference range** screen to handle various ranges of the selected test.

NOTICE

Interrupted application data flow

Automatically validated results may not be sent to the HIS/LIS, if tests are deactivated without removing the test mapping first.

▶ Prior to deactivating a test, do the following:Remove the test from materials defined in **Configuration** > **System** > **Material definition**.Deactivate the test mapping in **Configuration** > **Devices** > **Configuration** > **Driver configuration**.

Reference Range

Use this option to define and manage various ranges that are to be applied to the associated test parameter during validation of test results.

- This screen is accessed by choosing the **Reference ranges** button on the **System** > **Test definition** screen.
- All the actions taken on this screen affect the test record selected in the previous screen, **System** > **Test definition** screen.
- Only one range set for each gender can be defined.
- When a test result is received, it is checked against the configured range set.
- If a test result falls outside the specified ranges, it is appropriately flagged when sent to the Host.
- For results belonging to a patient record that has no specific entry in the **Gender** field, the **Generic** range set is used.

If the reference range of a test is modified, all past results will automatically update to the new reference range without displaying the previous one.

Tasks

- Creating range sets to be used with a specific test
- Querying for range sets to see its details
- Modifying existing range sets
- Deleting range sets

Test Group Definition

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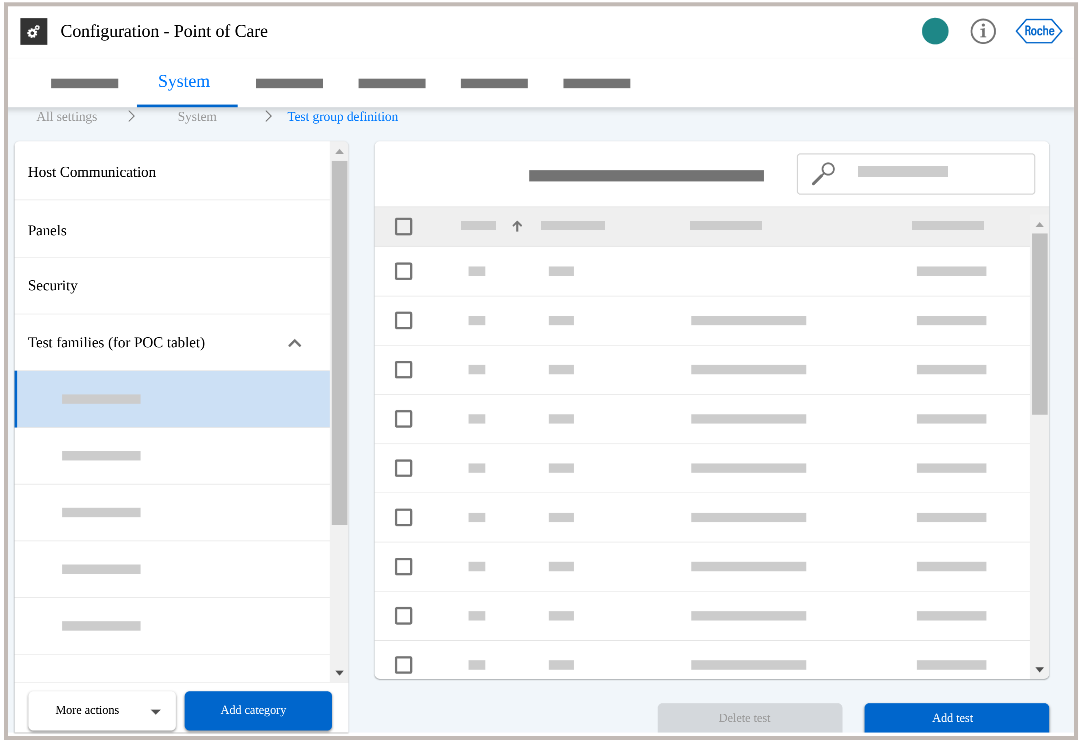
Test Group Definition

Use this screen to manage the various tests, which have been defined in **Configuration** > **System** > **Test definition**, and classify tests into test groups and test group categories. There are 4 predefined test group categories:

- **Host Communication**: test groups in this category are used to define groups of tests that need to be sent to certain hosts (e.g. all glucose results go to a diabetes information system, all blood gas results go to an ICU information system, or certain results alike Baro are not supposed to be sent to a Host system at all). The test groups defined in this category are mapped to specific hosts by Roche Service.

- **Panels**: test groups in this category are used as grouping criterion, e.g. for reports.
- **Security**: Retained for backwards compatibility.
- **Test families (for POC tablet)**: The QC families used in **Devices** > **Quality control**.



Color-coding Black text indicates an active group or category.

Gray text indicates an inactive group or category.

Querying for group definitions

You query for group definitions to see the tests assigned to a specific group.

1. Choose **Configuration** > **System** > **Test group definition**.

   → All categories are displayed in the Group structure to the left of the List area.

2. Expand the category and select the group to be queried.

   → All tests of this group are displayed in the List area on the right.

3. If necessary, use the search field to find a specific test.

Adding a new category

You add a new category to classify tests in different groups.

Category	It groups together test groups. Non-default categories are of no significant use within the current version of the application.
Complementary	If this option is selected, the same test cannot be included in more than one group of the same category.

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Choose the **Add category** button, fill in the fields, and choose the **Save** button.

→ The updated group information is displayed in the Group structure.

Adding a new group

Add a new test group so that tests are classified and it is easier to work with them.

Field definitions

Category	Category to which the test group belongs.
Group name	Full name of the test group.

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Select the category to which a group is to be added, from the **More actions** drop-down list, choose the **Add group** option, fill in the fields, and choose the **Save** button.

→ The updated group information is displayed in the Group structure.

Adding a new test to a group

You can add a new test to an already existing group to adapt test group definitions to the institution's working needs.

1. Choose **Configuration** > **System** > **Test group definition**.

   → All categories are displayed in the Group structure to the left of the List area.

2. Select the group to which a test is added, and choose the **Add test** button.

   → The **Test selection** screen is displayed.

3. Select the test or tests to be added from the list by selecting the check box, and then choose the **Add tests** button.

   → The **Test selection** screen is closed and the added tests appear on the grid.

Adding a new group through a copy

You can copy an already existing group to create a new one with similar properties.

1. Choose **Configuration** > **System** > **Test group definition**.

   → All categories are displayed in the Group structure to the left of the List area.

2. Select the group to be copied, from the **More actions** drop-down list, choose the **Copy group** option, fill in the field, and choose the **Save** button.

   → The updated group information is displayed in the Group structure.

Editing group or category information

Use this option to modify the name of an existing group or category, or to move a group to another category.

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Ensure that the group or category to be edited is active.

3 Select it, from the **More actions** drop-down list, choose the **Edit name** option, edit the field, and choose the **Save** button.

→ The updated group information is displayed in the List area.

Deleting a test from a group

You delete a test from a group when a test is removed from a group, the test definition itself is not deleted.

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Select the group from which a test is to be removed and select the test to be removed.

3 Choose the **Delete test** option.

→ The updated information is displayed in the List area.

Deactivating a group

You deactivate a group so that it is not available in other screens where test groups may be required.

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Select the active group you want to deactivate and choose the **Deactivate group** option from the **More actions** drop-down list.

→ The updated group information is automatically displayed in the Group structure.

Activating a group

You activate a group so that it is available in other screens where test groups may be required.

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Select the inactive group you want to activate and choose the **Activate group** option from the **More actions** drop-down list.

→ The updated group information is automatically displayed in the Group structure.

Activating a category

You activate a category so that its test groups can be used.

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Select the inactive category you want to activate, and choose the **Activate category** option from the **More actions** drop-down list.

→ The updated category information is automatically displayed in the Group structure.

Deactivating a category

1 Choose **Configuration** > **System** > **Test group definition**.

→ All categories are displayed in the Group structure to the left of the List area.

2 Select the active category you want to deactivate, and choose the **Deactivate category** option from the **More actions** drop-down list.

Tube Definition

This component is mainly intended for offline instruments, where results are entered manually rather than being sent by an instrument.

Prerequisites

Sample types must have been previously defined.

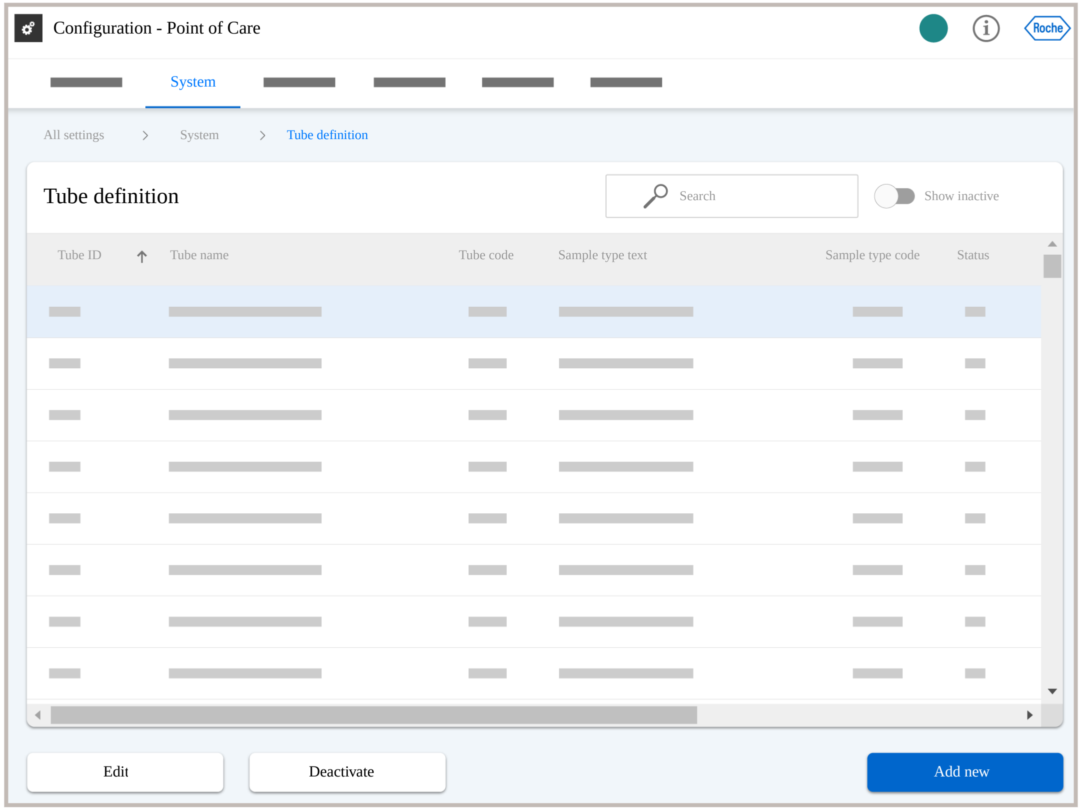
Interdependency with Sample Type Definition

If you add a sample type, you are required to create the corresponding tube.

Currently supported instruments do not physically use tubes for sample analysis. Tubes have no direct functionality in the current version of the application. However, the concept of tubes has been implemented in anticipation of future versions where more instruments will be managed by the application.

Default tubes

The application comes with a set of default tubes for use with connectable instrument types.



Tasks

- Defining a new tube to specify the tube used for analysis.
- Querying for tube definitions to see the details assigned to a specific tube identifier.
- Activating a tube definition so that it is available in other screens supporting tube information.
- Deactivating a tube definition so that it is not available in other screens supporting tube information.

- Modifying an existing tube definition. All fields are editable except for the tube identifier.

Sample Type Definition

Use this component to define and manage the various types of samples to be tested.

Used for manual entry of QC results for offline instruments

The functionality of the **Sample type definition** component in the current version of the application is mainly intended for use with offline instruments, where QC results are entered manually rather than being sent by a device.

Sample type added with instrument driver

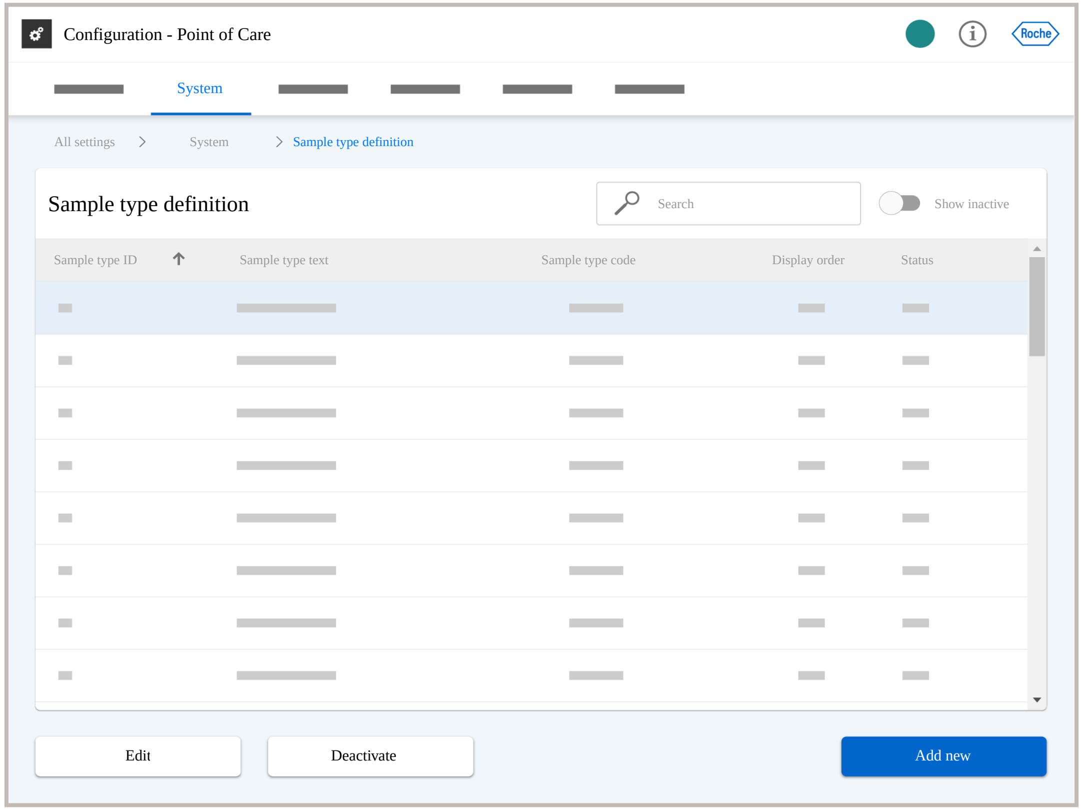
No additional definitions are necessary for instruments that can be connected to the application. Most sample type definitions are added during installation of the respective instrument driver.

Default sample types

The application comes with a set of default sample types to be used with connectable instrument types.

Test definition

For the definition of a test, both sample types and tubes need to be defined.



Tasks

- Creating a new sample type definition to determine the physiological samples on which tests can be conducted.
- Querying for sample type definitions to see the details assigned to a specific sample type.
- Activating a sample type definition so that it is available in other screens supporting sample type data.
- Deactivating a sample type definition so that it is not available in other screens supporting sample type data.
- Modifying an existing sample type definition. All fields are editable except for the **Sample type ID**.

Comment Definition

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Comment Definition

Use this component to set up comments and comment codes, and to simplify and standardize commenting. Comments defined in this screen are displayed in **Configuration** > **Devices** > **Comment assignment**, but are not applicable when commenting on patient or QC results inside the application.

The application provides a selection of predefined comments for the **Accu-Chek**® **Inform** - and possibly other device types depending on the respective drivers that are installed. If you do not want to use predefined comments, deactivate them in **Configuration** > **Devices** > **Comment assignment**.

Tasks

- Creating a new predefined comment to optimize the institution's workflow.
- Activating a comment so that it can be used in other screens supporting the use of predefined comments having comment codes..
- Deactivating a comment so that it cannot be used in other screens supporting the use of predefined comments having comment codes..

Predefined comments (for Accu-Chek® Inform)

Comment Code	Comment Text
ACC	Acceptable Control
ASY	Asymptomatic
CLM	Cleaned Meter
DON	Doctor Notified
LAD	Lab Draw
NCL	New Control Lot
NKL	New Strip/Kit Lot
NLN	New Lot Number
NOA	No Action
NSL	New Strip Lot
PRE	Procedure Error

Comment Code	Comment Text
PRT	Proficiency Test
RCT	Repeat Control Test
RNN	RN Notified
SQV	Switched QC Vial
WRT	Will Repeat Test

Abbreviations of the predefined comments

Message broker monitor

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Message broker monitor

The **Message broker monitor** is used as an internal system analysis tool, and can provide Roche Service with quick system information in case of problems with application data flow. The **Message broker monitor** is only to be used by Roche Service or trained administrators of the application.

This screen is for consultation only. However, messages that are no longer relevant can be deleted.

Message delivery status

A message can have one of the following delivery statuses:

Message delivery status	Description
Yes	The message has been processed without errors.
No	Processing of the message has not yet started.

Message delivery status	Description
Processing	The message is being processed.
With errors	The message has been processed with errors and could be reopened.

Delivery status of messages

Reopening a message

Use this function to reprocess data that the receiving system component has not processed correctly and is has status “`With errors`”.

1 Choose **Configuration** > **System** > **Message broker monitor**, select the namespace to be reviewed, and select the flagged record you want to reopen.

2 Choose the **Reopen** button.

→ The record is reprocessed and information delivery to the respective system component is attempted.

Viewing message details

1 Choose **Configuration** > **System** > **Message broker monitor**, and open the desired message.

2 Choose the **View details** button.

3 View the message details.

- To copy the message text to the *clipboard*, choose the **Copy message body** button.

Orderable panels

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Orderable panels

Orderable panels allow the LIS/host to order a predefined selection of tests from a specific instrument type.

Ensure that the orderable panels are complete and correctly configured before the application accepts orders from the LIS/host. Otherwise, test results may not be sent to the LIS/host.

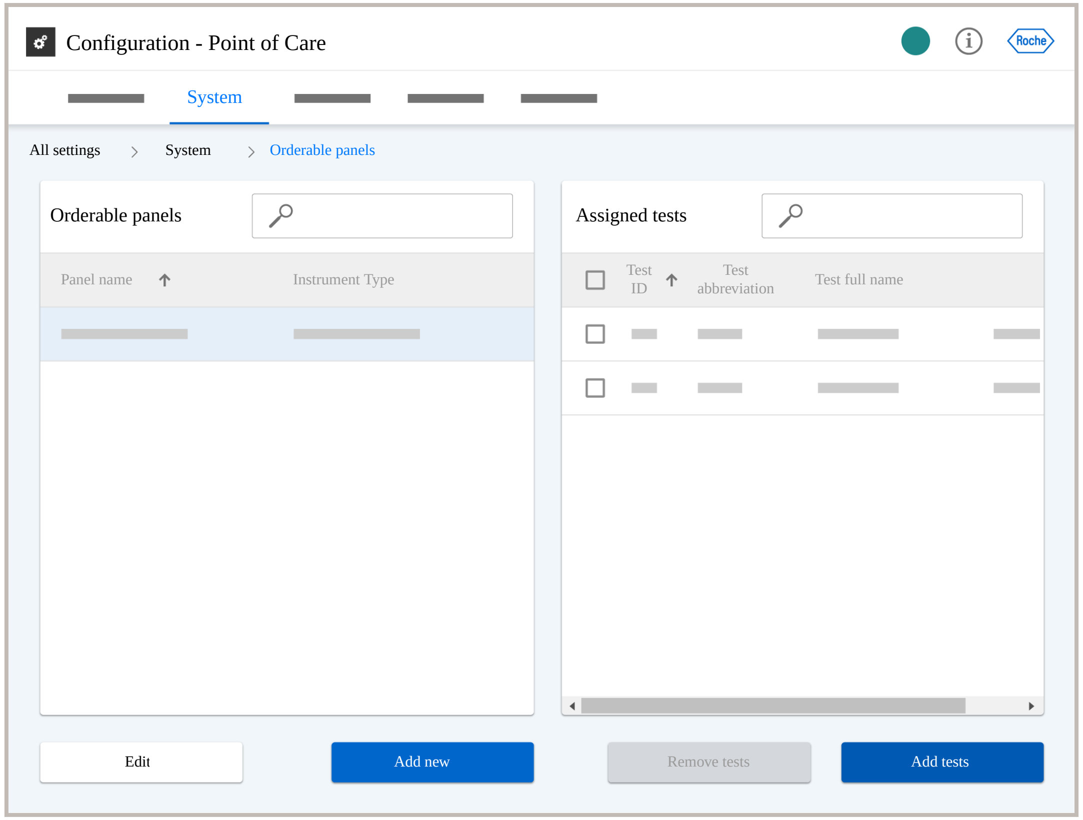
Orderable panels are not supported for instrument types that communicate with the application via the ICA interface.

CAUTION

Entry of incorrect data

Entering incorrect panel codes and/or assigning incorrect tests can affect the data transfer to the LIS as follows: the application sends unordered results, or fails to send expected results.

- Ensure that the entered panel codes are correct.
- Ensure that the correct tests have been assigned.



You can perform the following tasks:

Creating and editing orderable panels

You can create and edit panels according to the following rules:

- A panel must have a panel name and an instrument type. A panel code is optional.
- The combination of panel code and instrument type must be unique.
- You can assign an instrument type to multiple panels.
- You can assign a test to multiple panels of the same instrument type.
- You can change the panel name and panel code of an existing panel.
- You can remove tests from a panel, or add additional tests.
- You cannot change the instrument type of an existing panel.
- You cannot copy, delete, or deactivate a panel.

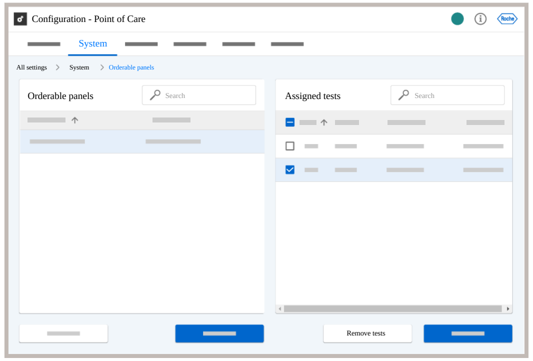
▶ To create an orderable panel

1. Choose **System** > **Orderable panels**.
2. Choose the **Add new** button and do the following:
   - Enter a panel name and assign an instrument type.
   - Enter an optional panel code.
3. Choose the **Save** button.
4. Select the panel and choose the **Add tests** button to assign available tests.
5. Select the tests you wish to assign, and choose the **Add tests** button.

▶ To edit an orderable panel

1. Choose **System** > **Orderable panels**.
2. Select the panel you want to change and choose the **Edit** button.
3. Change the panel name, and/or panel code, and choose the **Save** button.

▶ To remove tests from an orderable panel



1 Choose **System** > **Orderable panels**.

2 Choose a panel.

3 Select the tests you wish to remove, and choose the **Remove tests** button.

▶ To add tests to an orderable panel

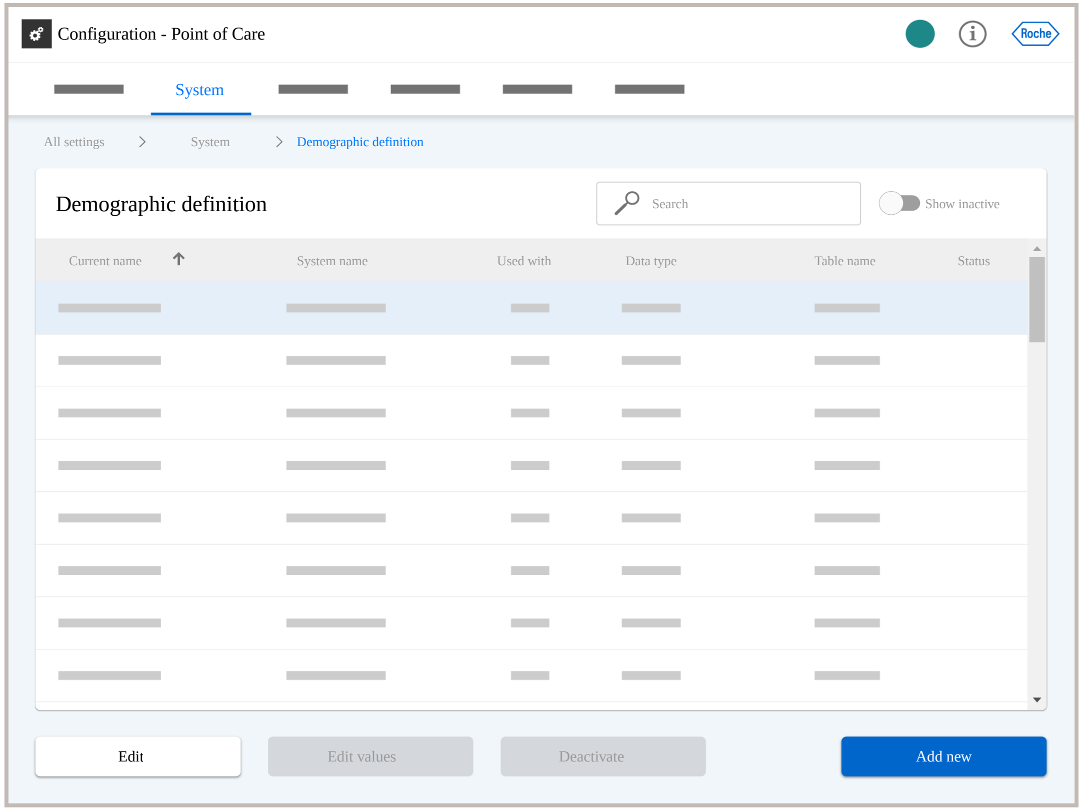
1 Choose **System** > **Orderable panels**.

2 Choose a panel, and choose the **Add tests** button.

3 Select the tests you wish to add, and choose the **Add tests** button.

Demographic Definition

This component enables you to set up the patient demographics within the application. In this screen, you can deactivate/activate demographic fields, and add new patient demographic fields.



Term definition The demographic information is related to patients (such as patient ID, name, gender, and date of birth).

Configuration Depending on the hospital requirements, administrators can define new, customized demographic fields.

Roche predefined fields cannot be edited, but can be renamed.

Field definitions

Current name	Name of the demographic field as it is displayed here. **Note:** When you change the Current name and save it, the change applies to screens in the Patient administration module only.
System name	Name of the demographic field as it is recorded in the system. The System Name is predefined and cannot be changed.
Data type	Type of data or information that can be introduced in this specific demographic field. The different types include **Free Text**, **Predefined**, **Free Text / Predefined**, **Image**, **URL**, **Table**, **Date**, **Time**, **General Text**.

Table name	Only active if data type is set as **Table**. Specific system table from which the respective information is retrieved. Enter the name as it is defined in the database.
Status	It indicates whether the demographic field is activated or deactivated.

Tasks From this screen, you can perform the following tasks:

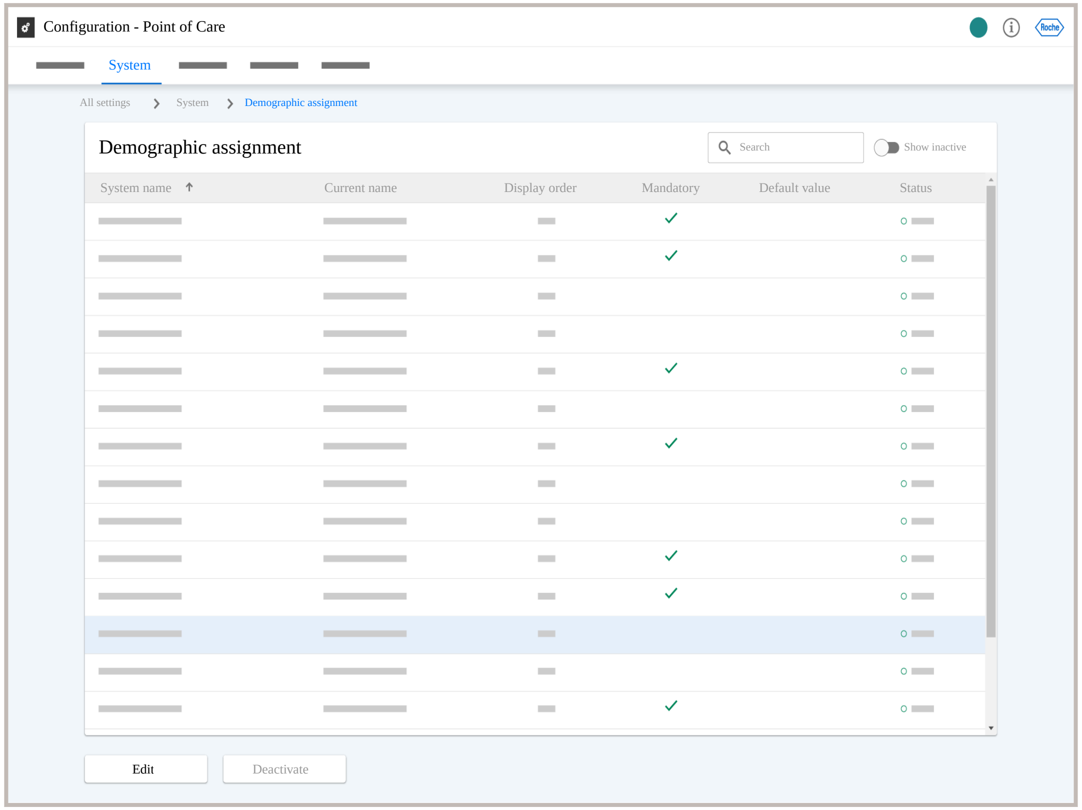
- Create a new demographic field to be used later in patient assignment and data entry.
- Query for demographic fields to see the details assigned to a specific demographic field.
- Activate/deactivate a demographic field so that it is visible/not visible in other screens where patient demographic fields are used.
- Modify an existing demographic field.
- Access the **Values** screen to manage fields of the data type **Predefined** values.

Demographic assignment

Use this component to configure which demographics are available in the **Patient administration** module. Define the order and the default value of each demographic field.

All fields in this component must previously be defined in the **Configuration** > **System** > **Demographic definition** component.

You must have administrator rights to make changes in this screen.



Configuration

Depending on the hospital requirements, administrators can define which demographic fields are available in the **Patient administration** module.

Mandatory fields cannot be deactivated.

Field definitions

System name	Name of the demographic field as it is recorded in the system. The **System name** is predefined and cannot be changed.
Current name	Name of the demographic field as it is displayed in the **Patient administration** module. Use the **Demographic definition** component to change the **Current name**.
Display order	It indicates the order of appearance of patient demographic fields in the **Patient administration** module. Enter a number to define the display order.
Mandatory	It indicates whether the field is mandatory or not. Mandatory fields cannot be deactivated.
Default value	Value that is displayed by default when a new record is added.
Status	It indicates whether the demographic field is activated or deactivated. Newly created demographics are deactivated by default. By default, only active demographics are displayed. To display active and inactive demographics, enable the **Show inactive** toggle button.

Tasks

From this screen, you can perform the following tasks:

- Editing a demographic field (**Display order**, **Default value**, **Mandatory**).
- Activating/deactivating a demographic field so that it is visible/not visible in the **Patient administration** component.

Material definition

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Material definition

Use the **Configuration** > **System** > **Material definition** component to define types of material used for each test. Specific test materials used by each instrument type must be defined in order to receive QC results from that instrument type.

The available fields change according to the material type that it is being defined.

For Radiometer instruments, not all results sent by the instrument are defined for the related materials. This is on purpose as some results (Temp and Baro) do not have ranges, and other values are considered calculated values.

For a QC material that does not reference a specific test material as “Reference Test”, (indicating that the QC ranges are maintained with the test lots), it is possible to select QC ranges as:

- dependent on test lots, or
- independent of test lots.

This is set by selecting or clearing the check box labeled **Test lot independent**. Materials for connectable instrument types are defined by default.

Remove test in Material definition prior to deactivating test in Test definition

Prior to deactivating a test, it should be removed from materials defined in **Configuration** > **System** > **Material definition** as well as (depending on the instrument type):

- **Configuration** > **Devices** > **Configuration** > **Driver configuration**

Prerequisites

- Instrument type and tests must have been previously defined.
- Test materials must be defined before defining QC or linearity materials.
- The combination of 'Material Name' and 'Instrument Type' must be unique.

Field Descriptions

Material name	Name applied to the material (max. 10 characters).
Long name	Brief description of the material (max. 50 characters).
Manufacturer	Name of the material manufacturer (max. 30 characters).
Instrument Type	Type of instrument with which the material is used.
Material type	Types of material: ▪ **Test** ▪ **Control** ▪ **Linearity**

Number of levels	Number of levels to be provided with this material.
Test material reference	Test taken as a reference. The check box **Test lot independent** on the right must be deselected to activate this field.
Test lot independent	For a QC material that does not reference a specific test material as "Reference Test", (indicating that the QC ranges are maintained with the test lots), selecting this check box indicates that the QC range is independent of the test lot.
Lot length	Number of alphanumeric or numeric characters that make up the lot number for this material (max. value is 15).
Alphanumeric	It indicates whether alphanumeric lot numbers are allowed.

Instrument Material ID	Material identifier used in instrument communications (max. 30 characters). **Note:** Users should not change this setting without consulting Roche Service – as changes will affect instrument communication
Peer reviewer material ID	Reserved for future use
Number of SDs	SD range that is covered by the given target ranges. Default value is 3. **Note:** Do not change the default value. If you wish to change the number of SDs, contact your Roche Service representative.
Not applicable for certification	Allows the user to set for each QC material (type **Control**), if certification tasks shall be counted or not. When adding a new material and selecting **Control** as the material type, this material is not marked automatically to be excluded from counting.

Tests	Tests related to the specific material.
Test materi-als	Name of test materials of that specific material type.

Tasks

- Creating a new material definition.
- Viewing the details of a particular material.
- Activating a material definition record so that it is available or visible in other screens within the application where it may be required.
- Deactivating a material definition record so that it is not available or visible in other screens within the application.
- Modifying an already existing material definition to change its properties.

NOTICE

Be careful when changing material configuration

Changing material configuration can affect the result evaluation.

▶ For material configuration changes, contact your Roche Service representative.

CAUTION

Existing material definitions

▶ Existing material definitions should only be changed after consulting Roche Service representative, as changes will affect instrument communication.

Defining a new material

1 In the **Configuration** > **System** > **Material definition** screen, choose the **Add new** button.

2 In the **Add new** dialog box, enter the required material description and choose the **Next** button.

3 Select the tests that are related to the material and choose the **Next** button.

4 Select the material names and choose the **Finish** button.

Organization

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Organization module components

It enables you to store and manage information on instrument trainings, as well as the ability of the user to work with an instrument.

You can import information on users and HIS locations into the **Organization** module from external files which provides easy interaction with other electronic systems in the hospital.

It includes organizational information regarding sites, locations and HIS locations. Each site consists of locations (also called wards) and HIS patient locations. To each location, instruments, instrument users, patient visits (via their HIS location assignment), and patient test results are assigned. Settings in this module affect the other modules in the application, such as users' access permissions.

Sites

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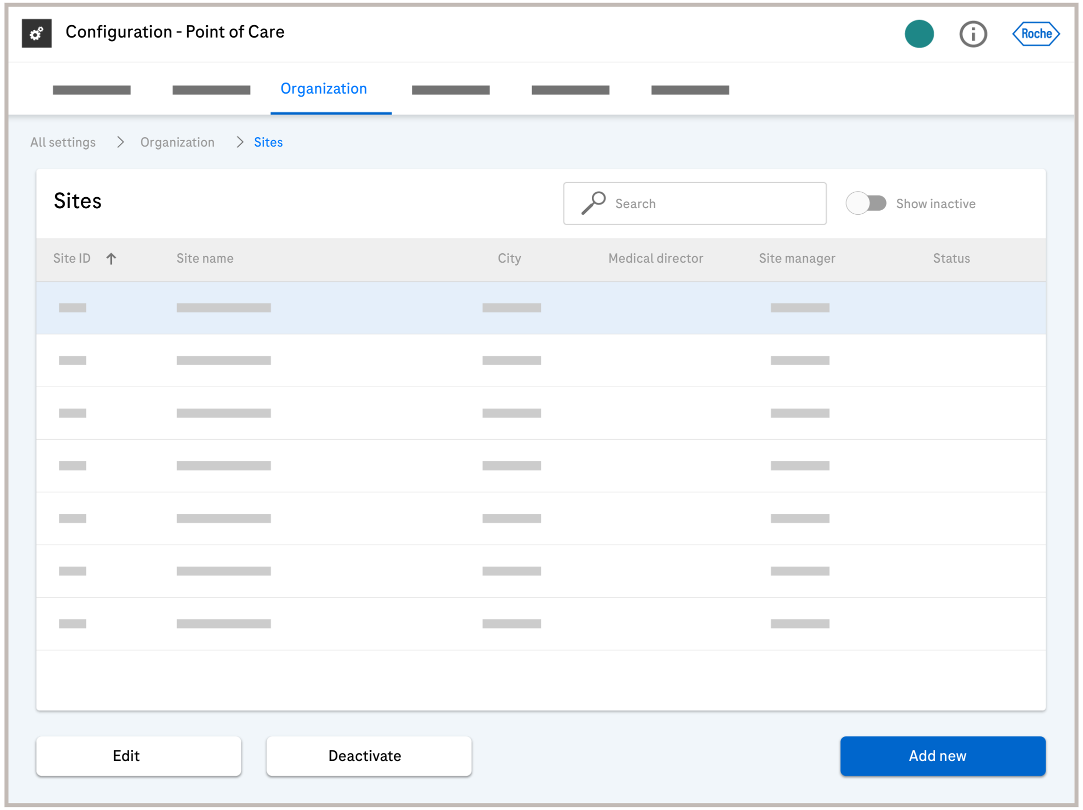
Sites

Use this component to create new sites, or to maintain administrative information for existing sites.

Within the application's data hierarchy, a site is the highest level to reflect an institution's organizational set-up. In this structure, each location belongs to a specific site.

Users and instruments (and therefore their QC and test results) can be assigned to locations and therefore to sites.

- Sites are unique.
- Except for the **Site name** and **Site ID**, the data on this screen for each category and site serves as additional information used in reports.
- Make sure that the site name is unique.
- You will be prompted to choose the site during the logon process.



Field definitions

Site ID	Identifier of the site. For usability reasons, restrict site ID to max. 5 characters. In case of Consilia workflow, all site IDs have to must the same length.
Site name	Name of the site.
Medical director	Name of the medical director of the site (Max. 30 alphanumeric characters).
Site manager	User ID of the administrator who is responsible for this particular site.
Role	Administrator’s profession (e.g. nurse, POCC, etc.).

Tasks

- Creating a site to start a new hierarchical organization in which the site is the highest level (e.g. if the application is used by two different hospitals).
- Querying for site information to see the details assigned to a specific site.
- Modifying site information of an already existing site.
- Activating a site so that it is visible and/or available in other screens within the application where it may be required.
- Deactivating a site so that it is not visible or available in other screens within the application.

Deactivating a site

If you deactivate a site, you must also deactivate the locations assigned to that site. Otherwise, you can still assign the locations from the deactivated site to an instrument in the **Instrument Assignment** menu. Patient and QC results from instruments assigned to a deactivated site are not displayed in the application.

1 Choose **Organization** > **Sites**, and select a site.
2 Choose the **Deactivate** button.
3 In the **Confirmation** callout, choose the **Confirm** button.

→ The site is deactivated.

Multisite concept

The multisite configuration enables you to manage more than one site (e.g. hospitals, laboratories) from one single installation of the application.

Settings applied to all sites

If the application is managing multiple sites, some information/settings are site-specific or operator-specific whereas other apply to all sites.

Operator-specific vs. site-specific vs. global components

In the following sections, the images show the various application modules. The tables list the tabs of the modules:

- The global column indicates that the information/settings in this tab affect all sites (global settings). The tabs display all information associated with the installation of the application.
- The site-specific column indicates that the information/settings in this tab are site-specific. The tabs display only the information of the site the current operator is assigned to.
- The user-specific column indicates that the information/settings in this tab are operator-specific. The tabs display only the information of the current operator.

Devicesmodule

Global	Site-specific	User-specific
Lots	**Devices**	**Settings**

Global	Site-specific	User-specific
	Quality control	

Device management module

Global	Site-specific
Certifications	Operators

Operator managementmodule

The **Certifications** tab displays all certificates associated with the installation of the application. But you can only certify operators that belong to the sites you have access to.

Site-specific
Result validation

Result validationmodule

Global
Patient administration

Patient administrationmodule

Global	Site-specific
Other reports	QC reports

Reportsmodule

Some reports on the **Other reports** tab show information across sites, some reports show only information of the site you have access to.

Configurationmodule

All settings in the **Configuration** module affect all sites.

Configuration > **System**:

Global
General settings
Instrument alarm assignment
System alarm assignment
Instrument alarm definition
Test definition

Global
Test group definition
Tube definition
Sample type definition
Comment definition
Message broker monitor
Orderable panels
Demographic definition
Material definition

Configuration > Organization:

Global
Sites
Locations

Global
HIS locations
User profiles
Certificates
Automated user import log
User import
HIS location import
Notification templates
LMS import errors
Workgroup definition

Configuration > Devices:

Global
Configuration

Global
EAP settings
EAP settings import
EAP settings assignment
Comment assignment

Configuration > **Quality control**:

Global
QC test mapping
Multi rules

Configuration > **Quality control** (Rili-BÄK):

Global
Rili-BÄK test assignment
QC test mapping (RB)

Implications

- Each site must have its own user ID range which cannot be used in any other site.
- Each site must have its own patient ID range which cannot be used in any other site.
- All sites will have the same general settings.
- All sites must work with the same materials.
- All sites will have the same severities assigned to the alarms.
- It is recommended for each site to maintain its own certifications, e.g. by using the site ID as prefix for the certificate type, in order to avoid issues when assigning certifications to a large group of users.

Shared responsibility of involved POCCs

A Point of Care coordinator of a particular site is solely responsible for the following components:

- **Devices** > **Devices**
- **Devices** > **Quality control**
- **Operator management** > **Operators**
- **Result validation**
- **Reports** > **QC reports**

For all other components, the Point of Care coordinators of the involved sites share responsibility.

Overlapping patient IDs on multiple sites / Consilia workflow

If two or more hospital information systems using the same patient ID ranges are to be connected, the application will add the site ID as a prefix to the received patient ID. In this way, patient IDs are unique in the application, if all site IDs are equally long.

This scenario is referred to as the Consilia workflow.

The setup is done by Roche Service. If the Consilia workflow is implemented, the site ID is displayed as a prefix to the patient ID and visit ID in the application. The site ID is not displayed on the instruments. LIS and instruments will only display the patient ID or visit ID, without prefix.

> If your site requires Consilia workflow, please contact Roche Service.

Use the following procedure to make sure that measurements are assigned to the correct patient.

1. Dock the instrument to synchronize it with the application. Make sure that all measurements have been successfully transferred to the application. Undock the instrument.
2. In **Devices** > **Devices**, assign the new configuration and location to the instrument.
3. Dock the instrument again to synchronize it. Undock the instrument.
4. Log on to the instrument and verify that it is assigned to the correct site and runs with the desired configuration.
5. Move the instrument to the new site.
6. Run a dummy measurement. Dock the instrument to synchronize it.
7. In **Result validation** > **Unreviewed patient results**, verify that the result was assigned to the correct patient.

Locations

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Locations

Use this component to create new locations, or to maintain administrative information for existing locations, such as a unique name, description, nurse, manager and phone number. Furthermore, existing locations can be copied and edited.

Locations are the fundamental link between user, instrument, patient and result data. Due to the important role as a unique identifier, location names are provided as query criteria in most areas of the application.

- Users are assigned to a location using the **Assign users** function.
- HIS locations can be assigned to locations either in the **Configuration** > **Organization** > **Locations** screen or in the **Configuration** > **Organization** > **HIS locations** screen.
- A location can be deactivated even though an instrument is assigned to it. In this case, assign the instrument to a different location in the **Devices** > **Devices** screen.
- A location can be deactivated even though an HIS location is assigned to it. However, in this case, you cannot assign the same

HIS location to a different location in this screen directly. Reassign the HIS location in the **Organization** > **HIS locations** screen.

Prerequisites

- Sites must have been previously defined.
- You must have administrator rights to make changes in this screen.

Field definitions

Location ID	Identifier of the location. For usability reasons, restrict location ID to max. 5 characters.
Initial sample ID	Initial number of the sample ID range for the location. If an initial sample ID is entered, a final sample ID must also be entered. (Max. 5 numeric characters.)
Final sample ID	Final number of the sample ID range for the location. It is mandatory if initial sample ID has been introduced. Max. 5 numeric characters.)
Spec. auto ID	If activated, samples from instruments assigned to this location receive an auto generated sample ID, whose incremental part is between the initial and final sample ID of this location.

Tasks

- Creating a location to be able to relate it to the users of this application, the instruments used and the patients.
- Copy a location to reuse it and edit the information of the designated new location in the Detail area.
  The location will be copied with all assigned instrument users, all assigned instrument comments, all assigned material lots and all active and inactive records including their status.
- Querying for location information to see the details assigned to a specific location.
- Editing the information of an existing location.
- Activating a location so that it is visible and/or available in other screens within the application where it may be required.
- Deactivating a location so that it is not visible or available in other screens within the application. Deactivated locations are not listed in the filter options of other grids.

Patient locations can also be assigned to a location in the **Organization** > **HIS locations** screen.

Deactivating a location

If you deactivate a site, also deactivate the locations assigned to that site.

1 Choose **Organization** > **Locations**.

2 Select a location and choose the **Deactivate** button.

3 In the **Deactivate element** dialog box, choose the **Confirm** button.

→ The location is deactivated.

Copying a location

Only locations of the active site can be copied.

1 Choose **Organization** > **Locations**.

2 Select a location and choose the **Copy from** option from the **More actions** drop-down list.

3 Fill in the fields and choose the **Save** button to save the new location.

Assigning multiple users to a single location

Use the **Assign users** option to allocate multiple users to a single location.

The **Assign home location** function allows you to set the home location for the assigned users in one step to the current location.

Prerequisites

Locations and users must have been previously defined.

> You must have administrator rights to make changes in this screen.

1 Choose **Organization** > **Locations**.

2 Select the location to which you want to assign users and choose the **Assign users** button.

> → The **Users assignment** window is displayed. Currently assigned users are shown.

3 Select a site from the **Site** drop-down list and choose the **Filter** button.

4 If required, choose a filter from the drop-down list to limit the number of listed users and choose the **Filter** button. The following filters are available:

- **All**
- **Home location**
- **Locations**
- **Unassigned**
- **Workgroups**

5 In the list of available users, select the users that you want to assign to the location.

6 Choose the **Confirm** button.

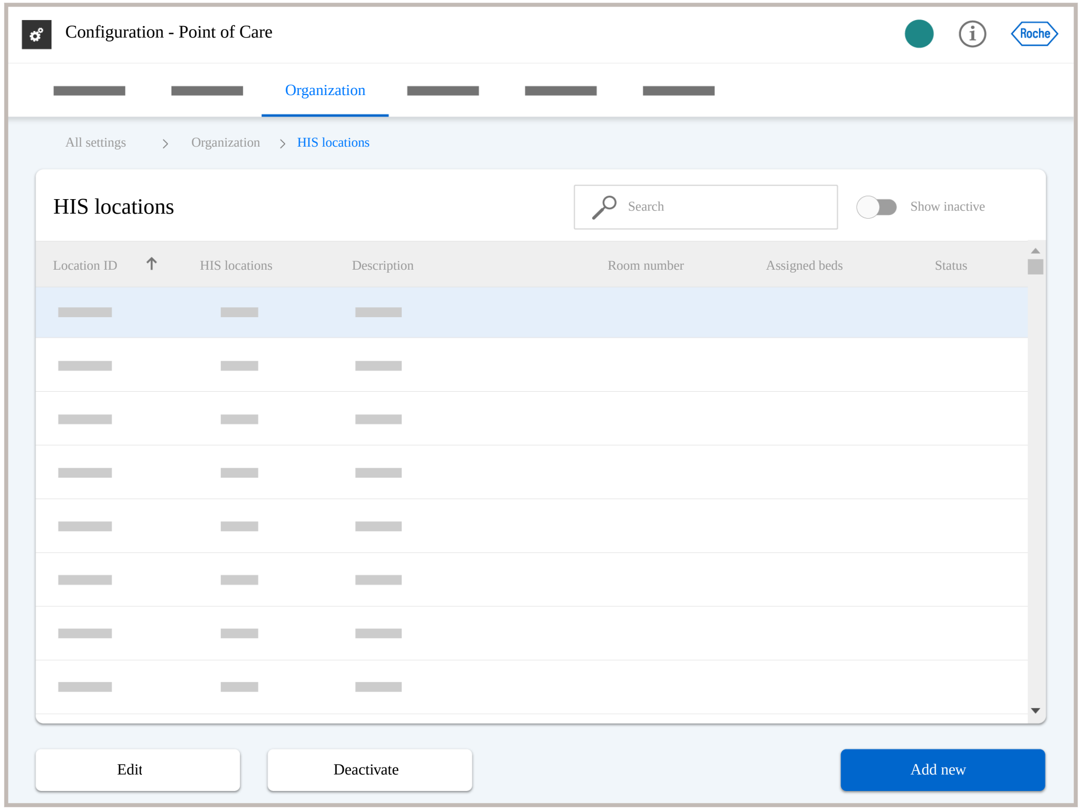
HIS Locations

This component enables you to create new HIS locations, or to maintain information for existing HIS locations, such as the room name and assigned beds. The HIS location is used as the link between patient visits and locations, based on the ADT information, received from the administrative system connected to the application.

Assignment of the HIS location to the proper location is essential to distribute patient visits correctly, thus providing appropriate patient information to individual instruments used in different locations.

> If you enter a description for the HIS Location manually, the text will be overwritten by the value received in the patient message from the host.

Prerequisites Sites and locations must have been previously defined.



Tasks

- Creating a new HIS location to be able to maintain accurate information about where patients are located.
- Querying for HIS location information to see the details assigned to a specific location.
- Editing HIS location information of an already existing one.
- Activating an HIS location so that it can be visible and/or available in other screens within the application where it may be required.
- Deactivating an HIS location so that it is not visible or available in other screens within the application

Description of HIS Location If you enter a description for the HIS location manually, the application overwrites the entered text when it receives an ADT message from the HIS/LIS.

User Profiles

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User Profiles

Use this component to create new user profiles, or to maintain information for existing user profiles.

Prerequisites You must have administrator rights to make changes in this screen.

Main function Defining to which modules and components a user is granted access. A user can only see the modules that are assigned to his user profile.

Profile definition Typical user profiles are defined based on job functions, such as Point of Care coordinator, nurse or physician.

Assigning profiles to users Profiles are assigned to users in **Operator management** > **Operators**.

Default setting By default, new user profiles do not have access to any menu options. Access rights must be granted.

Predefined profiles The following profiles are predefined:

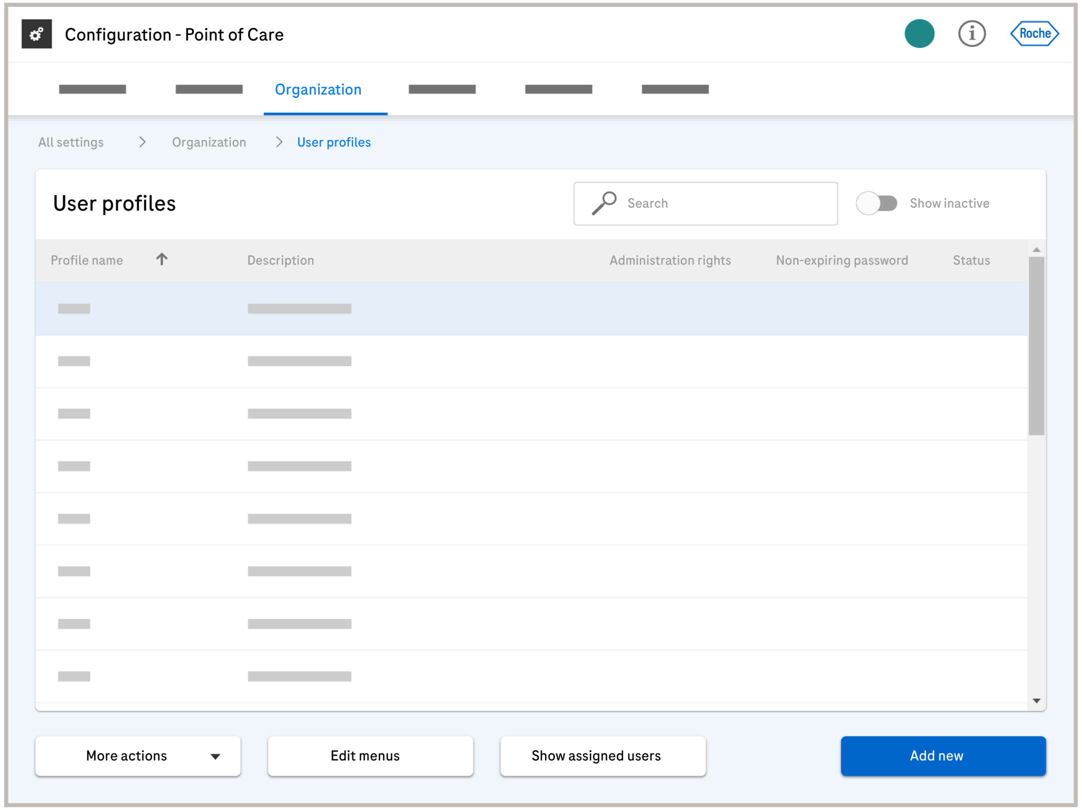
Predefined profile	Description	Administrator rights
IT Manager	For managing EAP Settings	yes

Predefined profile	Description	Administrator rights
Manual Entry	For manual entry of quality control results	yes
Nurse Educator	For end users with training responsibilities	no
Operator	For users of instruments without access to the application	no
POCC	For the POCC	yes
POCT Admin	For the system administrator	yes
Respiratory Therapist	For matching orders and results	no
Service	For Roche service	yes

Predefined profiles

Administrators, Non-administrators Users with administrator rights will be able to edit and manage those pages that necessarily require administrator rights as indicated in this manual (mainly configuration screens).

Deactivating a profile If a profile is deactivated, none of the users with this profile can log on to the system.



Field definitions

Profile name	Full name of the profile. (Max. 15 alphanumeric characters.)
Description	Brief description of the profile. (Max. 40 alphanumeric characters.)
Administration rights	It indicates whether the user has administrator rights.

Non-expiring password	Select this option when the password of the profile members should never expire.
Data volume	Number of rows displayed on the grid. For performance reasons, it is recommended to set the volume to **Low**. The number of data rows assigned to a level can be adapted by Roche Service. Default settings for **Low** and **High** are 800 and 5000 rows respectively.
Assigned users	The users assigned to this profile will be listed here. Users are assigned to a profile in the **User Management** module.

Tasks

- Creating a new user profile to define a new group of permissions and settings that can be assigned to users.
- Querying for profile information to see the details assigned to each individual profile.
- Activating a profile so that it can be assigned to new users and be available from other application screens.
- Deactivating a profile so that users with that profile cannot access the application and it cannot be assigned to new users or be available in other screens.
- Editing profile information to change any of the features of a profile.
- Accessing the **Tests** screen to handle test access level of the selected profile.
- Accessing the **Menus** screen to handle menu access level of the selected profile.

Menus

Menus

Use this option to grant access to individual screens of the application to a user profile. Access can either be granted or denied. If access is granted, the related screen can be accessed from the Navigation area. If denied access, the related module or component will not appear within the Navigation area.

- To access the **Menus** screen, choose the **Edit menus** button on the **User profiles** screen.

Prerequisites You must have administrator rights to make changes in this screen

Certifications

View certifications	To view and search certifications.
Update training and export list of operators	To export a list of operators needing a certification update. To update the certification status of operators and to certify operators.

Devices

View devices	To view and search devices.
Replace, edit, assign device	To add, assign, synchronize, deactivate, activate, edit, and replace devices.
Edit application-wide device model settings	To configure automatic material lot distribution. To configure the absolute QC target deviations per QC lot version (for device types supporting lot versions).

Instruments

Configuration	To define the configurations for the instruments connected to the application.
Comment assignment	To define the comments that are made available for selection on connected instruments.
EAP settings	To manage EAP settings definitions. By default, only the IT Manager and the Service user profiles have access to this screen.

EAP settings assignment	To manage the assignment of EAP settings to instruments with a configuration where WLAN security type is EAP. By default, only the IT Manager and the Service user profiles have access to this screen.
EAP settings import	To import EAP settings definitions. By default, only the IT Manager and the Service user profiles have access to this screen.
Device import	To import multiple devices to **navify**® POC Operations.
Off-Line Instruments	To configure types of instruments not connectable to the application. To add configurations for and individual instruments of such types.

Materials

Material definition	To define the various types of materials used with each instrument type.
View materials	To view and search materials.
Add and edit material	To create new material lots and edit existing material lots.
Activate and deactivate material	To activate and deactivate existing material lots.

Operators

View operators	To view and search operators.
Edit operator, add operator, update training	To add, edit, activate, and deactivate operators. To reset an operator's password. To add certifications to an operator.

Organization

Sites	To configure the sites where the application is used.

Locations	To configure the locations that make up each site.
HIS locations	To configure the HIS locations associated with each location.
Workgroup definition	To configure the workgroups that users may belong to.
User profiles	To configure user profiles and grant or restrict access to certain screens or data.
HIS location import	To import information about HIS locations from a text format file.
User import	To import information about users from a text format file.
Certification	To define and manage user certification for instrument access.
Notification templates	To configure notification templates for emails and letters.

LMS import errors	To view and search the errors during LMS exam results import.
Automated user import log	To view and search the automated user import log entries.

Patient administration

Delete patient and visit	To delete patients and visits.
View patient and visit information	To view patients and visit details.
Modify patient and visit information	To edit patients and visit details.

Result validation

View patient results	To view patient results.
Modify patient results	To accept, reject, undo validation, send to the host, and change the associated patient.

QC

View QC results	To view the QC results received from instruments and check their status.
Accept and reject QC results	To review the QC results received from instruments, and accept or reject them.
Add QC Results Manually	To enter QC results manually.

Quality control

Multirules	To define the multirules that are to be used with QC test results to detect instrument errors.
QC test mapping	To map individual tests with generic tests in instruments that run several types of tests but use one QC test only.

Reports

View reports	To view reports
Create and delete report	To create reports by using the predefined types of reports. To delete reports
Export report	To export reports.

Result and order matcher

Manage unmatched test results/orders	To match results to orders manually. To send results to the LIS without a matching order. Access is granted for the following default user profiles: POCC POCT Admin Respiratory Therapist Service

System

General settings	To define the basic workflow parameters of the application.
Demographic definition	To define or customize patient demographic fields.

Instrument alarm definition	To configure alarms with which test and QC results can be flagged.
Instrument alarm assignment	To map the configured alarms with flags sent by the instrument and set their severity.
System alarm assignment	To set the severity of the alarms generated by the system.
Comment definition	To predefine comments.
Sample type definition	To define types of samples used in **Tube definition**.
Tube definition	To define the types of tubes used in **Test definition** (for sample collection/storage).
Test definition	To define the tests that can be run with the instruments connected to this application and their settings.
Test group definition	To make test groups for preset test panels for manual entry / Sample Statistics or to restrict access to certain data or define rules for sending results to host systems
Message broker monitor	To check for possible errors in the communication between the application and the instruments and hosts.
Orderable panels	To create and edit orderable panels.

Quality Control (Rili-BÄK)

QC test mapping	To map individual tests with generic tests in instruments that run several types of tests but use one QC test only.
Rili-BÄK test assignment	To map QC (Rili-BÄK) tests with system tests.

Granting access to menus

Set access to the application menus to determine which are the menus to which users with this profile should have access.

1 Choose **Configuration** > **Organization** > **User profiles**.

2 Select a user profile and choose the **Edit menus** button.

3 If necessary, sort the list of available menus on the **Menus** screen.

4 Select a menu and grant or deny access to it, and then choose the **Save** button.

→ All the information is automatically stored into the database.

Copying settings from another profile

Use this option to set the same or similar settings to another profile.

1 Choose **Configuration** > **Organization** > **User profiles**.

2 Select a user profile and choose the **Edit menus** button.

3 On the **Menus** screen, choose the **Copy from profile** button.

→ A dialog box with a list of the existing user profiles is displayed.

4 Select the profile of which you want to make a copy and choose the **OK** button.

5 If necessary, edit the access settings.

6 Choose the **Save** button.

Certificates

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Certificates

The application uses certification to enable the management of user competence.

A certificate is a template defining tasks which a user has to complete to become “certified”. Certification is the process of assigning a certificate to a user.

One of the important governance aspects of running a Point of Care service is to ensure that staff is adequately trained to use the instruments.

The **Certificates** menu offers you a tool to manage user certifications for all supported instrument types from one place. The main features are:

- Define certificate details such the expiry date and the tasks required to become certified.
- Assign certificates to users.

Certifying operators

Prerequisites

You must have administration rights to create or edit certificates.

Certificate types

You can create 3 different types of certificates:

- *Non-expiring:*
  certificate with no expiry date, once assigned the user remains always certified.
  Note:
  Use non-expiring certificates as template when creating new certificates.
- *Expiring without auto-recertifying*:
  certificates has a fixed expiry date.
  When expired, the POOC must renew the certificates.
- *Expiring with auto-recertifying*: certificates consists of 3 stages. In each stage, the user must fulfill specific certification criteria. the certification/recertification process is automated by the application.

About auto-recertifying certificates

Auto-recertifying certificates consist of 3 certification stages (pending, interim, and certified). In each stage, the user must complete certain certification tasks to move to the next stage. If the user passes the pending and interim stages within the predefined time frame, the user is certified. To remain certified, the user must periodically repeat the assigned tasks in the certified stage (auto-recertifying).

Certification stages

The different stages allows you to manage certificates for users with different levels of experience.

- *Pending stage:*
  The pending stage is for new users undergoing their first certification. In this stage, users can only perform QC tests and *observed test sequences* (OTS) on their assigned instrument. Patient tests are not permitted. If a device misses specific privileges for patient and QC tests, users are restricted to performing only observed test sequences (OTS) on the assigned instrument.
  After meeting the certification criteria, users progress to either the interim (optional) or certified stage.
- *Interim stage (optional):* Intended for new users to be certified for the first time. The user is allowed to make patient measurements but cannot act as observer for an OTS. After achieving the certification criteria, the instrument user advances to the certified stage.
- *Certified stage:* intended for fully certified instrument users with full certification duration. After achieving the certification criteria, the user stays in certified stage.

Certification tasks

For each certification stage, you can assign different types of tasks to be performed by the user:

- QC and patient measurement on the instrument
- Competency task by classroom training
- Exam by a learning management system (LMS)

A competency task can be added by choosing **Configuration** > **Organization** > **Certificates**, choosing **Edit** for the desired certificate and choosing the check box **Competency required**(Optional) in the pending tab, for the desired certificate stages.

You can configure the certification tasks for each certification stage in the respective certification section.

QC level 1(#) **QC level 2(#)** **QC level 3(#)** **QC level 4(#)**	For reach QC level (1-4), you can define how many QC measurements must be performed.

Patient samples (#)	Number of required patient sample measurements. No patient measurements are allowed in the pending stage.
OTS	If selected, patient measurements must be performed under supervision.
Free text criteria	Number of tasks (max. 3).
Freetext Criterion 1 **Freetext Criterion 2** **Freetext Criterion 3**	Definition of task (criterion).
Competency required	You can define that a competency is required by classroom training.

Exam required	If selected, exam by a LMS is required. If a third-party LMS is installed: You must select the check box manually to link the training course to the certification. A linked training course enables the import of exam results data from the LMS. This field is only visible if the **Exam required** check box is selected.
Course name	Name of training course as defined in third-party LMS. This field is only visible if the **Exam required** check box is selected.

Maximum attempts	Maximum number of permitted exam attempts as defined in the LMS. - If a third-party LMS is installed: you must enter the number manually to link the training course to the certification (Supported values: integer >1). For exams with unlimited attempts, enter a high number (e.g. 9999). If the user fails the last exam attempt, the user's supervisor becomes the owner of this certification task.

Certificate validity The validity is used to calculate a user's certification expiry date in the certified stage. You enter the validity in months in the **Valid (months)** field in the **General** section.

Duration The duration is used to calculate a user's certification expiry date in the interim stage. You enter the duration in months in the **Duration (months)** field in the **Interim** section.

Expiry date calculation After completing all tasks in one stage, the user enters the next stage. For each stage transition, the expiry date is calculated as follows:

Stage transition	Certification expiry date
pending → interim	Date when all tasks in pending stage were completed + duration (Months) - 1 day
pending → certified	Date when all tasks in pending stage were completed + certificate validity (months) - 1 day
interim → certified	Date when interim stage started + certificate validity (months) - 1 day

Stage transition	Certification expiry date
certified → certified (for recertifying)	Date when all tasks in certified stage were completed + certificate validity (months) - 1 day

Expiry date calculation across stages

Observation time frame (OTF)

For each certification stage, the OTF defines the time frame during which all certifications tasks must be completed. You enter the OTF in days in the respective sections (**Pending**, **Interim**, **Certified**).

Pending stage:

- The OTF starts when the user has completed the first certification task. The pending stage does not have a fixed duration and the user advances to the next stage (interim or certified) immediately after completing all tasks.
- If the user does not complete all tasks within the OTF, the OTF is restarted. The start date of the new OTF is the date when the second task was completed. All tasks performed before the start of the new OTF must be repeated.

Interim and certified stage:

- The start date of the OTF is calculated as follows: Expiry date - OTF (days) +1.
  Example: **Expiration date** = April 30, OTF = 10 days f start of OTF = April 21
  Additional consequences depend on the type of instrument or test.
  If a user does not complete all tasks in the OTF, their certification status changes to "Expired" or "Long Term Expired".
  If certain tasks (e.g., running a patient sample) can't be completed, the system may create a new Observed Test Sequence (OTS) to replace them. The specific replacement tasks vary by device or test type.
  Generally, users with expired certifications aren't sent to devices. However, exceptions depend on device type, roles, and privileges. For some devices, users may still run Quality Control (QC) tests even if they can't perform patient tests.
  In such cases, users with expired certifications may still be routed to devices to complete tasks needed for recertification.

In-stage checks

In the certified stage, you can specify that QC tests are required not only at the end of the stage but also periodically within it. By selecting a **QC every**shorter than the overall validity period, you enable these in-stage checks. This setting determines how often, in months, the defined set of QC tests must be performed.

For example, if you set a **QC every**of 3 months for a certificate with a 12-month validity, three additional expiration dates (and corresponding OTFs) will be added within that 12-month period. These 'QC-only' OTFs have the same duration as the final OTF at the end of the validity period, during which all defined task types, including QC, must be completed.

Failing to meet a 'QC-only' expiration date has the same consequences as missing the final expiration date.

User role

If an instrument supports role-based access control, you can define the user role in the **General** section of the certificate.

The role defines which actions a user can perform on an instrument. The field is only displayed for instruments that support role-based access control. If supported, it is a mandatory field.

The roles can vary between different instrument types. For example, the **cobas** ® **Liat**® Analyzer supports the following roles: *User*, *Supervisor*, and *Administrator*.

If multiple roles are assigned to a user for a certain instrument type, the most privileged role is applied for that instrument type.

You can assign roles via certificates as follows:

- Define a certificate for each role.
- Name the certificate after the roles.
- Assign the roles to the users via certificates.

OTS - Observed test sequence

One way of demonstrating competence to use an instrument is to perform a test under supervision of an expert user (observer). This is known as OTS.

> Not all instrument types support OTS.

About privileges

Depending on the instrument type, a set of user privileges can be defined in the **Privileges** section.

Privileges determine what functions a certified user can access on a certain instrument.

Not all instruments support privileges. If the selected instrument type does not support privileges, the **Privileges** section is disabled.

The supported privileges can vary among different instrument types. To assign all privileges, select the **All** check box.

> Instrument access without privileges:
>
> - Instrument types supporting user roles:Users with certificates without any privileges are uploaded to the instrument.*Example:* The role *Administrator* for the **cobas®Liat®** Analyzer allows you to perform administrative tasks but no tests. To perform any tests, you require the respective privileges.
> - Instrument types not supporting user roles:Users with certificates without privileges are not uploaded to the instrument.

If a user holds different certificates, the effective privileges are derived from the combination of privileges.

Example for Glu2 tests on **Accu-Chek®**:

- One certificate for general instrument users with checked **Glucose Results** check boxes, but unchecked **Proficiency Test** check box
- One certificate for certain instrument users with unchecked **Glucose Results** check boxes, but checked **Proficiency Test** check box

An instrument user holding both certificates can do all Glucose tests on the instrument, including Glucose Proficiency Tests.

Predefined certificates for an instrument type define a subset of supported tests and privileges.

Examples for **Accu-Chek**® Inform II:

Predefined certificate	Privileges	Supported tests
ACI2 Glucose	Glucose patient test, Glucose control test, ...	Glu2
ACI 2 Pregnancy	Pregnancy test, Pregnancy control	Preg
ACI2 Visual UA	Visual UA test, Visual UA con-trol	Col VUA, Clar VUA, ...
ACI2 Rapid Strep	Rapid Strep test, Rapid Strep control	Rstrep

Examples of predefined certificate

Notification about certification tasks

Notifications are a mechanism for informing users or supervisors of certification events such as certification expiry or renewal tasks.

In the **Reminder** section, you can configure the notifications about certification tasks.

by email	Defines whether the user is notified about certification tasks by email Allows selection of the notifi-cation template to be used
cc Su-pervi-sor	Defines whether the supervi-sor will receive the user noti-fication email as well
By let-ter	Defines whether a letter is printed out that can be sent to the user to notify about certification tasks Allows selection of the notifi-cation template to be used
On de-vice	Defines whether the user is notified about certification tasks on the instrument (only available for instruments supporting user notification) Allows selection of the notifi-cation template to be used

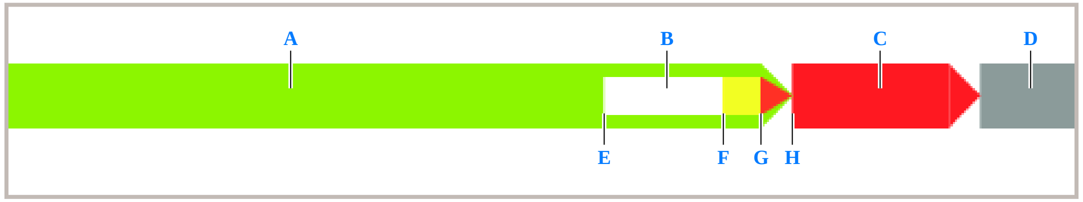
Notification period (days)	Defines how many days before the certification expiry date the user receives notifications on the instrument. To activate the field, you must check the **On Instrument** check box. ▪ Default value = not defined (empty). ▪ Supported values: integers from 1 to 999. ▪ If you use interim period, the notification period applies to all stages. ▪ No notifications are sent before the OTF starts (applies to all stages). If you update the application to a newer version, the value set for the notification period is replaced by the following parameters: ▪ For auto-recertifying certificates: **Observation time (days)** from **Organization** > **Certification** > **Certified**. ▪ For expiring certificates: **Certification Expiration Warning Time (Days)** from **System** > **General settings**.

Remind supervisor	Defines whether the supervisor is notified about certification tasks by email Allows selection of the notification template to be used
Supervisor is	Defines who the supervisor for the user is: ▪ **Supervisor of users home location** ▪ **Supervisor of all locations to which the user is assigned.** ▪ **Administrator of users workgroup** A user can be a supervisor, if the user is assigned to a **User profile** having administrator rights.

If the **by email** option is selected for a certification, the following four notifications are successively sent to the instrument user:

- Upcoming certification: when entering the observation time frame, i.e. on the start date of the observation time frame
- Pending certification: when entering the medium observation time frame
- Urgent certification: when entering the narrow observation time frame
- Expired certification: when entering the recently expired observation time frame

The medium observation time frame, narrow observation time frame and expired observation time frame are calculated from the general settings defined under **System** > **General settings**.



Example for observation timeframe

A Duration of validity of certification

B Observation timeframe

C Recently expired certification timeframe

D Certification in status long-time expired

E Start of observation timeframe: notification for upcoming certification

F Start of medium observation timeframe: notification for pending certification

G Start of narrow observation timeframe: notification for urgent certification

H Start of recently expired observation timeframe: notification for expired certification

In the pending stage, the notification for upcoming certification is sent after the user passed the first certification activity, starting the observation timeframe.

NOTICE

Misleading notifications

The application does not differentiate between an in-stage OTF triggered by the use of the **QC every** field in the **Certified** section and the end-stage OTF. Therefore, in-stage notifications can be misleading in two ways: First, in-stage notifications remind about all certification criteria, even those that can only be fulfilled later during the end-stage OTF (e.g. patient samples). Second, in-stage certifications are sent even after the instrument user has performed the mandatory QC tests because of the other “unfulfilled” certification criteria. These notifications only stop after the in-stage OTF is over.

- If such in-stage checks are of no interest to the facility’s certification policy, create certifications without use of the **QC every** field in the **Certified** section.

Creating a non-expiring certificate

A non-expiring certificate has no expiry date. If assigned, the user stays certified.

1 Choose **Organization** > **Certificates**, and choose the **Add new** button.

2 On the **General** section, select the **Non-expiring** check box.

3 Fill in the mandatory fields.

- **Device type**: type of instrument for which the certificate is valid
- **Certificate type**: name of certificate. For multi-site locations, prefix with site ID.
- **Role**: user-access control, not supported for all instrument types

4 If you wish to assign observer rights for this certificate, select the **Is observer** check box.

5 Enter an optional comment in the **Comment** field.

6 Do one of the following:

- If the instrument type support privileges, choose the **Next** button and assign the user privileges in the **Privileges** section. Then choose the **Finish** button to save the certificate.
- If the instrument type does not support privileges, choose the **Finish** button to save the certificate.

Creating an expiring certificate without auto-recertification

Expiring certificates without auto-recertification have a defined expiry. If a user's certification expires, the POCC must renew the certification.

1 Choose **Organization** > **Certificates**, and choose the **Add new** button.

2 On the **General** section, fill in the mandatory fields.

- **Device type**: type of instrument for which the certificate is valid
- **Valid (months)**: used to calculate the expiry date of the user's certification
- **Certificate type**: name of certificate. For multi-site locations, prefix with site ID.
- **Role**: user-access control, not supported for all instrument types

Non-expiringAuto-recertifying

3 If you wish to assign observer rights for this certificate, select the **Is observer** check box.

4 Enter an optional comment in the **Comment** field.

5 If the instrument type support privileges, choose the **Next** button and assign the user privileges in the **Privileges** section.

6 Choose the **Next** button.

7 On the **Reminder** section, configure the notifications for the certification tasks.

8 To save the certificate, choose the **Finish** button.

Creating an expiring certificate with auto-recertification

You can save an auto-recertifying certificate when you set the number of patient and QC samples to 0 and define no other tasks. However, when you assign such a certificate to a user, an internal error occurs.

1 Choose **Organization** > **Certificates**, and choose the **Add new** button.

2 On the **General** section, select the **Auto-recertifying** check box.

Non-expiringAuto-recertifying

3 Fill in the mandatory fields.

- **Device type**: type of instrument for which the certificate is valid
- **Valid (months)**: used to calculate the expiry date of the user's certification
- **Certificate type**: name of certificate. For multi-site locations, prefix with site ID.
- **Role**: user-access control, not supported for all instrument types

4 If you wish to assign observer rights for this certificate, select the **Is observer** check box.

5 Enter an optional comment in the **Comment** field.

6 If you wish to use the optional interim stage, select the **Use interim stage** check box.

Use interim stageInterim

7 If the instrument type support privileges, choose the **Next** button and assign the user privileges in the **Privileges** section.

8 Choose the **Next** button.

9 On the **Pending** section, define the observation timeframe and certification tasks for the pending stage. Then choose the **Next** button.

10 On the **Interim** section, define observation timeframe, duration, and certification tasks for the interim stage. Then choose the **Next** button.

11 On the **Certified** section, define observation time, in-stage check (QC every), and certification tasks for the certified stage. Then choose the **Next** button.

12 On the **Reminder** section, configure the notifications for the certification tasks.

13 To save the certificate, choose the **Finish** button.

Editing a certificate

Use this option to change an existing certificate.

NOTICE

Possible change of certification status

Editing expiring certifications (e.g. prolonging the validity plus adding a criterion) may change the certification status of assigned users from expired to certified.

- Do not edit an active certificate. Instead, copy the certificate and make the changes to the copy.

> If you edit privileges of a certification that is assigned to a big number of users, an **Internal error** message may pop up.
>
> This error message merely indicates a time-out on the user interface. The process continues in the background. No action is required.

1 Choose **Organization** > **Certificates**, select the certificate to be edited, and choose the **Edit** button.

2 Make the necessary changes and choose the **Save** button.

→ If users of the application are affected by the changes, the **Affected users** screen is displayed.

3 If necessary, to confirm the **Affected users** screen, choose the **OK** button.

→ The edited certification information is displayed in the List area and stored into the database.

Copying a certificate to create new certificate

You can copy an existing certificate to create a new one.

You can use a non-expiring certificate as a template when creating a new certificate for the same instrument type.

1 Choose **Organization** > **Certifications**, select the certificate to be copied, and choose the **Copy** button.
2 In the **Certificate type** field, change the name of the copied certificate and change the other fields where necessary.
3 Choose the **Save** button.

→ The new certificate is saved.

Automated user import log

The **Automated user import log** window logs 2 types of messages:

- Error messages: report errors why the user import has failed.
- Information messages: report any event that led to a change of user information stored in the application.

Log entries are kept for 120 days.

Field definitions

Import date	Date and time when the row was imported
User identifier	Displays either User ID or network ID (if it is a user deactivation event, the user ID is displayed)
Message	Content of information or error message Green dot = information, red dot = error
Original import file entry	Line of processed record from import file

User Import

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User Import

You can import user information from an external import file. You can import the file manually, or you can do an automated import from a Human Resources Management System (HRMS).

Manual vs automated user import

Manual import is mainly used to add new users to the application. For example when users are introduced the first time during a new installation of the application. The manual import is performed in the **User import** menu where the column headers of the import file are mapped to the respective fields in the application.

Automated user import is used to synchronize the application with the customer's HRMS. Users in the application are automatically updated by an extract file from the HRMS that is periodically created. This ensures that user information in the application is always up-to-date and in synchronization with the HRMS. In particular, it ensures the deactivation of users who have left the hospital.

Manual user import

Manual user import

Use this component to import records with user profile data from an external file into the application. Using the import function saves the time it would otherwise take to enter each record individually. When re-importing individual user data, existing user information is updated.

Apart from general user information, such as user ID or name, this function enables you to import the location or locations that are to be assigned to the user and instrument certifications.

NOTICE

No upload of operator list to instruments

If users without an associated location are imported, there is no upload of new or updated users to instruments requiring operator lists. This could result in unauthorized use of instruments.

▶ Always import users with associated location for instruments requiring operator lists.

Required file format: Imported files must be csv or txt files

- If the text contains headings, assign each heading to the corresponding field; if it does not contain headings, assign the data belonging to the first record to the corresponding field. The application automatically identifies the rest of the records.
- More than one location can be assigned to each user in one record by using a specific separator for locations in the source file.

- For the possible certificate stage values see the entry in the **User information in the import file** table of the Automated User import section.
- Only one certification can be imported for each user in one record. If more than one certification is to be imported, the source file must contain one record for each certification.
- Mandatory fields must be included in the source file.
- If no value is to be added in a field, the field may be left blank, but the field delimiter must be set in the record.
- When importing a location, profile, workgroup or certification name, they must already exist in the database. Otherwise, the record will not be imported.
- It is recommended that the import files should not contain any special characters or country specific letters. Alternatively, use UTF-8 encoding to ensure special characters are imported properly.

Column header (field name)	Usage	Description
FirstName	Mandatory	First name of the user ▪ If the first name is longer than 25 characters, it will be truncated to the first 25 characters in the database.
LastName	Mandatory	Last name of the user ▪ If the last name is longer than 25 characters, it will be truncated to the first 25 characters in the database.
SecondLastName	Optional	Second last name of the user
UserID	Conditional	The User ID is mandatory if defined as user identifying value in **System** > **General settings**. Supported characters: ▪ Uppercase or lowercase letters (A to Z) ▪ Numbers (0 to 9) ▪ Special characters + / _ - $ \ @ **Note:** When imported, the user ID is not converted to uppercase. It is therefore possible to import user IDs which only differ in letter case. However, the user ID is case sensitive. Users with the ID "A123" and "a123" are considered as two different users by the application. Therefore, it is strongly recommended to use consistent casing (i.e. only uppercase or only lowercase letters) for all user IDs on the application and on all associated devices.

Column header (field name)	Usage	Description
NetworkID	Conditional	If defined as user identifying value in **System** > **General settings**, the network ID is mandatory. **Note:** The exisiting user ID is deactivated and the Network ID is cleared under the following condition: ▶ The new imported Network ID is equal to the user ID or Network ID of an existing user.
UserProfile	Optional	User profile can only be imported if already defined in the application.
Workgroup	Optional	Workgroup can only be imported if already defined in the application.
Locations	Optional	▪ You can assign multiple locations to the same user. The locations have to be separated by the location separator set for the import. ▪ Locations can only be imported if already defined in the database. ▪ If an asterisk (*) is entered in the location field, the users will be added to all system-wide active location. ▪ If a site name is entered in the location field, the users will be added to all active locations of that site.

Column header (field name)	Usage	Description
HomeLocation	Optional	Home location is imported in addition to the locations.If all available locations are already selected for **Fixed** import, the **Home location** location selection field remains empty.
Password	Optional	The password can only consist of letters (a to z, and A to Z) and numbers (0 to 9). Do not use special characters or country specific letters. **Note:** Read the instrument specific guidelines for passwords.
UserDescription	Optional	
Role	Optional	
Email	Optional	
Phone	Optional	
Mobile	Optional	
Comment	Optional	

Column header (field name)	Usage	Description
LDAP	Conditional	If no value is added to the **LDAP** field when importing a record, LDAP will be disabled by default.
Status	Optional	Supported values: ▪ 0 = deactivated ▪ 1 = activated During an import, the user status is changed as follows: ▪ Users present in the import file but not in the application are added as new users to the application. ▪ Users present in the import file as well as the application are updated according to the information provided the import file. ▪ If a user is activated in the import file but deactivated in the application, the user is activated in the application. ▪ If a user is deactivated in the import file but activated in the application, the user is deactivated in the application.
Certificate stages	Optional	Supported values: ▪ 1 = Certified stage ▪ 2 = Interim stage ▪ 3 = Pending stage
RFID tag	Optional	RFID tags are unique and can not be associated with more than one user.

User information in the import file

Importing user information manually

1 Choose **Organization** > **User import**.

2 Choose the **Browse** button and select the file to be imported.

3 Fill in the fields in the upper part, and then choose the **Analyze** button.

→ The data from the file is added into the drop-down lists in the detail area.

4 Choose the **Import** button.

To remove the prefilled information, choose the **Clear filter** button.

→ The **User Import Summary** screen is displayed, showing a report of duplicated and erroneous records found during the import process.

Automated user import

Automated user import

The application supports automated import of user information from a HRMS.

The application automatically performs a user import when a new import file from the HRMS is available.

Delta vs. full import

You can choose between 2 different import modes: *delta* and *full*. The 2 modes differ in how the application updates users who are present in the application but are not included in the import file.

- Delta mode (default): users who are not included in the import file, but are present (and active) in the application are not updated during an import, and remain active.
- Full mode: users who are not included in the import file, but are present (and active) in the application are deactivated during an import.

> Use full import with caution: it may deactivate existing users who can no longer perform measurements on instruments connected to the application.

Import file

Required format and applied processing rules for the import file are as follows:

- The import file must be UTF-8 encoded.
- The first row contains the column headers.
- The column separator is `TAB`.
- The location separator is “`,`” (comma).
- The file format is fixed.
- The import file is processed only once. Processed files are renamed.
- If multiple import files exist, the files are processed in alphabetical order.

> Do not use double-quotes to encapsulate spaces or escape the column delimiter. The application does not recognize quoted/escaped field values in the import file.

> To clear existing values for certain fields, enter a single space in the corresponding field of the import file. The field will then be set to empty (NULL) in the database. This does not apply to fields like HomeLocation, Workgroup, or Locations, as these will result in an error.

Default values

The following user information can be imported via the import file.

Column header (field name)	Usage	Description
FirstName	Mandatory	First name of the user ▪ If the first name is longer than 25 characters, it will be truncated to the first 25 characters in the database.
LastName	Mandatory	Last name of the user ▪ If the last name is longer than 25 characters, it will be truncated to the first 25 characters in the database.

Column header (field name)	Usage	Description
UserID	Conditional	The User ID is mandatory if defined as user identifying value in **System** > **General settings**. Supported characters: ▪ Uppercase or lowercase letters (A to Z) ▪ Numbers (0 to 9) ▪ Special characters + / _ - $ \ @ **Note:** When imported, the user ID is not converted to uppercase. It is therefore possible to import user IDs which only differ in letter case. However, the user ID is case sensitive. Users with the ID "A123" and "a123" are considered as two different users by the application. Therefore, it is strongly recommended to use consistent casing (i.e. only uppercase or only lowercase letters) for all user IDs on the application and on all associated devices.
NetworkID	Conditional	If defined as user identifying value in **System** > **General settings**, the network ID is mandatory. **Note:** If a new user has the Network ID of an existing user (or it is the same as an existing user's user ID), the existing user's Network ID gets cleared and the existing user gets deactivated.
UserProfile	Optional	User profile can only be imported if already defined in the application.
Workgroup	Optional	Workgroup can only be imported if already defined in the application.

Column header (field name)	Usage	Description
Locations	Optional	▪ You can assign multiple locations to the same user. The location separator must be a comma (,). ▪ Locations can only be imported if already defined in the database. ▪ If an asterisk (*) is entered in the location field, the users will be added to all system-wide active location. ▪ If a site name is entered in the location field, the users will be added to all active locations of that site.
HomeLocation	Optional	Home location is imported in addition to location.
SecondLastName	Optional	
Password	Optional	The password can only consist of letters (a to z, and A to Z) and numbers (0 to 9). Do not use special characters or country specific letters. **Note:** Read the instrument specific guidelines for passwords.
UserDescription	Optional	
Role	Optional	

Column header (field name)	Usage	Description
Email	Optional	
Phone	Optional	
Mobile	Optional	
Comment	Optional	
LDAP	Conditional	If the LDAP field is empty when importing a record, LDAP is disabled by default.

Column header (field name)	Usage	Description
Status	Optional	Supported values: ▪ 0 = deactivated ▪ 1 = activated During an import, the user status is changed as follows: ▪ Users present in the import file but not in the application are added as new users to the application. ▪ Users present in the import file as well as the application are updated according to the information provided the import file. ▪ If a user is activated in the import file but deactivated in the application, the user is activated in the application ▪ If a user is deactivated in the import file but activated in the application, the user is deactivated in the application. ▪ If a user is present and active in the application but not present in the import file, the user is deactivated (only in full update mode).
RFID tag	Optional	RFID tags are unique and can not be associated with more than one user.

User information in the import file

The default values for *UserProfile*, *Workgroup*, and *Certifications* are defined in **System** > **General settings**.

The default values are the following:

User information	Default value
Profile	Operator
Workgroup	No value (empty)
Certification 1 - Certificate Type:	No value (empty)
Certification 1 - Stage	No value (empty)
Certification 1 - Certified By	No value (empty)
Certification 2- Certificate Type:	No value (empty)
Certification 2- Stage	No value (empty)
Certification 2- Certified By	No value (empty)
Certification 3- Certificate Type:	No value (empty)
Certification 3- Stage	No value (empty)
Certification 3- Certified By	No value (empty)

To change a default value, do the following:

- Double-click a setting, and choose a value from the **Current value** drop-down list.

During automated user import, the following rules apply:

- The current system date is assigned as certification date.
- The values defined in the configuration settings are only assigned for new users, but not when existing users are updated.
- If the user profile is specified in the import file, the application imports that value and ignores the value defined in the general setting.
- If the workgroup is specified in the import file, the application imports that value and ignores the value defined in the general setting.

Logging

Each user import is logged in **Organization** > **Automated user import log**.

Alternative user identifiers

The application can identify users in the import file either by the user ID or the network ID. In **System** > **General settings**, you can define the user identifier to be used.

If the network ID is used as identifier, the import file must contain an existing network ID.

Otherwise, the user import fails, and an error is logged in the **Automated user import log** menu.

Safelist In **System** > **General settings**, you can exclude users from being updated automatically. During an automated user import, excluded users are not updated by the information provided in the import file.

User settings New users are imported with the following settings:

- Language: `English`
- Date format: `MM/dd/yyyy`
- Time format: `-x,xxx.xx`
- Number format: `HH:mm`

If a user is updated, the settings are not changed.

Importing user information automatically

Place the user import file in the following folder on the drive where the application is installed:<drive:>\cobasIT1000\Services\Hrms\Im port

“`Archives`”

Defining the user import mode

You can chose between “`full`” and “`delta`” import mode.

1 Choose **System** > **General settings**.

2 Search for the **Auto User Import: Run import in delta or full mode?** option, select it, and choose the **Edit** button.

3 From the **Current value** drop-down list, choose the import mode.

4 Choose the **Save** button.

Excluding users from automated user update

You can exclude users from being updated automatically.

1 Choose **System** > **General settings**.

2 Search for the **Auto User Import: Exclude users** option, select it, and choose the **Edit** button.

3 In the **Current value** field, enter users to be excluded by a comma-separated list of the corresponding user IDs as defined in the application.

4 Choose the **Save** button.

Defining the user identifier for automated user import

1. Choose **System** > **General settings**.
2. Search for the **Auto user import: user identifying value** option, select it, and choose the **Edit** button.
3. From the **Current value** drop-down list, choose the user identifier.
4. Choose the **Save** button.

HIS Location Import

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HIS Location Import

Use this component to import location records from an external file into the application.

Import files must be plain text format files.

If the text contains headings, assign each heading to the corresponding field; if it does not contain headings, assign the data belonging to the first record to the corresponding field. The application automatically identifies the rest of the records.

Importing location information

Import location definitions from an external file to introduce new data fast and easily.

1 Choose **Organization** > **HIS location import**.

2 Choose the **Browse** button and choose the file to be imported.

3 Fill in the fields in the upper part, and then choose the **Analyze** button.

→ The data from the file is added into the detail area.

4 Complete in the fields in the detail area with the imported data, and then choose the **Import** button.

→ The **Locations Import Summary** screen is displayed reporting duplicated and erroneous records found during the import process.

Notification Templates

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Notification Templates

Use this component to create notification templates for emails and letters, and notifications displayed on instruments.

By default the application provides the following predefined notification templates:

- Instrument user notification by email
- Instrument user notification by letter
- Supervisor notification by email
- Instrument user notification on instrument (for each instrument type)

Templates for instrument user notifications are provided by the driver installers.

- For a new notification template it is recommended to create a copy of a predefined template.
- Only active notification templates can be copied.
- Do not to change or overwrite a predefined notification template.

Notification templates are edited in a template editor. According to the selected notification type the template editor provides different settings.

Email attachments

If the **Attach list** option is enabled, the application creates a CSV file and attaches it to the email. The CSV file always uses a semi-colon (;) as column delimiter. If your local Windows setting uses a different column delimiter, you cannot double-click the CSV file and open it directly from the email. Instead, import the CSV file in Excel, which allows you to choose the column delimiter.

Field descriptions

Notification type	The notification type of the notification template. Possible values: **User E-mail**, **Supervisor E-mail**, **User Letter**, **User Instrument**
Name	The name of the notification template.
Last updated	The date of the last modification.
Status	The status of the notification template. If checked the notification template is active.
Copied from	The copied predefined notification template is displayed for manually created notification templates.
Attach list	A list of the certification tasks is send as attachment.
Send interval	Possible values: ▪ **Daily** ▪ **Weekly** ▪ **Bi-Weekly**

Tasks

From this screen, you can perform the following tasks:

- Copy a selected notification template.
- Edit the properties of an existing notification template.
- Edit the content of an existing notification template.
- Activate or deactivate a notification template.

Copying notification templates

> Only active notification templates can be copied.

1 Choose **Organization** > **Notification templates**, select the template to be copied, and choose the **Copy from** button.

2 Change the name of the new notification template.

3 Make the necessary changes and choose the **Save copy** button to save the new notification template.

> → The new notification template is now available and can be edited using the template editor.

Editing notification templates

Basic editing of notification templates, e.g., renaming, can be done in the application. Any further editing of notification templates is done in the template editor.

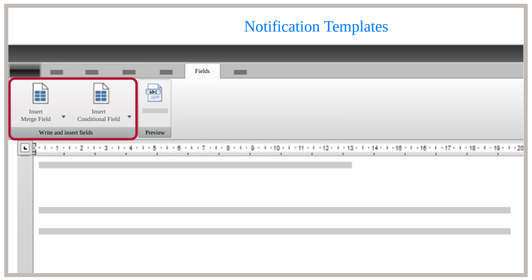
> Only active notification templates can be edited.

About the template editor

The content of notification templates is edited in the template editor.

The most important functions of the template editor are:

- Definition of subject and body of the message
- Introduction of system parameter placeholders
- Definition of conditions to show/hide lines
- Preview of the template with examples of existing "real" data from the database
- For emails intended for supervisors, definition of a table containing a list of certification tasks related with supervised user



There are predefined placeholders and conditions available in the template editor via the **Fields** menu:

- **Fields** > **Insert Merge Field** :
  Placeholders are used to fill the template with the correct information. In the actual notification message, the application replaces the placeholders with the corresponding values out of the database.
- **Fields** > **Insert Conditional Field** :

Conditions are used to show or to hide information. One condition per paragraph can be added in the template editor. In the actual notification message, the application only displays the conditional paragraphs if the corresponding condition is fulfilled.

Depending on the template type, different placeholders and conditions are available.

▶ To edit a notification template in the application

1 Choose **Organization** > **Notification templates** and choose the **Edit** button.

2 Enter the changes to the notification template, e.g., enter a new name for the notification template in the **Name** field.

3 Choose the **Save** button.

▶ To edit the content of a notification template in the template editor

1 Choose **Organization** > **Notification templates** and choose the **Download editor** button.

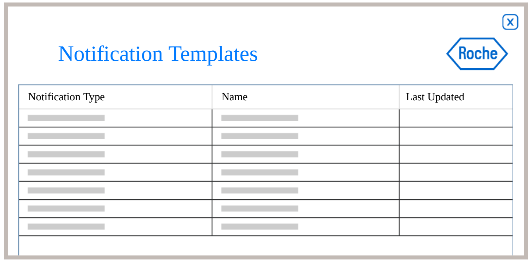
→ The *NotificationTemplateEditor.msi* file is downloaded to the download folder of your computer.

2 Copy the *NotificationTemplateEditor.msi* file to a local drive on your computer.

3 Double-click the *NotificationTemplateEditor.msi* file. If required, choose the **Run** button.

4 Enter the server address, the **navify**® POC Operations user name and password, and choose the **Login** button.

In such a case, contact your IT department helpdesk.



5 Double-click the desired notification type.

6 Modify the notification template with the functions of the template editor. According to the notification type, the template editor provides different functions and settings. In general, enter free text and predefined fields.

7 Choose **Fields** > **Preview** to get an on-screen preview of the notification template, filled with examples of existing “real” data from the database.

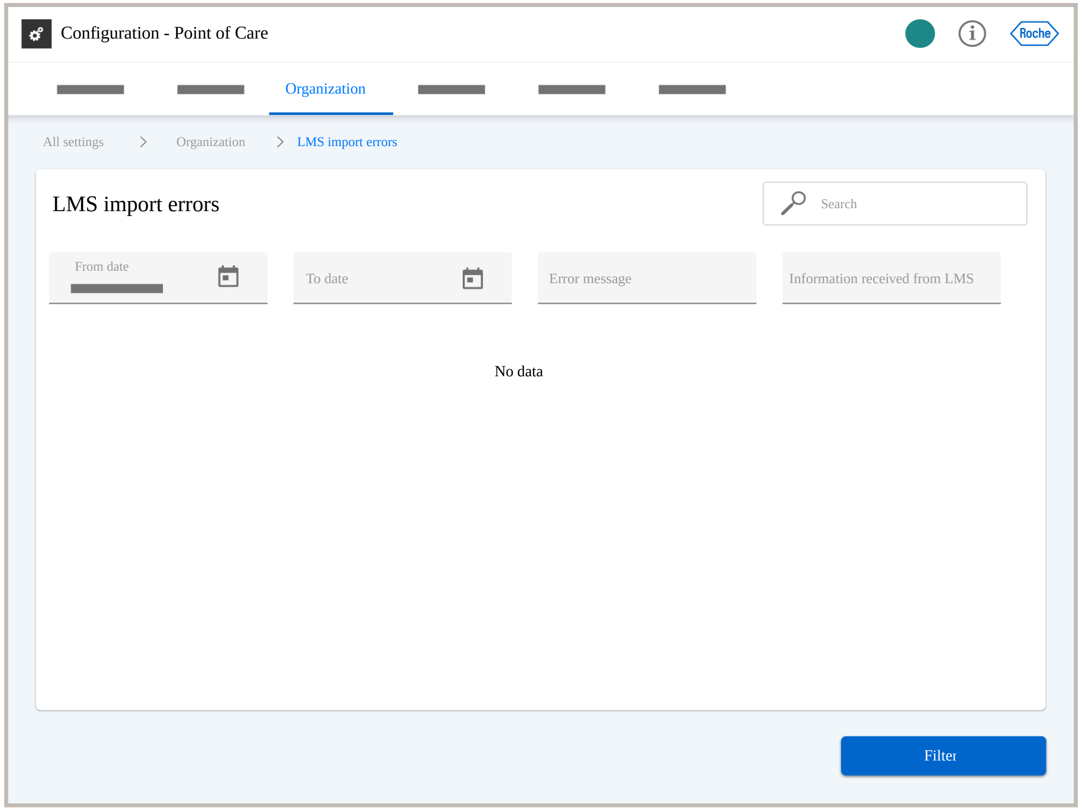
8 For email templates, choose the **Preview** button to get an off-screen preview of the notification template, filled with examples of existing “real” data from the database.

→ The application sends an email to the user who is currently editing.

9 Choose the **Save** button to save the notification template.

LMS Import Errors

Use the **LMS import errors** option to create a report of the errors that occurred during the import of LMS exam results.



Default settings

- The application shows errors from the last 7 days.
- The application stores errors for 120 days.

Error messages

The application supports the following errors messages.

Error message	Description
First Name is mandatory	Incomplete information: The LMS has not sent the user's first name.
Last Name is mandatory	Incomplete information: The LMS has not sent the user's last name.
Course Name is mandatory	Incomplete information: The LMS has not sent the course name.
Invalid exam date/timestamp format	Invalid information: The expected time stamp format does not match the used time stamp format.
Invalid exam result	Invalid information: The expected value for passed/failed exams does not match the used value.
User not unique	Multiple users with the same first/last name or LMS ID exist.
User not found	No user exists with the assigned first/last name or LMS ID exists.
Duplicate entry	The information received from the LMS has been received and successfully processed before.
Course not found	The user does not have the required certification to attend a course and pass an exam.
User certification not in observation timeframe	The user took the exam before the start date of the observation timeframe.
Exam date not within certification's validity	The user took the exam before the start date of the user's certification.

Error message	Description
Maximum number of attempts exceeded	The user failed the exams and has no attempts left.
Internal Error	An internal error occurred during processing of the LMS import.

Workgroup Definition

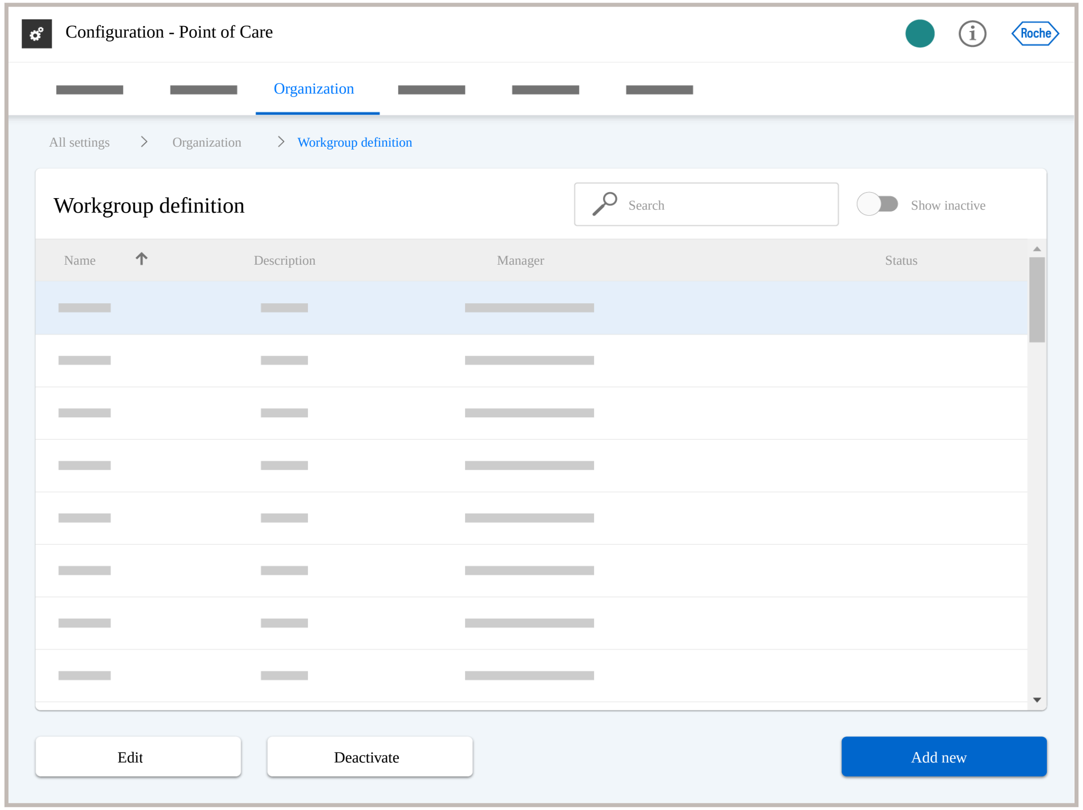
Use this component to define new workgroups, or to maintain information for existing workgroups, such as workgroup name, manager and description of the workgroup.

With the workgroup attribute, you can group individuals by a common criterion (e.g. profession or roles).

This criterion is independent of the location they are assigned to.

Prerequisites Users must have been previously setup so that a user can be defined as a manager of the workgroup. The selection of a manager is mandatory.

> When updating from an earlier version of the application, the application automatically assigns the user ROCHE as the manager of this workgroup if no manager is assigned.



Tasks

- Adding a workgroup to group together users of this application.
- Querying for workgroup information to see the details assigned to a specific workgroup.
- Editing workgroup information to change any of the features of a specific workgroup
- Activating a workgroup so that it can be used in other screens within the application.
- Deactivating a workgroup so that it cannot be used from other screens.

Devices

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Configuration

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Configuration

Use this component to create new instrument configurations, or to maintain information for existing instrument configurations.

Instrument profiles

Instrument profiles represent a group of settings, which control the behavior of a certain instrument type (e.g. such as **Accu-Chek®** Inform II).

Such typical instrument configurations are defined based on the requirements of the location where individual instrument is to be used.

A predefined configuration has to be assigned during the setup of each individual instrument.

By changing a single instrument configuration, the behavior of all instruments assigned to this configuration can be modified.

NOTICE

Be careful when changing instrument configuration

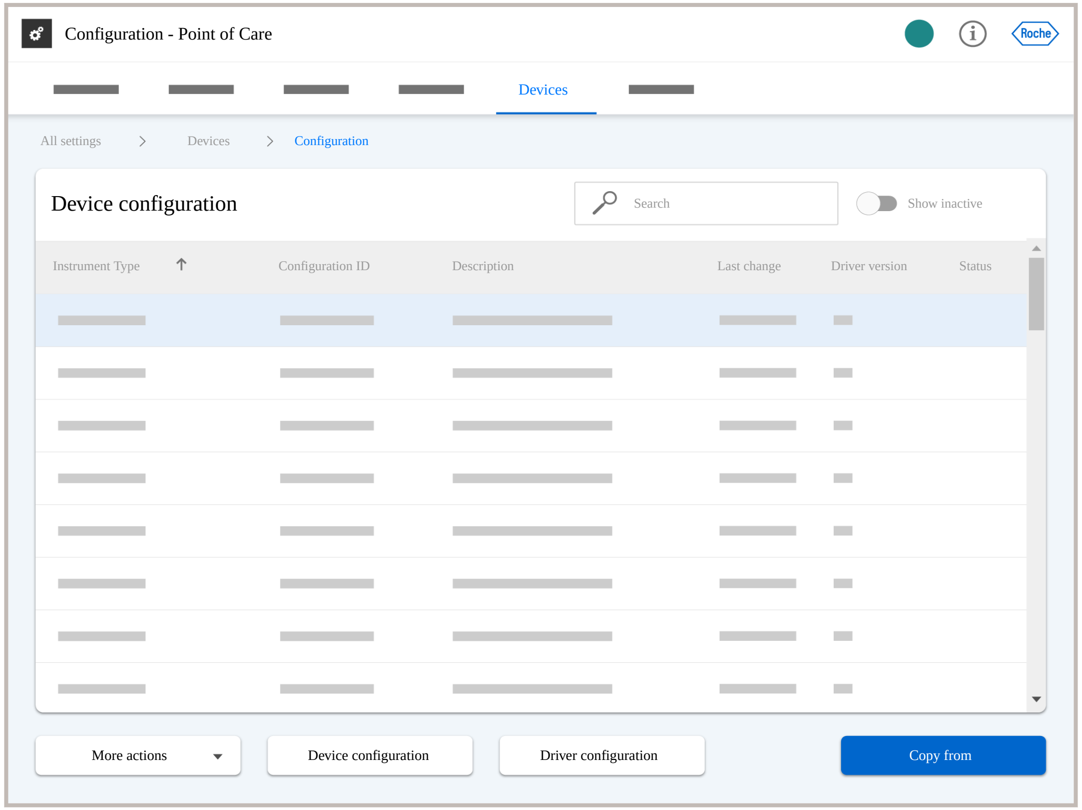
Changing instrument configuration can result in loss of data and malfunctioning of the application.

- Create a copy of the predefined configuration and modify the copy, if necessary. For instrument configuration changes, contact Roche Service.

> Activate the option in the **Instrument** tab before deletion in order to avoid test result loss in the event of communication problems between the instrument and the application.

> Configure instruments to enter patient IDs manually in the **Patient** tab, in case instrument connectivity failure happens. To do so, select **Keyboard Prompt** on the list.

The **Last change** and **Changed by** fields are automatically filled in by the system. Depending on the instrument type and the driver version, a different set of fields will be activated in the Detail area.



Field definitions

Instrument Type	Type of instrument to which the configuration is applied.
Configuration ID	ID given to the configuration.
Driver version	Driver version used in the configuration.
Description	Short text describing the configuration.
Last change	Last time when changes were done on the configuration.
Changed by	Define the user which is displayed by default in instruments that do not support the sending of user ID. Users that appear in the list are only those in the same site as the analyzer.
Default configuration	Select this box if the configuration shall be considered the default configuration for the instruments of that same instrument type.
Firmware file path	Write the path and the file name of a *firmware* update file. The path entered must be an absolute path on the computer running the PCommunicator component. Back slashes have to be used ("\\"), not forward slashes ("/"). File name must be included. This field is only available for instruments for which firmware updates are supported, e.g. **CoaguChek**® XS Plus and **cobas**® h 232.
Requires orders (Optional)	Select this check box if for results from devices assigned to this configuration the use of orders is required. For device configurations that require orders, the application sends a result to the LIS only when it can be matched to an order. If the application cannot match an order automatically, the result is not sent and kept in the **Result and order matcher** application.

Repeat time interval	Define for each instrument type configuration the amount of minutes when, if a repeated result is sent, it will be marked as repeated. Repeated results are flagged with the system alarm **TRR**and set to status **Pending**. The setting **Detect repeat results** has to be activated in the **General settings**.
User	Name of the user that is in charge of the instrument.
POCT	Select this box if the instruments you are configuring are to be operated in Point of Care testing (POCT) mode. If you select this, the QC interval and RMSE evaluation fields appear. This field is only available in Rili-BÄK mode for instruments that can be operated in POCT mode.
QC interval (h)	The maximum number of hours between QC tests. This field is only available in Rili-BÄK mode. By default, the QC interval is set to 168 hours. A maximum of 999 hours can be set.
RMSE evaluation	Select this check box if instruments configured this way will also be evaluated using RMSE calculations. This field is only available in Rili-BÄK mode.

Start of daily routine	This field appears when the POCT check box is not selected or the associated instrument type cannot be operated in POCT mode. You can set the time at which instruments theoretically start their daily routine. This time may for example be 7am or 1am depending on your ward's shift schedule. It is used as a reference for when the first QC test run is expected to be performed for the day. This field is only available in Rili-BÄK mode. **Note:** To allow for reception and evaluation of the QC results in the application, configure the start of the daily routine at least 5 minutes before the time the QC tests are done on the instruments.

Tasks

- Creating an instrument configuration to determine the setup features for a certain instrument type. Several configurations are allowed for one instrument type.
- Querying for instrument configurations to see the setup details of the selected configuration.
- Activating an instrument configuration so that this configuration can be used with the suitable devices for data transmission.
- Deactivating an instrument configuration so that this configuration cannot be assigned to any instruments.
- Modifying an instrument configuration to change the data transmission setup features.
- Configuring tasks to assign or un-assign predefined maintenance actions to the selected instrument profile.
- If operating in Rili-BÄK quality control mode, you can define whether an instrument is operated in POCT mode. If you select the **POCT** check box, you see different fields compared to if you do not select the **POCT** check box.

Copying an instrument configuration from an existing one

Copy an existing instrument configuration to create a new one with similar features.

1 Choose **Devices** > **Configuration** and select the configuration you want to copy.

2 Choose the **Copy from** button.

3 Change the configuration ID.

4 Choose the **Save** button.

→ The updated instrument configuration information is stored in the database.

5 Select the copied configuration and choose the **Edit** button.

6 Make the necessary changes and choose the **Save** button.

Driver configuration

Devices > **Configuration** > **Driver configuration**

Use this option to configure specific settings for the software driver controlling the data exchange between the selected diagnostic instrument type and the software application.

The **Driver configuration** button is not available for ICA device types.

Modifications made in this screen apply to the record selected on the previous screen (**Devices** > **Configuration**).

Prior to deactivating a test it should be removed from materials defined in **System** > **Material definition** as well as (depending on the instrument type) in **Devices** > **Configuration** > **Driver configuration**.

CAUTION

Do not change driver configuration yourself

Changing driver configuration can result in loss of data and malfunctioning of the application.

▶ For driver configuration changes, contact Roche Service.

Actions

- Choose the **More actions** > **Edit** button to handle the basic conditions of instrument configuration.
- Choose the **Alphanumeric results** button to manage alphanumeric result modification.
- Choose the **Numeric results** button to manage numeric result modification.
- Choose the **Range based results** button to manage range result modification.

Device Configuration

Use this option to configure specific settings of instruments, for example the **Accu-Chek**® Inform II, **CoaguChek**® XS Plus, **CoaguChek**® Pro II or **cobas**® h 232.

Modifications made on this screen apply to the record selected on the previous screen (**Devices** > **Configuration**).

The sending of patient and user IDs to the instrument depend also on proper settings of parameters in the **System** > **General settings** screen.

> There is the option to select the **Visit ID** as the source of IDs to be used in the patient upload to the instruments.
>
> - For instrument types that support **Device configuration**, it can be configured in the **Patient ID Mapping** section.
> - For other instrument types, the general setting **Instrument Patient ID** will apply.

⚠ CAUTION

Incorrect assignment of patient results

Changing the patient ID mapping for an instrument configuration can lead to incorrect assignment of patient results. When changing the patient ID mapping, always do the following:

- Before changing the patient ID mapping, connect the affected instruments with the application to synchronize pending results with the current patient ID mapping.
- Change the patient ID mapping.
- Re-dock the affected instruments to implement the new patient ID mapping.
- Ensure correctness of patient result assignment according to the new patient ID mapping.
- Make a patient measurement with an instrument.
- Dock the instrument and synchronize the result.
- Check whether the patient result is correctly assigned.

> For the **CoaguChek**® XS Plus or **cobas**® h 232 POC system, the terminology used in this manual is British English. The terminology may change when you configure the systems to American English.

Tasks

- Modifying an instrument configuration to meet the requirements of the POCC.

- Restoring an instrument configuration to its default configuration by choosing the **Reset factory settings** button.

> If a user sets his password on an **Accu-Chek**® Inform II instrument, the user record is updated in the application accordingly.

Accu-Chek Inform II

EAP security

You can enable this instrument to use EAP security when communicating with the application via WLAN.

> For the **Accu-Chek**® Inform II to support EAP security, the instrument needs to have software version 03.05.00 or higher installed.

To enable EAP security, do the following:

1. Choose **Instrument** > **WLAN Communication** and select the **WLAN Communication** check box.
2. Choose **WLAN** > **Security** and choose the **EAP** entry from the **Security Type** drop-down list.

The settings in the **Device configuration** screen enable EAP security on the instrument.

For an instrument to effectively use EAP security for WLAN communication with the application, the following needs to be done additionally:

- Define an EAP setting in the **Devices** > **EAP settings** screen.
- Assign the instrument to the EAP setting in the **Devices** > **EAP settings assignment** screen.

> Only active instruments assigned to a configuration with EAP security enabled are displayed in the **Devices** > **EAP settings assignment** screen and can be assigned to an EAP setting.

Bayer 1200

Disable Parameters for analysis

You can prevent this instrument from performing specific analysis by disabling them here. Select the check box to disable a parameter. Clear it to enable it.

pH/H+

PCO_2

PO_2

Na+

K+

Ca++

Cl-

Glucose

tHb

Lactate

Patient

Patient ID mapping. Select the patient ID mapping from the drop down list.

Bayer 400

Disable Parameters for analysis

You can prevent this instrument from performing specific analysis by disabling them here. Select the check box to disable a parameter. Clear it to enable it.

pH/H+

PCO_2

PO_2

Na+

K+

Ca++

Cl-

Glucose

tHb

HCT

Patient

Patient ID mapping. Select the patient ID mapping from the drop down list.

ABX Micros CRP

Patient Patient ID mapping. Select the patient ID mapping from the drop down list.

Sysmex KX-21N

Patient Patient ID mapping. Select the patient ID mapping from the drop down list.

QC lot mapping QC lot number matching the instrument QC file number.

Offline devices

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Offline devices

An offline instrument does not have an online connection to the application, and QC test results are entered manually. Use this component to add new offline instruments, configure them and assign tests.

- The offline instrument license must be correctly installed (to be acquired separately).
- All offline device types and their configurations (especially test assignment) are managed in this screen.

Offline instrument type IDs (5 characters) must start with “X_”. Individual offline instruments are always regarded as POCT instruments in Rili-BÄK mode.

Field definitions

Instrument types

Instrument Type ID	ID code for the offline instrument type that is being created or edited.
Instrument Type Name	Name of the currently active offline instrument type.

Configurations

Configuration ID	Name of the configuration for the selected offline instrument.
Configuration description	Description of the selected offline instrument configuration.
Last changed	Date when the selected configuration was last changed (automatically filled in by the system.)
Status	The status of the instrument.

Tasks

- Add a new Instrument Type by choosing the **Add new** button on the right.
- Add a new Instrument Type Configuration by selecting an instrument type, choosing the **Configuration** button, and then choosing the **Add new** button on the right.
- Edit the individual instrument configuration by choosing the **Edit** button.
- Configure the tests associated with the selected Instrument Type Configuration.

Creating a new configuration from an existing one

1 Choose **Devices** > **Offline devices**.

→ The **Offline devices** screen is displayed.

2 Select an instrument type and choose the **Configuration** button.

→ The configurations defined for the selected instrument type are displayed.

3 Select the configuration to be copied and choose the **Copy from** button.

4 Change the **Configuration ID** field and **Configuration description** field for the new configuration, and choose the **Save copy** button.

→ The new configuration is saved with the same details as the original one. The tests associated to the newly-created configuration can be changed.

EAP settings

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EAP settings

An EAP setting defines the security used by instruments assigned to this setting when communicating with the application via WLAN.

Basic terms and concepts

EAP

EAP (Extensible Authentication Protocol) is an authentication framework. It enables the communication between server and client in wireless networks for authentication purposes.

RADIUS server

The authentication process is handled by a dedicated RADIUS server (Remote Authentication Dial-In User Service). The use of a RADIUS server allows centralized management of user information and authentication processes.

EAP types

The EAP types are the specific methods used for authentication. They differ in what is demanded for authentication, i.e. authentication via certificates and/or passwords, and from which side it is demanded, i.e. from the server and/or the client. The following EAP types are supported by the application:

- PEAP (Protected Extensible Authentication Protocol):This EAP type uses the TLS protocol only to authenticate the server to the client, so only the server has to have a certificate. The client only needs a user name and password to validate to the server.
- TLS (Transport Layer Security):TLS is based on the authentication of both sides, so each client has to have its own certificate as well. Also the client private key and the client private key password have to be set when using TLS.
- TTLS (Tunneled Transport Layer Security):TTLS is very similar to the PEAP type. It will typically be used instead of PEAP if the RADIUS server should not be directly accessible through the WLAN. Therefore, an EAP-TTLS server will be used as a proxy, forwarding the messages in a non-wireless bounded way to the RADIUS server itself. If the first step of authentication to the EAP-TTLS server should be separated to the final user name/password to the RADIUS server, a general anonymous user name can be specified. So the user name/password will only be used to authenticate against the RADIUS server when a secure communication has already been established.

Digital certificate

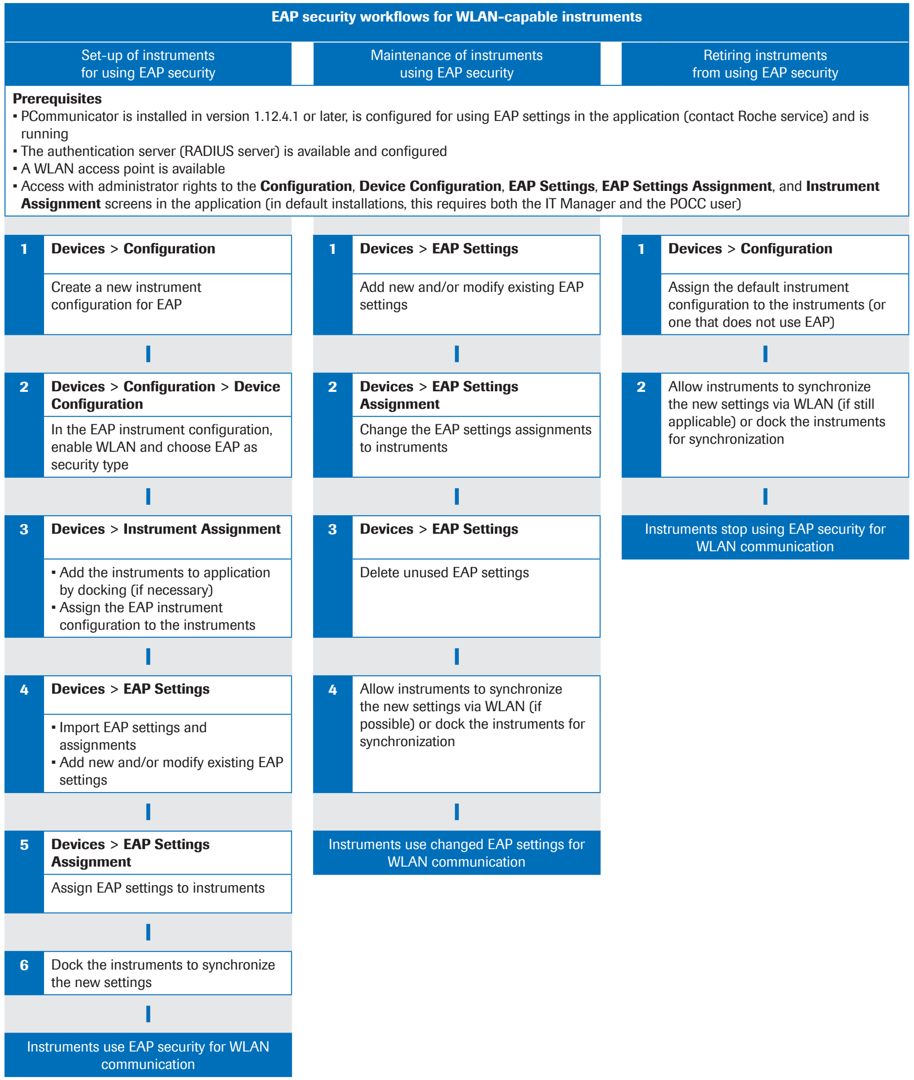
A digital certificate is an electronic document proving ownership of a public key, which is used during information encryption. Digital certificates are issued by a certification authority (CA). The application supports X.509 certificates in a Base64 .pem format.

Cipher type

The cipher type is the algorithm used for encrypting and decrypting. During the encryption procedure a key is applied to information converting it from plain text into cipher text. The following cipher types are supported by the application:

- CCMP (WPA2 AES):Counter Cipher Mode with Block Chaining Message Authentication Code ProtocolCCMP is the standard encryption protocol for use with the current WPA2 standard specifying security mechanisms for wireless networks.
- TKIP (WPA TKIP):Temporal Key Integrity ProtocolTKIP is a security protocol used with the older WPA standard allowing to keep hardware not suitable for use with CCMP.
- CCMP TKIP (allow both):Enables both CCMP and TKIP.

Overview of common EAP workflows



EAP settings screen functionalities

EAP settings screen functionalities

Use the **Devices** > **EAP settings** screen to set up, maintain and delete EAP settings.

> You must have administrator rights to make changes in this screen.

Field definitions for all EAP types

Name	Name of the EAP setting. Name is mandatory and must be unique. It is used in the **Devices** > **EAP settings assignment** screen to identify the EAP setting.
SSID	SSID (service set identifier) to be used for the communication, i.e. the name of the wireless network, via which PCommunicator can be connected to
Cipher type	Possible cipher types: ▪ CCMP (WPA2 AES) ▪ TKIP (WPA TKIP) ▪ CCMP TKIP (allow both)

EAP type	The EAP type to be used for the communication: ▪ PEAP (Protected Extensible Authentication Protocol) ▪ TLS (Transport Layer Security) ▪ TTLS (Tunneled Transport Layer Security)
User name	User name for the RADIUS server
CA certificate	Browse for the digital certificate file of the certification authority (CA) used to authenticate the server to the client Supported are X.509 certificates in a Base64 .pem format

Depending on the chosen EAP type (PEAP, TLS or TTLS), different additional fields are enabled in the Details area and have to be filled with the corresponding information.

Field definitions for PEAP

Password	Password to authenticate the user name on the RADIUS server
Password confirmation	Type in the same Password again

Field definitions for TLS

Client certificate	Browse for the digital certificate file used to authenticate the client to the server Supported are X.509 certificates in a Base64 .pem format.
Client private key	The digital certificate file containing the private key of the client Supported are X.509 certificates in a Base64 .pem format
Client private key password	Password needed to access the content of the client private key file
Client private key password confirmation	Type in the same Password again

Field definitions for TTLS

Password	Password to authenticate the user name on the RADIUS server
Password confirmation	Type in the same Password again
Anonymous user	Anonymous user name used by EAP-TTLS as outer identity to rout the user's logon request to the RADIUS server before authenticating with the inner identity (the user name and password) through an encrypted connection. The anonymous user shall provide the domain name to rout the request to the RADIUS server (e.g. “`anonymous@domain.com`” or “`domain\anonymous`”).

Tasks

- Adding a new EAP setting
- Searching for an EAP setting
- Modifying an EAP setting
- Deleting an EAP setting

Adding an EAP setting

Add a new EAP setting.

1 Choose **Devices** > **EAP settings**.

2 Choose the **Add new** button.

3 Fill in the required fields.

4 Choose the **Save** button.

→ The new EAP setting is stored in the database.

Searching for an EAP setting

Search for an existing EAP setting.

1 Choose **Devices** > **EAP settings**.

2 Enter the search criteria in the search field.

Modifying an EAP setting

Modify an existing EAP setting.

NOTICE

Instrument connection to the application

When modifying an existing EAP setting, all instruments assigned to this setting will use the modified setting after the next synchronization with the application.

▶ Edit and validate the modified EAP setting before synchronization.

1 Choose **Devices** > **EAP settings**.

2 Select an EAP setting from the list and choose the **Edit** button.

3 Edit the details of the EAP setting.

4 Choose the **Save** button.

→ The modified EAP setting is stored in the database. The modified setting is transferred to the assigned instruments with the next synchronization.

Deleting an EAP setting

Delete an existing EAP setting.

NOTICE

Instrument connection to the application

When deleting an existing EAP setting, all instruments previously assigned to this setting are unassigned in the application and marked for required synchronization.

- Check that no instruments still need the EAP setting before deleting it and reassign them if needed.

1 Choose **Devices** > **EAP settings**.

2 Choose one or multiple EAP settings from the list.

3 Choose the **Delete** button.

→ A confirmation message is displayed. Choose the **Confirm** button.

If any of the EAP settings to be deleted is still assigned to instruments, that are configured to use EAP security, a warning message is displayed as those instruments will be unassigned once the EAP settings are deleted and will require a new one. Choose the **Confirm** button to confirm the deletion, or the **Cancel** button to be able to review the assigned instruments before deleting.

The database is updated. Instruments previously assigned to a deleted EAP setting are unassigned and marked as not applied in the **EAP settings assignment** screen.

EAP settings import

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EAP settings import

Use this component to import EAP settings from an external file into the application.

You must have administrator rights to make changes in this screen.

About EAP settings import

The following rules apply to the EAP settings import:

- Import files must be PRN, TXT, or CSV files.
- If the text contains headings, map each heading to the corresponding field; if it does not contain headings, map the data belonging to the first record to the corresponding field. The application automatically identifies the rest of the records.
- Mandatory fields must be included in the source file or must be provided with a fixed value.
- If a fixed prefix is provided for the EAP names, the application automatically generates unique names for the imported EAP settings using the prefix followed by consecutive numbers. E.g., prefix “EAP” generates names like “EAP001”, “EAP002”, ... (the number of digits depends on the number of different EAP settings imported).
- Importing EAP settings overwrites existing settings if the EAP name, that is provided in the import file, matches one in the database. Automatically generated names do not overwrite existing settings. The generated numbers will continue an existing sequence.
- If the same EAP setting shall be assigned to multiple devices during import, the import file has to contain one row for each device serial number. Such duplicate EAP settings in the import file are imported as a single EAP setting definition and assigned to multiple devices.
- If no value is to be added in a field, the field may be left blank, but the field delimiter must be set in the row of the import file.

- It is recommended that the import files should not contain any special characters or country specific letters. Alternatively, use UTF-8 encoding to ensure special characters are imported properly.
- The columns CA Certificate, Client Certificate and Client Private Key can contain a file name (and possibly the path).

Importing EAP settings

You can import EAP settings from an external file into the application and assign these settings to devices that are identified by serial number.

Using the import function saves the time it would otherwise take to enter each data set individually. When re-importing individual data, existing information is updated in the database.

EAP settings screen functionalities

Devices must have been previously added to the application by docking them.

1 Choose **Devices** > **EAP settings import**.
2 Choose the **Browse** button and choose the file to be imported.
3 Fill in the fields in the upper part. Choose the **Analyze** button.

   → The data from the file is added into the detail area.

4 Complete in the fields in the detail area with the imported data, and then choose the **Import** button.
5 If the import includes digital certificates, the **Upload the following files.** dialog box lists the required certificate files. Do the following:
   - Choose the **Browse** button and choose the certificate files to be imported.
   - Choose the **Confirm** button.

6 In the **Warning** dialog box, choose the **Confirm** button.

→ The EAP settings are imported and listed on the **EAP settings** screen.

EAP settings assignment

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EAP settings assignment

Use this component to assign EAP settings to instruments. Instruments configured to use EAP security in WLAN communication with the application need to have an EAP setting assigned, that will be transferred to the instrument at the next synchronization.

- The instrument needs to be docked on a connected base station to be able to synchronize for the first time.
- WLAN communication using EAP security is not available for all instruments.

Prerequisites

- Instruments must have been previously added to the application by docking them.
- An instrument configuration enabling EAP must have been previously configured (**Configuration** > **Devices** > **Configuration** > **Device configuration**).
- Instruments must have been previously assigned to the instrument configuration enabling EAP security (**Devices** > **Devices**).
- An EAP setting must have been previously defined (**Configuration** > **Devices** > **EAP**

settings or Configuration > Devices > EAP settings import).

You must have administrator rights to make changes in this screen.

Field definitions

Selected	Check box to choose instruments for assigning EAP setting
Serial number	Serial number of the instrument
Location	Location the instrument is assigned to
Site	Site the instrument is assigned to
EAP setting (column)	Name of the previously assigned EAP setting
Applied	Indicates if the instrument has synchronized and acknowledged it has got the latest changes in its EAP setting
EAP setting (field)	Name of the EAP setting to be assigned newly

Task

Assigning an EAP setting to instruments

It is not possible to “unassign” an instrument from its EAP setting in the **EAP settings assignment** screen. In order to stop an instrument from using EAP security (retiring an instrument), the instrument has to be assigned to a different, previously defined instrument configuration that does not use EAP security (**Devices** > **Instrument Assignment**).

Assigning an EAP setting to instruments

Assign an EAP setting to selected instruments for the first time or change the EAP setting assigned to certain instruments.

> Only active instruments assigned to a configuration with EAP security enabled are displayed in the **EAP settings assignment** screen.

1 Choose **Devices** > **EAP settings assignment**.

2 To find the respective instruments, use the search function.

3 Select the instruments to be assigned. Do one of the following:

- Use the **Select all** and **Deselect All** buttons.The **Select all** and **Deselect All** buttons will (de-)select all instruments listed on the **EAP settings assignment** screen. Not only the ones visible in the current page of the screen.
- Select one or multiple entries by selecting the corresponding check boxes in the **Selected** column.

4 From the **EAP setting** drop-down list, select the EAP setting to be assigned to the selected instruments.

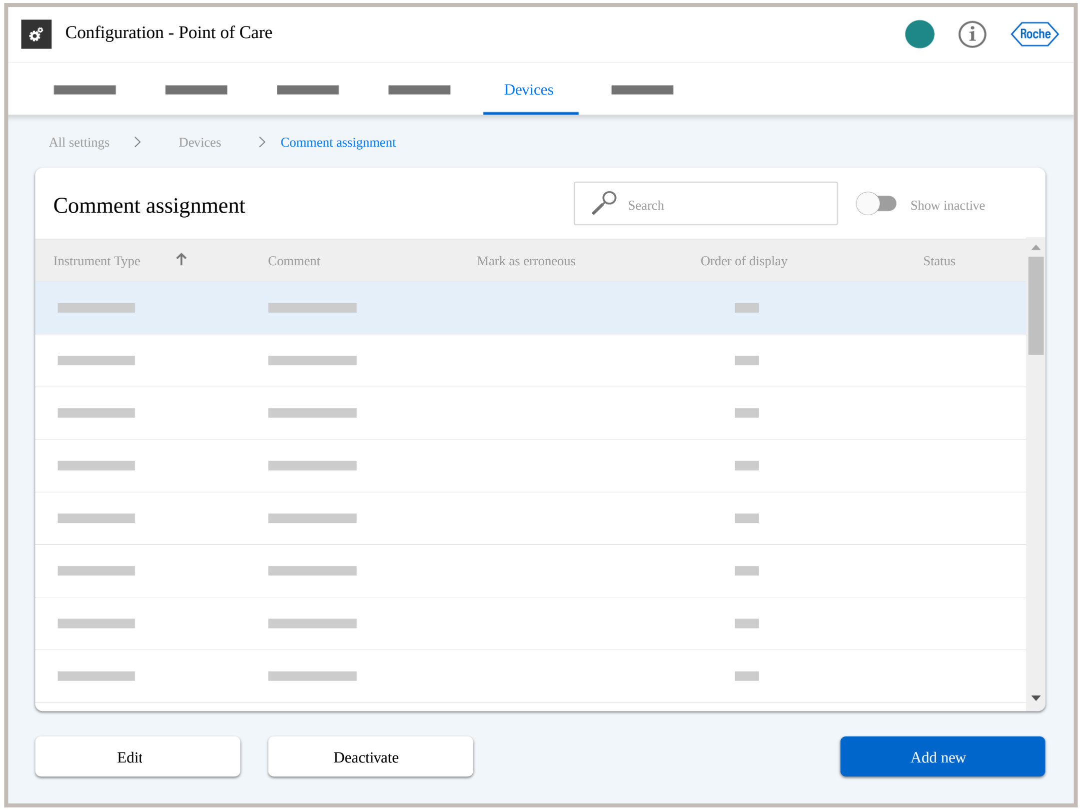
5 Choose the **OK** button.

> → If the instruments are already assigned to a different EAP setting, a confirmation message is displayed. Choose the **OK** button.
>
> The EAP setting assignment is stored in the database. The assignment is transferred to the instruments with the next synchronization. After successful synchronization, the instruments are marked as applied in the **Applied** column in the **EAP settings assignment** screen.

Comment Assignment

Enables you to create new instrument comments, or to maintain information for existing instrument comments (predefined comments). The setup of instruments within the application is used to align the selection of comments for different instrument types. If supported by the instrument type, comments will be made available on individual instruments to provide standardized comments with QC or patient test results reporting.

- For each comment it can be defined, if it is an “error” comment by selecting the **Mark as erroneous** check box. If the comment is marked as erroneous, the system displays the **Marked as erroneous by operator (comment)** (**OCE**) alarm. The alarm must be mapped for this specific instrument type for the system to display it.
- The allowed comment length for raising the **Marked as erroneous by operator (comment)** alarm varies depending on the device model, firmware version, and memory constraints. The comment length must be configured in accordance with the capabilities of the device. The default length is 20 characters unless otherwise specified in the connector user manual.
- More than one comment can be defined for each instrument type and result type combination.
- Comments must be unique.
- The **Order of display** is specific for each instrument type. When changing it for one comment, all other comments will be automatically rearranged.
- The predefined comments from **System** > **Comment definition** screen come already mapped and with predefined display sort order for the **Accu-Chek® Inform**.



Prerequisites

- Instrument types and comments must have been previously defined.
- Comments can only be set up for instruments supporting the use of comments.

Field definitions

Instrument Type	Type of instrument to which the comment applies.

Comment	Comment that will appear on screen related to the instruments.
Order of display	Order in which the comments should appear on screen.
Mark as erroneous	Indicates that the comment is of type error.
Result type	Type of result to which the comment is applied.
Assigned Location(s)	The locations in which the selected comment will be used.

Tasks

From the **Devices** > **Comment assignment** screen, you can perform the following tasks:

- Create a new instrument comment that can be later used within the instrument module.
- Query for instrument comments to see the details of the selected comment.
- Activate an instrument comment so that it is available.
- Deactivate an instrument comment so that it is not available.
- Modify an instrument comment to change any of the settings of an existing comment.
- Assign location(s) in which the selected comment will be used.

Device import

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Adding multiple devices to navify® POC Operations

1. Choose **Configuration** > **Devices** > **Device import**.
2. Choose the **Browse** button and select the file to be imported.

   If a device can not be imported as active due to license restrictions, it will be added as inactive.

   → In the **Import summary** overview, a message is displayed to inform you whether the action was successful or to indicate any errors.
3. To be able to import the file, it is mandatory to provide the following fields:
   - **Device name**
   - **Serial number**
   - **Status**
   - **Type**
   - **Location**
4. Optionally, you can also provide the following fields:
   - **Configuration**
   - **Host name**/**IP address**

5 If you want to replace a device, ensure that it is first deactivated.

Quality Control

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QC Test Mapping

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QC Test Mapping

This screen is used to map individual tests with generic tests. Generic tests are used in instrument types that can run several types of tests (such as blood gas instruments) but use one QC test only. By mapping these tests to an already defined generic test of an instrument type, the QC result is valid for all – the generic and the mapped tests.

One test can only be assigned to an instrument type once, but it can be assigned to different instrument types more than once.

Prerequisites

- Instruments must be previously defined in the database.
- Tests must be previously defined in the database. Generic tests must be created in order to map instrument tests to them.
- A test material must be previously created for the generic test. A QC material must be previously associated with the test material and the generic test.
- One test can only be assigned to an Instrument type once, but it can be assigned to different Instrument types more than once.

Querying for an existing mapping configuration

You check existing mapping definitions to see the tests assigned to a specific generic test.

1. Choose **Quality control** > **QC test mapping**.

   → All instrument types are displayed on the tree structure on the left side.

2. Expand the instrument type and select the generic test to be checked.

   → All the tests associated to this generic test are displayed in the list area on the right.

Adding a generic test

1 Choose **Quality control** > **QC test mapping**.

> → All instrument types are displayed on the tree structure on the left side.

2 Choose the **Add** button.

3 Choose the instrument type from the list and choose the magnifying glass to access the **Test selection** screen.

4 In the **Test selection** screen, select the generic test and choose the **Confirm** button.

> → The **Test selection** screen is closed and the system goes back to the **QC master test** screen.

5 Choose the **Save** button.

Mapping tests to existing generic tests

It enables you to add a new test to an already existing generic test.

1 Choose **Quality control** > **QC test mapping**.

> → All instrument types are displayed on the tree structure on the left side.

2 Expand the instrument type branch where you find the generic test, select the generic test to which a test is to be added, and then choose the **Add tests** button.

> → The **Test selection** screen is displayed.

3 Select the test or tests to be added from the list, and choose the **Add tests** button.

Deleting a generic test

It enables you to delete a generic test associated with an instrument type and all the associated test mappings. This does not erase the test definition from the system.

1 Choose **Quality control** > **QC test mapping**.

→ All instrument types are displayed on the tree structure on the left side.

2 Expand the instrument type branch where you find the generic test, select the generic test to be deleted, and then choose the **Delete** button.

3 On the confirmation message, choose the **Confirm** button.

→ The test is removed from the tree structure and is no longer a generic test for the current instrument type.

Deleting a test mapping

It enables you to delete a test associated to a generic test if it is no longer to be mapped with that generic test. This does not delete the test definition from the system

1 Choose **Quality control** > **QC test mapping**.

→ All instrument types are displayed on the tree structure on the left side.

2 Expand the instrument type branch where you find the generic test, select the generic test from which a test mapping will be removed, select the test to be removed and then choose the **Delete test** button.

3 On the confirmation message, choose the **Confirm** button.

→ The test mapping is deleted.

Multirules

Multirules are used to considerably extend the significance of QC test results. By combining QC test results, multirules allow a sensitive detection of multiple kinds of instrument QC error trends and variations. A wide range of common QC rules, such as 1-2s, 2-2s, R-4s, 10x and others are supported by the application, and can be combined with multirules at an instrument and parameter level. Use this component to assign new multirules, or to change existing multirule definitions.

Use this component to assign new multirules, or to change existing multirules definitions.

- If a rule is not to be executed, leave it inactive.
- Rules must be unique.
- Choose the field once to activate a warning and the warning icon will appear.
- Choose the field twice to activate an error and the error icon will appear.

You must have administrator rights to make changes in this screen.

Icons in this screen

⚠	Active warning. An MRW system alarm is raised if the rule is violated. Depending on the mapping of the MRW system alarm in the **System** > **System alarm assignment** screen, the QC test result is flagged or blocked.
⚠	Active error. An MRE system alarm is raised if the rule is violated. Depending on the mapping of the MRE system alarm in the **System** > **System alarm assignment** screen, the QC test result is flagged or blocked.

Alarms with patient results:

- A flagged QC result, i.e. a QC result with associated warnings, will raise an QCW system alarm with related patient results.
- A blocked QC result, i.e. a QC result with associated errors, will raise an QCE system alarm with related patient results.
- Depending on the mapping of the QCW and QCE system alarms in the **System** > **System alarm assignment** screen, the patient results will be flagged or blocked.

Selection of **Apply rules to** option determines which fields are enabled:

Apply Rules to option	Instrument Type	Instrument Name	Test
Selected instrument test	Yes	Yes	Yes

Apply Rules to option	Instrument Type	Instrument Name	Test
All tests of selected instrument	Yes	Yes	No
All instruments of selected type	Yes	No	No
Selected test	No	No	Yes

Fields are enabled according to the option in **Apply Rules to**

Prioritization of rules

More specific rules have priority over less specific ones, meaning if more than one rule can be applied to one single result, only the most specific rule is applied. Priority is set as follows:

- **Instrument Type**, **Instrument Name** and **Test**
- **Instrument Type** and **Instrument Name**
- **Instrument Type**
- **Test**

Quality control > **Multi rules**

Tasks

- Adding a new instrument type rule, instrument rule or instrument test rule so that they can be used with the different instrument types connected to the system.
- Querying for rule definitions to see the details assigned to one specific rule.
- Modifying an existing rule definition in the database.
- Deleting a rule definition stored in the database that shall no longer be used.

Field definitions

1x2s	The rule is violated if one control measurement exceeds $\bar{\chi} \pm 2s$. This rule is not used as a Pass/Fail rule. It is used to alert the operator to review current and subsequent QC material measurements to detect failure of other rules in force.
1xKs	The rule is violated if one control measurement exceeds $\bar{\chi} \pm Ks$, where K is a multiplier of the SD. This rule is often implemented where K = 3.
K (1xKs)	A factor K has to be defined which indicates the rule's violation limit in units of *standard deviation* (SDI). This value will be used with the 1xKs rule for evaluation purposes. A value from 2 to 4 in increments of 0.01 can be entered (e.g. 2.88).

RxKs (Within Control)	The rule is violated if the difference between high and low control measurements within R results in the past exceeds Ks. The value of K is usually set to 4. Also, a factor K has to be defined which indicates the rule's violation limit in units of standard deviation (SDI). This value will be used with the RxKs rule for evaluation purposes. A value from 2 to 6 in increments of 0.01 can be entered (e.g., 5.88).
R (RxKs)	A factor R has to be defined which indicates the rule's violation limit in units of standard deviation (SDI). This value will be used with the RxKs rule for evaluation purposes. A value from 2 to 6 in increments of 0.01 can be entered (e.g., 5.88).
K (RxKs)	A factor K has to be defined which indicates the rule's violation limit in units of standard deviation (SDI). This value will be used with the RxKs rule for evaluation purposes. A value from 2 to 6 in increments of 0.01 can be entered (e.g., 5.88).

2x2s	The rule is violated if two consecutive control measurements exceed the same limit, which is either $\bar{\chi}+2s$ or $\bar{\chi}-2s$. The definition of consecutive measurements for across controls shall be the current measurement of any control and the previous measurement if it is of any other control within 2 results in the past.
2of3x2s	The rule is violated if two control measurements out of the last three consecutive control measurements exceed the same limit, which is either $\bar{\chi}+2s$ or $\bar{\chi}-2s$. This rule is violated when two out of three “consecutive” control measurements are >2s from their respective target values in the same direction.

3x1s	The rule is violated if the last three consecutive control measurements exceed the same limit, which is either $\bar{\chi}+1s$ or $\bar{\chi}-1s$. This rule is violated when the last 3 “consecutive” control measurements are greater than 1s from their respective target values in the same direction.
4x1s	The rule is violated if the last four consecutive control measurements exceed the same limit, which is either $\bar{\chi}+1s$ or $\bar{\chi}-1s$. This rule is violated when or the last 4 “consecutive” control measurements are greater than 1s from their respective target values in the same direction.
Mxbar	The Mxbar rule is used to identify when a specific number of consecutive control measurements (M) fall on one side of the mean. Also, a factor M has to be defined which indicates the number of consecutive control measurements that are required to fall on one side of the mean for the Mxbar rule to be violated.
M (Mxbar)	A factor M has to be defined which indicates the number of consecutive control measurements that are required to fall on one side of the mean for the Mxbar rule to be violated.
Mmono	The Mmono rule is used to identify when a specific number of consecutive control measurements (M) are all monotonically increasing or decreasing in value.

M (Mmo-no)	A factor M has to be defined which indicates the number of consecutive control measurements that are required to be all monotonically increasing or decreasing in value for the Mmono rule to be violated.
LotExp	This rule indicates whether the system should take lot expiration into account or not, and if this has to be considered a warning or an error.

Quality Control (Rili-BÄK)

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Rili-BÄK test assignment

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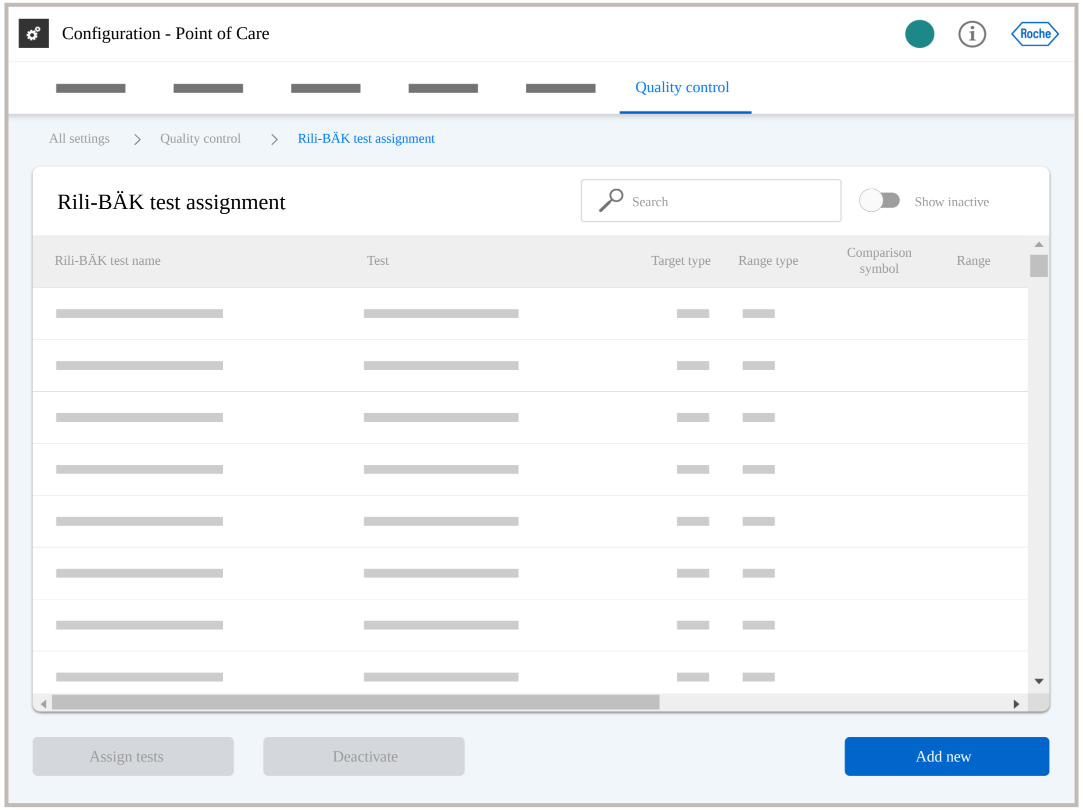
Rili-BÄK test assignment

You must first map QC (*Rili-BÄK*) tests with tests defined in the system, so that the quality control process using the materials defined for each test can have an associated range that is the basis for QC validation.



Use this component to map QC (Rili-BÄK) tests with system tests. Some tests are already defined by default, however, users may change these definitions and add new ones from this screen. Here, you can also define the parameters for Rili-BÄK quality control process.

- System tests must be previously defined.
- Ranges can be defined in absolute values (ABS) or in percentage (%).
- Each QC (Rili-BÄK) test can have more than one system test assigned.
- But each system test can only belong to one QC (Rili-BÄK) test.
- QC (Rili-BÄK) test ranges cannot be deactivated, but should be deleted instead.
- If two defined ranges overlap, the system applies the most specific one in the validation process.
- When a QC (Rili-BÄK) test mapping is activated or deactivated the mapping and all the associated ranges are activated or deactivated.

In order for the quality control process to succeed, the unit in the **Rili-BÄK test assignment** has to match either the unit defined under **Primary unit** or **Secondary unit** on the **Test definition** screen under **Configuration** > **System**.

Field definitions

Rili-BÄK test name	Name of the Rili-BÄK test. The term “Rili-BÄK test” refers to the tests for which an allowed deviation is specified in the Rili-BÄK Table B 1.

Target type	Target type as defined in the Rili-BÄK regulations. SW: **Verfahrensabhängiger Sollwert.** RMW: **Referenzmethodenwert**
Col. 3	The maximum extent to which the calculated RMSE is allowed to deviate from the target. (Called Column 3 because it refers to column 3 for tests listed in table B 1 of the Rili-BÄK regulations.
Col. 3a	Limit for effective QC range, applied to QC results for raising alarms.
Range type	Specifies whether the defined range is absolute (**ABS**) or relative (**%**). **Note:** If the target is set to zero, then the corresponding Rili-BÄK test needs to be specified as absolute (**ABS**) range type.

Range *(upper fields in Detail area)*	Define the lower limit (= from value) of the QC range. Inserted values are assigned to the **Op.** and **Range**columns in the **Rili-BÄK test assign-ment**screen. **Note:** If the QC range has no lower limit, leave these fields empty and specify the upper limit of the QC range as “< x” or “<= x” (see below). Do not set both rang-es to 0.

To range *(lower fields in Detail area)*	Define the upper limit (= to value) of the QC range. Inserted values are assigned to the **To Op.** and **To range** columns in the **Rili-BÄK test as-signment** screen. **Note:** If the QC range has no upper limit, leave these fields empty and specify the lower limit of the QC range as “> x” or “>= x” (see above). Do not set both rang-es to 0.
Lab unit	The field is cur-rently not applica-ble.

Measurement	Measurement unit of the Rili-BÄK test.

Make sure to insert the upper and lower limit of the QC range into the correct **Range** field.

Make sure that for each unit of a test the definitions always cover the entire numerical range.

Tasks

- Making a query to view the different mappings and their values.
- Creating new test mappings (Rili-BÄK) for the existing system tests.
- Modifying existing test mappings and ranges to adapt them to changing needs.
- Activating test assignments or ranges so that the system uses them in the quality control process.
- Deactivating test assignments or ranges if you do not want them to be considered in the QC observation process.

After adding, editing, or deleting range entries, you must temporarily modify the Col. 3a value to ensure that range values are correctly displayed. Once the temporarily modified Col. 3a value has been stored, change it back to the previous value.

Target values outside Rili-BÄK ranges

If the target value of a QC test lies outside the ranges defined by Rili-BÄK, the application behaves as follows:

Tests with 1 Rili-BÄK range

The application ignores the limits of the Rili-BÄK range and uses the deviation defined for the Rili-BÄK range also for target values above or below the range.

Tests with multiple Rili-BÄK ranges

The application uses the deviation defined for the lowest Rili-BÄK range for target values below all defined Rili-BÄK ranges.

The application uses the deviation defined for the highest Rili-BÄK range for target values above all defined Rili-BÄK ranges.

Example with 2 ranges

For the *analyte* Bilirubin, Rili-BÄK defines the following 2 ranges:

Allowed deviation	Range	Unit
13.0%	2 < target value ≤ 30	mg/dl
22.0%	0.1 < target value < 2	mg/dl

Bilirubin allowed target value deviations according to range

The application uses the deviations as follows (for unit mg/dl):

- 22.0% for target values < 2
- 13.0% for target values > 2

Example with 3 ranges

For the *analyte* PO_2 Rili-BÄK defines the following 3 ranges:

Allowed deviation	Range	Unit
5.5%	125 < target value ≤ 350	mmHg
7.0%	80 < target value ≤ 125	mmHg
22.0%	30 < target value < 80	mmHg

PO_2 allowed target value deviations according to range

The application uses the deviations as follows:

- 5.5% for target values > 125
- 7.0% for target values > 80 and < 1125
- 11.0% for target values < 2

Adding a new range for an existing test assignment

After adding, editing, or deleting range entries, you must temporarily modify the Col. 3a value to ensure that range values are correctly displayed. Once the temporarily modified Col. 3a value has been stored, change it back to the previous value.

1 Choose **Quality control** > **Rili-BÄK test assignment**.

2 Select the mapping to which you want to add an additional range and choose the **Add range** button.**Info:** Do not set both **Range** and **To range** values to 0.

3 Enter the required information, and then choose the **Save** button.

→ Updated information is displayed under the corresponding QC (Rili-BÄK) test.

Deleting a range from a test mapping

After adding, editing, or deleting range entries, you must temporarily modify the Col. 3a value to ensure that range values are correctly displayed. Once the temporarily modified Col. 3a value has been stored, change it back to the previous value.

1 Choose **Quality control** > **Rili-BÄK test assignment**.

2 Select the range to be deleted and choose the **Delete range** button.

→ The Detail area is enabled with the test assignments information already filled in.

3 In the confirmation dialog, choose the **OK** button.

→ The range is deleted.

QC test mapping (RB)

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QC test mapping (RB)

This screen is used to map individual tests with generic tests. These generic tests are used in instrument types that can run several types of tests (such as blood gas instruments) but use one QC test only. Thus, by mapping these tests to an already defined generic test of an instrument type, the QC result is valid for all, the generic and the mapped tests.

- Instruments must be previously defined in the database.
- Tests must be previously defined in the database. Generic tests must be created in order to map instrument tests to them.
- A test material must be previously created for the generic test. A QC material must be previously associated with the test material and the generic test.
- One test can only be assigned to an Instrument type once, but it can be assigned to different Instrument types more than once.

Field definitions

Test ID	The system test identification number assigned to a Rili-BÄK test.
Test abbreviation	The abbreviated name of the test. e.g. Glu2 for Accu-Chek® Inform II glucose tests.
Full test name	The fully displayed name for the test. e.g. Accu-Chek® Inform II glucose tests for Glu2.
Sample type	The type of sample on which the test is performed.
Instrument Type	The type of instrument on which the test can be performed

Querying for an existing mapping configuration

You check existing mapping definitions to see the tests assigned to a specific generic test.

1. Choose **Quality control** > **QC test mapping (RB)**.

   → All instrument types are displayed on the tree structure to the left.

2. Expand the instrument type and select the generic test to be checked.

   → All the tests associated to this generic test are displayed on the right.

Adding a generic test

1 Choose **Quality control** > **QC test mapping (RB)**.

→ All instrument types are displayed on the tree structure to the left

2 Choose the **Add** button.

3 Choose the instrument type from the list and choose the magnifying glass to access the **Test selection** screen.

4 In the **Test selection** screen, select the generic test and choose the **Confirm** button.

→ The **Test selection** screen is closed and the system goes back to the **QC master test** screen.

5 Choose the **Save** button.

Mapping tests to existing generic tests

It enables you to add a new test to an already existing generic test.

1 Choose **Quality control** > **QC test mapping (RB)**.

→ All instrument types are displayed on the tree structure to the left.

2 Expand the instrument type branch where you find the generic test, select the generic test to which a test is to be added, and then choose the **Add tests** button.

→ The **Test selection** screen is displayed.

3 Select the test or tests to be added from the list, and choose the **Add tests** button.

→ The **Test selection** screen is closed and the added tests appear on the grid.

Deleting a generic test

It enables you to delete a generic test that is associated with an instrument type and all the associated test mappings. This does not erase the test definition from the system.

1 Choose **Quality control** > **QC test mapping (RB)**.

→ All instrument types are displayed on the tree structure to the left.

2 Expand the instrument type branch where you find the generic test, select the generic test to be deleted, and then choose the **Delete** button.

3 On the confirmation message, choose the **Confirm** button.

→ The test is removed from the tree structure and is no longer a generic test for the current instrument type.

Deleting a test mapping

It enables you to delete a test associated to a generic test if it is no longer to be mapped with that generic test. This does not delete the test definition from the system.

1 Choose **Quality control** > **QC test mapping (RB)**.

→ All instrument types are displayed on the tree structure to the left.

2 Expand the instrument type branch where you find the generic test, select the generic test from which a test mapping will be removed, select the test to be removed and then choose the **Delete test** button.

3 On the confirmation message, choose the **Confirm** button.

→ The test mapping is deleted.

Glossary

acknowledgment

Message that is passed between communicating processes or computers to indicate a positive transmission or receipt of response.

activate

To make something active or functional.

alarms notification

Type of notification that informs about abnormal events happening on the system which might lead to failure and for which immediate user attention is required.

alphanumeric results

Result that is expressed in a combination of numbers and letters.

analyte

Constituent of a sample with a measurable property.

application

Software program that performs a group of coordinated functions, tasks, or activities.

archive

To move or copy an item to another location for storage.

auto logoff

Logoff where an inactive user is automatically logged off by the system.

auto QC

QC that is carried out by the system.

barcode

Optical machine-readable representation of data.

clipboard

Area of temporary storage for objects, data, or their references after a user carries out a cut or copy command.

connection status

Status that indicates whether the connection between a device and a host was successful.

critical range

Range of test results that indicates that a patient's life is in danger or that the patient will suffer some form of harm.

data alarms

Additional piece of information associated with a test, QC or calibration result. It is caused by the abnormal condition of measurement.

deactivate

To make something that was previously active not active or functional anymore.

firmware

Software that is embedded in a hardware device and controls how the device interacts with the operating system.

fluid pack

Consumable that contains all system reagents that are needed for measurement, calibration, and QC.

host name

Name that is assigned to a device connected to a computer network and that is used to identify the device in various forms of electronic communication.

learning management system

Software that is used for the administration, documentation, tracking, reporting, delivery, and consumption of training courses.

Levey-Jennings chart

Chart that shows QC results in relation to the expected mean and standard deviation.

observed test sequences

Workflow where a sequence of testing done by an operator is performed under supervision of an observer.

result and order matcher

Feature that is used to manage and manually match test results sent to a host for which no matching order can be found automatically.

Rili-BÄK

Quality guideline of the German Federal Medical Council that includes acceptable standard deviation for laboratory tests.

standard deviation

Positive square root of the variance.

test definition

Definition of a test in the analysis package. It contains the information about the process sequences and calculation parameters.

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